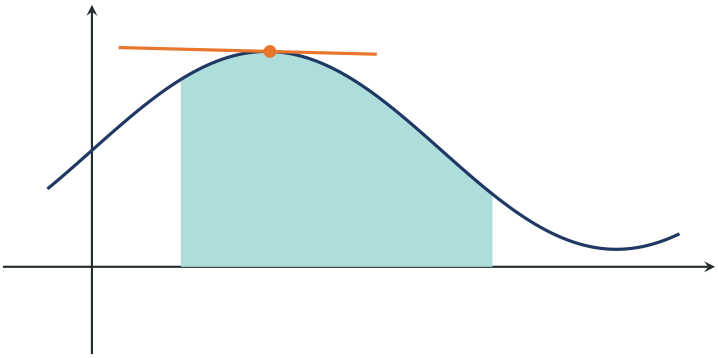
Dar Alfikr Schools – Mathematics Department

AP Calculus AB

The Complete Student Booklet

Explanations · Worked Examples · Official-Style Practice

Units 1–8, every Course and Exam Description topic



Mr Malek Thiab

Part of the Mathematics Department course materials, 2026–27

How to use this booklet

Each unit follows the same pattern

1. Read the topic. Every topic opens with an explanation of the idea, why it works and how the exam asks about it, followed by a box of definitions and formulas.
2. Copy the worked examples. Cover the solution, try each example, then compare. The examples use the language of the AP scoring guides.
3. Check yourself. Short questions close each topic; solutions are in the separate Solutions Booklet.
4. Sit the unit practice. Part A is multiple choice without a calculator, Part B is multiple choice with a graphing calculator, and Part C is two free-response questions worth 9 points each.

The AP Calculus AB exam

Section	Format	Questions	Time	Weight
I A	Multiple choice, no calculator	30	60 min	33.3%
I B	Multiple choice, graphing calculator	15	45 min	16.7%
II A	Free response, graphing calculator	2	30 min	16.7%
II B	Free response, no calculator	4	60 min	33.3%

Each free-response question is worth 9 points. Scores are reported on a 1–5 scale.

Justification language that earns points

Scoring guides reward a *reason* tied to a named theorem or sign condition: "$f'(c) = 0$ and f' changes from positive to negative at c, so f has a relative maximum at c"; "f is continuous on $[a, b]$ and differentiable on (a, b), so by the Mean Value Theorem…". On calculator parts, write the integral or equation you entered and give decimals to three places.

Contents

Unit	Title	Exam weighting
1	Limits and Continuity	10–12%
2	Differentiation: Definition and Fundamental Properties	10–12%
3	Differentiation: Composite, Implicit, and Inverse Functions	9–13%
4	Contextual Applications of Differentiation	10–15%
5	Analytical Applications of Differentiation	15–18%
6	Integration and Accumulation of Change	17–20%
7	Differential Equations	6–12%
8	Applications of Integration	10–15%

Topics that appear only on the AP Calculus BC exam are not included.

AP Calculus AB – Unit 1 Exam weighting: 10–12% of the AB exam

Limits and Continuity

In this unit

Calculus is built on one idea that algebra never needed: the *limit*. A limit describes what a function is doing as its input *approaches* a value, which is different from what the function does *at* that value. Limits let us define instantaneous rates of change (derivatives, Unit 2) and accumulated change (integrals, Unit 6), and they let us say precisely what it means for a graph to have no breaks (continuity). Unit 1 has two halves: first you learn to find limits from graphs, tables and algebra; then you use limits to classify discontinuities, to describe asymptotes, and to prove existence statements with the Intermediate Value Theorem.

Topics.

1.1 Introducing calculus: can change occur at an instant?
1.2 Defining limits and using limit notation
1.3 Estimating limit values from graphs
1.4 Estimating limit values from tables
1.5 Determining limits using algebraic properties of limits
1.6 Determining limits using algebraic manipulation
1.7 Selecting procedures for determining limits
1.8 Determining limits using the Squeeze Theorem
1.9 Connecting multiple representations of limits
1.10 Exploring types of discontinuities
1.11 Defining continuity at a point
1.12 Confirming continuity over an interval
1.13 Removing discontinuities
1.14 Connecting infinite limits and vertical asymptotes
1.15 Connecting limits at infinity and horizontal asymptotes
1.16 Working with the Intermediate Value Theorem (IVT)

Big ideas. (1) A limit describes *approach*, not arrival: $\lim_{x\to a} f(x)$ ignores the value $f(a)$. (2) A two-sided limit exists only when both one-sided limits exist and agree. (3) Direct substitution is the first move; when it gives $\frac{0}{0}$ you must rewrite the expression, and when it gives $\frac{k}{0}$ you are looking at an asymptote. (4) f is continuous at a exactly when $\lim_{x\to a} f(x) = f(a)$. (5) Continuity on a closed interval is the hypothesis of both the Intermediate Value Theorem and (in Unit 5) the Extreme Value Theorem.

Topic 1.1 Introducing Calculus: Can Change Occur at an Instant?

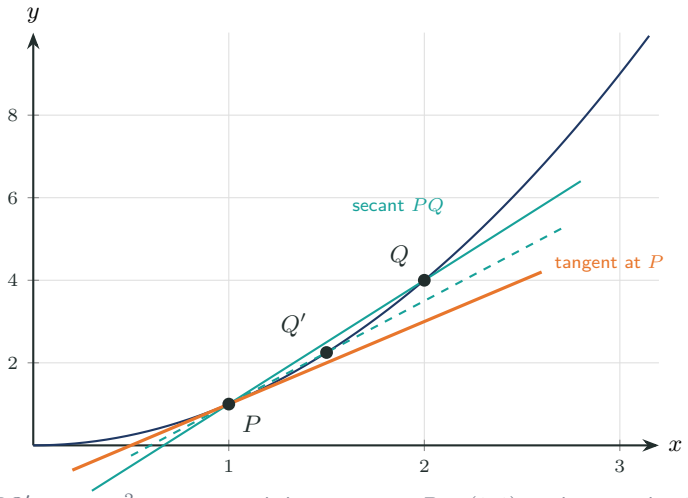
Average change versus change at an instant

Suppose a car travels 120 kilometres in 2 hours. Its *average* speed is 60 km/h, but the speedometer never has to read 60: it may read 0 at a red light and 100 on the highway. The speedometer reports a different quantity, the speed *at one instant*. Algebra can compute the first quantity with a single division, $\dfrac{\text{change in position}}{\text{change in time}}$. It cannot compute the second, because at a single instant the elapsed time is 0 and the change in position is 0, which produces the meaningless quotient $\frac{0}{0}$.

Calculus resolves this by never dividing by zero. Instead, it computes the average rate of change over shorter and shorter intervals and asks what number those averages are *approaching*. Geometrically, the average rate of change of f between $x = a$ and $x = b$ is the slope of the *secant line* through $(a, f(a))$ and $(b, f(b))$. As b moves toward a, the secant line rotates and settles toward the *tangent line* at $x = a$. The slope of that tangent line is the instantaneous rate of change. The word "approaching" in that sentence is exactly what a limit makes precise, which is why limits come first in the course.

Key ideas: average and instantaneous rate

- **Average rate of change** of f on $[a, b]$: $\dfrac{f(b) - f(a)}{b - a}$, the slope of the secant line.
- **Instantaneous rate of change** at $x = a$: the value the average rates approach as the interval shrinks to a,



Secants PQ and PQ' on $y = x^2$ rotate toward the tangent at $P = (1, 1)$ as the second point slides toward P.

$\lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$, the slope of the tangent line (made precise in Unit 2).

- Units of a rate: (units of output) per (unit of input), for example metres per second.
- A limit is the tool that lets us talk about "the value approached" without dividing by 0.

Example 1 — Average velocities that close in on an instantaneous velocity

A ball moves along a line so that its position is $s(t) = t^2$ metres at time t seconds. Estimate the velocity at $t = 1$.

Step 1. The average velocity on $[1, 1+h]$ is $\dfrac{s(1+h)-s(1)}{h} = \dfrac{(1+h)^2-1}{h} = \dfrac{2h+h^2}{h} = 2+h$ for $h \neq 0$.

Step 2. Evaluate for shrinking intervals.

h	1	0.5	0.1	0.01	0.001
Average velocity on $[1, 1+h]$ (m/s)	3	2.5	2.1	2.01	2.001

Step 3. The averages approach 2, so the instantaneous velocity at $t = 1$ is 2 m/s. Note that $h = 0$ is never used; the value 2 is where the averages are *heading*.

Example 2 — Estimating a rate of change from a table

The distance $D(t)$ in kilometres of a train from its starting station is recorded t minutes after departure.

t (min)	0	2	4	6	8	10
$D(t)$ (km)	0	0.4	1.6	3.6	6.4	10

(a) Find the average speed from $t = 2$ to $t = 6$. (b) Estimate the speed at $t = 6$.

(a) $\dfrac{D(6)-D(2)}{6-2} = \dfrac{3.6-0.4}{4} = 0.8$ km/min.

(b) The speed at $t = 6$ is not a single secant slope, but the secants on either side bracket it. On $[4, 6]$: $\dfrac{3.6-1.6}{2} = 1.0$ km/min. On $[6, 8]$: $\dfrac{6.4-3.6}{2} = 1.4$ km/min. The speed at $t = 6$ lies between these, and their mean, 1.2 km/min, is a reasonable estimate (equivalently, the secant slope on $[4, 8]$ is $\dfrac{6.4-1.6}{4} = 1.2$).

Common errors

- **Using one point to get a slope.** A slope needs two points. "The slope at $x = 1$" is the *limit* of secant slopes, not $\frac{f(1)-f(1)}{1-1}$.
- **Reporting a secant slope as an instantaneous rate.** Say "average rate on $[a, b]$" for $\frac{f(b)-f(a)}{b-a}$; reserve "at $t = a$" for the limiting value.

- **Dropping units.** Both kinds of rate carry units of output per input.

Check yourself

1. A position function is $s(t) = 3t^2 + t$ metres, t in seconds. Find the average velocity on $[1, 3]$, with units.
2. For $s(t) = t^2 + t$ find the average velocity on $[2, 2+h]$ in terms of h, evaluate it for $h = 0.1$ and $h = 0.01$, and state what value the averages approach.

Topic 1.2 Defining Limits and Using Limit Notation

What a limit says

The statement $\lim_{x \to a} f(x) = L$ is read "the limit of $f(x)$ as x approaches a equals L." It means that $f(x)$ can be made as close to L as we like by taking x close enough to a (on both sides) but not equal to a. Two features of that sentence matter on every exam. First, the value $f(a)$ plays no role: f may be undefined at a, or defined with a completely different value, and the limit can still exist. Second, the number L must be a single real number that $f(x)$ approaches from *both* directions.

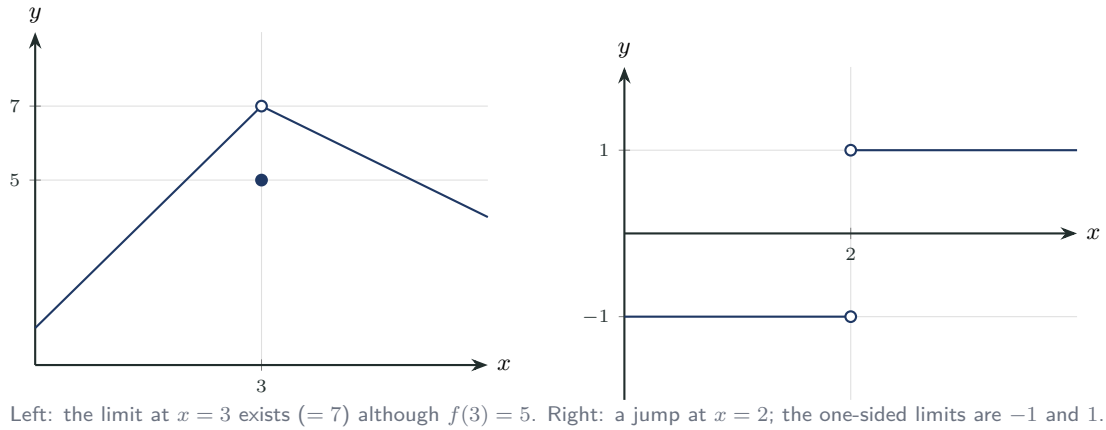
One-sided limits and when the limit exists

Approaching a from the left (values $x < a$) gives the *left-hand limit* $\lim_{x \to a^-} f(x)$; approaching from the right gives the *right-hand limit* $\lim_{x \to a^+} f(x)$. The two-sided limit exists if and only if both one-sided limits exist *and are equal*. If they differ, or if either fails to exist, we write $\lim_{x \to a} f(x)$ *does not exist* (DNE).

Sometimes the function values grow without bound. If $f(x)$ increases without bound as $x \to a$ we write $\lim_{x \to a} f(x) = \infty$ (similarly $-\infty$). Strictly, an infinite limit is a way of describing *why* the limit fails to exist, but it is useful information, and the AP exam accepts and expects the notation $= \infty$ or $= -\infty$ when that is what happens. A third way for a limit to fail is wild oscillation, as for $\sin \frac{1}{x}$ near 0 (Topic 1.3).

Key ideas: limit notation

- $\lim_{x \to a} f(x) = L$: $f(x)$ approaches L as x approaches a from either side, $x \neq a$.
- $\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = L$ **and** $\lim_{x \to a^+} f(x) = L$.
- The limit at a does *not* depend on $f(a)$.
- DNE if the one-sided limits differ, if f oscillates, or if f grows without bound (then $\pm\infty$ may be written).



Left: the limit at $x = 3$ exists ($= 7$) although $f(3) = 5$. Right: a jump at $x = 2$; the one-sided limits are -1 and 1.

Example 1 — A limit that differs from the function value

Let $f(x) = \begin{cases} 2x + 1, & x < 3 \\ 5, & x = 3 \\ 10 - x, & x > 3 \end{cases}$ (left graph above). Find $\lim_{x \to 3^-} f(x)$, $\lim_{x \to 3^+} f(x)$, $\lim_{x \to 3} f(x)$ and $f(3)$.

Step 1. For $x < 3$, $f(x) = 2x + 1$, so $\lim_{x \to 3^-} f(x) = 2(3) + 1 = 7$.

Step 2. For $x > 3$, $f(x) = 10 - x$, so $\lim_{x \to 3^+} f(x) = 10 - 3 = 7$.
Step 3. The one-sided limits agree, so $\lim_{x \to 3} f(x) = 7$. **Step 4.** $f(3) = 5$, which is different and irrelevant to the limit.

Example 2 — A limit that does not exist

Let $g(x) = \dfrac{|x-2|}{x-2}$ for $x \neq 2$ (right graph above). For $x > 2$, $|x-2| = x - 2$ so $g(x) = 1$; for $x < 2$, $|x-2| = -(x-2)$ so $g(x) = -1$. Thus $\lim_{x \to 2^-} g(x) = -1 \neq 1 = \lim_{x \to 2^+} g(x)$, and $\lim_{x \to 2} g(x)$ does not exist.

Example 3 — Infinite limits

Consider $\dfrac{1}{x^2}$ and $\dfrac{1}{x}$ near $x = 0$. Squares are positive, so $\dfrac{1}{x^2}$ is a large *positive* number on both sides: $\lim_{x \to 0} \dfrac{1}{x^2} = \infty$. But $\dfrac{1}{x}$ is large positive for small $x > 0$ and large negative for small $x < 0$: $\lim_{x \to 0^+} \dfrac{1}{x} = \infty$, $\lim_{x \to 0^-} \dfrac{1}{x} = -\infty$, so the two-sided limit does not exist, and the symbol ∞ may *not* be used for it.

Common errors

- **Evaluating instead of approaching.** In Example 1 the answer is 7, not $f(3) = 5$.
- **Averaging one-sided limits.** If the left limit is -1 and the right is 1, the limit is not 0; it does not exist.
- **Saying "the limit is ∞" for $1/x$ at 0.** The two sides disagree.

Check yourself

1. Let $f(x) = \begin{cases} x^2 - 1, & x < 2 \\ 6, & x = 2 \\ 7 - x, & x > 2. \end{cases}$ Find the left limit, the right limit and the limit of f at $x = 2$. Does the limit exist?
2. Evaluate $\lim_{x \to 0^-} \dfrac{1}{x}$ and $\lim_{x \to 0} \dfrac{1}{x^2}$.
3. If $\lim_{x \to 4^-} g(x) = 7$, $\lim_{x \to 4^+} g(x) = 7$ and $g(4) = 2$, what is $\lim_{x \to 4} g(x)$?

Topic 1.3 Estimating Limit Values from Graphs

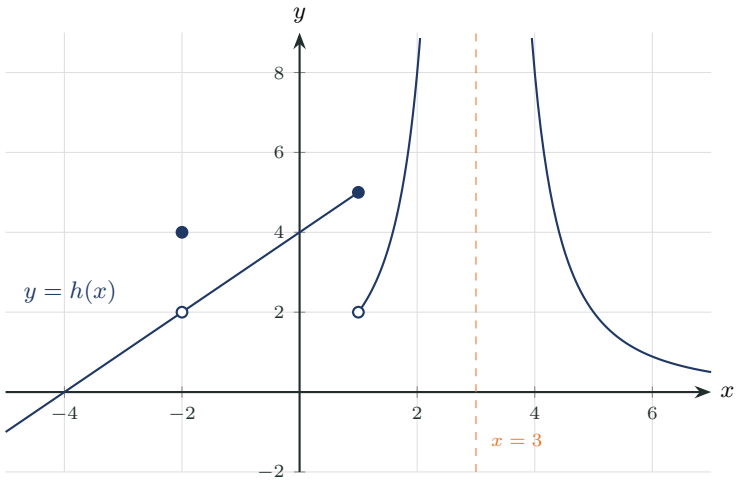
Following the graph with your eyes

A graph is the most direct picture of a limit. To find $\lim_{x \to a} f(x)$, place your finger on the curve far to the left of $x = a$ and slide it toward $x = a$; note the height y it is heading toward. Do the same from the right. If both fingers are heading to the same height L, then $\lim_{x \to a} f(x) = L$. If they head to different heights, the limit does not exist; if either runs off the top or bottom of the graph, the limit is infinite (or does not exist).

Three details need care. An *open circle* marks a point that is *not* on the graph, but the curve is still heading there, so the open circle gives the limit. A *filled dot* at a different height marks the actual value $f(a)$, which has nothing to do with the limit. And an arrow or a dashed asymptote tells you what happens beyond the window of the picture, so a limit as $x \to \infty$ is read from the right-hand end of the graph.

Key ideas: limits from a graph

- Compare the heights the curve approaches from the left and from the right; the limit exists only if they match.
- Open circle = missing point (still counts as where the curve is heading). Filled dot = the function value, which may differ from the limit.
- Curve rising or falling without bound at a vertical asymptote: the limit is ∞ or $-\infty$ (or DNE if the two sides differ).
- End behaviour: $\lim_{x \to \infty} f(x)$ is the height the right-hand end settles to.



The graph of h used in Examples 1 and 2 (it is used again in Topics 1.10, 1.14 and 1.15). The dashed line is $x = 3$.

Example 1 — Reading limits from the graph of h

Use the graph of h above.

Quantity	Value and reason
$\lim_{x\to -2} h(x)$	2: both sides head to the open circle at $(-2, 2)$
$h(-2)$	4: the filled dot
$\lim_{x\to 1^-} h(x)$, $\lim_{x\to 1^+} h(x)$	5 and 2
$\lim_{x\to 1} h(x)$	does not exist (one-sided limits differ)
$\lim_{x\to 3} h(x)$	∞: both branches rise without bound at $x = 3$
$\lim_{x\to \infty} h(x)$, $\lim_{x\to -\infty} h(x)$	0 and $-\infty$

Example 2 — A limit of a composite function from a graph

Find $\lim_{x\to -2} h\big(h(x)\big)$ and $\lim_{x\to 1^-} h\big(h(x)\big)$.

Step 1 – inner function. As $x \to -2$, $h(x) \to 2$ (never equal to 2 for $x \neq -2$, which does not matter). So we need the behaviour of $h(y)$ as $y \to 2$.

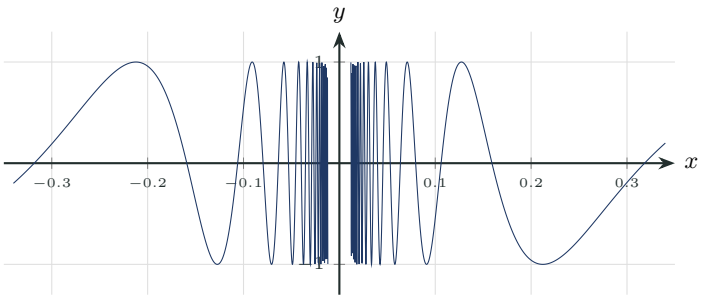
Step 2 – outer function. Near $y = 2$ the graph of h is the right-hand branch $h(y) = \dfrac{8}{(y-3)^2}$, which is continuous there, so $h(y) \to \dfrac{8}{(2-3)^2} = 8$. Hence $\lim_{x\to -2} h(h(x)) = 8$.

Step 3. As $x \to 1^-$, $h(x) \to 5$, and $h(y) \to \dfrac{8}{(5-3)^2} = 2$ as $y \to 5$. Hence $\lim_{x\to 1^-} h(h(x)) = 2$.

Notice that the inner function is replaced by its *limit* (2), not by its value $h(-2) = 4$. Here $h(4) = 8$ happens to give the same number, but for other functions it would not.

Example 3 — A limit that fails by oscillation

The graph of $f(x) = \sin\dfrac{1}{x}$ near $x = 0$ crosses every height between -1 and 1 infinitely often.



At $x = \dfrac{1}{n\pi}$ the function equals $\sin(n\pi) = 0$, while at $x = \dfrac{1}{\pi/2 + 2n\pi}$ it equals 1. Both kinds of x approach 0, so $f(x)$ does not settle on one value and $\lim\limits_{x\to 0} \sin\dfrac{1}{x}$ does not exist. (There is no vertical asymptote and the values stay bounded; the limit fails for a different reason.)

Common errors

- **Reading the dot instead of the curve.** At $x = -2$ the answer is 2 (open circle), not 4 (filled dot).
- **Using $h(a)$ in a composite limit.** Use the limit of the inner function, then evaluate the outer behaviour there.
- **Calling every unbounded limit "DNE" without direction.** State ∞ or $-\infty$ when both sides agree.

Check yourself

1. Using the graph of h, find $\lim\limits_{x\to 1^+} h(x) + \lim\limits_{x\to -2} h(x)$.
2. Find $\lim\limits_{x\to 0} h(x)$ and $\lim\limits_{x\to\infty} h(x)$.
3. Is $\lim\limits_{x\to 3} h(x)$ equal to ∞, to $-\infty$, or does it fail to be infinite? Explain from the graph.

Topic 1.4 Estimating Limit Values from Tables

Numerical evidence for a limit

When a function is given by a table (or when a calculator generates one), you estimate a limit by looking at the outputs for inputs that get *closer and closer* to a from *both sides*. If the outputs from the left and from the right both settle toward the same number, that number is the estimate for the limit. If the left-hand outputs settle toward one number and the right-hand outputs toward another, the limit does not exist. If the outputs grow without bound in size, the limit is infinite or does not exist.

Always examine both sides. A one-sided table cannot detect a jump. The sequence of inputs should be *systematic*, for example $a \pm 0.1$, $a \pm 0.01$, $a \pm 0.001$, so that you can see the trend rather than guess it. Notice also that the table need not contain the input $x = a$ at all; if it does, the output there is ignored when estimating the limit.

A table gives *evidence*, not proof. It can be fooled by functions that oscillate faster than the table samples them (see the error box below), so on the exam it is used for estimates and for deciding between candidate answers, while algebraic limits (Topics 1.5–1.7) give exact values.

Key ideas: limits from tables

- Use inputs on *both* sides of a that approach a.
- Trend to one number L from both sides $\Rightarrow \lim\limits_{x\to a} f(x) \approx L$.
- Left trend $\neq$ right trend $\Rightarrow$ the limit does not exist.
- Outputs growing in size $\Rightarrow \pm\infty$ or DNE; check the signs on each side.
- A table is an estimate; match it with algebra when you can.

Example 1 — A calculator table

Estimate $\lim_{x\to 0}\dfrac{\sqrt{x+4}-2}{x}$.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$\dfrac{\sqrt{x+4}-2}{x}$	0.25158	0.25016	0.25002	0.24998	0.24984	0.24846

The left-hand values decrease toward 0.25 and the right-hand values increase toward 0.25. The limit is approximately $0.25 = \dfrac{1}{4}$. (The function is undefined at $x = 0$; that does not matter.)

Example 2 — Table of values for a function f

Selected values of f are given.

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	11.41	11.9401	11.994001	12.006001	12.0601	12.61

Estimate $\lim_{x\to 2} f(x)$. From the left the values increase toward 12; from the right they decrease toward 12. So $\lim_{x\to 2} f(x) \approx 12$.

Example 3 — Left and right trends disagree

x	2.9	2.99	2.999	3.001	3.01	3.1
$g(x)$	4.8	4.98	4.998	7.002	7.02	7.2

As $x \to 3^-$, $g(x) \to 5$; as $x \to 3^+$, $g(x) \to 7$. The one-sided limits are different, so $\lim_{x\to 3} g(x)$ does not exist.

Common errors

- **One-sided table.** Values only from the right say nothing about the left. Always test both sides.
- **Stopping too early.** $0.9,\ 0.99$ may not reveal the trend; add 0.999.
- **Table sampling can hide oscillation.** For $f(x) = \sin\dfrac{\pi}{x}$, the inputs $x = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ all give $f(x) = 0$, which falsely suggests a limit of 0 at $x = 0$. The inputs $x = \dfrac{2}{4n+1}$ give 1. The limit does not exist.

Check yourself

1. A table gives $f(4.9) = 9.9$, $f(4.99) = 9.99$, $f(4.999) = 9.999$, $f(5.001) = 10.001$, $f(5.01) = 10.01$, $f(5.1) = 10.1$. Estimate $\lim_{x\to 5} f(x)$.
2. A table gives $g(0.9) = 2.9$, $g(0.99) = 2.99$, $g(0.999) = 2.999$ and $g(1.001) = -2.999$, $g(1.01) = -2.99$, $g(1.1) = -2.9$. What can you say about $\lim_{x\to 1} g(x)$?
3. Explain why evaluating $f(x) = \sin\dfrac{\pi}{x}$ at $x = \dfrac{1}{n}$ $(n = 1, 2, 3, \ldots)$ is misleading.

Topic 1.5 Determining Limits Using Algebraic Properties of Limits

Limit laws

Graphs and tables estimate; algebra *determines*. The foundation is a set of limit laws that let you break a complicated limit into simple ones. Two simple limits come first: $\lim_{x\to a} c = c$ for a constant c, and $\lim_{x\to a} x = a$. Everything else is built by combining them. If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ both exist (as real numbers), then the limit of a sum, difference, constant multiple, product, power or root can be computed from L and M, and so can the limit of a quotient *provided* $M \neq 0$.

An important consequence is **direct substitution**: if f is a polynomial, a rational function (with non-zero denominator at a), a root, an exponential, a logarithm, or a trigonometric function, and a is in its domain, then $\lim_{x\to a} f(x) = f(a)$.

That is the first thing to try. (The reason is continuity, studied from Topic 1.11.) If direct substitution produces a real number, you are finished.

Key ideas: limit laws

If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ exist, then

- $\lim_{x\to a}\big[f(x) \pm g(x)\big] = L \pm M$ and $\lim_{x\to a}\big[k\,f(x)\big] = kL$.
- $\lim_{x\to a}\big[f(x)\,g(x)\big] = LM$ and $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{L}{M}$ if $M \neq 0$.
- $\lim_{x\to a}\big[f(x)\big]^n = L^n$ and $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L}$ (with $L \geq 0$ if n is even).
- Direct substitution: for polynomial, rational (denominator $\neq 0$), root, exponential, logarithmic and trigonometric functions, $\lim_{x\to a} f(x) = f(a)$ when a is in the domain.

Example 1 — Combining given limits

Suppose $\lim_{x\to 2} f(x) = 3$ and $\lim_{x\to 2} g(x) = -4$. Find (a) $\lim_{x\to 2}\big[f(x) + 2g(x)\big]$, (b) $\lim_{x\to 2}\big[f(x)g(x)\big]$, (c) $\lim_{x\to 2}\dfrac{f(x)}{g(x)}$, (d) $\lim_{x\to 2}\Big(\big[f(x)\big]^2 - g(x)\Big)$.

(a) $3 + 2(-4) = -5$. **(b)** $3 \cdot (-4) = -12$. **(c)** $M = -4 \neq 0$, so the limit is $\dfrac{3}{-4} = -\dfrac{3}{4}$. **(d)** $3^2 - (-4) = 13$.

Example 2 — Direct substitution

Find $\lim_{x\to 2}\dfrac{x^2+1}{x^3-3}$ and $\lim_{x\to 3}\dfrac{\sqrt{x^2+7}}{x-1}$.

Step 1. For the first, the denominator at $x = 2$ is $8 - 3 = 5 \neq 0$, so substitute: $\dfrac{4+1}{8-3} = \dfrac{5}{5} = 1$.

Step 2. For the second, $\sqrt{9+7} = 4$ and $3 - 1 = 2 \neq 0$, so the limit is $\dfrac{4}{2} = 2$.

Example 3 — When the laws do *not* apply

Let $\lim_{x\to 2} f(x) = 3$ and $\lim_{x\to 2} g(x) = 3$. Can we find $\lim_{x\to 2}\dfrac{1}{f(x) - g(x)}$?

The limit of the denominator is $3 - 3 = 0$, so the quotient law does not apply, and the limit cannot be determined from the given information. For instance, if $f(x) = 3 + (x-2)$ and $g(x) = 3$, then $\dfrac{1}{f-g} = \dfrac{1}{x-2}$ has no limit at 2, whereas $\dfrac{x-2}{f(x)-g(x)}$ would tend to 1.

Common errors

- **Using the quotient law when $M = 0$.** $\frac{L}{0}$ is not a number; the form $\frac{0}{0}$ is *indeterminate*, not "0" and not "DNE" (Topic 1.6).
- **Splitting limits that do not exist.** The product/sum laws require *each* limit to exist as a real number.
- **Assuming $\lim f(g(x)) = f(\lim g(x))$ always.** That is valid only when f is continuous at the inner limit (Topic 1.3, Example 2).

Check yourself

1. If $\lim_{x\to 3} f(x) = 4$ and $\lim_{x\to 3} g(x) = -2$, find $\lim_{x\to 3}\big[2f(x) + 3g(x)\big]$, $\lim_{x\to 3}\dfrac{f(x)}{g(x)}$ and $\lim_{x\to 3}\big[g(x)\big]^2$.
2. Evaluate $\lim_{x\to -1}\dfrac{x^3+2x}{x^2+4}$.
3. Evaluate $\lim_{x\to 4}\dfrac{\sqrt{x^2-7}}{x-2}$.

Topic 1.6 Determining Limits Using Algebraic Manipulation

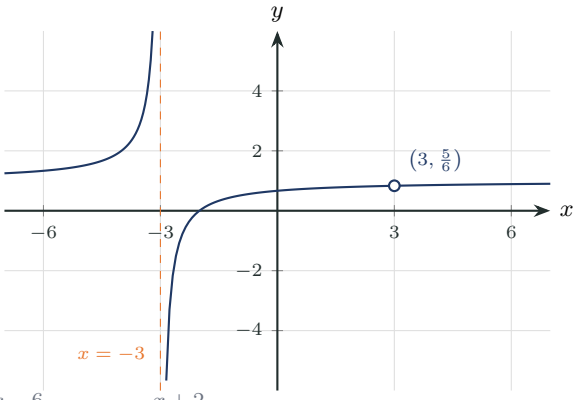
When substitution gives $\frac{0}{0}$

Direct substitution sometimes produces $\frac{0}{0}$. This form is called *indeterminate*: it tells you nothing yet. The limit may be a number, $\pm\infty$, or may not exist; the expression must be rewritten. The key observation is that a limit ignores the value at $x = a$, so you may replace $f(x)$ by any *equivalent* expression that is valid for all $x \neq a$ near a, and then substitute. The goal is to cancel the factor that makes both numerator and denominator vanish.

There are four standard moves. (1) **Factor and cancel**: if a polynomial vanishes at a, then $(x - a)$ is a factor. (2) **Multiply by the conjugate** when a square root is involved: $(\sqrt{A} - B)(\sqrt{A} + B) = A - B^2$ removes the root and often creates the factor to cancel. (3) **Combine fractions** over a common denominator in a complex fraction. (4) **Use a trigonometric limit or identity**, for example $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$ (proved in Topic 1.8).

Key ideas: algebraic manipulation

- $\frac{0}{0}$ means "rewrite", not "zero" and not "DNE".
- Factor and cancel the common factor $(x - a)$; then substitute. Write "=" only between equal expressions for $x \neq a$.
- Conjugate: multiply numerator and denominator by the conjugate of the root expression.
- Complex fraction: get a common denominator in the numerator, then simplify.
- Trig: $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$, $\lim_{x\to 0} \dfrac{1 - \cos x}{x} = 0$; adjust constants (for example $\dfrac{\sin kx}{x} = k \cdot \dfrac{\sin kx}{kx} \to k$).



Graph of $f(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$: it equals $\dfrac{x+2}{x+3}$ except for a hole at $\left(3, \frac{5}{6}\right)$ and a vertical asymptote at $x = -3$.

Example 1 — Factor and cancel

Find $\lim_{x\to 3} \dfrac{x^2 - x - 6}{x^2 - 9}$.

Step 1. Substitute: $\dfrac{9 - 3 - 6}{9 - 9} = \dfrac{0}{0}$, indeterminate.

Step 2. Factor: $\dfrac{(x-3)(x+2)}{(x-3)(x+3)} = \dfrac{x+2}{x+3}$ for $x \neq 3$.

Step 3. $\lim_{x\to 3} \dfrac{x+2}{x+3} = \dfrac{5}{6}$. This is the height of the open circle in the figure.

Example 2 — Conjugate

Find $\lim_{x\to 0} \dfrac{\sqrt{x+9} - 3}{x}$.

Step 1. Substitution gives $\frac{0}{0}$.

Step 2. Multiply by $\dfrac{\sqrt{x+9}+3}{\sqrt{x+9}+3}$: $\dfrac{(x+9)-9}{x\left(\sqrt{x+9}+3\right)} = \dfrac{x}{x\left(\sqrt{x+9}+3\right)} = \dfrac{1}{\sqrt{x+9}+3}$ for $x \neq 0$.

Step 3. The limit is $\dfrac{1}{\sqrt{9}+3} = \dfrac{1}{6}$.

Example 3 — Complex fraction

Find $\lim_{x\to 0} \dfrac{\dfrac{1}{x+2}-\dfrac{1}{2}}{x}$.

Step 1. Common denominator in the numerator: $\dfrac{1}{x+2}-\dfrac{1}{2}=\dfrac{2-(x+2)}{2(x+2)}=\dfrac{-x}{2(x+2)}$.

Step 2. Divide by x: $\dfrac{-x}{2x(x+2)}=\dfrac{-1}{2(x+2)}$ for $x\neq 0$.

Step 3. The limit is $\dfrac{-1}{2\cdot 2}=-\dfrac{1}{4}$.

Example 4 — Trigonometric

Find $\lim_{x\to 0}\dfrac{\sin 5x}{3x}$ and $\lim_{x\to 0}\dfrac{\tan x}{x}$.

Step 1. $\dfrac{\sin 5x}{3x}=\dfrac{5}{3}\cdot\dfrac{\sin 5x}{5x}$, and as $x\to 0$, $5x\to 0$, so $\dfrac{\sin 5x}{5x}\to 1$. The limit is $\dfrac{5}{3}$.

Step 2. $\dfrac{\tan x}{x}=\dfrac{\sin x}{x}\cdot\dfrac{1}{\cos x}\to 1\cdot\dfrac{1}{\cos 0}=1$.

AP Exam Tip: showing work for limits

Write the substitution that gives $\frac{0}{0}$ (or state it), then the algebra, then the answer. Never write $\frac{0}{0}=0$ or $\frac{0}{0}=1$. Scoring guides give the answer point only when the work shows the algebraic step that removes the indeterminacy.

Common errors

- **"$\frac{0}{0}=0$" or "$\frac{0}{0}$ is undefined so the limit does not exist."** Both are wrong; rewrite the expression.
- **Cancelling terms instead of factors.** In $\dfrac{x^2-9}{x-3}$ you cannot cancel the x's; factor first.
- **Forgetting to multiply the whole numerator and denominator by the conjugate.**

Check yourself

1. Evaluate $\lim_{x\to 5}\dfrac{x^2-25}{x^2-3x-10}$.
2. Evaluate $\lim_{x\to 0}\dfrac{\sqrt{x+25}-5}{x}$.
3. Evaluate $\lim_{x\to 0}\dfrac{\tan 4x}{8x}$.

Topic 1.7 Selecting Procedures for Determining Limits

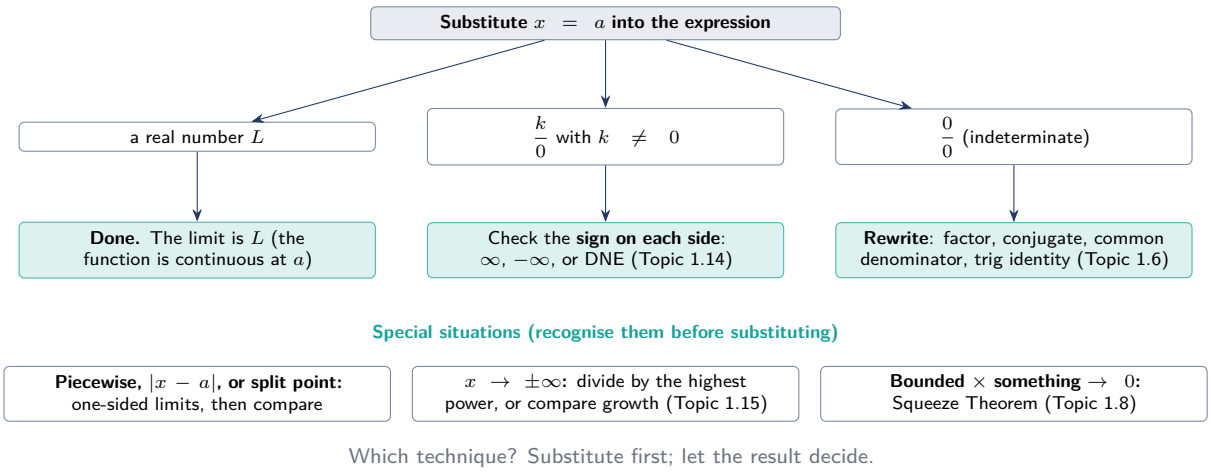
A decision process

By now you have four ways to attack a limit: read it from a graph or table, substitute, rewrite algebraically, or squeeze. The AP exam does not tell you which; part of the skill is choosing. A reliable habit is to *always substitute first*, because the result classifies the problem. A real number is the answer. A non-zero number over 0 means an infinite limit or a limit that fails because the two sides disagree, so you check the sign on each side. $\frac{0}{0}$ means the expression must be rewritten.

Some situations announce themselves before you substitute. A piecewise definition, an absolute value, or a point where the formula changes forces you to compute the two one-sided limits separately. A limit as $x\to\pm\infty$ calls for comparing the highest powers (Topic 1.15). A bounded oscillating factor multiplied by something that tends to 0 calls for the Squeeze Theorem (Topic 1.8). The chart summarizes the process.

Key ideas: choosing a procedure

- Substitute first. Real number: done. $\frac{k}{0}$: sign analysis. $\frac{0}{0}$: rewrite.
- Piecewise or absolute-value expressions need one-sided limits.
- Limits at infinity: highest powers. Bounded oscillation: Squeeze.



Which technique? Substitute first; let the result decide.

- Graphs and tables are for estimates; algebra is for exact values.

Example 1 — Two ways to remove the indeterminacy

Find $\lim\limits_{x\to 4} \dfrac{x-4}{\sqrt{x}-2}$.

Step 1. Substitution gives $\frac{0}{0}$.

Step 2 (conjugate). $\dfrac{(x-4)(\sqrt{x}+2)}{(\sqrt{x}-2)(\sqrt{x}+2)} = \dfrac{(x-4)(\sqrt{x}+2)}{x-4} = \sqrt{x}+2$ for $x \neq 4$.

Step 3. The limit is $\sqrt{4}+2 = 4$. (Alternatively, factor $x-4 = (\sqrt{x}-2)(\sqrt{x}+2)$ and cancel.)

Example 2 — Absolute value forces one-sided limits

Find $\lim\limits_{x\to 2} \dfrac{x^2-4}{|x-2|}$.

Step 1. Substitution gives $\frac{0}{0}$ and an absolute value, so split.

Step 2. For $x > 2$, $|x-2| = x-2$ and $\dfrac{(x-2)(x+2)}{x-2} = x+2 \to 4$.

Step 3. For $x < 2$, $|x-2| = -(x-2)$ and $\dfrac{(x-2)(x+2)}{-(x-2)} = -(x+2) \to -4$.

Step 4. The one-sided limits are 4 and -4, so the limit does not exist.

Example 3 — Finding a constant so that a limit exists

Find the value of k for which $\lim\limits_{x\to 1} \dfrac{x^2+kx-2}{x-1}$ exists, and evaluate the limit.

Step 1. The denominator tends to 0. For the limit to be a finite number the numerator must also tend to 0: $1+k-2=0$, so $k=1$.

Step 2. With $k=1$: $\dfrac{x^2+x-2}{x-1} = \dfrac{(x-1)(x+2)}{x-1} = x+2$ for $x \neq 1$.

Step 3. The limit is $1+2=3$.

Common errors

- **Cancelling before checking.** Substitute first so you know which technique is needed.
- **Ignoring the absolute value.** $\dfrac{|x-2|}{x-2}$ is -1 on the left and $+1$ on the right.
- **Concluding DNE from $\frac{k}{0}$ without signs.** If both sides tend to $+\infty$ the limit is $+\infty$.

Check yourself

1. Evaluate $\lim\limits_{x\to 2}\dfrac{x-2}{\sqrt{x+7}-3}$.
2. Evaluate $\lim\limits_{x\to -3^+}\dfrac{x+1}{x+3}$.
3. Find k such that $\lim\limits_{x\to 2}\dfrac{x^2+kx-10}{x-2}$ exists, and find the limit.

Topic 1.8 Determining Limits Using the Squeeze Theorem

Trapping a function between two others

Some limits resist every algebraic manipulation. A classic example is $x^2\sin\dfrac{1}{x}$ as $x\to 0$: there is no cancellation to perform, and $\sin\frac{1}{x}$ has no limit of its own. Yet the product is easy to understand: $\sin\frac{1}{x}$ always lies between -1 and 1, so $x^2\sin\frac{1}{x}$ always lies between $-x^2$ and x^2, and both of those bounds shrink to 0. A function trapped between two functions that approach the same number must approach it as well. That is the **Squeeze Theorem** (also called the Sandwich Theorem).

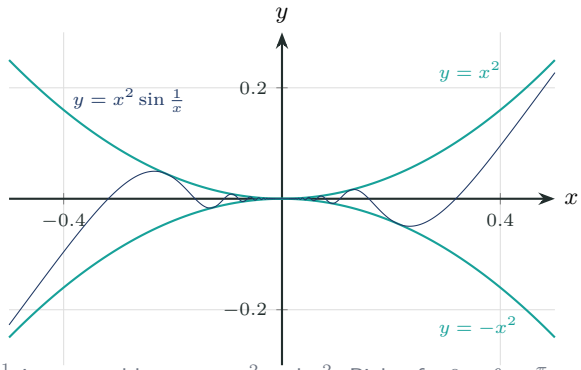
The inequality only needs to hold for x *near* a (it need not hold at a itself, and it need not hold far away), and the two outer functions must have the *same* limit. If their limits differ, the theorem says nothing.

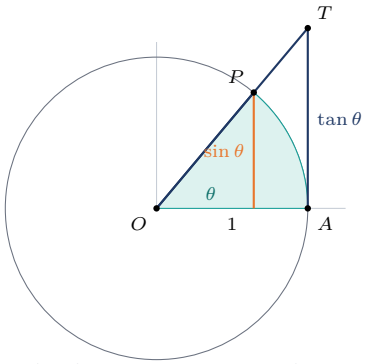
Squeeze (Sandwich) Theorem

If $g(x)\le f(x)\le h(x)$ for all x near a (except possibly at $x=a$) and $\lim\limits_{x\to a}g(x)=\lim\limits_{x\to a}h(x)=L$, then $\lim\limits_{x\to a}f(x)=L$.

Two fundamental trigonometric limits (proved by squeezing):

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\qquad \lim_{x\to 0}\frac{1-\cos x}{x}=0.$$





Left: $x^2\sin\frac{1}{x}$ is squeezed between $-x^2$ and x^2. Right: for $0<\theta<\frac{\pi}{2}$, area of $\triangle OAP$ < area of sector OAP < area of $\triangle OAT$.

Example 1 — A bounded oscillation times a vanishing factor

Show that $\lim\limits_{x\to 0}x^2\sin\dfrac{1}{x}=0$.

Step 1. For all $x\neq 0$, $-1\le\sin\dfrac{1}{x}\le 1$. Multiplying by $x^2>0$ preserves the inequalities: $-x^2\le x^2\sin\dfrac{1}{x}\le x^2$.

Step 2. $\lim\limits_{x\to 0}(-x^2)=0$ and $\lim\limits_{x\to 0}x^2=0$.

Step 3. By the Squeeze Theorem, $\lim\limits_{x\to 0}x^2\sin\dfrac{1}{x}=0$.

Example 2 — Given bounds

For all x, $-x^2+4x-1 \le f(x) \le x^2-4x+7$. Find $\lim_{x\to 2} f(x)$.

Step 1. $\lim_{x\to 2}(-x^2+4x-1) = -4+8-1 = 3$.

Step 2. $\lim_{x\to 2}(x^2-4x+7) = 4-8+7 = 3$.

Step 3. Both bounds approach 3, so by the Squeeze Theorem $\lim_{x\to 2} f(x) = 3$.

Example 3 — Why $\lim_{x\to 0}\frac{\sin x}{x} = 1$

In the unit-circle figure let $0 < \theta < \frac{\pi}{2}$. Then $\triangle OAP$ has area $\frac{1}{2}\sin\theta$, the sector has area $\frac{1}{2}\theta$, and $\triangle OAT$ has area $\frac{1}{2}\tan\theta$. So $\frac{1}{2}\sin\theta < \frac{1}{2}\theta < \frac{1}{2}\tan\theta$.

Step 1. Divide by $\frac{1}{2}\sin\theta > 0$: $1 < \dfrac{\theta}{\sin\theta} < \dfrac{1}{\cos\theta}$, so $\cos\theta < \dfrac{\sin\theta}{\theta} < 1$.

Step 2. As $\theta \to 0^+$, $\cos\theta \to 1$ and the upper bound is 1. By the Squeeze Theorem $\dfrac{\sin\theta}{\theta} \to 1$ from the right. Because $\dfrac{\sin\theta}{\theta}$ is an even function the left-hand limit is the same, so $\lim_{\theta\to 0}\dfrac{\sin\theta}{\theta} = 1$.

Step 3 (consequence). $\dfrac{1-\cos x}{x} = \dfrac{\sin^2 x}{x(1+\cos x)} = \dfrac{\sin x}{x}\cdot\dfrac{\sin x}{1+\cos x} \to 1\cdot\dfrac{0}{2} = 0.$

AP Exam Tip: Squeeze Theorem justification

Write the three ingredients: the inequality, the two outer limits, and the words *"by the Squeeze Theorem"*. When the oscillating factor is a sine or cosine, begin with $-1 \le \sin(\cdot) \le 1$.

Common errors

- **Different outer limits.** If $g \to 2$ and $h \to 5$, the Squeeze Theorem gives no conclusion.
- **Multiplying an inequality by a quantity that can be negative** (reverses the inequality). Multiply by x^2 or $|x|$, which are non-negative.
- **Using $\sin x \approx x$ without justification** on a limit that is not at 0 (for example $\lim_{x\to\pi}\dfrac{\sin x}{x} = 0$).

Check yourself

1. Use the Squeeze Theorem to show that $\lim_{x\to 0} x\cos\dfrac{1}{x^2} = 0$.
2. Suppose $3-(x-3)^2 \le f(x) \le 3+2(x-3)^2$ for all x. Find $\lim_{x\to 3} f(x)$.
3. Evaluate $\lim_{x\to 0}\dfrac{\sin 7x}{\sin 3x}$.

Topic 1.9 Connecting Multiple Representations of Limits

One limit, four languages

A limit can be presented as a *graph*, a *table*, a *formula*, or a *sentence in a context*. The exam deliberately mixes these: you may be given the graph of one function and the table of another and asked for a limit of a combination, or a limit statement and asked what it says about a real situation. The unifying idea is the same in every case: a limit describes where $f(x)$ is *heading* as x approaches a value.

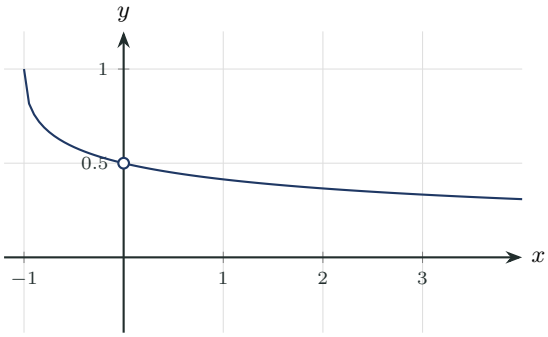
For a single function the three mathematical representations should agree, and checking one against another is a good safeguard. The graph shows the hole or jump; the table shows the numerical trend; the algebra gives the exact value. When the representations are different functions (one given graphically, the other numerically), apply the limit laws of Topic 1.5 to the one-sided limits read from each representation separately.

Key ideas: representations

- **Graph:** read heights the curve approaches. **Table:** read the trend from both sides. **Formula:** substitute or rewrite. **Context:** translate the limit into a statement about the quantity.
- To combine limits from different representations, find each limit (or one-sided limit) first, then use the limit laws.
- $\lim_{t\to\infty} Q(t) = L$ in context: Q levels off near L in the long run. $\lim_{t\to c^-} Q \neq \lim_{t\to c^+} Q$: Q jumps at time c.

Example 1 — The same limit from a graph, a table and algebra

Let $f(x) = \dfrac{\sqrt{x+1}-1}{x}$.



x	-0.1	-0.01	0.01	0.1
$f(x)$	0.51317	0.50126	0.49876	0.48809

Graph: an open circle at $\left(0, \frac{1}{2}\right)$. **Table:** both sides approach 0.5. **Algebra:** multiply by the conjugate, $\dfrac{(x+1)-1}{x\left(\sqrt{x+1}+1\right)} = \dfrac{1}{\sqrt{x+1}+1} \to \dfrac{1}{2}$. All three agree: $\lim_{x\to 0} f(x) = \dfrac{1}{2}$.

Example 2 — Combining a table and a graph

Let h be the function graphed in Topic 1.3 and let f be given by the table

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	2.9	2.99	2.999	3.001	3.01	3.1

Find $\lim_{x\to 1^-}\big[f(x)h(x)\big]$, $\lim_{x\to 1^+}\big[f(x)h(x)\big]$ and $\lim_{x\to 1}\big[f(x)h(x)\big]$.

Step 1. From the table, $\lim_{x\to 1} f(x) \approx 3$ (both sides). From the graph, $\lim_{x\to 1^-} h(x) = 5$ and $\lim_{x\to 1^+} h(x) = 2$.

Step 2. Product law on each side: $3 \cdot 5 = 15$ from the left and $3 \cdot 2 = 6$ from the right.

Step 3. The one-sided limits differ, so $\lim_{x\to 1}\big[f(x)h(x)\big]$ does not exist.

Example 3 — Limits in context

The number of fish in a lake is $N(t)$, where t is in months. It is known that $\lim_{t\to\infty} N(t) = 1200$, $\lim_{t\to 5^-} N(t) = 300$ and $\lim_{t\to 5^+} N(t) = 450$. Interpret each statement.

(1) In the long run the population levels off near 1200 fish. **(2)–(3)** Just before $t = 5$ months the population is about 300 and just after it is about 450: the population jumps by about 150 fish at $t = 5$ (for example because the lake was restocked). Because the one-sided limits differ, N is not continuous at $t = 5$.

Common errors

- **Mixing up which function a limit belongs to.** Label each limit f or h before combining.
- **Using the table's value at $x = a$** when estimating a limit (if it is present, ignore it).
- **Interpreting a limit as the value at an instant.** $\lim_{t\to 5^-} N(t) = 300$ describes behaviour as t approaches 5.

Check yourself

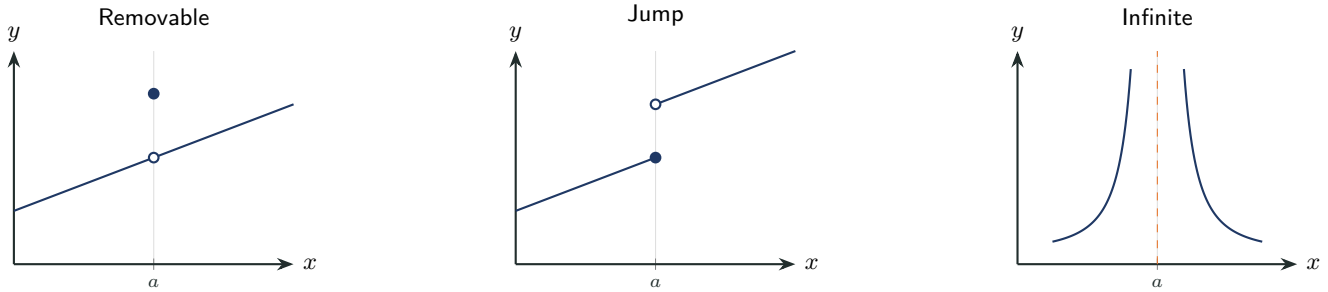
1. A table gives $f(2.9) = 5.9$, $f(2.99) = 5.99$, $f(2.999) = 5.999$, $f(3.001) = 6.001$, $f(3.01) = 6.01$, $f(3.1) = 6.1$. The function is $f(x) = \dfrac{x^2 - 9}{x - 3}$. Confirm the table's estimate of $\lim\limits_{x \to 3} f(x)$ by algebra.
2. Is it possible for a function to have $\lim\limits_{x \to a^-} f(x) = 2$, $\lim\limits_{x \to a^+} f(x) = 2$ and $f(a) = 7$? Describe its graph.

Topic 1.10 Exploring Types of Discontinuities

Where a graph breaks, and how

A function is *discontinuous* at $x = a$ when its graph cannot be drawn through $x = a$ without lifting the pencil. Not all breaks are alike, and the exam expects you to name them. Three types matter in Unit 1.

A **removable discontinuity** (a hole) occurs when $\lim\limits_{x \to a} f(x)$ exists as a real number but either $f(a)$ is undefined or $f(a)$ is a different number. The break can be "repaired" by redefining a single point (Topic 1.13). A **jump discontinuity** occurs when both one-sided limits exist (as real numbers) but are different. An **infinite discontinuity** occurs when $f(x)$ grows without bound near a, that is, when at least one one-sided limit is $\pm\infty$; the graph has a vertical asymptote there. (A fourth behaviour, oscillation like $\sin\frac{1}{x}$ at 0, is also a discontinuity but has no special name in the course.) Jump and infinite discontinuities are *non-removable*.



The three discontinuities of Unit 1: removable (hole), jump, and infinite (vertical asymptote at $x = a$).

Key ideas: types of discontinuity at $x = a$

Type	Condition	Typical cause
Removable	$\lim\limits_{x \to a} f(x) = L$ exists, but $f(a)$ is undefined or $\neq L$	common factor cancels
Jump	$\lim\limits_{x \to a^-} f(x) = L_1$, $\lim\limits_{x \to a^+} f(x) = L_2$, $L_1 \neq L_2$ (both real)	piecewise definition, $\dfrac{\lvert x - a \rvert}{x - a}$
Infinite	at least one one-sided limit is $\pm\infty$	factor remains in the denominator

Example 1 — Classifying the discontinuities of h

Using the graph of h (Topic 1.3): at $x = -2$ the limit is 2 but $h(-2) = 4$, so the discontinuity is **removable**. At $x = 1$ the one-sided limits are 5 and 2: a **jump** (of size $2 - 5 = -3$, a drop of 3). At $x = 3$ the function is unbounded and the graph has a vertical asymptote: an **infinite** discontinuity.

Example 2 — Classifying from a formula

Classify the discontinuities of $f(x) = \dfrac{x^2 - 9}{(x - 3)(x + 1)}$.

Step 1. f is undefined where the denominator is 0: $x = 3$ and $x = -1$.

Step 2 ($x = 3$). Cancel $x - 3$: $f(x) = \dfrac{x + 3}{x + 1}$ for $x \neq 3$, so $\lim\limits_{x \to 3} f(x) = \dfrac{6}{4} = \dfrac{3}{2}$, a real number. Removable.

Step 3 ($x = -1$). After cancelling, $\dfrac{x + 3}{x + 1}$ has numerator $2 \neq 0$ and denominator $\to 0$, so f is unbounded: infinite

discontinuity (vertical asymptote $x = -1$).

Example 3 — A piecewise function

Let $g(x) = \begin{cases} x^2, & x < 2 \\ 8 - x, & x \geq 2. \end{cases}$ Then $\lim_{x \to 2^-} g(x) = 4$ and $\lim_{x \to 2^+} g(x) = 8 - 2 = 6$. Both are real and different: a **jump** of size 2 at $x = 2$. Notice $g(2) = 6$, so g is continuous *from the right* at 2 but not from the left.

Common errors

- **Calling a hole an asymptote.** Check whether the troublesome factor cancels. $\dfrac{x^2 - 9}{x - 3}$ has a hole at 3, $\dfrac{1}{x - 3}$ has an asymptote.
- **Calling every break a "jump".** A jump needs finite, different one-sided limits.
- **Thinking a defined value cures a discontinuity.** $f(0) = 5$ for $f(x) = \dfrac{1}{x^2}$ does not remove the infinite discontinuity.

Check yourself

1. Classify the discontinuities of $f(x) = \dfrac{x - 2}{(x - 2)(x + 2)}$.
2. For $g(x) = \begin{cases} x^2, & x < 2 \\ 8 - x, & x \geq 2 \end{cases}$, is the discontinuity at $x = 2$ removable? What is the size of the jump?
3. Let $f(x) = \dfrac{1}{x^2}$ for $x \neq 0$ and $f(0) = 5$. What type of discontinuity is at $x = 0$?

Topic 1.11 Defining Continuity at a Point

What "continuous" means

In everyday language a function is continuous if you can draw its graph without lifting your pencil. Calculus turns this into a test that uses limits and can be applied to a formula, a table or a graph. A function f is **continuous at** $x = a$ when three things are true: the value $f(a)$ exists; the limit $\lim_{x \to a} f(x)$ exists; and the two are equal. Put compactly,

$$\lim_{x \to a} f(x) = f(a).$$

If any of the three conditions fails, f is discontinuous at a, and the failure tells you which type of discontinuity it is (Topic 1.10). Condition 1 fails at a hole or an asymptote with a outside the domain; condition 2 fails at a jump or an infinite break; condition 3 fails at a hole with a displaced point.

At the endpoint of a domain, or at a point where the formula changes, we use one-sided versions. f is *continuous from the right* at a if $\lim_{x \to a^+} f(x) = f(a)$ and *continuous from the left* if $\lim_{x \to a^-} f(x) = f(a)$. Continuity at an interior point requires both.

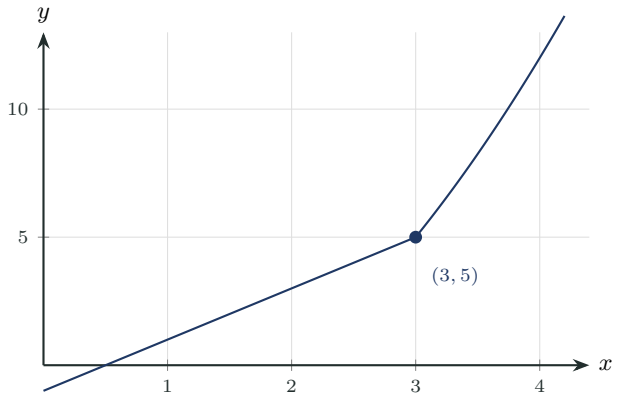
Key ideas: continuity at $x = a$

- f is continuous at a iff (1) $f(a)$ exists, (2) $\lim_{x \to a} f(x)$ exists, (3) $\lim_{x \to a} f(x) = f(a)$.
- Piecewise functions: compare $\lim_{x \to a^-} f$, $\lim_{x \to a^+} f$ and $f(a)$.
- To find a constant that makes a function continuous, set the one-sided limits equal to each other and to $f(a)$.

Example 1 — Testing continuity from the graph of h

For the graph of h in Topic 1.3, test the three conditions.

$x = 0$	$h(0) = 4$, $\lim_{x\to 0} h(x) = 4$: all three hold, so h is continuous at 0.
$x = -2$	$h(-2) = 4$ exists, $\lim h = 2$ exists, but $2 \neq 4$: condition 3 fails.
$x = 1$	$\lim_{x\to 1} h(x)$ does not exist (one-sided limits $5, 2$): condition 2 fails. ($h(1) = 5 = \lim_{x\to 1^-} h$, so h is continuous from the left.)
$x = 3$	$h(3)$ is undefined: condition 1 fails.



Graph of $g(x) = 2x - 1$ for $x < 3$ and $x^2 - 4$ for $x \geq 3$ (Example 2, $k = -1$): the pieces meet at $(3, 5)$.

Example 2 — Finding a constant for continuity

Let $g(x) = \begin{cases} 2x + k, & x < 3 \\ x^2 - 4, & x \geq 3. \end{cases}$ Find k so that g is continuous at $x = 3$.

Step 1. $g(3) = 3^2 - 4 = 5$.
Step 2. $\lim_{x\to 3^-} g(x) = 6 + k$ and $\lim_{x\to 3^+} g(x) = 9 - 4 = 5$.
Step 3. Continuity requires $6 + k = 5 = g(3)$, so $k = -1$.
Step 4 (justification). With $k = -1$: $g(3) = 5$, $\lim_{x\to 3} g(x) = 5$ because both one-sided limits equal 5, so $\lim_{x\to 3} g(x) = g(3)$ and g is continuous at 3.

Example 3 — Defined, but not continuous; and one-sided continuity

(a) Let $f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1 \\ 3, & x = 1. \end{cases}$ Here $f(1) = 3$ exists and $\lim_{x\to 1} f(x) = \lim_{x\to 1}(x + 1) = 2$ exists, but $2 \neq 3$. So f is not continuous at 1 (condition 3 fails).
(b) Let $p(x) = \sqrt{x - 1}$. Its domain is $[1, \infty)$ and $\lim_{x\to 1^+} \sqrt{x - 1} = 0 = p(1)$, so p is continuous from the right at $x = 1$. There is no left-hand limit, and at an endpoint right-continuity is the correct notion of continuity.

AP Exam Tip: justifying continuity

A complete justification states the three conditions with their values: "$f(a) = \ldots$, $\lim_{x\to a} f(x) = \ldots$, and they are equal, so f is continuous at $x = a$." To show a discontinuity, name the condition that fails and give the numbers.

Common errors

- **Checking only $f(a)$ (or only the limit).** All three conditions are required.
- **Equating one side only.** For a piecewise function both one-sided limits must equal $f(a)$.
- **Sign errors when solving for the constant.** Write both limits as expressions in k before equating.

Check yourself

1. Let $f(x) = \dfrac{x^2 - 1}{x - 1}$ for $x \neq 1$ and $f(1) = 3$. Is f continuous at $x = 1$? Justify using limits.
2. Find k so that $g(x) = \begin{cases} 2x + k, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$ is continuous at $x = 3$.
3. Which of the three conditions fails for $f(x) = \dfrac{1}{x - 2}$ at $x = 2$?

Topic 1.12 Confirming Continuity over an Interval

Continuity on an interval

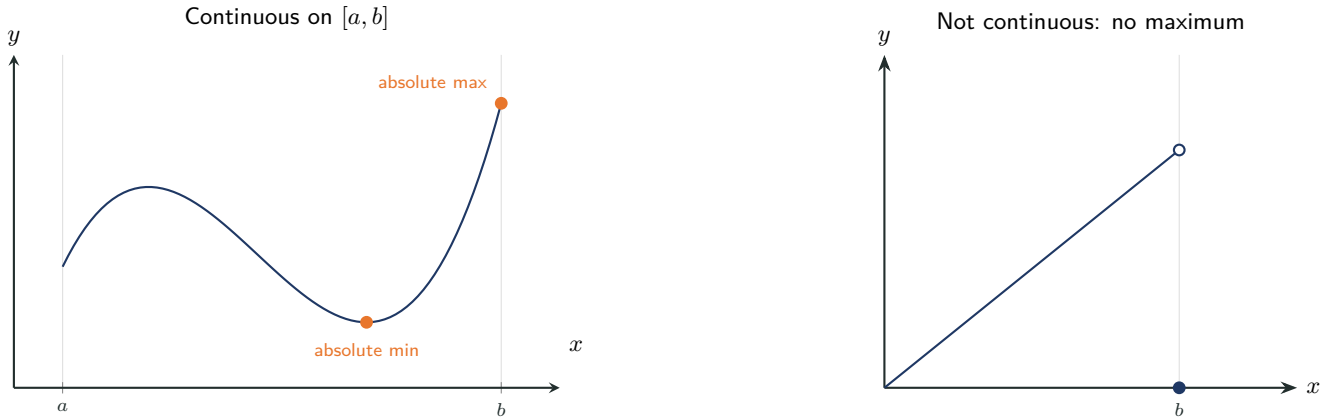
A function is *continuous on an open interval* (a, b) if it is continuous at every point in it. It is *continuous on a closed interval* $[a, b]$ if it is continuous on (a, b), continuous from the right at a, and continuous from the left at b. Geometrically, you can draw the graph over the whole interval without lifting the pencil.

You rarely check every point. Instead you use two facts. First, the standard families are continuous on their domains: polynomials everywhere; rational functions wherever the denominator is not zero; $\sqrt[n]{x}$, $\ln x$, e^x, $\sin x$, $\cos x$, $\tan x$ (except where it is undefined) on their domains. Second, sums, differences, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous. So to find where a formula is continuous, find its domain and see which pieces of that domain are intervals.

The Extreme Value Theorem. Continuity over a *closed* interval has a powerful consequence: the graph cannot escape to infinity or approach a height it never reaches. Such a function must attain a highest and a lowest value. The hypotheses matter: the function must be continuous *and* the interval must be closed (Unit 5 builds on this theorem).

Key ideas: continuity over an interval

- Continuous on $[a, b]$: continuous at every interior point, right-continuous at a, left-continuous at b.
- Polynomials, e^x, $\sin x$, $\cos x$ are continuous on $(-\infty, \infty)$; rational functions on their domains; $\sqrt{x}$ on $[0, \infty)$; $\ln x$ on $(0, \infty)$.
- Sums, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous.
- **Extreme Value Theorem:** if f is continuous on the closed interval $[a, b]$, then f attains an absolute maximum and an absolute minimum on $[a, b]$.



Left: a function continuous on a closed interval attains an absolute maximum and minimum. Right: with a discontinuity the highest value is approached but never reached.

Example 1 — Intervals of continuity from a formula

Determine where $f(x) = \dfrac{\sqrt{x + 2}}{x - 1}$ is continuous.

Step 1. The numerator requires $x + 2 \geq 0$, i.e. $x \geq -2$. The denominator requires $x \neq 1$.

Step 2. f is a quotient of continuous functions on its domain, so it is continuous on $[-2, 1) \cup (1, \infty)$ (right-continuous at the endpoint -2). At $x = 1$ it has an infinite discontinuity.

Example 2 — May the Extreme Value Theorem be used?

Let $f(x) = \dfrac{1}{x-2}$.
(a) On $[3,6]$: f is continuous (the only break, $x=2$, is outside the interval) and the interval is closed, so f has an absolute maximum and minimum on $[3,6]$ (they are $f(3)=1$ and $f(6)=\frac{1}{4}$).
(b) On $[1,3]$: f is not continuous on the interval (it is undefined at 2), so the theorem does not apply, and in fact f has neither a maximum nor a minimum there because it is unbounded.

Example 3 — Continuity of a piecewise function on an interval

Is $f(x) = \begin{cases} x^2, & 0 \le x < 2 \\ 6-x, & 2 \le x \le 4 \end{cases}$ continuous on $[0,4]$?

Step 1. On $[0,2)$ and $(2,4]$ the pieces are polynomials, hence continuous (one-sided at the endpoints 0 and 4).
Step 2. At $x=2$: $\lim\limits_{x\to 2^-} x^2 = 4$, $\lim\limits_{x\to 2^+}(6-x) = 4$, and $f(2)=4$. All equal, so f is continuous at 2.
Step 3. Hence f is continuous on $[0,4]$.

Common errors

- **Applying the Extreme Value Theorem on an open interval or to a discontinuous function.** Both hypotheses are required.
- **Declaring a rational function continuous "everywhere".** It is continuous only where the denominator is non-zero.
- **Forgetting the domain of a root or logarithm** when stating intervals.

Check yourself

1. State the intervals on which $f(x) = \dfrac{\sqrt{x+5}}{x-2}$ is continuous, and do the same for $g(x) = \ln(x-3)$.
2. Does the Extreme Value Theorem apply to $f(x) = \dfrac{1}{x-2}$ on $[1,3]$? On $[3,6]$? Explain.
3. Is $f(x) = \begin{cases} x^2, & 0 \le x < 2 \\ 6-x, & 2 \le x \le 4 \end{cases}$ continuous at $x=2$? Justify.

Topic 1.13 Removing Discontinuities

Repairing a hole

A removable discontinuity is called removable because it can be repaired by changing the function at a single point. If $\lim\limits_{x\to a} f(x) = L$ exists but $f(a)$ is undefined or not equal to L, define a new function

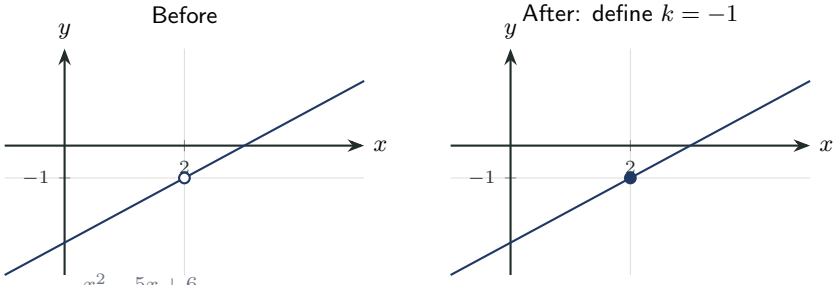
$$g(x) = \begin{cases} f(x), & x \ne a \\ L, & x = a. \end{cases}$$

Then g agrees with f everywhere except at a, and $\lim\limits_{x\to a} g(x) = L = g(a)$, so g is continuous at a. Jump and infinite discontinuities cannot be repaired this way: no single value can fill a gap between two different one-sided limits, and no real number can fill an asymptote.

A common exam task chooses a parameter. For a function defined in pieces with an unknown constant k (or several constants), you write the continuity condition at each break as an equation and solve. With one break you get one equation; with two breaks and two constants you get a 2×2 system, often solved one equation at a time.

Key ideas: removing a discontinuity

- Removable at a: $\lim\limits_{x\to a} f(x) = L$ exists (finite). Fix: redefine $f(a) = L$.
- Jump or infinite: not removable.
- To make a piecewise function continuous at a break c: left limit = right limit = value at c.



$f(x) = \dfrac{x^2-5x+6}{x-2} = x-3$ for $x \neq 2$ has a hole at $(2,-1)$; defining $f(2) = -1$ fills it.

Example 1 — Which discontinuities can be removed?

For $f(x) = \dfrac{x^2-x-6}{x^2-9} = \dfrac{(x-3)(x+2)}{(x-3)(x+3)}$ the discontinuities are at $x=3$ and $x=-3$.

At $x=3$: $\lim\limits_{x\to 3} f(x) = \dfrac{5}{6}$ exists, so the discontinuity is removable; define $f(3) = \dfrac{5}{6}$.

At $x=-3$: $\dfrac{x+2}{x+3}$ has numerator -1 and denominator $\to 0$, so f is unbounded. This is an infinite discontinuity and cannot be removed.

Example 2 — Choosing k

Let $f(x) = \begin{cases} \dfrac{x^2-5x+6}{x-2}, & x \neq 2 \\ k, & x = 2. \end{cases}$ Find k so that f is continuous at $x=2$.

Step 1. For $x \neq 2$: $\dfrac{(x-2)(x-3)}{x-2} = x-3$, so $\lim\limits_{x\to 2} f(x) = 2-3 = -1$.

Step 2. Continuity requires $f(2) = \lim\limits_{x\to 2} f(x)$, so $k = -1$.

Example 3 — Two constants, two breaks

Find a and b so that $f(x) = \begin{cases} ax+2, & x \leq 1 \\ x^2+b, & 1 < x < 3 \\ 2x+5, & x \geq 3 \end{cases}$ is continuous for all x.

Step 1 ($x=3$). $\lim\limits_{x\to 3^-} f(x) = 9+b$ and $f(3) = \lim\limits_{x\to 3^+} f(x) = 11$. So $9+b=11$, $b=2$.

Step 2 ($x=1$). $f(1) = \lim\limits_{x\to 1^-} f(x) = a+2$ and $\lim\limits_{x\to 1^+} f(x) = 1+b = 3$. So $a+2=3$, $a=1$.

Check. With $a=1, b=2$: at $x=1$ both sides give 3; at $x=3$ both give 11.

Common errors

- **Trying to "remove" an asymptote** by defining a value there.
- **Using the wrong piece for a one-sided limit.** For $x \to 1^+$ use the formula valid for $x > 1$.
- **Stopping after finding one constant** when there are two breaks.

Check yourself

1. The function $f(x) = \dfrac{x^2-16}{x-4}$ is undefined at $x=4$. What value should be assigned to $f(4)$ to make f continuous there?
2. Find k so that $f(x) = \begin{cases} \dfrac{x^3-1}{x-1}, & x \neq 1 \\ k, & x = 1 \end{cases}$ is continuous at $x=1$.
3. Find a and b so that $f(x) = \begin{cases} ax+1, & x < 2 \\ 9, & x = 2 \\ x^2+b, & x > 2 \end{cases}$ is continuous at $x=2$.

Topic 1.14 Connecting Infinite Limits and Vertical Asymptotes

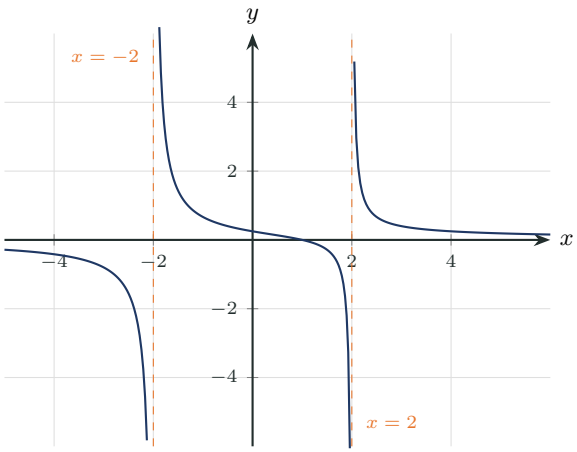
Why a vertical asymptote appears

When a rational function has a factor in the denominator that is not cancelled by the numerator, then near the zero of that factor the denominator is tiny while the numerator stays away from 0. A fixed non-zero number divided by a tiny number is huge, so $|f(x)| \to \infty$. The *sign* of the huge number depends on the signs of numerator and denominator on each side of the zero, so each side must be analysed separately.

The line $x = a$ is a **vertical asymptote** of the graph of f if at least one of the one-sided limits $\lim_{x\to a^-} f(x)$, $\lim_{x\to a^+} f(x)$ is ∞ or $-\infty$. To find vertical asymptotes of a rational function, cancel common factors first (a cancelled factor gives a hole, Topic 1.10) and then find zeros of the remaining denominator. Other functions have asymptotes too: $\ln x$ at $x = 0$ (from the right), $\tan x$ at odd multiples of $\frac{\pi}{2}$.

Key ideas: infinite limits and vertical asymptotes

- $x = a$ is a vertical asymptote if $\lim_{x\to a^{\pm}} f(x) = \pm\infty$ for at least one side.
- For $\dfrac{p(x)}{q(x)}$ with $q(a) = 0$ and $p(a) \neq 0$: determine the sign of p and of q just left and just right of a.
- Even power of a vanishing factor, such as $(x-a)^2$: the sign does not change, so both sides give the same infinity. Odd power: the sign changes, so the sides disagree.
- $\lim_{x\to 0^+} \ln x = -\infty$, $\lim_{x\to(\pi/2)^-} \tan x = \infty$, $\lim_{x\to(\pi/2)^+} \tan x = -\infty$.



$f(x) = \dfrac{x-1}{x^2-4}$ has vertical asymptotes $x = -2$ and $x = 2$; the branches go in opposite directions on the two sides of each.

Example 1 — Sign analysis

Find $\lim_{x\to 2^-} f(x)$, $\lim_{x\to 2^+} f(x)$, $\lim_{x\to -2^-} f(x)$, $\lim_{x\to -2^+} f(x)$ for $f(x) = \dfrac{x-1}{x^2-4} = \dfrac{x-1}{(x-2)(x+2)}$.

Interval	$(-\infty,-2)$	$(-2,1)$	$(1,2)$	$(2,\infty)$
sign of $x-1$	$-$	$-$	$+$	$+$
sign of x^2-4	$+$	$-$	$-$	$+$
sign of $f(x)$	$-$	$+$	$-$	$+$

At $x = 2$ the numerator is $1 > 0$, so $\lim_{x\to 2^-} f(x) = -\infty$ and $\lim_{x\to 2^+} f(x) = \infty$. At $x = -2$ the numerator is $-3 < 0$: $\lim_{x\to -2^-} f(x) = -\infty$ and $\lim_{x\to -2^+} f(x) = \infty$. Neither two-sided limit exists; both lines are vertical asymptotes.

Example 2 — Hole or asymptote?

Find the vertical asymptotes of $f(x) = \dfrac{x^2-4}{x^2-5x+6}$.

Step 1. Factor: $\dfrac{(x-2)(x+2)}{(x-2)(x-3)}$.

Step 2. The factor $x-2$ cancels, giving a hole at $x=2$ (limit $\dfrac{4}{-1}=-4$). The factor $x-3$ remains, and at $x=3$ the numerator of $\dfrac{x+2}{x-3}$ is $5\neq 0$, so $\lim\limits_{x\to 3^+} f(x)=\infty$ and $\lim\limits_{x\to 3^-} f(x)=-\infty$.

Step 3. The only vertical asymptote is $x=3$.

Example 3 — An infinite limit in context

The cost, in thousands of dollars, of removing p percent of the pollutant from a factory's waste water is $C(p)=\dfrac{80p}{100-p}$ for $0\le p<100$. Find $\lim\limits_{p\to 100^-} C(p)$ and interpret.

As $p\to 100^-$ the numerator tends to $8000>0$ and the denominator $100-p$ tends to 0 through positive values, so $\lim\limits_{p\to 100^-} C(p)=\infty$. Removing a larger and larger percentage costs more and more; removing all of the pollutant is impossible at any finite cost. (For comparison $C(90)=720$ and $C(99)=7920$.)

Common errors

- **Declaring every zero of the denominator an asymptote.** If the numerator vanishes there too, check for cancellation.
- **Ignoring the sign on one side.** $\lim\limits_{x\to 2}\dfrac{1}{x-2}$ does not exist, while $\lim\limits_{x\to 2}\dfrac{1}{(x-2)^2}=\infty$.
- **Writing $\frac{k}{0}=0$ or "undefined" as the answer.** Give the limit, ∞ or $-\infty$, with the sign.

Check yourself

1. Find $\lim\limits_{x\to 3^+}\dfrac{x+2}{x-3}$ and $\lim\limits_{x\to 3^-}\dfrac{x+2}{x-3}$.
2. Find all vertical asymptotes of $f(x)=\dfrac{x-1}{x^2-x-6}$.
3. Evaluate $\lim\limits_{x\to 0^+}\ln x$ and $\lim\limits_{x\to(\pi/2)^+}\tan x$.

Topic 1.15 Connecting Limits at Infinity and Horizontal Asymptotes

End behaviour

A *limit at infinity*, $\lim\limits_{x\to\infty} f(x)$ or $\lim\limits_{x\to-\infty} f(x)$, describes where the graph goes as x moves far to the right or far to the left. If the limit is a finite number L, the line $y=L$ is a **horizontal asymptote** of the graph. A graph can have at most two horizontal asymptotes, one for each end, and it may cross a horizontal asymptote at finite x.

Rational functions. Divide numerator and denominator by the highest power of x in the denominator. Terms such as $\frac{1}{x}$ and $\frac{1}{x^2}$ tend to 0, and what remains is a ratio of leading coefficients (when degrees match) or 0 (when the denominator has the higher degree). If the numerator has the higher degree, the function is unbounded and there is no horizontal asymptote.

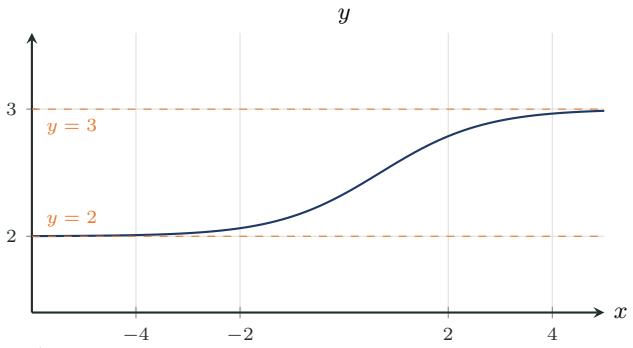
Roots. Be careful with $\sqrt{x^2}=|x|$. As $x\to\infty$, $|x|=x$; as $x\to-\infty$, $|x|=-x$. The two ends can therefore give different asymptotes with opposite signs.

Exponentials. As $x\to\infty$, $e^x\to\infty$ and $e^{-x}\to 0$; as $x\to-\infty$ the roles swap. Exponentials dominate polynomials, which dominate logarithms: $\ln x \ll x^n \ll e^x$ as $x\to\infty$.

Key ideas: limits at infinity

- $y=L$ is a horizontal asymptote if $\lim\limits_{x\to\infty} f(x)=L$ or $\lim\limits_{x\to-\infty} f(x)=L$.
- For $\dfrac{a_nx^n+\cdots}{b_mx^m+\cdots}$ as $x\to\pm\infty$: $n<m\Rightarrow 0$; $n=m\Rightarrow\dfrac{a_n}{b_m}$; $n>m\Rightarrow\pm\infty$ (no horizontal asymptote).

- $\lim_{x\to\infty} \dfrac{1}{x^p} = 0$ for $p > 0$. $\sqrt{x^2} = |x|$.
- Growth: $\ln x \ll x^n \ll e^x$.



$f(x) = \dfrac{3e^x + 4}{e^x + 2}$ has two horizontal asymptotes: $y = 2$ on the left and $y = 3$ on the right.

Example 1 — A rational function

Find $\lim_{x\to\infty} \dfrac{3x^2 - 5x}{2x^2 + 1}$.

Divide every term by x^2: $\dfrac{3 - \frac{5}{x}}{2 + \frac{1}{x^2}}$. As $x \to \infty$, $\frac{5}{x} \to 0$ and $\frac{1}{x^2} \to 0$, so the limit is $\dfrac{3}{2}$. The line $y = \dfrac{3}{2}$ is a horizontal asymptote.

Example 2 — Different asymptotes at the two ends

Find $\lim_{x\to\infty} \dfrac{\sqrt{4x^2+1}}{x+3}$ and $\lim_{x\to-\infty} \dfrac{\sqrt{4x^2+1}}{x+3}$.

Step 1. Factor x^2 from the root: $\sqrt{4x^2+1} = \sqrt{x^2}\sqrt{4 + \frac{1}{x^2}} = |x|\sqrt{4 + \frac{1}{x^2}}$.

Step 2 ($x \to \infty$). $|x| = x$: $\dfrac{x\sqrt{4 + 1/x^2}}{x(1 + 3/x)} = \dfrac{\sqrt{4 + 1/x^2}}{1 + 3/x} \to \dfrac{2}{1} = 2.$

Step 3 ($x \to -\infty$). $|x| = -x$: $\dfrac{-x\sqrt{4 + 1/x^2}}{x(1 + 3/x)} = \dfrac{-\sqrt{4 + 1/x^2}}{1 + 3/x} \to -2.$

The horizontal asymptotes are $y = 2$ and $y = -2$.

Example 3 — Exponentials and the graph

For $f(x) = \dfrac{3e^x + 4}{e^x + 2}$ (figure above):

As $x \to \infty$: divide by e^x: $\dfrac{3 + 4e^{-x}}{1 + 2e^{-x}} \to \dfrac{3}{1} = 3.$

As $x \to -\infty$: $e^x \to 0$, so $f(x) \to \dfrac{0 + 4}{0 + 2} = 2.$

The graph has horizontal asymptotes $y = 3$ and $y = 2$. Also, $f(0) = \dfrac{7}{3} \approx 2.33$, which lies between them.

Common errors

- **Treating ∞ as a number** ("$\frac{\infty}{\infty} = 1$"). Divide by the highest power instead.
- $\sqrt{x^2} = x$**.** For $x < 0$ it is $-x$.
- **Saying a graph can never cross its horizontal asymptote.** It can; the asymptote describes the far ends.

Check yourself

1. Evaluate $\lim_{x\to\infty} \dfrac{4x^2 - x}{1 - 2x^2}$.
2. Find the horizontal asymptotes of $f(x) = \dfrac{5x + 1}{\sqrt{x^2 + 1}}$.
3. Evaluate $\lim_{x\to\infty} \dfrac{x^3 - 2x}{5x^2 + 1}$ and $\lim_{x\to\infty} \dfrac{6x + 1}{x^2 + 4}$.

Topic 1.16 Working with the Intermediate Value Theorem (IVT)

A continuous function cannot skip values

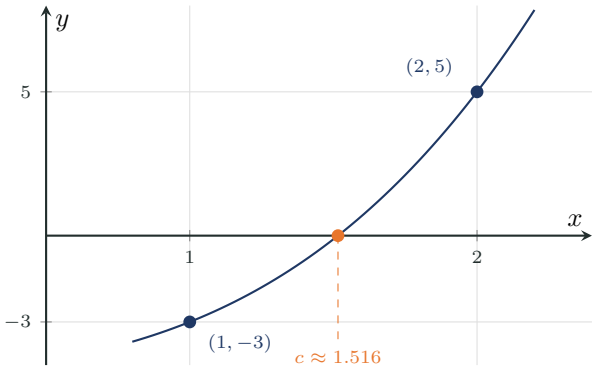
If you walk from a point at height 2 m to a point at height 8 m along a path with no teleporting, you must pass through every height between 2 and 8. The **Intermediate Value Theorem** says the same about a continuous function on a closed interval: it takes every value between its two endpoint values. The theorem is an *existence* theorem. It tells you a number c exists with $f(c) = k$; it does not tell you how many there are or where. To find c you use algebra or a calculator.

The most common use is to show that an equation has a solution: choose an interval $[a, b]$ where f is continuous and $f(a)$, $f(b)$ lie on opposite sides of the target value k (for a zero, one is negative and one is positive). Both hypotheses are essential. Continuity on $[a, b]$ must be *stated*, and the number k must lie strictly between $f(a)$ and $f(b)$.

Intermediate Value Theorem

If f is continuous on the closed interval $[a, b]$ and k is a number strictly between $f(a)$ and $f(b)$, then there exists at least one number c in the open interval (a, b) such that $f(c) = k$.

Special case: if f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then $f(c) = 0$ for some c in (a, b).



$f(x) = x^3 + x - 5$ is continuous, $f(1) = -3 < 0 < 5 = f(2)$, so the graph crosses the x-axis at some c in $(1, 2)$.

Example 1 — Showing a zero exists

Show that $x^3 + x - 5 = 0$ has a solution in $(1, 2)$.
Let $f(x) = x^3 + x - 5$. **Step 1.** f is a polynomial, so it is continuous on $[1, 2]$.
Step 2. $f(1) = -3 < 0$ and $f(2) = 5 > 0$, so 0 lies between $f(1)$ and $f(2)$.
Step 3. By the Intermediate Value Theorem there is a c in $(1, 2)$ with $f(c) = 0$, i.e. c solves the equation. (A calculator gives $c \approx 1.516$.)

Example 2 — A table of values

A continuous function g has the values below.

x	0	2	5	8
$g(x)$	3	−1	6	2

(a) What is the minimum number of solutions of $g(x) = 0$ guaranteed on $(0, 8)$? (b) Is $g(x) = 4$ guaranteed to have a solution on $(2, 8)$?
(a) $g(0) > 0 > g(2)$ gives a zero in $(0, 2)$; $g(2) < 0 < g(5)$ gives a zero in $(2, 5)$. The two intervals do not overlap,

so at least 2 zeros. (On $[5,8]$ both values are positive, so no zero is guaranteed there.)
(b) $g(2) = -1 < 4 < 6 = g(5)$, so by the IVT $g(c) = 4$ for some c in $(2,5)$, hence in $(2,8)$. Also $g(5) = 6 > 4 > 2 = g(8)$ gives another in $(5,8)$.

Example 3 — When the theorem does not apply

Let $f(x) = \dfrac{1}{x}$ on $[-1,1]$. Then $f(-1) = -1 < 0 < 1 = f(1)$, but $f(x) = 0$ has no solution. The IVT does not contradict this: f is not continuous on $[-1,1]$ (it is undefined at $x = 0$), so the hypothesis fails.

AP Exam Tip: writing an IVT justification

Include all four pieces: (1) f *is continuous on* $[a,b]$; (2) the values $f(a)$ and $f(b)$; (3) that k is between them; (4) the conclusion "by the Intermediate Value Theorem, there is a c in (a,b) with $f(c) = k$." Omitting the continuity statement loses the point.

Common errors

- **Forgetting to state continuity** or applying the theorem to a function with a break.
- **Concluding exactly one solution.** The IVT guarantees at least one.
- **Using a target value outside the endpoint values.** For k not between $f(a)$ and $f(b)$ the theorem gives no information.

Check yourself

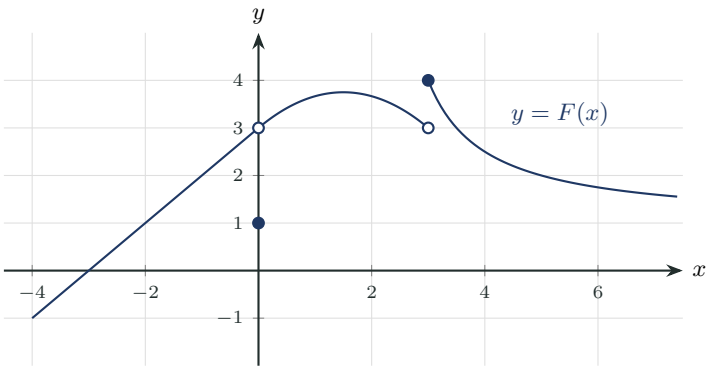
1. Show that $x^5 + x - 1 = 0$ has a solution in $(0,1)$.
2. f is continuous on $[2,6]$ with $f(2) = 5$ and $f(6) = -3$. Must $f(c) = 0$ for some c in $(2,6)$? Must $f(c) = 7$ for some c in $(2,6)$?
3. A continuous function f has $f(1) = 2$, $f(3) = -1$, $f(4) = 4$, $f(8) = 4$. What is the minimum number of zeros of f that are guaranteed on $(1,8)$?

Unit 1 Practice: Limits and Continuity

Part A: Multiple Choice

No calculator · Questions 1–14 · Choose the best answer

Questions 1–3 refer to the graph of the function F below. The graph of F has an open circle at $(0,3)$ and a filled dot at $(0,1)$; an open circle at $(3,3)$ and a filled dot at $(3,4)$. For $x>3$ the graph decreases toward $y=1$.



1 What is $\lim_{x\to 0} F(x)$?

(A) 1 **(B)** 3

(C) 4 **(D)** does not exist

2 What is $\lim_{x\to 3} F(x)$?

(A) 3 **(B)** 4

(C) does not exist **(D)** $\dfrac{7}{2}$

3 F is continuous from the right at $x=c$ (that is, $\lim_{x\to c^+} F(x)=F(c)$) but F is not continuous at $x=c$. What is c?

(A) 3 **(B)** 0

(C) -2 **(D)** 5

4 $\lim_{x\to -2} \dfrac{x^2+x-2}{x+2} =$

(A) 0 **(B)** 3

(C) does not exist **(D)** -3

5 $\lim_{x\to 0} \dfrac{\sin 4x}{2x} =$

(A) 2 **(B)** 4

(C) $\dfrac{1}{2}$ **(D)** 1

6 A function f satisfies $|f(x)-4|\le (x-1)^2$ for all real x. What is $\lim_{x\to 1} f(x)$?

(A) 0 **(B)** 1

(C) 4 **(D)** It cannot be determined from the information given.

7 $\lim_{x\to\infty} \dfrac{5x^3-2x}{2x^3+7x^2} =$

(A) 0 **(B)** $\dfrac{5}{2}$

(C) $\dfrac{2}{5}$ **(D)** ∞

8 The graph of $f(x)=\dfrac{x^2-4}{x^2-5x+6}$ has vertical asymptote(s) at

(A) $x=2$ only **(B)** $x=2$ and $x=3$

(C) $x=-2$ and $x=3$ **(D)** $x=3$ only

9 $\lim_{x\to 1^+} \dfrac{x+2}{x^2-1} =$

(A) ∞ **(B)** $-\infty$

(C) 0 **(D)** $\dfrac{3}{2}$

10 Let $f(x)=\begin{cases} kx-1, & x<2 \\ x^2+k, & x\geq 2. \end{cases}$ For what value of k is f continuous at $x=2$?

(A) 4 **(B)** 5

(C) $\dfrac{5}{3}$ **(D)** 3

11 The function f is continuous on the closed interval $[1,5]$ with $f(1)=-3$ and $f(5)=8$. Which of the following must be true?

(A) $f(3)=2.5$

(B) f is increasing on $[1,5]$.

(C) $f(c)=0$ for some c in the open interval $(1,5)$.

(D) $f(c)=10$ for some c in the open interval $(1,5)$.

12 Which of the following functions has a removable discontinuity at $x=1$?

(A) $\dfrac{1}{x-1}$ **(B)** $\dfrac{|x-1|}{x-1}$

(C) $\dfrac{x-1}{x^2+1}$ **(D)** $\dfrac{x^2-1}{x-1}$

13 $\lim_{x\to 0} \dfrac{\sqrt{x+16}-4}{x} =$

(A) $\dfrac{1}{8}$ **(B)** $\dfrac{1}{4}$

(C) 0 **(D)** 8

14 $\lim_{x\to -\infty} \dfrac{\sqrt{9x^2+x}}{2x+1} =$

(A) $\dfrac{3}{2}$ **(B)** 3

(C) $-\dfrac{3}{2}$ **(D)** 0

Part B: Multiple Choice

Graphing calculator allowed · Questions 15–21

15 Using a graphing calculator, what is $\lim_{x\to 0}\dfrac{e^{3x}-1}{x}$?

(A) 1 **(B)** 3

(C) $\dfrac{1}{3}$ **(D)** 0

16 Which of the following is the value of $\lim_{x\to 2}\dfrac{\sin(x-2)}{x^2-4}$ (calculator in radian mode)?

(A) 0.004 **(B)** 0.5

(C) 1 **(D)** 0.25

17 Let $g(x)=e^x-4x$. By the Intermediate Value Theorem, on which of the following closed intervals is the equation $g(x)=0$ guaranteed to have a solution?

(A) $[0,1]$ **(B)** $[1,2]$

(C) $[3,4]$ **(D)** $[-1,0]$

18 What is the average rate of change of $f(x)=xe^{-x}$ on the interval $[1,3]$?

(A) 0.109 **(B)** -0.219

(C) -0.109 **(D)** -0.073

19 $\lim_{x\to 0}\dfrac{\tan 3x}{\sin 5x}=$

(A) $\dfrac{5}{3}\approx 1.667$ **(B)** 0.6

(C) 1 **(D)** 0

20 The horizontal asymptote(s) of the graph of $f(x)=\dfrac{2x-1}{\sqrt{x^2+4}}$ is (are)

(A) $y=2$ only **(B)** $y=-2$ only

(C) $y=0$ **(D)** $y=2$ and $y=-2$

21 A continuous function g has $g(0)=-2$, $g(4)=3$, $g(7)=-1$ and $g(10)=5$. What is the minimum number of solutions of $g(x)=0$ guaranteed on the interval $(0,10)$?

(A) 3 **(B)** 2

(C) 1 **(D)** 4

Part C: Free Response

Question 22: graphing calculator allowed · Question 23: no calculator · 9 points each

Show the work that leads to your answers. Unless otherwise indicated, give numerical answers correct to three decimal places.

22 **[9 points]** (Calculator allowed.) A phone is charged in two stages. The charge level $C(t)$, in percent, t hours after charging begins, is modeled by

$$C(t)=\begin{cases}40t, & 0\le t\le 1\\ 100-k\,e^{-0.5(t-1)}, & t>1,\end{cases}$$

where k is a positive constant.

(a) Find $\lim_{t\to 1^-}C(t)$ and $\lim_{t\to 1^+}C(t)$. Your answer for the second limit will involve k. **[2 points]**

(b) Find the value of k for which C is continuous at $t=1$. Justify your answer using the definition of continuity. **[2 points]**

(c) For the value of k found in part (b), find $\lim_{t\to\infty}C(t)$. Interpret this limit in the context of the problem. **[2 points]**

(d) For the value of k found in part (b), show that there is a time t with $2<t<5$ at which $C(t)=80$. Justify your answer, and then find that value of t. **[3 points]**

23 **[9 points]** (No calculator.) Let g be the function defined by $g(x) = \dfrac{x^2 - 2x - 8}{x^2 - 16}$ for $x \neq \pm 4$.

(a) Find $\lim_{x \to 4} g(x)$. Show the work that leads to your answer. **[2 points]**

(b) The graph of g has a vertical asymptote at exactly one value of x. Find that value. Justify your answer using one-sided limits. **[3 points]**

(c) Let h be the function defined by $h(x) = g(x)$ for $x \neq \pm 4$ and $h(4) = k$. For what value of k is h continuous at $x = 4$? Justify your answer using the definition of continuity. **[2 points]**

(d) Find $\lim_{x \to \infty} g(x)$ and write an equation for the horizontal asymptote of the graph of g. **[2 points]**

AP Calculus AB – Unit 2 | Exam weighting: 10–12% of the AB exam

Differentiation: Definition and Fundamental Properties

In this unit

Unit 1 asked what happens to $f(x)$ as x approaches a value. Unit 2 turns that question into the single most important idea of calculus: *how fast is a quantity changing at one exact instant?* You will see that the answer is a limit of slopes of secant lines, give it a name (the **derivative**), learn what it means graphically, numerically and in context, learn when it fails to exist, and then build a toolkit of rules that lets you differentiate most functions you meet on the exam in a few lines.

Topics.

2.1 Average and instantaneous rates of change
2.2 The derivative and derivative notation
2.3 Estimating derivatives from tables and graphs
2.4 Differentiability and continuity
2.5 The Power Rule
2.6 Constant, sum, difference and constant-multiple rules
2.7 Derivatives of $\sin x$, $\cos x$, e^x, $\ln x$
2.8 The Product Rule
2.9 The Quotient Rule
2.10 Derivatives of $\tan x$, $\cot x$, $\sec x$, $\csc x$

Big ideas. (1) The derivative $f'(a)$ is the limit of difference quotients; it is the slope of the tangent line and the instantaneous rate of change. (2) A function must be *continuous* to be differentiable, but continuity is not enough. (3) Differentiation rules are consequences of the limit definition, so every rule can be justified—and on the exam, justification matters.

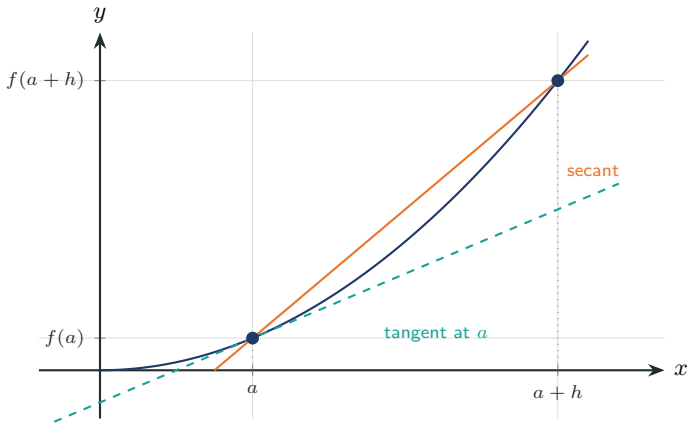
Topic 2.1 Defining Average and Instantaneous Rates of Change at a Point

From slope of a line to rate of change

For a straight line the steepness is the same everywhere, so "the slope" is a single number. For a curve the steepness changes from point to point, and the first job of calculus is to make sense of *the steepness of a curve at one point*. We build it in two stages.

The **average rate of change** of a function f over an interval $[a, b]$ compares how much the output changes with how much the input changes. It is the slope of the *secant line* through the points $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$. If $f(t)$ is a position, the average rate of change is the average velocity; if $f(t)$ is the amount of water in a tank in litres and t is in minutes, it is the average number of litres gained per minute. Always carry the units: *(units of output) per (unit of input)*.

To capture the rate of change *at the instant* $x = a$ we cannot use an interval of length zero—that would give $\frac{0}{0}$. Instead we let the second point slide toward the first: take the interval $[a, a+h]$ and let $h \to 0$. The secant lines swing toward a limiting line, the *tangent line*, and the limit of the slopes is the **instantaneous rate of change** of f at a. This is exactly the kind of limit you evaluated in Unit 1, and it nearly always has the indeterminate form $\frac{0}{0}$ that must be simplified algebraically (factor, expand, or use a conjugate) before the limit can be taken.



Secant slope $= \dfrac{f(a+h) - f(a)}{h}$ for $f(x) = x^2$, $a = 1$, $h = 2$. As $h \to 0$ the secant turns into the tangent.

Key definitions: rates of change

Average rate of change of f on $[a,b]$: $\dfrac{f(b)-f(a)}{b-a}$ (slope of the secant line).

Instantaneous rate of change of f at $x=a$ (when the limit exists):

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} \qquad \text{or equivalently} \qquad \lim_{x\to a}\frac{f(x)-f(a)}{x-a}.$$

This limit is the slope of the tangent line to the graph of f at $\big(a,f(a)\big)$. For position $s(t)$ it is the (instantaneous) velocity $v(t)$.

Example 1 Average versus instantaneous (with a graph)

Let $f(x)=x^2$ (the graph above). (a) Find the average rate of change of f on $[1,3]$. (b) Find the instantaneous rate of change of f at $x=1$.

Step 1. (a) $\dfrac{f(3)-f(1)}{3-1}=\dfrac{9-1}{2}=4$. This is the slope of the secant line drawn in the figure.

Step 2. (b) Write the limit: $\lim\limits_{h\to 0}\dfrac{f(1+h)-f(1)}{h}=\lim\limits_{h\to 0}\dfrac{(1+h)^2-1}{h}$.

Step 3. Expand and cancel: $\dfrac{1+2h+h^2-1}{h}=\dfrac{h(2+h)}{h}=2+h$ for $h\neq 0$.

Step 4. Take the limit: $\lim\limits_{h\to 0}(2+h)=2$. The tangent line at $x=1$ has slope 2 (dashed line in the figure), noticeably flatter than the secant of slope 4.

Example 2 Average rate of change from a table, with units

A cyclist's distance from the start of a time trial is $D(t)$ metres, where t is in seconds.

t (s)	0	10	25	40	60
$D(t)$ (m)	0	35	110	215	400

Find the average velocity on $[10,40]$ and on $[40,60]$, and say what the comparison tells you.

Step 1. $[10,40]$: $\dfrac{D(40)-D(10)}{40-10}=\dfrac{215-35}{30}=6$ metres per second.

Step 2. $[40,60]$: $\dfrac{D(60)-D(40)}{60-40}=\dfrac{400-215}{20}=9.25$ metres per second.

Step 3. The average velocity is larger on the later interval, so the cyclist is, on average, going faster late in the ride.

Example 3 Evaluating an instantaneous rate of change with a conjugate

Let $f(x)=\sqrt{x}$. Find the instantaneous rate of change of f at $x=4$.

Step 1. $f'(4)=\lim\limits_{h\to 0}\dfrac{\sqrt{4+h}-2}{h}$. Direct substitution gives $\frac{0}{0}$.

Step 2. Multiply by the conjugate: $\dfrac{\sqrt{4+h}-2}{h}\cdot\dfrac{\sqrt{4+h}+2}{\sqrt{4+h}+2}=\dfrac{(4+h)-4}{h\big(\sqrt{4+h}+2\big)}=\dfrac{1}{\sqrt{4+h}+2}$.

Step 3. Take the limit: $\lim\limits_{h\to 0}\dfrac{1}{\sqrt{4+h}+2}=\dfrac{1}{4}$.

At $x=4$, $\sqrt{x}$ is increasing at the instantaneous rate $\frac{1}{4}$ unit of output per unit of input.

Common errors

- **Substituting $h=0$ too early.** The quotient is $\frac{0}{0}$; simplify *first*, then take the limit.
- **Subtracting in different orders**: $\frac{f(b)-f(a)}{a-b}$ flips the sign. Keep "later minus earlier" in numerator and denominator.
- **Forgetting units**, or giving the units of f rather than "units of f per unit of x".
- **Treating the average rate as the instantaneous rate.** They agree only for linear functions.

AP Exam Tip

When asked to *interpret* a rate, write a complete sentence with units and the context: "At $t = 25$ seconds the cyclist's distance is increasing at about 6 metres per second." Use the word *approximately* when the value came from a table.

Check yourself 2.1

Let $f(x) = x^2 - 3x$ and $s(t) = t^3 - 6t$ (metres, t in seconds).

1. Find the average rate of change of f on $[1, 4]$.

2. Using the limit definition, find the instantaneous rate of change of f at $x = 2$.

3. Find the average velocity of s on $[1, 3]$, then write and evaluate a limit giving the velocity at $t = 2$.

Topic 2.2 Defining the Derivative of a Function and Using Derivative Notation

From a number to a function

The limit in Topic 2.1 gave one number for one input a. If we do not fix a but keep x general, the same limit produces a *new function*: its input is a number x and its output is the slope of the tangent to f at x. That function is the **derivative** of f, written f' (read "f prime"). Its domain consists of all x at which the limit exists, which may be smaller than the domain of f.

There are several notations for the same object, and the exam uses all of them. $f'(x)$ is Lagrange's notation. $\dfrac{dy}{dx}$ is Leibniz's notation for $y = f(x)$; it is read "the derivative of y with respect to x" and is *not* a fraction of two numbers (yet), but it reminds you that the derivative is a limit of $\frac{\Delta y}{\Delta x}$ and it carries the units "units of y per unit of x". $\dfrac{d}{dx}\big[f(x)\big]$ is the operator form: "differentiate what is inside the brackets". The value at a point is written $f'(a)$ or $\left.\dfrac{dy}{dx}\right|_{x=a}$.

The derivative has three equivalent faces, and AP questions move between them constantly: **graphical** (slope of the tangent line), **numerical/contextual** (instantaneous rate of change with units), and **analytic** (the limit of the difference quotient). Once you know $f(a)$ and $f'(a)$ you can write the tangent line immediately: $y - f(a) = f'(a)(x - a)$. The *normal line* is perpendicular to the tangent, with slope $-\dfrac{1}{f'(a)}$ when $f'(a) \neq 0$.

Key definitions: the derivative

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}, \qquad f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

Notation: $f'(x) = \dfrac{dy}{dx} = y' = \dfrac{d}{dx}\big[f(x)\big]$. Tangent line at $x = a$: $y - f(a) = f'(a)\,(x - a)$.

Meaning of $f'(a)$: slope of the tangent at $x = a$; instantaneous rate of change of f with respect to x at a, in (units of f) per (unit of x).

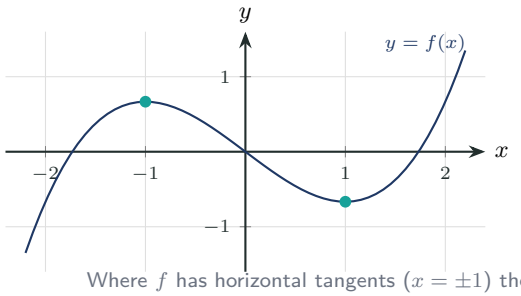
Example 1 The definition on a cubic, and the graphs of f and f'

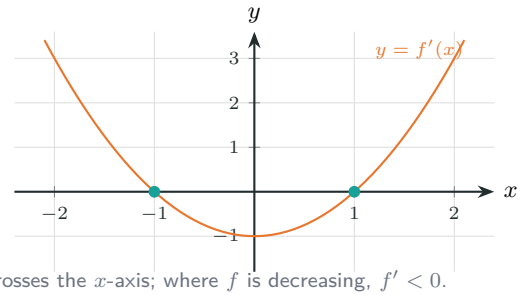
Use the definition to find $f'(x)$ for $f(x) = \dfrac{x^3}{3} - x$.

Step 1. $f(x+h) - f(x) = \dfrac{(x+h)^3 - x^3}{3} - h = \dfrac{3x^2h + 3xh^2 + h^3}{3} - h = x^2h + xh^2 + \dfrac{h^3}{3} - h.$

Step 2. Divide by h: $\dfrac{f(x+h) - f(x)}{h} = x^2 + xh + \dfrac{h^2}{3} - 1.$

Step 3. Let $h \to 0$: $f'(x) = x^2 - 1.$





Where f has horizontal tangents ($x = \pm 1$) the graph of f' crosses the x-axis; where f is decreasing, $f' < 0$.

Example 2 Derivative of $\sqrt{x}$ and a tangent line

(a) Use the definition to find $f'(x)$ for $f(x) = \sqrt{x}$, $x > 0$. (b) Write an equation of the tangent line at $x = 9$.

Step 1. $f'(x) = \lim_{h\to 0} \dfrac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \dfrac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} = \lim_{h\to 0} \dfrac{h}{h\left(\sqrt{x+h} + \sqrt{x}\right)} = \dfrac{1}{2\sqrt{x}}$.

Step 2. $f(9) = 3$ and $f'(9) = \dfrac{1}{6}$.

Step 3. Tangent line: $y - 3 = \dfrac{1}{6}(x - 9)$.

Example 3 Recognising a limit as a derivative; interpreting f' in context

(a) The limit $\lim_{h\to 0} \dfrac{\cos\left(\frac{\pi}{3} + h\right) - \frac{1}{2}}{h}$ is the derivative of which function at which point? Evaluate it using the fact $\dfrac{d}{dx}\cos x = -\sin x$.

(b) $A(t)$ is the number of litres of water in a tank at time t minutes, and $A'(5) = -12$. Interpret the statement.

Step 1. (a) Compare with $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$: here $f(x) = \cos x$, $a = \frac{\pi}{3}$, and $f(a) = \cos\frac{\pi}{3} = \frac{1}{2}$. The limit is $f'\left(\frac{\pi}{3}\right) = -\sin\frac{\pi}{3} = -\dfrac{\sqrt{3}}{2}$.

Step 2. (b) At $t = 5$ minutes the amount of water in the tank is decreasing at a rate of 12 litres per minute.

Common errors

- Writing $f'(x) = \dfrac{f(x+h) - f(x)}{h}$ *without the limit*. The difference quotient is only an average rate.
- Writing the tangent line as $y = f'(a)(x - a)$ and leaving out $f(a)$.
- Confusing $f'(a)$ (a number) with $\big[f(a)\big]'$: the derivative of the constant $f(a)$ is 0. Differentiate first, then substitute.
- Reporting $f'(a)$ with the units of f instead of units of f per unit of x.

AP Exam Tip

For "Write an equation of the tangent line" the scoring guide gives one point for the slope $f'(a)$ and one for a correct equation using the point $\big(a, f(a)\big)$. Leave the answer in point-slope form; simplifying is not required and can only introduce arithmetic errors.

Check yourself 2.2

1. Use the limit definition to find $f'(x)$ for $f(x) = 3x^2 - x$.

2. Use the definition to show that if $f(x) = \dfrac{1}{x}$ then $f'(x) = -\dfrac{1}{x^2}$, and write an equation of the tangent line at $x = 2$.

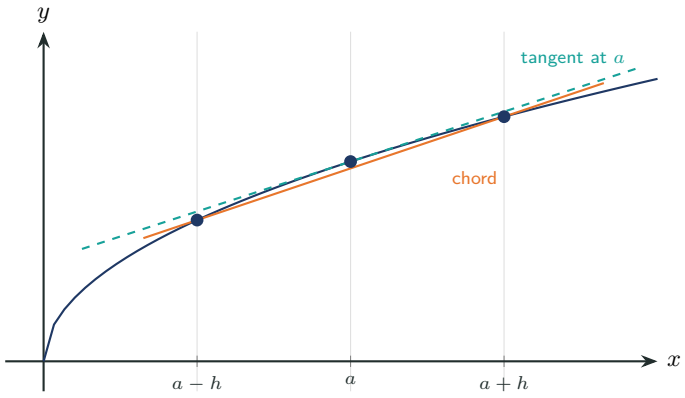
3. $\lim_{x\to 2} \dfrac{x^4 - 16}{x - 2}$ equals $f'(2)$ for $f(x) = x^4$. Evaluate the limit.

Topic 2.3 Estimating Derivatives of a Function at a Point

When you have data instead of a formula

Many exam questions give you a *table of values* or a *graph* rather than a formula. You cannot take a limit then, but you can estimate it: choose a short interval around the point of interest and compute the average rate of change over it. The closer the two points are to a, the better the estimate, so use the **two table entries closest to a on either side** of it.

When the table has an entry on each side of a, the best single estimate is $\dfrac{f(b)-f(c)}{b-c}$ for the nearest $c<a<b$. If the table is evenly spaced around a ($c=a-h$, $b=a+h$) this is the **symmetric difference quotient** $\dfrac{f(a+h)-f(a-h)}{2h}$, which is usually more accurate than the one-sided quotient $\dfrac{f(a+h)-f(a)}{h}$ because the errors on the two sides tend to cancel. On a graphing calculator the same idea underlies the numerical-derivative command `nDeriv` / `d/dx`, which is allowed on calculator-active parts when you report the result to three decimal places and say it is a calculator result. From a **graph**, estimate the slope of the tangent by reading two points on the tangent line (or by estimating rise over run over a small interval).



Chord from $a-h$ to $a+h$ for $f(x)=\sqrt{x}$, $a=4$, $h=2$: its slope ≈ 0.259 is already very close to $f'(4)=0.25$.

Key idea: estimating $f'(a)$

Table: $f'(a)\approx\dfrac{f(b)-f(c)}{b-c}$, using the entries $c<a<b$ *closest* to a. **Evenly spaced:** $f'(a)\approx\dfrac{f(a+h)-f(a-h)}{2h}$.

Graph: $f'(a)\approx$ slope of the tangent line, read from two points on that line. **Calculator:** `nDeriv` / `d/dx` (state it; three decimals).

Example 1 Estimating from an unevenly spaced table

Use the cyclist's table from Topic 2.1 to estimate $D'(30)$ and interpret it.

t (s)	0	10	25	40	60
$D(t)$ (m)	0	35	110	215	400

Step 1. The entries closest to $t=30$ on either side are $t=25$ and $t=40$.

Step 2. $D'(30)\approx\dfrac{D(40)-D(25)}{40-25}=\dfrac{215-110}{15}=7$.

Step 3. At $t=30$ seconds the cyclist's distance is increasing at approximately 7 metres per second.

Example 2 Shrinking h: why the estimate improves

Estimate $f'(2)$ for $f(x)=\ln x$ using $\dfrac{f(2+h)-f(2)}{h}$ for smaller and smaller h.

h	0.1	0.01	0.001
$\dfrac{\ln(2+h)-\ln 2}{h}$	0.4879	0.4988	0.4999

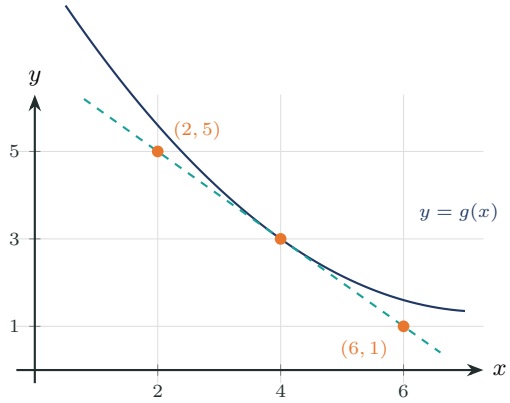
The quotients approach 0.5, so $f'(2) = 0.5$ (indeed $\frac{d}{dx}\ln x = \frac{1}{x}$ by Topic 2.7). With the symmetric quotient and $h = 0.01$: $\dfrac{\ln 2.01 - \ln 1.99}{0.02} \approx 0.50000$, already correct to five decimals.

Example 3 Estimating from a graph

The line tangent to the graph of g at $x = 4$ passes through $(2,\,5)$ and $(6,\,1)$. Estimate $g'(4)$ and write an equation of the tangent line at $x = 4$, given $g(4) = 3$.

Step 1. $g'(4) = \dfrac{1-5}{6-2} = -1$, read off the tangent line.

Step 2. Tangent line: $y - 3 = -1\,(x - 4)$.



Common errors

- Using table values that are *not* the closest ones (for example the endpoints of the table) when nearer entries exist.
- Dividing by the wrong interval length when the table is unevenly spaced; always divide by $b-c$, never by a fixed step.
- Giving a table-based estimate as an exact value. Say "approximately".
- Reading the slope of the *curve* between two plotted points rather than of the *tangent line*.

AP Exam Tip

An estimate should show the arithmetic: write $\dfrac{f(b)-f(c)}{b-c}$ with the numbers in it. A bare decimal receives no credit if it is slightly off; a correct setup with an arithmetic slip usually earns the setup point.

Check yourself 2.3

Table of a differentiable function f:

x	0	2	3	6	10
$f(x)$	5	9	10	4	-6

1. Estimate $f'(4)$ using the closest entries. **2.** Estimate $f'(8)$.
3. For $f(x) = x^3$, compute the symmetric difference quotient at $x = 2$ with $h = 0.1$ and compare with the exact value.

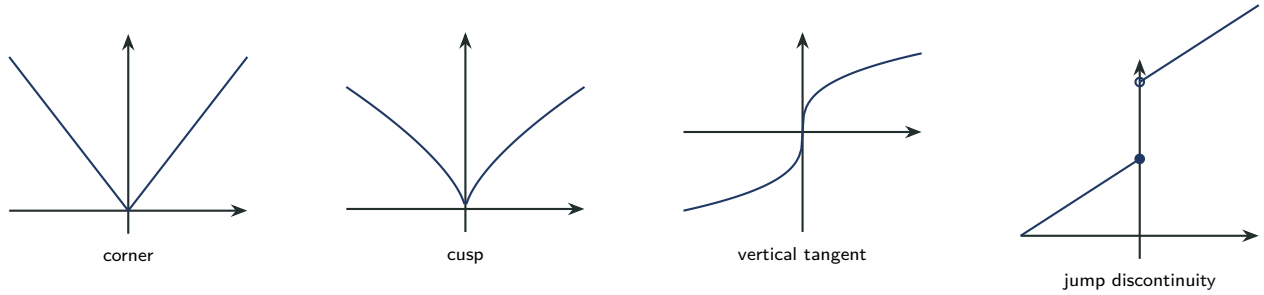
Topic 2.4 Connecting Differentiability and Continuity: Determining When Derivatives Do and Do Not Exist

Differentiable means "smooth enough to have a tangent"

A function is **differentiable at** $x = a$ if $f'(a)$ exists, that is, if the limit of the difference quotient exists as a finite number. Geometrically, if you zoom in on the graph at $(a, f(a))$ far enough, a differentiable function looks like a straight line (it is *locally linear*). The derivative exists exactly when the secant slopes from the left and from the right approach the *same finite number*. These two one-sided limits are the **left-hand and right-hand derivatives**.

There are four standard ways for $f'(a)$ to fail to exist, and the exam expects you to recognise each from a graph, a table or a formula:

- a **corner** (for example $|x|$ at 0): the one-sided slopes are different finite numbers;
- a **cusp** (for example $x^{2/3}$ at 0): the one-sided slopes tend to $+\infty$ and $-\infty$;
- a **vertical tangent** (for example $x^{1/3}$ at 0): the slopes tend to $+\infty$ (or $-\infty$) from both sides;
- a **discontinuity** (jump, removable or infinite): there is no tangent line at all.



The four ways a derivative can fail to exist at $x = 0$.

Differentiability implies continuity. If $f'(a)$ exists then $f(x) - f(a) = \dfrac{f(x) - f(a)}{x - a} \cdot (x - a) \to f'(a) \cdot 0 = 0$, so f is continuous at a. The statement is one-way: $|x|$ is continuous at 0 but not differentiable there. Hence "not continuous $\Rightarrow$ not differentiable" is always a valid justification, whereas "continuous" alone never proves differentiability.

For a **piecewise function** whose rule changes at $x = a$, check two things in order: (1) *continuity at* a: left value $=$ right value $= f(a)$; (2) *matching slopes*: the derivative of the left piece at a equals the derivative of the right piece at a. Both are needed, and skipping (1) is the most common mistake.

Key facts: differentiability

- f differentiable at $a \implies f$ continuous at a. (The converse is *false*.)
- $f'(a)$ exists $\iff \lim\limits_{h \to 0^-} \dfrac{f(a+h) - f(a)}{h} = \lim\limits_{h \to 0^+} \dfrac{f(a+h) - f(a)}{h}$ (finite).
- Failure types: corner, cusp, vertical tangent, discontinuity.
- Piecewise test at the boundary a: (1) continuous; (2) left slope $=$ right slope.

Example 1 A piecewise function: corner or smooth?

Let $f(x) = \begin{cases} x^2, & x \le 1 \\ 2x - 1, & x > 1 \end{cases}$ and $g(x) = \begin{cases} x^2, & x \le 1 \\ x, & x > 1. \end{cases}$ Decide whether each is differentiable at $x = 1$.

Step 1. Continuity. For f: left value $1^2 = 1$, right value $2(1) - 1 = 1$. For g: 1 and 1. Both are continuous at 1.
Step 2. Slopes. For f: left $\frac{d}{dx}x^2 = 2x = 2$ at $x = 1$, right $\frac{d}{dx}(2x - 1) = 2$. Equal, so f **is differentiable** at 1 with $f'(1) = 2$.
Step 3. For g: left slope 2, right slope 1. Different, so g has a corner and **is not differentiable** at $x = 1$.

Example 2 Continuous but not differentiable

Let $f(x) = \begin{cases} kx^2, & x \le 2 \\ 3x - 2, & x > 2. \end{cases}$ (a) Find k so that f is continuous at $x = 2$. (b) With this k, is f differentiable at $x = 2$? Justify.

Step 1. (a) Left limit $4k$; right limit $3(2) - 2 = 4$. Continuity requires $4k = 4$, so $k = 1$.
Step 2. (b) Left-hand derivative: $\frac{d}{dx}x^2 = 2x = 4$ at $x = 2$. Right-hand derivative: $\frac{d}{dx}(3x - 2) = 3$.

Step 3. Since $4 \neq 3$, f is **not differentiable at** $x = 2$: the graph has a corner there, although f is continuous.

Example 3 A vertical tangent from the difference quotient

Show that $f(x) = x^{1/3}$ is not differentiable at $x = 0$.

Step 1. $\dfrac{f(0+h) - f(0)}{h} = \dfrac{h^{1/3}}{h} = h^{-2/3} = \dfrac{1}{h^{2/3}}$.

Step 2. As $h \to 0^+$ and as $h \to 0^-$ this quotient increases without bound ($h^{2/3} > 0$ and tends to 0), so the limit is $+\infty$: it does not exist as a finite number.

Step 3. f is continuous at 0 but has a vertical tangent there, so $f'(0)$ does not exist.

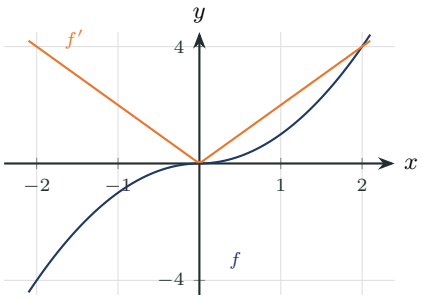
Example 4 An absolute value that does *not* break differentiability

Let $f(x) = x\,|x|$. Is f differentiable at $x = 0$? Find $f'(x)$.

Step 1. $\dfrac{f(0+h) - f(0)}{h} = \dfrac{h|h|}{h} = |h|$ for $h \neq 0$.

Step 2. $\lim\limits_{h \to 0^-} |h| = 0 = \lim\limits_{h \to 0^+} |h|$, so $f'(0) = 0$: f **is** differentiable at 0. (The graph has no corner: it passes through the origin with a horizontal tangent.)

Step 3. For $x > 0$, $f(x) = x^2$ and $f'(x) = 2x$; for $x < 0$, $f(x) = -x^2$ and $f'(x) = -2x$. Hence $f'(x) = 2|x|$ for all x.



Common errors

- **"It is continuous, so it is differentiable."** False ($|x|$, $x^{1/3}$, $x^{2/3}$ at 0).
- Comparing only the *slopes* of a piecewise function and forgetting to check that the pieces meet.
- Matching the *function values* of the pieces and thinking that is enough for differentiability.
- Believing that a vertical tangent means "derivative is ∞, so it exists". On the AP exam the derivative *does not exist* there.

AP Exam Tip

Use the language of the scoring guides: "f is not differentiable at $x = 2$ because the left-hand derivative 4 is not equal to the right-hand derivative 3", or "f is not continuous at $x = 2$, so it is not differentiable there". Do not just say "there is a corner"—say what you computed.

Check yourself 2.4

1. Is $f(x) = |x - 3|$ differentiable at $x = 3$? Use one-sided slopes.

2. Find a and b so that $g(x) = \begin{cases} x^2, & x \leq 1 \\ ax + b, & x > 1 \end{cases}$ is differentiable at $x = 1$.

3. Is $h(x) = x^{2/3}$ continuous at 0? Differentiable at 0? Explain with the difference quotient.

Topic 2.5 Applying the Power Rule

The first shortcut

Computing a derivative from the limit definition every time would be hopelessly slow. The first and most useful shortcut covers powers of x. Where does it come from? For a positive integer n the binomial theorem gives $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n$, so $\dfrac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1} \longrightarrow nx^{n-1}$ as $h \to 0$. The result is valid for *every real exponent*, so it also handles negative and fractional powers.

The practical difficulty on the exam is not the rule itself but **rewriting** the function first. Roots become fractional powers ($\sqrt[3]{x^2} = x^{2/3}$), and reciprocals of a power become negative exponents ($\dfrac{1}{x^4} = x^{-4}$). Only after rewriting do you bring down the exponent and subtract 1. Remember that you subtract 1 from the exponent *algebraically*: x^{-2} becomes $-2x^{-3}$, and $x^{1/2}$ becomes $\frac{1}{2}x^{-1/2}$.

Power Rule

For any real number n: $\dfrac{d}{dx}\left[x^n\right] = n\,x^{n-1}$.

Rewriting: $\sqrt[m]{x^k} = x^{k/m}$, $\dfrac{1}{x^n} = x^{-n}$, $\dfrac{c}{\sqrt{x}} = c\,x^{-1/2}$. The derivative of x is 1.

Example 1 Rewrite, then differentiate

Function	Rewritten	Derivative
$f(x) = x^7$	x^7	$7x^6$
$g(x) = \dfrac{1}{x^3}$	x^{-3}	$-3x^{-4} = -\dfrac{3}{x^4}$
$h(x) = \sqrt[3]{x^2}$	$x^{2/3}$	$\frac{2}{3}x^{-1/3} = \dfrac{2}{3\sqrt[3]{x}}$
$k(x) = \dfrac{1}{\sqrt{x}}$	$x^{-1/2}$	$-\frac{1}{2}x^{-3/2} = -\dfrac{1}{2x\sqrt{x}}$

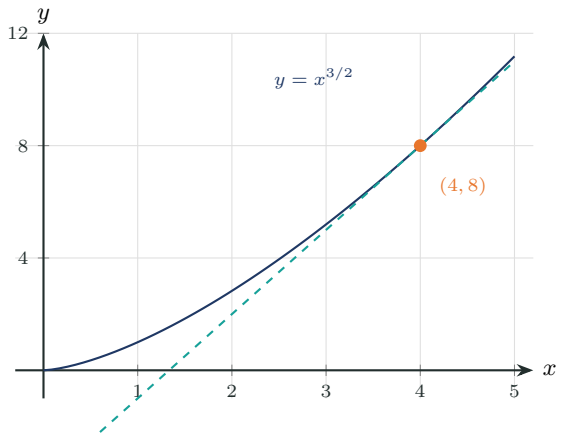
Example 2 Tangent line to $y = x^{3/2}$ (with a graph)

Write an equation of the line tangent to $y = x^{3/2}$ at $x = 4$.

Step 1. $y(4) = 4^{3/2} = 8$.

Step 2. $\dfrac{dy}{dx} = \frac{3}{2}x^{1/2}$, so $\left.\dfrac{dy}{dx}\right|_{x=4} = \frac{3}{2}\cdot 2 = 3$.

Step 3. Tangent line: $y - 8 = 3(x-4)$, that is $y = 3x - 4$.



Example 3 A tangent line with a prescribed slope

At which points on the graph of $f(x) = x^3$ is the tangent line parallel to the line $y = 12x - 5$?

Step 1. Parallel lines have equal slopes, so we need $f'(x) = 12$.

Step 2. $f'(x) = 3x^2 = 12 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$.
Step 3. The points are $(2, 8)$ and $(-2, -8)$.

Common errors

- Differentiating $\dfrac{1}{x^3}$ as $\dfrac{1}{3x^2}$: you must rewrite as x^{-3} first (answer $-3x^{-4}$).
- Exponent arithmetic: $\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$, not $\frac{1}{2}x^{1/2}$ and not $\frac{1}{2}x^{-3/2}$.
- Differentiating a constant c (or π, or e^2) as c or 1: it is 0.
- Using $\dfrac{d}{dx}x^n = nx^{n-1}$ for 2^x or e^x—the variable is in the *exponent*, not the base.

AP Exam Tip

Do not simplify a correct derivative into a messy form unless asked; the scoring guide accepts $\frac{3}{2}x^{1/2}$ and $\frac{3}{2}\sqrt{x}$ alike. But when you *substitute* a value, show the substitution.

Check yourself 2.5

1. Differentiate $f(x) = x^7$, $g(x) = x^{-3}$, $h(x) = \sqrt[3]{x^2}$ (use the Power Rule after rewriting).
2. If $f(x) = \dfrac{1}{\sqrt{x^3}}$, find $f'(4)$.
3. At what value of x is the tangent line to $y = x^4$ of slope 32?

Topic 2.6 Derivative Rules: Constant, Sum, Difference, and Constant Multiple

Differentiation is linear

Most functions on the exam are built from simpler pieces by adding, subtracting and scaling. Differentiation behaves very well with these operations, because limits do: the limit of a sum is the sum of the limits, and constants factor out of limits. This gives the **linearity** of the derivative. It means you may differentiate a polynomial (or any sum of power terms) *term by term*, keeping each coefficient as it is.

Remember two things. First, the derivative of a constant is 0, since a horizontal line has slope 0. Second, these rules are for *sums and constant multiples only*. The derivative of a product is not the product of the derivatives; that is the subject of Topic 2.8. When the denominator is a single term, you may split first: $\dfrac{x^3 + 4}{x} = x^2 + 4x^{-1}$ and differentiate term by term, which avoids the Quotient Rule altogether.

A very common exam task is **horizontal tangent lines**: set $f'(x) = 0$ and solve. The solutions are candidate high and low points—a theme that Unit 5 develops fully. Another is the connection with motion: if $s(t)$ is position then velocity is $v(t) = s'(t)$, and the particle is momentarily at rest when $v(t) = 0$.

Basic differentiation rules

For constants c, k and differentiable f, g:

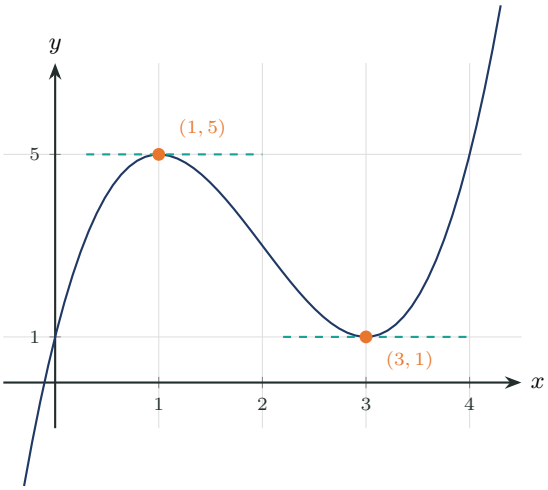
$$\frac{d}{dx}[c] = 0, \qquad \frac{d}{dx}\big[kf(x)\big] = k\,f'(x), \qquad \frac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x).$$

Example 1 Term by term

Find $f'(x)$ for $f(x) = 4x^3 - 6\sqrt{x} + \dfrac{5}{x^2} + 7$.
Step 1. Rewrite: $f(x) = 4x^3 - 6x^{1/2} + 5x^{-2} + 7$.
Step 2. Differentiate each term: $f'(x) = 12x^2 - 6 \cdot \frac{1}{2}x^{-1/2} + 5(-2)x^{-3} + 0$.
Step 3. Simplify: $f'(x) = 12x^2 - \dfrac{3}{\sqrt{x}} - \dfrac{10}{x^3}$.

Example 2 Horizontal tangents (with a graph)

Find the points on the graph of $f(x) = x^3 - 6x^2 + 9x + 1$ at which the tangent line is horizontal.
Step 1. $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x-1)(x-3)$.
Step 2. $f'(x) = 0$ when $x = 1$ or $x = 3$.
Step 3. $f(1) = 1 - 6 + 9 + 1 = 5$ and $f(3) = 27 - 54 + 27 + 1 = 1$. The points are $(1, 5)$ and $(3, 1)$.



Example 3 Position, velocity and a table

(a) A particle moves with position $s(t) = 2t^3 - 9t^2 + 12t$ for $t \geq 0$. When is it at rest?
(b) Let $h(x) = 2f(x) - 3g(x) + x^2$ where $f'(2) = 5$ and $g'(2) = -1$. Find $h'(2)$.
Step 1. (a) $v(t) = s'(t) = 6t^2 - 18t + 12 = 6(t-1)(t-2)$. The particle is at rest at $t = 1$ and $t = 2$.
Step 2. (b) $h'(x) = 2f'(x) - 3g'(x) + 2x$, so $h'(2) = 2(5) - 3(-1) + 2(2) = 10 + 3 + 4 = 17$.

Common errors

- Multiplying a constant factor instead of keeping it: $\dfrac{d}{dx}(5x^3) = 15x^2$, not $5 \cdot x^2$ and not $15x^3$.
- Differentiating $\dfrac{x^3 + 4}{x}$ "top and bottom" separately; split it into terms or use the Quotient Rule.
- Dropping a sign: $\dfrac{d}{dx}(-6\sqrt{x}) = -\dfrac{3}{\sqrt{x}}$.
- Solving $f'(x) = 0$ and stopping at x-values when the question asks for the *points* (you must also find $f(x)$).

AP Exam Tip

"Find the x-coordinates at which the tangent line is horizontal" earns one point for setting $f'(x) = 0$ and another for the correct solutions. Show both lines: $f'(x) = 3(x-1)(x-3)$ and $f'(x) = 0$.

Check yourself 2.6

1. If $f(x) = 4x^3 - 5x^2 + 7x - 9$, find $f'(1)$.
2. Differentiate $y = 3\sqrt{x} - \dfrac{6}{x^2} + \pi^2$ and evaluate $\dfrac{dy}{dx}$ at $x = 4$.
3. Find all x at which the tangent to $f(x) = x^3 - 3x^2 - 9x + 2$ is horizontal.

Topic 2.7 Derivatives of $\cos x$, $\sin x$, e^x, and $\ln x$

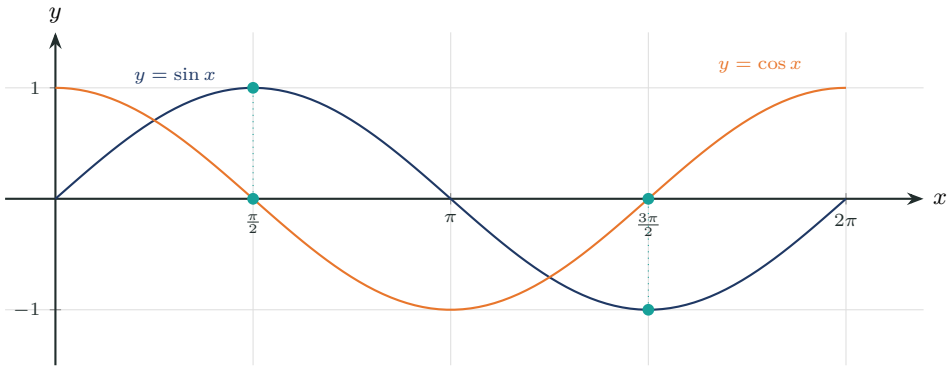
Four derivatives to know by heart

These four functions appear in almost every AP problem, and their derivatives must be recalled instantly. They are not guessed; each follows from the limit definition and two special limits from Unit 1, $\lim_{h\to 0}\dfrac{\sin h}{h}=1$ and $\lim_{h\to 0}\dfrac{\cos h-1}{h}=0$. For example,

$$\frac{\sin(x+h)-\sin x}{h}=\sin x\cdot\frac{\cos h-1}{h}+\cos x\cdot\frac{\sin h}{h}\ \longrightarrow\ \sin x\cdot 0+\cos x\cdot 1=\cos x.$$

The pattern for the sine and cosine is a cycle of period four: $\sin x\to\cos x\to-\sin x\to-\cos x\to\sin x$. Look at the graphs: where $\sin x$ is at a maximum its tangent is horizontal, and the graph of $\cos x$ (its derivative) is zero there; where $\sin x$ rises steepest ($x=0$) the slope is $1=\cos 0$.

The exponential function e^x is special because it is its own derivative: the slope of its graph at any point equals the height of the graph. The natural logarithm $\ln x$, defined only for $x>0$, has derivative $\dfrac{1}{x}$: it increases quickly near 0 and ever more slowly for large x, exactly as the slopes $\frac{1}{x}$ suggest. (The rule for $\ln x$ follows from the inverse relationship with e^x, which you will see in Unit 3.)



$\dfrac{d}{dx}\sin x=\cos x$: at $x=\frac{\pi}{2}$ and $x=\frac{3\pi}{2}$ the sine curve has a horizontal tangent and the cosine curve crosses zero.

Derivatives of the basic transcendental functions

$$\frac{d}{dx}[\sin x]=\cos x,\qquad \frac{d}{dx}[\cos x]=-\sin x,\qquad \frac{d}{dx}[e^x]=e^x,\qquad \frac{d}{dx}[\ln x]=\frac{1}{x}\ \ (x>0).$$

Angles are in radians. Combine with the rules of 2.5–2.6: constants factor out, sums are differentiated term by term.

Example 1 Term by term with the new functions

Let $f(x)=5\sin x-2\cos x+3e^x-4\ln x$. Find $f'(x)$ and $f'(\pi)$.

Step 1. $f'(x)=5\cos x-2(-\sin x)+3e^x-\dfrac{4}{x}=5\cos x+2\sin x+3e^x-\dfrac{4}{x}$.

Step 2. $f'(\pi)=5(-1)+2(0)+3e^{\pi}-\dfrac{4}{\pi}=-5+3e^{\pi}-\dfrac{4}{\pi}$.

Example 2 Horizontal tangent of $y=e^x-2x$

For what x is the tangent line to $y=e^x-2x$ horizontal, and what is the y-coordinate there?

Step 1. $\dfrac{dy}{dx}=e^x-2$.

Step 2. $e^x-2=0\ \Rightarrow\ e^x=2\ \Rightarrow\ x=\ln 2$.

Step 3. $y(\ln 2)=e^{\ln 2}-2\ln 2=2-2\ln 2\approx 0.614$.

Example 3 A tangent line to $y=\ln x$

Write an equation of the line tangent to $y=\ln x$ at $x=e$.

Step 1. $y(e)=\ln e=1$ and $y'=\dfrac{1}{x}$, so the slope is $\dfrac{1}{e}$.

Step 2. $y - 1 = \dfrac{1}{e}(x - e)$, which simplifies to $y = \dfrac{x}{e}$—a line through the origin.

Common errors

- **Sign of the cosine derivative**: $\dfrac{d}{dx}\cos x = -\sin x$, not $+\sin x$.
- Using the Power Rule on e^x: $\dfrac{d}{dx}e^x \neq xe^{x-1}$.
- Writing $\dfrac{d}{dx}\ln x = \ln x$ or $\dfrac{1}{\ln x}$ (it is $\frac{1}{x}$).
- Working in degrees: all derivative formulas above assume radian measure.

AP Exam Tip

Both $\sin$ and $\cos$ and e^x, $\ln x$ are on every formula list you must memorise—they are *not* given on the exam. For "Write an equation of the tangent line" evaluate trigonometric values exactly ($\cos\frac{\pi}{2} = 0$) unless the problem requires a decimal.

Check yourself 2.7

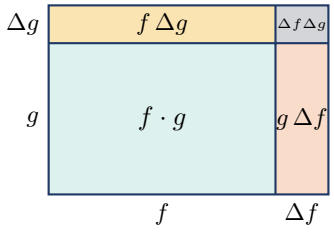
1. If $f(x) = 3\sin x - 2\cos x + 5e^x$, find $f'(0)$.
2. If $g(x) = 4\ln x - e^x$, find $g'(1)$.
3. Write an equation of the tangent line to $y = \cos x$ at $x = \dfrac{\pi}{2}$.

Topic 2.8 The Product Rule

Differentiating a product of two functions

Is the derivative of $f(x)g(x)$ equal to $f'(x)g'(x)$? Test it: $f(x) = g(x) = x$ gives a product x^2 whose derivative is $2x$, while $f'g' = 1 \cdot 1 = 1$. The rule fails, and the correct formula has two terms because the product can change in two ways at once.

Think of a rectangle with sides $f(x)$ and $g(x)$, so that its area is $f \cdot g$. When x increases by a small amount h, side f grows by about $f'h$ and side g grows by about $g'h$. The new area exceeds the old one by two thin strips, $g \cdot f'h$ and $f \cdot g'h$, plus a tiny corner $f'g'h^2$. Dividing by h and letting $h \to 0$, the corner vanishes and only the two strips remain.



Area picture of the Product Rule: $\Delta(fg) \approx g\,\Delta f + f\,\Delta g$; the corner $\Delta f\,\Delta g$ is negligible.

Use the Product Rule when you cannot multiply out (for instance x^2e^x or $\sin x \ln x$). When the product is a simple polynomial, such as $(x+1)(x+2)$, multiplying first is often quicker and gives the same answer. Always say clearly which factor you call f and which g.

Product Rule

$$\frac{d}{dx}\big[f(x)\,g(x)\big] = f'(x)\,g(x) + f(x)\,g'(x)$$

In words: "derivative of the first times the second, plus the first times the derivative of the second".

Example 1 Product of an algebraic and a transcendental factor

Find $\dfrac{dy}{dx}$ for (a) $y = x^3\sin x$ and (b) $y = \sqrt{x}\,e^x$.

Step 1. (a) Let $f = x^3$, $g = \sin x$. Then $\dfrac{dy}{dx} = 3x^2 \sin x + x^3 \cos x$.

Step 2. (b) Let $f = x^{1/2}$, $g = e^x$. Then $\dfrac{dy}{dx} = \frac{1}{2}x^{-1/2}e^x + x^{1/2}e^x = e^x\left(\dfrac{1}{2\sqrt{x}} + \sqrt{x}\right)$.

Example 2 A product from a table

The table gives values of differentiable functions f and g and their derivatives. Let $h(x) = f(x)\,g(x)$. Find $h'(2)$ and $h'(5)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	3	-1	4	5
5	-2	6	1	-3

Step 1. $h'(x) = f'(x)g(x) + f(x)g'(x)$.
Step 2. $h'(2) = (-1)(4) + (3)(5) = -4 + 15 = 11$.
Step 3. $h'(5) = (6)(1) + (-2)(-3) = 6 + 6 = 12$.

Example 3 A tangent line

Write an equation of the line tangent to $y = (x+2)e^x$ at $x = 0$.
Step 1. $y(0) = 2e^0 = 2$.
Step 2. $\dfrac{dy}{dx} = 1 \cdot e^x + (x+2)e^x = (x+3)e^x$, so the slope at $x = 0$ is 3.
Step 3. Tangent line: $y - 2 = 3x$, that is $y = 3x + 2$.

Common errors

- **Product of derivatives**: $\dfrac{d}{dx}(fg) = f'g'$ is false.
- Forgetting one of the two terms (differentiating only one factor).
- In a table problem, mixing up values: use $f'(a)\,g(a) + f(a)\,g'(a)$ with every entry taken at the *same* $x = a$.
- Writing the rule with a minus sign (that is the Quotient Rule).

AP Exam Tip

On table problems, write the formula first and substitute second. Even if an arithmetic error occurs, the scoring guide awards the point for the correct Product Rule setup, "$f'(2)g(2) + f(2)g'(2)$".

Check yourself 2.8

1. Find $\dfrac{d}{dx}\left[x^2 e^x\right]$ and evaluate it at $x = 1$.
2. Find $\dfrac{d}{dx}\left[x \ln x\right]$ and evaluate it at $x = e$.
3. If $h = fg$ with $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, $g'(2) = 5$, find $h'(2)$.

Topic 2.9 The Quotient Rule

Differentiating a ratio

For a quotient the derivative is again not the quotient of the derivatives. Using the Product Rule on $f \cdot \frac{1}{g}$ (or the limit definition), one obtains the **Quotient Rule**, whose numerator has a *minus* sign. Because the order of subtraction matters, memorise it as: "low d-high minus high d-low, over low squared", with "low" the denominator and "high" the numerator.

Before applying the rule, ask whether you need it. If the denominator is a single term, rewrite and use the Power Rule: $\dfrac{x^2+3x}{x} = x+3$. If the numerator is a constant, $\dfrac{5}{x^2+1}$ can be written $5(x^2+1)^{-1}$, but its derivative needs the Chain Rule (Unit 3), so for now use the Quotient Rule with $f=5$, $f'=0$. After using the rule, simplify the numerator—often a lot cancels—and factor if the question asks where the derivative is zero (a fraction is zero only where its *numerator* is zero and the denominator is not).

Quotient Rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{\big[g(x)\big]^2}, \qquad g(x) \neq 0.$$

Example 1 Algebraic quotient

Find $f'(x)$ for $f(x) = \dfrac{x^2-1}{x^2+1}$.

Step 1. Numerator $= x^2-1$ with derivative $2x$; denominator $= x^2+1$ with derivative $2x$.

Step 2. $f'(x) = \dfrac{(2x)(x^2+1) - (x^2-1)(2x)}{(x^2+1)^2}$.

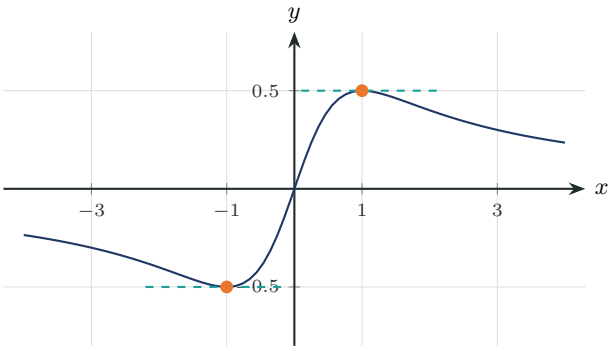
Step 3. Numerator: $2x^3 + 2x - 2x^3 + 2x = 4x$. Therefore $f'(x) = \dfrac{4x}{(x^2+1)^2}$.

Example 2 Horizontal tangents of $f(x) = \dfrac{x}{x^2+1}$ (with a graph)

Step 1. $f'(x) = \dfrac{1\cdot(x^2+1) - x\cdot 2x}{(x^2+1)^2} = \dfrac{1-x^2}{(x^2+1)^2}$.

Step 2. The denominator is never 0, so $f'(x)=0$ only when $1-x^2=0$, that is $x=\pm 1$.

Step 3. $f(1) = \frac{1}{2}$ and $f(-1) = -\frac{1}{2}$: the horizontal tangents are at $\left(1, \frac{1}{2}\right)$ and $\left(-1, -\frac{1}{2}\right)$.



Example 3 A quotient from a table

With $f(5)=10$, $f'(5)=-3$, $g(5)=2$, $g'(5)=4$, find $q'(5)$ for $q=\dfrac{f}{g}$ and $r'(5)$ for $r=\dfrac{g}{f}$.

Step 1. $q'(5) = \dfrac{f'(5)g(5) - f(5)g'(5)}{[g(5)]^2} = \dfrac{(-3)(2) - (10)(4)}{4} = \dfrac{-46}{4} = -11.5.$

Step 2. $r'(5) = \dfrac{g'(5)f(5) - g(5)f'(5)}{[f(5)]^2} = \dfrac{(4)(10) - (2)(-3)}{100} = \dfrac{46}{100} = 0.46.$

Common errors

- Reversing the subtraction: the numerator is *(derivative of top)(bottom)* minus *(top)(derivative of bottom)*.
- Forgetting to square the denominator, or squaring only part of it.
- Using the Quotient Rule where rewriting would do ($\frac{x^3+4}{x}$), then making an algebra error.
- Cancelling term-by-term in a sum: $\dfrac{x^2+1}{x^2+x} \neq \dfrac{1}{x}$.

AP Exam Tip

You do not need to expand $(x^2+1)^2$ in the denominator; leave it. But you *do* need to simplify the numerator enough to show when it is 0 if the question asks about horizontal tangents.

Check yourself 2.9

1. Differentiate $y = \dfrac{2x-1}{x+3}$ and simplify.

2. Find $\dfrac{d}{dx}\left[\dfrac{e^x}{x^2}\right]$ and evaluate it at $x = 2$.

3. If $f(1) = 6$, $f'(1) = 2$, $g(1) = 3$, $g'(1) = -1$ and $h = \dfrac{f}{g}$, find $h'(1)$.

Topic 2.10 Finding the Derivatives of Tangent, Cotangent, Secant, and/or Cosecant Functions

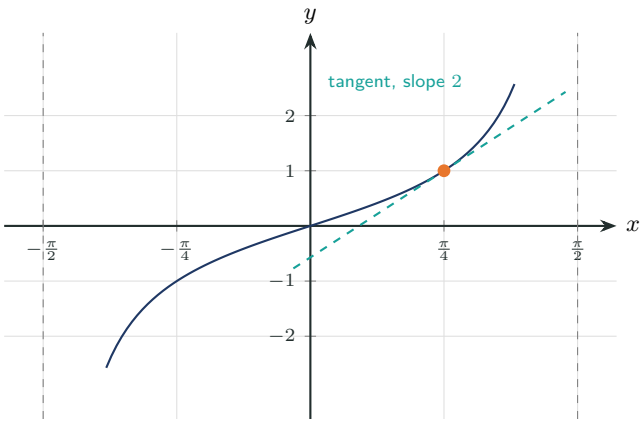
Four more trigonometric derivatives

The remaining four trigonometric functions are quotients of $\sin x$ and $\cos x$, so their derivatives follow from the Quotient Rule and the identity $\sin^2 x + \cos^2 x = 1$. For example

$$\frac{d}{dx}\tan x = \frac{d}{dx}\frac{\sin x}{\cos x} = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

Because the formulas are tested directly, learn them as a set. There is a helpful pattern: the derivatives of the three "co" functions ($\cos$, $\cot$, $\csc$) all carry a **minus sign**, and each derivative of a secant or cosecant keeps the original function and multiplies by the matching tangent or cotangent.

The graph of $\tan x$ explains $\sec^2 x \geq 1$: the tangent function is always increasing (where it exists) and its slope is at least 1, with slope exactly 1 at $x = 0$ and growing without bound as the graph approaches its vertical asymptotes at $x = \pm\frac{\pi}{2}$.



$y = \tan x$ and its tangent line at $x = \frac{\pi}{4}$; slope $\sec^2 \frac{\pi}{4} = 2$.

Derivatives of the other trigonometric functions

$$\frac{d}{dx}[\tan x] = \sec^2 x, \qquad \frac{d}{dx}[\cot x] = -\csc^2 x,$$
$$\frac{d}{dx}[\sec x] = \sec x \tan x, \qquad \frac{d}{dx}[\csc x] = -\csc x \cot x.$$

Example 1 Deriving $\frac{d}{dx}\sec x$

Use the Quotient Rule on $\sec x = \dfrac{1}{\cos x}$.

Step 1. $\dfrac{d}{dx}\dfrac{1}{\cos x} = \dfrac{0\cdot\cos x - 1\cdot(-\sin x)}{\cos^2 x} = \dfrac{\sin x}{\cos^2 x}$.

Step 2. Split: $\dfrac{\sin x}{\cos^2 x} = \dfrac{1}{\cos x}\cdot\dfrac{\sin x}{\cos x} = \sec x \tan x$.

Example 2 Tangent line to $y = \tan x$ (with a graph)

Write an equation of the line tangent to $y = \tan x$ at $x = \dfrac{\pi}{4}$.

Step 1. $y\left(\frac{\pi}{4}\right) = \tan\frac{\pi}{4} = 1$.

Step 2. $\dfrac{dy}{dx} = \sec^2 x$, so the slope is $\sec^2\frac{\pi}{4} = \left(\sqrt{2}\right)^2 = 2$.

Step 3. Tangent line: $y - 1 = 2\left(x - \dfrac{\pi}{4}\right)$.

Example 3 Mixed rules

Differentiate (a) $y = x\sec x$ and (b) $y = e^x \cot x$.

Step 1. (a) Product Rule: $\dfrac{dy}{dx} = 1\cdot\sec x + x\sec x\tan x = \sec x\,(1 + x\tan x)$.

Step 2. (b) Product Rule: $\dfrac{dy}{dx} = e^x\cot x + e^x(-\csc^2 x) = e^x\left(\cot x - \csc^2 x\right)$.

Common errors

- Dropping the minus sign for $\cot$ and $\csc$.
- Writing $\dfrac{d}{dx}\tan x = \sec x$ or $\sec x\tan x$. The correct answer is $\sec^2 x$.
- Writing $\sec^2 x$ as $\sec(x^2)$; $\sec^2 x$ means $(\sec x)^2$.
- Evaluating on a calculator in degree mode when the question uses radians.

AP Exam Tip

If you forget a formula in the exam, you can reconstruct it in seconds from $\tan x = \frac{\sin x}{\cos x}$ and the Quotient Rule—but do it on scratch paper, and then write only the result in your answer.

Check yourself 2.10

1. Find $\dfrac{d}{dx}\left[\tan x - 2\sec x\right]$ and evaluate at $x = \dfrac{\pi}{4}$.

2. Find $\dfrac{d}{dx}\left[x\cot x\right]$.

3. Find the slope of the tangent to $y = \sec x$ at $x = \dfrac{\pi}{3}$.

Unit 2 summary: rules to know without notes

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

Tangent line: $y - f(a) = f'(a)(x-a)$

$\sin' = \cos,\ \cos' = -\sin,\ (e^x)' = e^x,\ (\ln x)' = \frac{1}{x}$

$\tan' = \sec^2,\ \cot' = -\csc^2$

$\sec' = \sec\tan,\ \csc' = -\csc\cot$

$$\frac{d}{dx}x^n = nx^{n-1}$$

$(cf)' = cf',\ (f \pm g)' = f' \pm g'$

$(fg)' = f'g + fg'$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Differentiable $\Rightarrow$ continuous (not conversely)

Unit 2 Practice: Exam-Style Questions

Part A – Multiple choice, no calculator

14 questions. Choose the best answer.

1 $\lim_{h\to 0} \dfrac{(2+h)^3 - 8}{h}$ is

(A) 0 **(B)** 12

(C) 6 **(D)** 8

2 What is the average rate of change of $f(x) = x^3 - x$ on the interval $[0, 2]$?

(A) 2 **(B)** 6

(C) 3 **(D)** 11

3 If $f(x) = x^3 - 8\sqrt{x}$, then $f'(4) =$

(A) 46 **(B)** 40

(C) 50 **(D)** 44

4 $\dfrac{d}{dx}\left[x^3 \ln x\right]$ at $x = e$ is

(A) $3e$ **(B)** e^2

(C) $3e^2$ **(D)** $4e^2$

5 If $f(x) = \dfrac{x+1}{x-1}$, then $f'(3) =$

(A) 1 **(B)** -1

(C) $-\dfrac{1}{2}$ **(D)** $\dfrac{1}{2}$

6 Selected values of a differentiable function g are shown.

x	0	4	6	10
$g(x)$	3	10	16	20

Which of the following is the best estimate of $g'(5)$ from the table?

(A) 1.7 **(B)** 6

(C) $\dfrac{13}{6}$ **(D)** 3

7 Let $f(x) = \begin{cases} x^3, & x \le 1 \\ ax^2 + bx, & x > 1. \end{cases}$ For which values of a and b is f differentiable at $x = 1$?

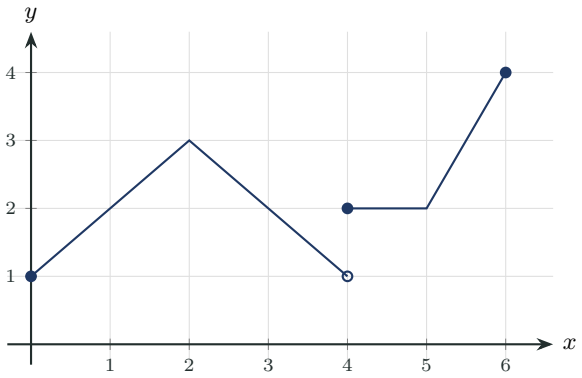
(A) $a = 2,\ b = -1$

(B) $a = 3,\ b = -2$

(C) $a = 1,\ b = 0$

(D) $a = 0,\ b = 3$

8 The graph of a function f on $0 \le x \le 6$ is shown. The graph consists of line segments; the open circle at $(4, 1)$ marks a point not on the graph, and $f(4) = 2$.



At how many values of x in the open interval $0 < x < 6$ is f *not* differentiable?

(A) 1 **(B)** 3
(C) 2 **(D)** 4

9 $\dfrac{d}{dx}\left[2\sec x - \tan x\right]$ at $x = \dfrac{\pi}{3}$ is

(A) $4\sqrt{3} + 4$ **(B)** 4
(C) $4\sqrt{3} - 4$ **(D)** $2\sqrt{3} - 4$

10 What is the slope of the line tangent to the graph of $y = \dfrac{e^x}{x}$ at $x = 2$?

(A) $\dfrac{e^2}{4}$ **(B)** $\dfrac{3e^2}{4}$
(C) $\dfrac{e^2}{2}$ **(D)** e^2

11 At what value of $x > 0$ does the graph of $f(x) = x^2 \ln x$ have a horizontal tangent line?

(A) 1 **(B)** $e^{-1/2}$
(C) $e^{1/2}$ **(D)** e^{-2}

12 The table gives values of a differentiable function f and its derivative f'. Let $k(x) = x^2 f(x)$. What is $k'(3)$?

x	$f(x)$	$f'(x)$
1	4	-1
3	-2	5

(A) 30 **(B)** 45
(C) 33 **(D)** 57

13 $\lim\limits_{x \to \pi/6} \dfrac{\sin x - \frac{1}{2}}{x - \frac{\pi}{6}}$ is

(A) 0 **(B)** $\dfrac{1}{2}$
(C) $-\dfrac{\sqrt{3}}{2}$ **(D)** $\dfrac{\sqrt{3}}{2}$

14 Let f be a function that is differentiable at $x = 3$. Which of the following must be true?

I. f is continuous at $x = 3$. II. $\lim_{x \to 3} f(x)$ exists. III. f' is continuous at $x = 3$.

(A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

Part B – Multiple choice, calculator allowed

7 questions. Radian mode; decimals to three places.

15 If $f(x) = e^x \ln x$, then $f'(2) =$

(A) 1.427 **(B)** 3.695
(C) 8.816 **(D)** 5.122

16 The temperature $H(t)$, in degrees Celsius, of a cup of coffee t minutes after it is poured is given at selected times.

t (min)	0	2	5	9	14
$H(t)$ (°C)	95	88	71	57	40

Using the average rate of change over the smallest interval of the table that contains $t = 7$, the best estimate of $H'(7)$, in degrees Celsius per minute, is

(A) -3.5 **(B)** -4.43
(C) -5.67 **(D)** 3.5

17 At what value of x is the line tangent to the graph of $y = x^2 + e^x$ of slope 5?

(A) 1.241 **(B)** 1.609
(C) 2.500 **(D)** 1.059

18 If $f(x) = x \tan x$, where x is in radians, then $f'(2) =$

(A) 2.037 **(B)** 9.364
(C) 3.589 **(D)** 5.774

19 For $x > 1$, let $g(x) = \dfrac{x^2}{\ln x}$. What is $g'(3)$?

(A) -2.976 **(B)** 3.269
(C) 2.976 **(D)** 18.000

20 The line tangent to the graph of $f(x) = x^3 \ln x$ at $x = 2$ crosses the y-axis at $y =$

(A) -19.090 **(B)** -24.636
(C) 5.545 **(D)** 12.318

21 How many points on the graph of $f(x) = x + 2\sin x$, for $0 < x < 10$, have a horizontal tangent line?

(A) 2 **(B)** 4
(C) 6 **(D)** 3

Part C – Free response

2 questions, 9 points each.

22 **[9 points] (Calculator active.)** The amount of water in a reservoir, in thousands of litres, is modelled by $W(t) = 3t + t^2 \ln t$, where t is measured in hours, $1 \leq t \leq 6$.

(a) Find the average rate of change of W over the interval $1 \leq t \leq 4$. Show the computation that leads to your answer, and include units.

(b) Find $W'(t)$. Using correct units, interpret the meaning of $W'(2)$ in the context of the problem. Is the amount of water in the reservoir increasing or decreasing at $t = 2$? Give a reason for your answer.

(c) Write an equation for the line tangent to the graph of W at $t = 2$.

(d) A second reservoir holds $V(t)$ thousand litres of water, where V is a differentiable function. Selected values of V are given.

t (hours)	0	1	2.5	4	6
$V(t)$ (thousand litres)	10	14	21	25	37

Use the data in the table to estimate $V'(3)$. Show the computation that leads to your answer, and indicate units.

Give decimal answers correct to three places.

23 **[9 points] (No calculator.)** Functions f and g are differentiable for all real numbers. Values of f, f', g and g' at selected values of x are given.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	4	-2	3	1
2	5	6	2	3
3	1	2	4	-1

(a) Let $h(x) = f(x)\,g(x)$. Find $h'(2)$.

(b) Let $p(x) = \dfrac{f(x)}{g(x)}$. Write an equation for the line tangent to the graph of p at $x = 3$.

(c) The function k is defined by $k(x) = \begin{cases} f(x), & x \leq 2 \\ a\,g(x) + b, & x > 2, \end{cases}$ where a and b are constants. If k is differentiable at $x = 2$, find the values of a and b. Justify your answer.

(d) Let $w(x) = f(x) - 2x$. Show that the line tangent to the graph of w at $x = 3$ is horizontal, and write an equation for this line.

AP Calculus AB – Unit 3 Exam weighting: 9–13% of the AB exam score

Differentiation: Composite, Implicit, and Inverse Functions

In this unit

Unit 2 gave you the basic rules for differentiating sums, products, quotients and the standard functions. Almost every function that appears on the exam, however, is *built from several pieces nested inside each other*, or is defined by an equation in which y is not isolated, or is the inverse of a function you know. This unit supplies the three tools that handle those situations—the chain rule, implicit differentiation and the derivative of an inverse—and then teaches you to choose between all of your rules quickly and to differentiate more than once.

Topics.

3.1 The Chain Rule 3.2 Implicit Differentiation 3.3 Differentiating Inverse Functions
3.4 Differentiating Inverse Trigonometric Functions 3.5 Selecting Procedures for Calculating Derivatives
3.6 Calculation of Higher-Order Derivatives

Big ideas. (1) A derivative of a composite function is a *product of rates*: the rate of the outer function times the rate of the inner function. (2) When y is tied to x by an equation, differentiating both sides with respect to x treats y as an unknown function of x, so every y-term brings a factor $\frac{dy}{dx}$. (3) Inverse functions swap inputs and outputs, so their slopes are *reciprocals* at corresponding points. (4) A derivative is itself a function, so it can be differentiated again; f'' describes concavity and acceleration.

Topic 3.1 The Chain Rule

The idea: rates multiply

Suppose a drone climbs so that its height is $u = g(x)$ metres after x seconds, and the air pressure at height u is $y = f(u)$. The pressure then depends on the time only *through* the height. If the height changes at $g'(x)$ metres per second and the pressure changes at $f'(u)$ pascals per metre, the pressure changes at the product of the two rates, in pascals per second. This is the chain rule: the derivative of a composite function is the derivative of the outer function, *evaluated at the inner function*, multiplied by the derivative of the inner function.

In the Leibniz notation introduced in Unit 2 the same statement looks like a cancellation of fractions, $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$. The cancellation is only a memory aid, not a proof, but it is a reliable one: the units on both sides agree, and it tells you immediately which derivative is which. Notice that every function you differentiated in Unit 2 with x as its input, such as $\sin x$, e^x or $\ln x$, now needs a companion rule for the case where the input is a more complicated expression u. For instance, $\frac{d}{dx}\sin x = \cos x$ becomes $\frac{d}{dx}\sin u = \cos u \cdot \frac{du}{dx}$.

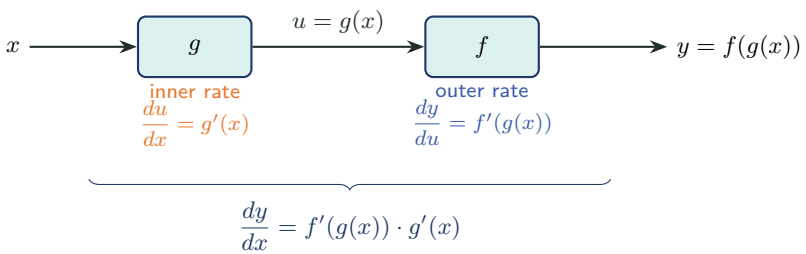


Figure 3.1. The chain rule multiplies the rate of the outer function, evaluated at $g(x)$, by the rate of the inner function.

The Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then $y = f(g(x))$ is differentiable at x and

$$\frac{d}{dx}\big[f(g(x))\big] = f'(g(x)) \cdot g'(x), \qquad \text{equivalently} \qquad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (u = g(x)).$$

Chain-rule forms of the standard derivatives (u is a differentiable function of x):

$$\frac{d}{dx}[u^n] = nu^{n-1}u' \qquad \frac{d}{dx}[e^u] = e^u\,u' \qquad \frac{d}{dx}[\ln u] = \frac{u'}{u}$$
$$\frac{d}{dx}[\sin u] = \cos u\;u' \qquad \frac{d}{dx}[\cos u] = -\sin u\;u' \qquad \frac{d}{dx}[\tan u] = \sec^2 u\;u'$$
$$\frac{d}{dx}[a^u] = a^u \ln a\;u' \qquad \frac{d}{dx}[\sec u] = \sec u\tan u\;u' \qquad \frac{d}{dx}[\cot u] = -\csc^2 u\;u'$$

How to apply it: work from the outside in

To differentiate a composite, first decide what the *outermost operation* is—the last thing you would do if you were evaluating the function at a number. Differentiate that operation, leaving its input untouched, and then multiply by the derivative of the input. If the input is itself a composite, repeat. The expression $\sin^3(4x)$ means $\big(\sin(4x)\big)^3$; the last operation is cubing, then taking the sine, and the first is multiplying by 4. The chain rule peels these layers in the reverse order, and each layer contributes one factor.

Example 1 A power of a polynomial, and its tangent line

Let $f(x) = (3x^2 - 5x + 1)^4$. (a) Find $f'(x)$. (b) Write an equation of the tangent line to the graph of f at $x = 1$.

(a) Outer: u^4. Inner: $u = 3x^2 - 5x + 1$, with $u' = 6x - 5$.

$$f'(x) = 4(3x^2 - 5x + 1)^3 \cdot (6x - 5).$$

(b) *Step 1:* the point. $f(1) = (3 - 5 + 1)^4 = (-1)^4 = 1$, so the point is $(1, 1)$.
Step 2: the slope. $f'(1) = 4(-1)^3 \cdot (1) = -4$.
Step 3: the line. $y - 1 = -4(x - 1)$.

Example 2 Several layers

Find $\dfrac{dy}{dx}$. (a) $y = \sin^3(4x)$ (b) $y = \ln(\cos 3x)$ and evaluate it at $x = \dfrac{\pi}{12}$.

(a) Write $y = v^3$ with $v = \sin u$ and $u = 4x$. The three layers give three factors:

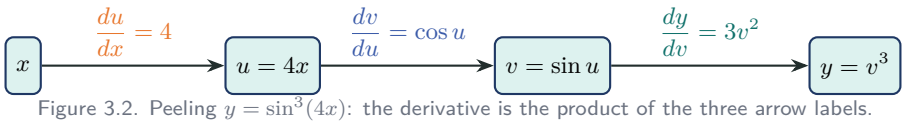


Figure 3.2. Peeling $y = \sin^3(4x)$: the derivative is the product of the three arrow labels.

$$\frac{dy}{dx} = 3\sin^2(4x) \cdot \cos(4x) \cdot 4 = 12\sin^2(4x)\cos(4x).$$

(b) Outer: $\ln u$ with derivative $\dfrac{1}{u}$. Inner: $u = \cos 3x$ with $u' = -3\sin 3x$.

$$\frac{dy}{dx} = \frac{-3\sin 3x}{\cos 3x} = -3\tan 3x. \qquad \text{At } x = \frac{\pi}{12}: \; -3\tan\frac{\pi}{4} = -3.$$

Example 3 The chain rule from a table

The differentiable functions f and g have the values below.

x	1	2	3
$f(x)$	2	3	1
$f'(x)$	5	-2	4
$g(x)$	3	1	2
$g'(x)$	-1	6	7

Find (a) $h'(2)$ where $h(x) = f(g(x))$; (b) $k'(2)$ where $k(x) = g(f(x))$; (c) $m'(1)$ where $m(x) = f(x^2)$; (d) $p'(3)$ where $p(x) = \big(g(x)\big)^3$.

(a) $h'(x) = f'(g(x))\,g'(x)$, so $h'(2) = f'(g(2))\,g'(2) = f'(1) \cdot 6 = 5 \cdot 6 = 30$.

(b) $k'(x) = g'(f(x))\,f'(x)$, so $k'(2) = g'(f(2))\,f'(2) = g'(3)\cdot(-2) = 7\cdot(-2) = -14$.
(c) The inner function is x^2 with derivative $2x$: $m'(x) = f'(x^2)\cdot 2x$, so $m'(1) = f'(1)\cdot 2 = 10$.
(d) Outer: cube. $p'(x) = 3\big(g(x)\big)^2 g'(x)$, so $p'(3) = 3\,(2)^2\cdot 7 = 84$.

Exponentials, logarithms and radicals with an inner function

The chain rule is the engine behind every "function of an expression" on the exam. A radical is a power, so $\sqrt{u} = u^{1/2}$ differentiates to $\dfrac{u'}{2\sqrt{u}}$. An exponential with an expression in the exponent keeps itself and multiplies by the derivative of the exponent, $\dfrac{d}{dx}e^u = e^u u'$. A logarithm of an expression gives the derivative of the expression divided by the expression, $\dfrac{d}{dx}\ln u = \dfrac{u'}{u}$. In each case the chain rule has not changed; only the outer function has.

Example 4 Exponential, radical and negative power

Differentiate (a) $y = e^{\sin 2x}$ (b) $y = \sqrt{\dfrac{x}{x+1}}$ (c) $y = (x^2+1)^{-3}$, and for (b) find $\dfrac{dy}{dx}$ at $x = 1$.

(a) Outer e^u, inner $u = \sin 2x$ with $u' = 2\cos 2x$: $\dfrac{dy}{dx} = e^{\sin 2x}\cdot 2\cos 2x$.

(b) Outer $u^{1/2}$, inner $u = \dfrac{x}{x+1}$. The inner derivative needs the quotient rule: $u' = \dfrac{(x+1)-x}{(x+1)^2} = \dfrac{1}{(x+1)^2}$. Hence

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x}{x+1}\right)^{-1/2}\cdot\frac{1}{(x+1)^2}. \qquad \text{At } x=1:\ \frac{1}{2}\left(\frac{1}{2}\right)^{-1/2}\cdot\frac{1}{4} = \frac{1}{2}\cdot\sqrt{2}\cdot\frac{1}{4} = \frac{\sqrt{2}}{8}.$$

(c) Outer u^{-3}, inner x^2+1: $\dfrac{dy}{dx} = -3(x^2+1)^{-4}\cdot 2x = -\dfrac{6x}{(x^2+1)^4}$.

Reading the chain rule from a graph and from a context

Exam questions often give a graph of a piecewise linear function instead of a formula. Slopes of the segments are the derivatives; where the graph has a corner the derivative does not exist, so such questions are always set at points inside a segment. The chain rule is applied exactly as with a table: find the inner value, read the outer slope at that value, and multiply.

Example 5 Chain rule from a graph

The graph of f consists of three line segments joining $(0,0)$, $(2,4)$, $(4,3)$ and $(6,5)$. Find (a) $h'(3)$ where $h(x) = \big(f(x)\big)^2$; (b) $k'(1.5)$ where $k(x) = f(2x)$; (c) $m'(5)$ where $m(x) = \sqrt{f(x)}$.

The slopes of the segments are $\dfrac{4-0}{2-0} = 2$ on $(0,2)$, $\dfrac{3-4}{4-2} = -\dfrac{1}{2}$ on $(2,4)$ and $\dfrac{5-3}{6-4} = 1$ on $(4,6)$.
(a) $h'(x) = 2f(x)f'(x)$. On the segment $(2,4)$, $f(3) = 4 - \frac{1}{2} = 3.5$ and $f'(3) = -\frac{1}{2}$, so $h'(3) = 2(3.5)\left(-\frac{1}{2}\right) = -3.5$.
(b) $k'(x) = f'(2x)\cdot 2$, so $k'(1.5) = f'(3)\cdot 2 = -1$.
(c) $m'(x) = \dfrac{f'(x)}{2\sqrt{f(x)}}$. On $(4,6)$: $f(5) = 4$, $f'(5) = 1$, so $m'(5) = \dfrac{1}{2\cdot 2} = \dfrac{1}{4}$.

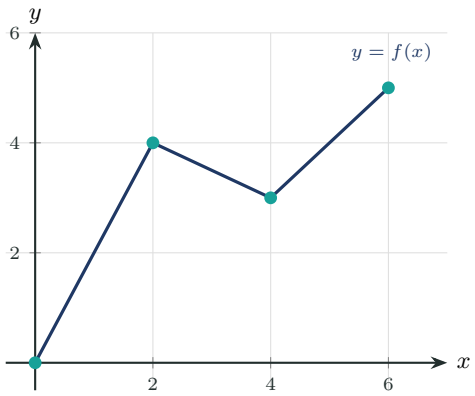


Figure 3.3. The piecewise linear function f of Example 5.

Example 6 Rates that multiply: a growing ripple

A circular ripple spreads on a pond so that its radius is $r(t) = 2t$ centimetres, t in seconds. The area is $A = \pi r^2$. Find the rate at which the area is changing at $t = 3$.

A depends on t only through r, so $\dfrac{dA}{dt} = \dfrac{dA}{dr} \cdot \dfrac{dr}{dt} = 2\pi r \cdot 2$. At $t = 3$: $r = 6$, so $\dfrac{dA}{dt} = 2\pi(6)(2) = 24\pi \approx 75.4$ square centimetres per second. The units agree: $\dfrac{\text{cm}^2}{\text{cm}} \cdot \dfrac{\text{cm}}{\text{s}} = \dfrac{\text{cm}^2}{\text{s}}$. (Topics 4.4 and 4.5 build whole problems from this idea.)

Example 7 A cooling drink (calculator allowed)

The temperature of a drink, in degrees Celsius, is $T(t) = 20 + 60e^{-0.05t}$, where t is in minutes. Find $T'(10)$ and interpret it.

The exponent $-0.05t$ is the inner function, with derivative -0.05. Hence $T'(t) = 60e^{-0.05t} \cdot (-0.05) = -3e^{-0.05t}$ and $T'(10) = -3e^{-0.5} \approx -1.820$. At $t = 10$ minutes the temperature of the drink is decreasing at about 1.820 degrees Celsius per minute. (The units of T' are the units of T divided by the units of t.)

Common errors

Dropping the inner derivative. $\dfrac{d}{dx}\sin(3x)$ is $3\cos 3x$, not $\cos 3x$. Every time the input of a function is anything other than plain x, ask "what multiplies this?"

Evaluating the outer derivative at the wrong place. In $h'(2) = f'(g(2))\,g'(2)$ the derivative of f is evaluated at $g(2)$, *not* at 2. Writing $f'(2)\,g'(2)$ is the most common table-question mistake.

Differentiating the inside twice. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$. The factor $\cos x$ comes from the derivative of the inner layer $\sin x$; the $\sin x$ is the inner layer itself and is *kept*.

AP Exam Tip

When a table or graph is given, show the structure before the numbers, e.g. "$h'(2) = f'(g(2))\cdot g'(2) = f'(1)\cdot g'(2) = \ldots$". Scoring guides award a point for the correct chain-rule setup even if a table value is misread, and they accept an unsimplified numerical answer such as $5 \cdot 6$.

Check yourself 3.1

1. Find $\dfrac{dy}{dx}$ for $y = \cos(x^3)$.
2. Find $f'(x)$ for $f(x) = \sqrt{4 - x^2}$, and state where it is defined.
3. Using the table in Example 3, find $h'(3)$ where $h(x) = f(g(x))$.

Topic 3.2 Implicit Differentiation

When y is not isolated

Many curves cannot be written as a single function $y = f(x)$. The circle $x^2 + y^2 = 25$ fails the vertical line test; the folium $x^3 + y^3 = 6xy$ is worse, because there is no tidy formula for y at all. Yet such a curve still has a tangent line at almost every point, and its slope is exactly what the derivative is supposed to measure. *Implicit differentiation* finds that slope without ever solving for y.

The idea is to *assume* that near the point of interest y is some differentiable function of x, even though you do not know its formula, and to differentiate both sides of the equation with respect to x. Terms containing only x are handled as usual. A term containing y is a composite function of x—an expression in y whose input y depends on x—so the chain rule applies and produces a factor $\dfrac{dy}{dx}$. For example, $\dfrac{d}{dx}[y^2] = 2y\,\dfrac{dy}{dx}$ and $\dfrac{d}{dx}[\sin y] = \cos y\,\dfrac{dy}{dx}$. A term such as xy is a product of two functions of x and needs the product rule: $\dfrac{d}{dx}[xy] = y + x\dfrac{dy}{dx}$.

Implicit differentiation: procedure

1. Differentiate both sides of the equation with respect to x. Remember: $\dfrac{d}{dx}[h(y)] = h'(y)\,\dfrac{dy}{dx}$.
2. Collect every term containing $\dfrac{dy}{dx}$ on one side, all other terms on the other.

3. Factor out $\dfrac{dy}{dx}$ and divide.
4. The answer generally contains *both* x and y. To find a slope, substitute the coordinates of the point.

A tangent line is **horizontal** where $\dfrac{dy}{dx} = 0$ (numerator $= 0$, denominator $\neq 0$) and **vertical** where the denominator is 0 (and the numerator is not).

Example 1 The circle $x^2 + y^2 = 25$

Find $\dfrac{dy}{dx}$, then an equation of the tangent line at $(3, 4)$ and the points where the tangent is horizontal or vertical.

Step 1. Differentiate: $2x + 2y\dfrac{dy}{dx} = 0$.

Step 2. Solve: $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

Step 3. At $(3, 4)$ the slope is $-\dfrac{3}{4}$, so the tangent line is $y - 4 = -\dfrac{3}{4}(x - 3)$.

Step 4. Horizontal: $-\dfrac{x}{y} = 0 \Rightarrow x = 0$, giving $(0, 5)$ and $(0, -5)$. Vertical: $y = 0$, giving $(5, 0)$ and $(-5, 0)$.

Check: the radius to $(3, 4)$ has slope $\dfrac{4}{3}$, and $\dfrac{4}{3}\cdot\left(-\dfrac{3}{4}\right) = -1$, so the tangent is perpendicular to the radius, as geometry demands.

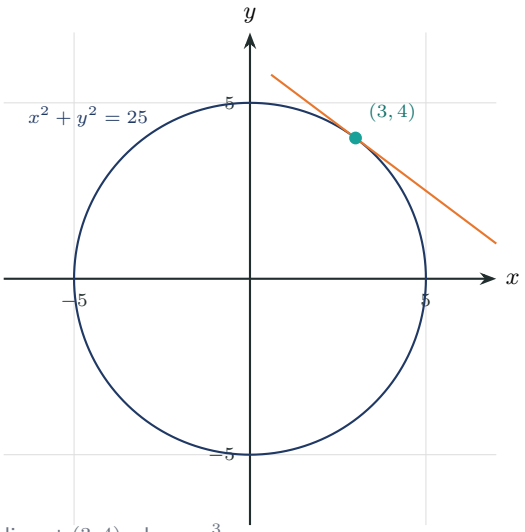


Figure 3.4. The circle and its tangent line at $(3, 4)$, slope $-\frac{3}{4}$.

Example 2 The folium of Descartes

The curve $x^3 + y^3 = 6xy$ passes through $(3, 3)$. (a) Find $\dfrac{dy}{dx}$. (b) Find the slope at $(3, 3)$. (c) Find the point in the first quadrant, other than the origin, where the tangent line is horizontal.

(a) Differentiate; the right side needs the product rule:

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}.$$

Collect: $(3y^2 - 6x)\dfrac{dy}{dx} = 6y - 3x^2$, hence

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

(b) At $(3, 3)$: $\dfrac{6 - 9}{9 - 6} = -1$.

(c) Horizontal tangent needs $2y - x^2 = 0$, i.e. $y = \dfrac{x^2}{2}$. Substitute into the equation of the curve: $x^3 + \dfrac{x^6}{8} = 3x^3$, so $\dfrac{x^6}{8} = 2x^3$ and $x^3 = 16$ (since $x \neq 0$). Then $x = 16^{1/3} = 2^{4/3}$ and $y = \dfrac{x^2}{2} = 2^{5/3}$, i.e. about $(2.520, 3.175)$. The denominator $y^2 - 2x$ is not 0 there, so the tangent really is horizontal.

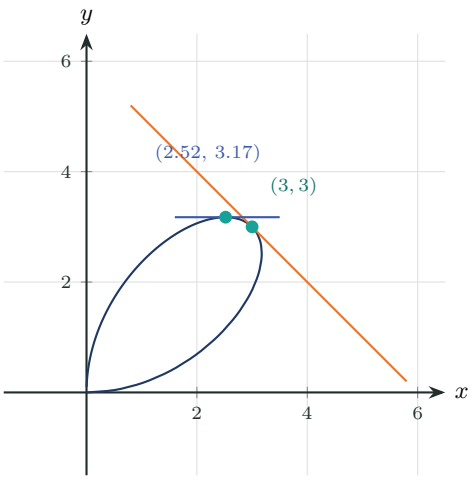


Figure 3.5. The loop of the folium with the tangent of slope -1 at $(3, 3)$ and the horizontal tangent.

Example 3 Slope at a point, and a horizontal tangent

Consider the curve $x^2 + xy + y^2 = 7$. (a) Verify that $(2, 1)$ lies on the curve and find the slope there. (b) Find the x-coordinates of all points where the tangent is horizontal.

(a) $2^2 + 2 \cdot 1 + 1^2 = 7$, so the point is on the curve. Differentiating, $2x + y + x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} = 0$, so

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}. \qquad \text{At } (2, 1): \; -\frac{5}{4}.$$

(b) A horizontal tangent needs $2x + y = 0$, so $y = -2x$. Substituting: $x^2 - 2x^2 + 4x^2 = 3x^2 = 7$, hence $x = \pm\sqrt{\dfrac{7}{3}}$. (There $x + 2y = -3x \neq 0$, so the slope is truly 0.)

Tangent lines, normal lines and vertical tangents

Once $\dfrac{dy}{dx}$ is known, everything you did with tangent lines in Unit 2 carries over. The *normal line* at a point is the line perpendicular to the tangent there, so its slope is the negative reciprocal $-\dfrac{1}{m}$ when $m \neq 0$. A vertical tangent occurs where the denominator of $\dfrac{dy}{dx}$ is 0 while the numerator is not; the points must also lie on the curve, so substitute back into the original equation.

Derivatives of common y-expressions

$$\frac{d}{dx}[y^2] = 2y\,\frac{dy}{dx} \qquad \frac{d}{dx}[\sin y] = \cos y\,\frac{dy}{dx} \qquad \frac{d}{dx}[e^y] = e^y\,\frac{dy}{dx}$$

$$\frac{d}{dx}[xy] = y + x\frac{dy}{dx} \qquad \frac{d}{dx}[x^2 y] = 2xy + x^2\frac{dy}{dx} \qquad \frac{d}{dx}[\ln y] = \frac{1}{y}\frac{dy}{dx}$$

$$\frac{d}{dx}\left[\frac{x}{y}\right] = \frac{y - x\dfrac{dy}{dx}}{y^2} \qquad \frac{d}{dx}[xe^y] = e^y + xe^y\frac{dy}{dx} \qquad \frac{d}{dx}[\arctan y] = \frac{1}{1 + y^2}\frac{dy}{dx}$$

Example 4 The ellipse $x^2 + xy + y^2 = 7$: normal line and vertical tangents

For the curve of Example 3, (a) write an equation of the normal line at $(2, 1)$; (b) find the points where the tangent line is vertical.

(a) We found $\dfrac{dy}{dx} = -\dfrac{2x + y}{x + 2y}$, so the slope of the tangent at $(2, 1)$ is $-\dfrac{5}{4}$. The normal slope is $\dfrac{4}{5}$:

$$y - 1 = \frac{4}{5}(x - 2).$$

(b) The tangent is vertical where $x + 2y = 0$ (and $2x + y \neq 0$), so $x = -2y$. Substitute into the curve: $4y^2 - 2y^2 + y^2 = 3y^2 = 7$, so $y = \pm\sqrt{\frac{7}{3}} \approx \pm 1.528$ and $x = -2y$. The points are $\left(-2\sqrt{\frac{7}{3}}, \sqrt{\frac{7}{3}}\right) \approx (-3.055,\, 1.528)$ and $\left(2\sqrt{\frac{7}{3}}, -\sqrt{\frac{7}{3}}\right) \approx (3.055,\, -1.528)$. At both, $2x + y = -3y \neq 0$, so the vertical tangents are genuine.

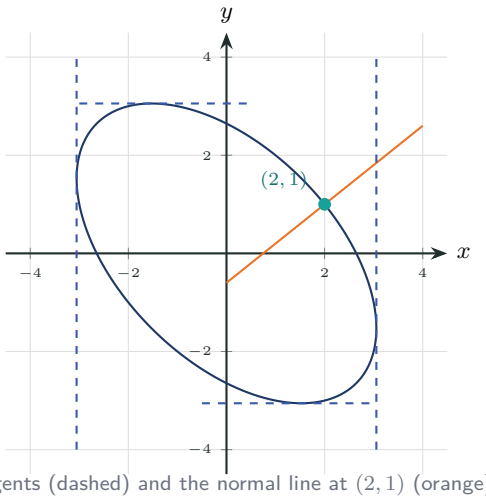


Figure 3.6. The curve $x^2 + xy + y^2 = 7$ with its horizontal and vertical tangents (dashed) and the normal line at $(2, 1)$ (orange).

Example 5 An exponential term

The curve $xe^y + y = 3$ passes through $(3, 0)$. Find $\dfrac{dy}{dx}$ and an equation of the tangent line at $(3, 0)$.

Step 1. Check: $3e^0 + 0 = 3$. *Step 2.* Differentiate, using the product rule on xe^y: $e^y + xe^y\dfrac{dy}{dx} + \dfrac{dy}{dx} = 0$.

Step 3. Solve: $\dfrac{dy}{dx}(xe^y + 1) = -e^y$, so $\dfrac{dy}{dx} = -\dfrac{e^y}{xe^y + 1}$.

Step 4. At $(3, 0)$: $-\dfrac{1}{3+1} = -\dfrac{1}{4}$. Tangent line: $y = -\dfrac{1}{4}(x - 3)$.

Example 6 Only x is given

The curve $y^3 + xy = 10$ contains a point with $x = 1$. Find the y-coordinate of that point and the slope of the curve there.

Step 1. Substitute $x = 1$: $y^3 + y = 10$. The left side is increasing, and $y = 2$ gives $8 + 2 = 10$, so the point is $(1, 2)$.

Step 2. Differentiate: $3y^2\dfrac{dy}{dx} + y + x\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{y}{3y^2 + x}$.

Step 3. At $(1, 2)$: $-\dfrac{2}{12+1} = -\dfrac{2}{13}$.

Common errors

Forgetting $\dfrac{dy}{dx}$. $\dfrac{d}{dx}[y^3] = 3y^2\dfrac{dy}{dx}$, not $3y^2$. Any term with y gets the factor.

Forgetting the product rule on xy. $\dfrac{d}{dx}[xy] = y + x\dfrac{dy}{dx}$, not $y\dfrac{dy}{dx}$ and not $x\dfrac{dy}{dx}$ alone.

Solving for y first when it is not needed (or not possible), and **not substituting the y-coordinate** when evaluating a slope: the point $(2, 1)$ supplies *both* $x = 2$ and $y = 1$.

AP Exam Tip

On the exam you are expected to leave $\dfrac{dy}{dx}$ in terms of x and y; do not attempt to solve for y. For a tangent line, write the slope as a number first and then the equation in point–slope form $y - y_1 = m(x - x_1)$. To justify a horizontal tangent, state that the numerator is zero *and* that the denominator is nonzero at that point.

Check yourself 3.2

1. Find $\dfrac{dy}{dx}$ if $e^y = x + y$.
2. Write an equation of the tangent line to $x^2 - xy + y^2 = 3$ at $(1, 2)$.
3. Find $\dfrac{dy}{dx}$ at $(1, 1)$ on the curve $xy + \ln y = 1$.

Topic 3.3 Differentiating Inverse Functions

Reflecting slopes

A function f has an inverse $g = f^{-1}$ when f is one-to-one (its graph passes the horizontal line test). The inverse undoes f: $g(f(x)) = x$ and $f(g(y)) = y$. If (a, b) lies on the graph of f, then (b, a) lies on the graph of g, so the graph of g is the reflection of the graph of f in the line $y = x$. Reflecting also swaps the roles of "run" and "rise", so a tangent line of slope m at (a, b) becomes a tangent line of slope $\dfrac{1}{m}$ at (b, a). That is the whole content of the inverse derivative rule.

To see it algebraically, differentiate $f(g(x)) = x$ with the chain rule: $f'(g(x)) \cdot g'(x) = 1$, hence $g'(x) = \dfrac{1}{f'(g(x))}$. The rule is practical because you can evaluate g' at a number a *without finding a formula for* g: you only need the number $g(a)$—the x-value that f sends to a—and the derivative of f there. On the exam this is how the question is asked: "Let g be the inverse of f. Find $g'(5)$."

Derivative of an inverse function

Let f be differentiable and one-to-one with inverse $g = f^{-1}$. If $f'(g(a)) \neq 0$, then g is differentiable at a and

$$\boxed{g'(a) = \frac{1}{f'\big(g(a)\big)}} \qquad \text{equivalently} \qquad (f^{-1})'(b) = \frac{1}{f'(a)} \quad \text{where } f(a) = b.$$

In Leibniz form: $\dfrac{dx}{dy} = \dfrac{1}{\dfrac{dy}{dx}}$. If $f'(g(a)) = 0$, the graph of g has a vertical tangent at $x = a$ and $g'(a)$ does not exist.

Example 1 An inverse we cannot write down

Let $f(x) = x^3 + x$ and $g = f^{-1}$. Find $g'(2)$ and $g'(10)$, and an equation of the tangent line to g at $x = 2$.

Step 1. $f'(x) = 3x^2 + 1 > 0$ for all x, so f is increasing, hence one-to-one, and g exists. Solving $x^3 + x = y$ for x is not possible by elementary algebra—this is why we need the rule.

Step 2. Find $g(2)$: which x satisfies $x^3 + x = 2$? Trying $x = 1$ gives 2, so $g(2) = 1$.

Step 3. $g'(2) = \dfrac{1}{f'(g(2))} = \dfrac{1}{f'(1)} = \dfrac{1}{4}$.

Step 4. Since $f(2) = 10$, $g(10) = 2$ and $g'(10) = \dfrac{1}{f'(2)} = \dfrac{1}{13}$.

Step 5. Tangent to g at $(2,1)$: $y - 1 = \dfrac{1}{4}(x - 2)$. It is the reflection of the tangent to f at $(1,2)$, which has slope 4.

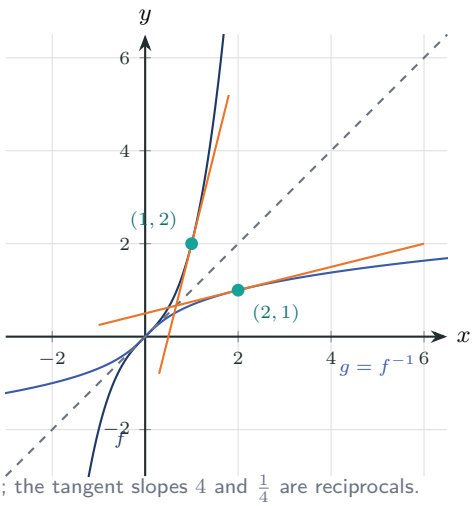


Figure 3.7. f (navy) and $g = f^{-1}$ (blue) are mirror images in $y = x$; the tangent slopes 4 and $\frac{1}{4}$ are reciprocals.

Example 2 Why $\frac{d}{dx}\ln x = \frac{1}{x}$

The function $\ln x$ is the inverse of $f(x) = e^x$. Let $g(x) = \ln x$. By the rule, $g'(x) = \dfrac{1}{f'(g(x))} = \dfrac{1}{e^{\ln x}} = \dfrac{1}{x}$ for $x > 0$. The rule has recovered the derivative of the natural logarithm. In the same way, $\dfrac{d}{dx}\sqrt[3]{x}$ can be obtained from $f(x) = x^3$: $\dfrac{1}{3(\sqrt[3]{x})^2} = \dfrac{1}{3}x^{-2/3}$.

Example 3 An inverse from a table

The function f is differentiable and increasing, with values in the table. Let $g = f^{-1}$.

x	0	1	2	3
$f(x)$	2	5	7	8
$f'(x)$	4	3	2	$\frac{1}{2}$

Find (a) $g'(5)$, (b) $g'(8)$, (c) $q'(7)$ where $q(x) = \big(g(x)\big)^2$.

(a) $f(1) = 5$, so $g(5) = 1$ and $g'(5) = \dfrac{1}{f'(1)} = \dfrac{1}{3}$.

(b) $f(3) = 8$, so $g(8) = 3$ and $g'(8) = \dfrac{1}{f'(3)} = \dfrac{1}{1/2} = 2$.

(c) $q'(x) = 2g(x)\,g'(x)$. Here $g(7) = 2$ and $g'(7) = \dfrac{1}{f'(2)} = \dfrac{1}{2}$, so $q'(7) = 2 \cdot 2 \cdot \dfrac{1}{2} = 2$.

Inverse derivatives from a graph

When the graph of f is given, you read the point (a,b) on it, read the slope of f there, and the inverse has the point (b,a) and the reciprocal slope. The order of coordinates is the thing to watch: the *input* of g is the *output* of f.

Example 4 Inverse of a piecewise linear function

The graph of f consists of segments joining $(0,1)$, $(2,4)$, $(5,6)$ and $(8,7)$. Let $g=f^{-1}$. Find (a) $g'(5)$, (b) $g'(2)$, (c) an equation of the tangent line to g at $x=5$.

The slopes are $\dfrac{4-1}{2-0}=\dfrac{3}{2}$ on $(0,2)$, $\dfrac{6-4}{5-2}=\dfrac{2}{3}$ on $(2,5)$ and $\dfrac{7-6}{8-5}=\dfrac{1}{3}$ on $(5,8)$.

(a) On the middle segment $f(x)=4+\frac{2}{3}(x-2)=5$ gives $x=3.5$, so $g(5)=3.5$ and $g'(5)=\dfrac{1}{f'(3.5)}=\dfrac{3}{2}$.

(b) On the first segment $1+\frac{3}{2}x=2$ gives $x=\frac{2}{3}$, so $g(2)=\frac{2}{3}$ and $g'(2)=\dfrac{1}{3/2}=\dfrac{2}{3}$.

(c) The tangent line passes through $(5,3.5)$ with slope $\dfrac{3}{2}$: $y-3.5=\dfrac{3}{2}(x-5)$.

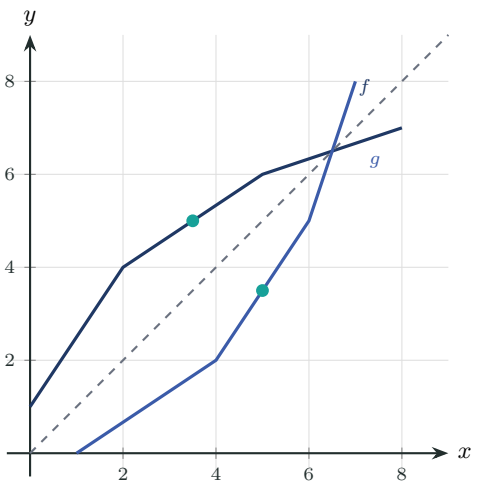


Figure 3.8. f (navy) and its inverse g (blue): reflect each vertex in $y=x$.

Example 5 Showing that an inverse exists

Let $f(x)=x+e^x$. (a) Explain why f has an inverse g. (b) Find $g'(1)$ and $g'(1+e)$.

(a) $f'(x)=1+e^x>0$ for every x, so f is increasing on $(-\infty,\infty)$ and therefore one-to-one.

(b) $f(0)=0+1=1$, so $g(1)=0$ and $g'(1)=\dfrac{1}{f'(0)}=\dfrac{1}{2}$. Also $f(1)=1+e$, so $g(1+e)=1$ and $g'(1+e)=\dfrac{1}{f'(1)}=\dfrac{1}{1+e}\approx 0.269$.

Example 6 An inverse in context, with units

Water is poured into a tank. After t minutes the depth is $d(t)=t^3+t$ centimetres, for $t\ge 0$. Let T be the inverse of d, so that $T(y)$ is the time at which the depth reaches y centimetres. Find $T'(10)$ and interpret it.

Since $d(2)=8+2=10$, $T(10)=2$. Also $d'(t)=3t^2+1$, so $d'(2)=13$ and $T'(10)=\dfrac{1}{d'(2)}=\dfrac{1}{13}$. The units of T' are minutes per centimetre: when the depth is 10 cm, each additional centimetre takes about $\dfrac{1}{13}$ minute (roughly 4.6 seconds) to fill. A fast-rising depth ($d'=13$ cm per minute) means a short time per centimetre.

Common errors

Reciprocal at the wrong input. $g'(5)=\dfrac{1}{f'(g(5))}$, not $\dfrac{1}{f'(5)}$. The input to f' is the *output of the inverse*.

Confusing the inverse with the reciprocal. $f^{-1}(x)$ is not $\dfrac{1}{f(x)}$, and $(f^{-1})'(x)$ is not $\dfrac{1}{f'(x)}$.

Missing the zero-slope case. If $f'(g(a))=0$ (for example $f(x)=x^3$ at 0), the inverse has a vertical tangent and is *not* differentiable there.

AP Exam Tip

Write the formula first, with the table values filled in step by step: "$g(5)=1$ because $f(1)=5$; $g'(5)=\dfrac{1}{f'(g(5))}=$

$\dfrac{1}{f'(1)} = \dfrac{1}{3}$." Finding $g(a)$ is its own point on many scoring guides.

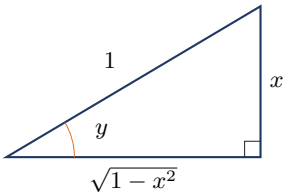
Check yourself 3.3

1. Let $f(x) = x^5 + x^3 + x$ and $g = f^{-1}$. Find $g'(3)$.
2. The tangent line to the graph of f at $(2, 5)$ has slope 3. Write an equation of the tangent line to the graph of f^{-1} at $x = 5$.
3. Explain why $g = f^{-1}$ is not differentiable at $x = 0$ for $f(x) = x^3$.

Topic 3.4 Differentiating Inverse Trigonometric Functions

The inverse trigonometric functions and where their derivatives come from

Sine, cosine and tangent are not one-to-one on their whole domains, so their inverses are defined on restricted pieces: $y = \arcsin x$ means $\sin y = x$ with $-\dfrac{\pi}{2} \le y \le \dfrac{\pi}{2}$; $y = \arccos x$ means $\cos y = x$ with $0 \le y \le \pi$; $y = \arctan x$ means $\tan y = x$ with $-\dfrac{\pi}{2} < y < \dfrac{\pi}{2}$. (The notation $\sin^{-1} x$ means the same as $\arcsin x$ and has nothing to do with $\dfrac{1}{\sin x}$.) Their derivatives follow from Topics 3.2 and 3.3. Let $y = \arcsin x$, so $\sin y = x$. Differentiating implicitly, $\cos y \dfrac{dy}{dx} = 1$, so $\dfrac{dy}{dx} = \dfrac{1}{\cos y}$. In the right triangle of Figure 3.9 the angle y has opposite side x and hypotenuse 1, so the adjacent side is $\sqrt{1 - x^2}$ and $\cos y = \sqrt{1 - x^2}$ (positive because $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$). Hence $\dfrac{d}{dx} \arcsin x = \dfrac{1}{\sqrt{1 - x^2}}$.



$\sin y = x$
$\cos y = \sqrt{1 - x^2}$
$\tan y = \dfrac{x}{\sqrt{1 - x^2}}$

Figure 3.9. The reference triangle for $y = \arcsin x$.

The same argument for $\arctan$ uses $\tan y = x \Rightarrow \sec^2 y \dfrac{dy}{dx} = 1$ and $\sec^2 y = 1 + \tan^2 y = 1 + x^2$, giving $\dfrac{dy}{dx} = \dfrac{1}{1 + x^2}$. The arccosine derivative is the negative of the arcsine derivative, because $\arcsin x + \arccos x = \dfrac{\pi}{2}$ is constant.

Derivatives of the inverse trigonometric functions

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}} \qquad (|x| < 1)$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2} \qquad \frac{d}{dx}\operatorname{arccot} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}} \qquad \frac{d}{dx}\operatorname{arccsc} x = -\frac{1}{|x|\sqrt{x^2-1}} \qquad (|x| > 1)$$

With a chain: $\dfrac{d}{dx}\arcsin u = \dfrac{u'}{\sqrt{1-u^2}}$, $\dfrac{d}{dx}\arctan u = \dfrac{u'}{1+u^2}$. The arcsine and arctangent forms are the ones used most often on the exam.

Example 1 Inverse trigonometric functions with the chain rule

Differentiate: (a) $y = \arcsin(x^2)$ (b) $y = \arctan\sqrt{x}$ (c) $y = x \arctan x$.

(a) $u = x^2$, $u' = 2x$: $\dfrac{dy}{dx} = \dfrac{2x}{\sqrt{1 - (x^2)^2}} = \dfrac{2x}{\sqrt{1 - x^4}}$.

(b) $u = \sqrt{x}$, $u' = \dfrac{1}{2\sqrt{x}}$: $\dfrac{dy}{dx} = \dfrac{1}{1 + x} \cdot \dfrac{1}{2\sqrt{x}} = \dfrac{1}{2\sqrt{x}\,(1 + x)}$.

(c) Product rule: $\dfrac{dy}{dx} = 1 \cdot \arctan x + x \cdot \dfrac{1}{1+x^2} = \arctan x + \dfrac{x}{1+x^2}$.

Example 2 A tangent line to the arcsine curve

Write an equation of the line tangent to $y = \arcsin x$ at $x = \dfrac{1}{2}$, and describe the behaviour of the slope near $x = 1$.

Point. $\arcsin \dfrac{1}{2} = \dfrac{\pi}{6}$ because $\sin \dfrac{\pi}{6} = \dfrac{1}{2}$.

Slope. $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 - \frac{1}{4}}} = \dfrac{1}{\frac{\sqrt{3}}{2}} = \dfrac{2}{\sqrt{3}}$.

Line. $y - \dfrac{\pi}{6} = \dfrac{2}{\sqrt{3}}\left(x - \dfrac{1}{2}\right)$.

Behaviour. As $x \to 1^-$, $\sqrt{1-x^2} \to 0^+$, so the slope grows without bound: the graph of $\arcsin$ becomes vertical at $x = \pm 1$, where the derivative does not exist. The right-hand graph shows f' with its vertical asymptotes.

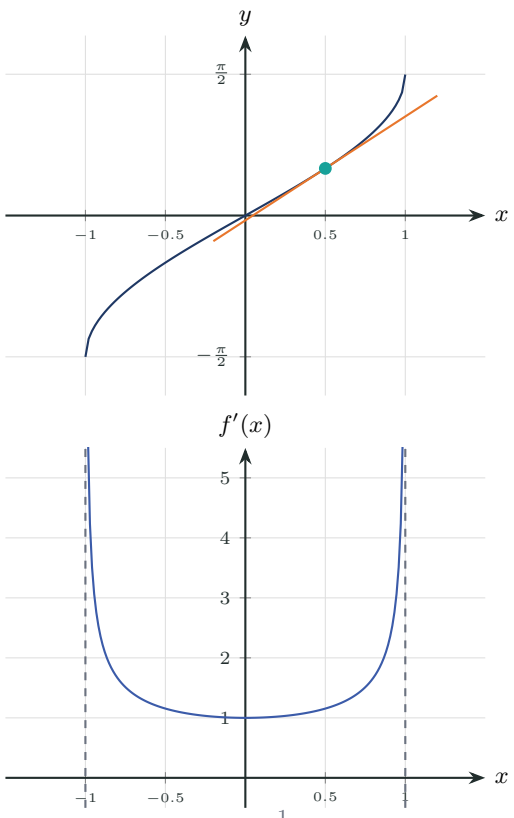


Figure 3.10. Top: $y = \arcsin x$ and its tangent at $x = \frac{1}{2}$. Bottom: $f'(x) = \dfrac{1}{\sqrt{1-x^2}}$.

Example 3 Angle of a camera

A camera is placed 20 m from the base of a vertical wall, and a drone rises along the wall so that its height after t seconds is $h(t) = 5t$ metres. The camera's elevation angle is $\theta(t) = \arctan\left(\dfrac{h(t)}{20}\right) = \arctan\left(\dfrac{t}{4}\right)$ radians. Find $\theta'(3)$ and interpret it.

With $u = \dfrac{t}{4}$, $u' = \dfrac{1}{4}$:

$$\theta'(t) = \frac{1/4}{1 + t^2/16} = \frac{4}{16 + t^2}, \qquad \theta'(3) = \frac{4}{25} = 0.16.$$

At $t = 3$ seconds the camera angle is increasing at 0.16 radian per second. (The rate shrinks as t grows: the drone is climbing at a constant speed but the camera has to tilt less and less for each extra metre.)

Example 4 Arccosine and a logarithm of an arctangent

Differentiate (a) $y = \arccos(2x)$ and evaluate at $x = \dfrac{1}{4}$; (b) $y = \ln(\arctan x)$ and evaluate at $x = 1$.

(a) $u = 2x$, $u' = 2$: $\dfrac{dy}{dx} = -\dfrac{2}{\sqrt{1 - 4x^2}}$. At $x = \dfrac{1}{4}$: $-\dfrac{2}{\sqrt{3/4}} = -\dfrac{4}{\sqrt{3}}$.

(b) Outer $\ln u$, inner $u = \arctan x$: $\dfrac{dy}{dx} = \dfrac{1}{\arctan x} \cdot \dfrac{1}{1+x^2}$. At $x = 1$: $\dfrac{1}{\pi/4} \cdot \dfrac{1}{2} = \dfrac{2}{\pi}$.

Example 5 The shape of $\arctan x$

Let $f(x) = \arctan x$. (a) Where is the slope of the graph greatest, and what is it? (b) What happens to the slope as $x \to \infty$? (c) Write an equation of the tangent line at $x = 1$.

(a) $f'(x) = \dfrac{1}{1+x^2}$. The denominator is smallest, namely 1, at $x = 0$, so the slope is greatest there: $f'(0) = 1$.
(b) $1 + x^2 \to \infty$, so $f'(x) \to 0$: the graph flattens out toward its horizontal asymptotes $y = \pm\dfrac{\pi}{2}$.
(c) $f(1) = \dfrac{\pi}{4}$ and $f'(1) = \dfrac{1}{2}$, so $y - \dfrac{\pi}{4} = \dfrac{1}{2}(x-1)$.

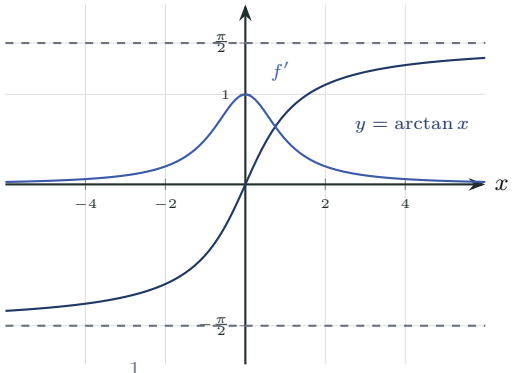


Figure 3.11. $y = \arctan x$ (navy) and its derivative $\dfrac{1}{1+x^2}$ (blue).

Common errors

Mixing up the forms. $\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$ (no radical), while $\dfrac{d}{dx}\arcsin x = \dfrac{1}{\sqrt{1-x^2}}$ (radical, minus sign). Changing $1 - x^2$ to $1 + x^2$, or vice versa, is a frequent slip.

Forgetting the chain-rule factor. $\dfrac{d}{dx}\arctan(3x) = \dfrac{3}{1+9x^2}$: the 3 is u' and x^2 becomes $(3x)^2 = 9x^2$.

Reading $\sin^{-1}$ as a reciprocal. $\sin^{-1} x = \arcsin x$; the reciprocal $\dfrac{1}{\sin x}$ is $\csc x$.

Check yourself 3.4

1. Find $\dfrac{d}{dx}\arctan(e^x)$.
2. Let $f(x) = \arcsin\dfrac{x}{2}$. Find $f'(1)$.
3. Find $\dfrac{dy}{dx}$ at $x = 0$ if $y = \arccos(2x)$.

Topic 3.5 Selecting Procedures for Calculating Derivatives

Choosing the right rule

By now you own a full toolbox: the power, product, quotient and chain rules, the derivatives of the exponential, logarithmic, trigonometric and inverse trigonometric functions, implicit differentiation, and the inverse rule. The skill tested on the exam is no longer "do you know the rule" but "which rule, in which order". Almost every difficulty can be resolved by the question from Topic 3.1: *what is the outermost operation?* If the function is a sum, differentiate term by term; if it is a product or quotient, use the product or quotient rule first, and use the chain rule inside the factors as needed; if it is a composite, apply the chain rule to peel one layer, then repeat on what is inside.

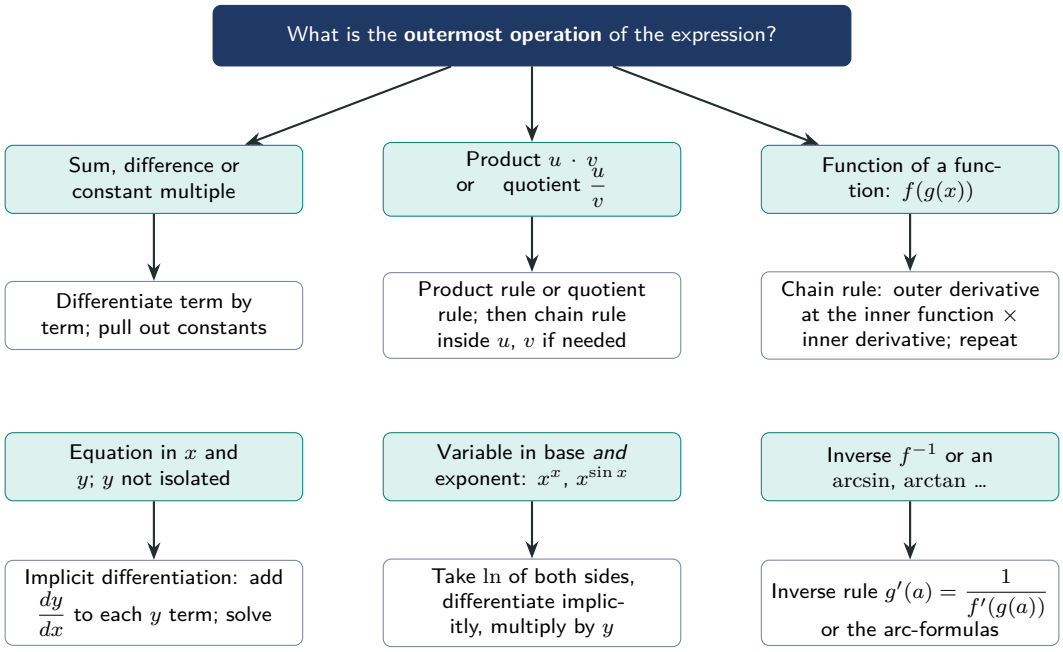


Figure 3.12. Which technique? Start at the top; always repeat the question on the inside of whatever you have just differentiated.

Logarithmic differentiation

Powers like x^x or $x^{\sin x}$ have a variable in both base and exponent; neither the power rule (x^n with n constant) nor the exponential rule (a^x with a constant) applies. Taking the natural logarithm turns the exponent into a factor, after which implicit differentiation works: if $y = x^x$, then $\ln y = x \ln x$, so $\frac{1}{y}\frac{dy}{dx} = \ln x + 1$ and $\frac{dy}{dx} = x^x(\ln x + 1)$.

Combination checklist

- **Product**: $(uv)' = u'v + uv'$. **Quotient**: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$. **Chain**: $[f(g)]' = f'(g)\,g'$.
- **Constant base**: $\frac{d}{dx}a^u = a^u \ln a\, u'$. **Constant exponent**: $\frac{d}{dx}u^n = nu^{n-1}u'$. **Both variable**: use ln.
- If a function can be simplified first (for instance $\ln(x^3\sqrt{x}) = \frac{7}{2}\ln x$, or $\frac{x^2+x}{x} = x + 1$), simplify *before* differentiating.

Example 1 Product with a chain

Differentiate $y = x^2 \sin(3x)$.

Outermost: a product of $u = x^2$ and $v = \sin(3x)$. Here $u' = 2x$ and, by the chain rule, $v' = 3\cos(3x)$.

$$\frac{dy}{dx} = 2x\sin(3x) + x^2 \cdot 3\cos(3x) = 2x\sin 3x + 3x^2\cos 3x.$$

Example 2 Quotient with a chain

Differentiate $y = \dfrac{e^{2x}}{1+x^2}$ and find where the tangent line is horizontal.

$u = e^{2x}$, $u' = 2e^{2x}$; $v = 1+x^2$, $v' = 2x$.

$$\frac{dy}{dx} = \frac{2e^{2x}(1+x^2) - e^{2x}\cdot 2x}{(1+x^2)^2} = \frac{2e^{2x}\,(x^2 - x + 1)}{(1+x^2)^2}.$$

The numerator is zero only if $x^2 - x + 1 = 0$, whose discriminant is $1 - 4 = -3 < 0$. So $\dfrac{dy}{dx} > 0$ for every x: there is no horizontal tangent; the function is always increasing.

Example 3 Both base and exponent vary

Find $\dfrac{dy}{dx}$ for $y = x^{\sin x}$ $(x > 0)$ and evaluate it at $x = \dfrac{\pi}{2}$.

Step 1. $\ln y = \sin x \, \ln x$.

Step 2. Differentiate implicitly (product rule on the right): $\dfrac{1}{y}\dfrac{dy}{dx} = \cos x \, \ln x + \dfrac{\sin x}{x}$.

Step 3. $\dfrac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \dfrac{\sin x}{x}\right)$.

Step 4. At $x = \dfrac{\pi}{2}$: $y = \left(\dfrac{\pi}{2}\right)^1 = \dfrac{\pi}{2}$, $\cos\dfrac{\pi}{2} = 0$ and $\dfrac{\sin(\pi/2)}{\pi/2} = \dfrac{2}{\pi}$. So $\dfrac{dy}{dx} = \dfrac{\pi}{2}\cdot\dfrac{2}{\pi} = 1$.

Example 4 Mixing rules with a table

Use the table of Topic 3.1, Example 3: $f(1)=2$, $f(2)=3$, $f(3)=1$; $f'(1)=5$, $f'(2)=-2$, $f'(3)=4$; $g(1)=3$, $g(2)=1$, $g(3)=2$; $g'(1)=-1$, $g'(2)=6$, $g'(3)=7$. Find (a) $p'(2)$ for $p(x) = f(x)\,g(x)$; (b) $q'(3)$ for $q(x) = \dfrac{f(x)}{g(x)}$; (c) $s'(1)$ for $s(x) = \ln\big(f(x)\big)$.

(a) Product rule: $p'(2) = f'(2)g(2) + f(2)g'(2) = (-2)(1) + (3)(6) = 16$.

(b) Quotient rule: $q'(3) = \dfrac{f'(3)g(3) - f(3)g'(3)}{\big(g(3)\big)^2} = \dfrac{4\cdot 2 - 1\cdot 7}{4} = \dfrac{1}{4}$.

(c) Chain rule: $s'(x) = \dfrac{f'(x)}{f(x)}$, so $s'(1) = \dfrac{5}{2}$.

Example 5 Simplify first

Find $\dfrac{dy}{dx}$ if $y = \ln\dfrac{x^2}{\sqrt{x^2+1}}$ and evaluate it at $x = 1$.

Differentiating the quotient inside the logarithm directly is long. Use the laws of logarithms first: $y = 2\ln x - \dfrac{1}{2}\ln(x^2+1)$. Then

$$\frac{dy}{dx} = \frac{2}{x} - \frac{1}{2}\cdot\frac{2x}{x^2+1} = \frac{2}{x} - \frac{x}{x^2+1}. \qquad \text{At } x = 1:\ 2 - \frac{1}{2} = \frac{3}{2}.$$

Which technique? Six worked specimens

Function	Outermost operation	Derivative
$y = xe^{-x}$	product	$e^{-x}(1-x)$
$y = \sqrt{x^2+4}$	power (chain)	$\dfrac{x}{\sqrt{x^2+4}}$
$y = \dfrac{\ln x}{x^2}$	quotient	$\dfrac{1-2\ln x}{x^3}$
$y = e^x \arctan x$	product, arctan	$e^x\left(\arctan x + \dfrac{1}{1+x^2}\right)$
$y = (\sin x)^x$	variable base *and* exponent	$(\sin x)^x\big(\ln(\sin x) + x\cot x\big)$
$x^2y + y^3 = 5$	implicit	$\dfrac{dy}{dx} = -\dfrac{2xy}{x^2+3y^2}$

Common errors

Wrong rule for a variable exponent. $\dfrac{d}{dx}2^x \neq x\,2^{x-1}$ (the power rule needs a constant exponent), and $\dfrac{d}{dx}x^x \neq x\cdot x^{x-1}$. Use $a^x \ln a$ or logarithmic differentiation.

Using the product rule where the chain rule is needed. $\sin(3x)$ is a composite, not the product $\sin\cdot 3x$. Conversely, $x\sin x$ is a product, not a composite.

Quotient rule with the terms reversed. The numerator is $u'v - uv'$ (bottom times derivative of top, minus top times derivative of bottom), not $uv' - u'v$.

AP Exam Tip

Do not waste time simplifying a derivative that the problem does not ask you to simplify. Scoring guides accept unsimplified correct expressions, and algebra errors made *after* a correct derivative are normally not penalised. But if you do simplify, check it: later parts (zeros of f') depend on it.

Check yourself 3.5

1. Differentiate $y = x^3\ln(2x)$.
2. Show that $\dfrac{d}{dx}\left[\dfrac{\cos x}{1+\sin x}\right] = -\dfrac{1}{1+\sin x}$.
3. Differentiate $y = 3^{x^2}$.

Topic 3.6 Calculation of Higher-Order Derivatives

Differentiating a derivative

The derivative f' is a function in its own right, so it can be differentiated. The result is the *second derivative* f'', and differentiating again gives f''', and so on. The notations are $f''(x)$, $\dfrac{d^2y}{dx^2}$, and for higher orders $f'''(x)$, $f^{(4)}(x)$, $\dfrac{d^ny}{dx^n}$. Note that $\dfrac{d^2y}{dx^2}$ is read "d two y, d x squared" and is *not* $\left(\dfrac{dy}{dx}\right)^2$.

The second derivative measures how the slope changes. If the graph of f is concave up on an interval, the slope is increasing and $f'' > 0$; if it is concave down, the slope is decreasing and $f'' < 0$ (this is developed in Unit 5). In motion along a line, with position $s(t)$, the velocity is $v = s'(t)$ and the *acceleration* is $a = v'(t) = s''(t)$. In practice, a second derivative is found by applying the rules of this unit twice; the important discipline is to simplify the first derivative into a form that is convenient to differentiate again.

Higher-order derivatives

$$f''(x) = \frac{d}{dx}\big[f'(x)\big], \qquad f'''(x) = \frac{d}{dx}\big[f''(x)\big], \qquad f^{(n)}(x) = \frac{d}{dx}\big[f^{(n-1)}(x)\big].$$

Implicit second derivative. After finding $\dfrac{dy}{dx} = F(x,y)$, differentiate F with respect to x *again using implicit*

differentiation, then substitute the formula for $\dfrac{dy}{dx}$ wherever $\dfrac{dy}{dx}$ appears, so that $\dfrac{d^2y}{dx^2}$ is expressed in x and y only.

Example 1 Successive derivatives of a cubic

Let $f(x) = x^3 - 6x^2 + 9x$. Find f', f'', f''', and use them to describe f.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$.
$f''(x) = 6x - 12 = 6(x-2)$.
$f'''(x) = 6$, and $f^{(4)}(x) = 0$.

Interpretation. f' is zero at $x = 1$ and $x = 3$ (horizontal tangents). $f'' < 0$ for $x < 2$ and $f'' > 0$ for $x > 2$, so the graph is concave down then concave up, changing at $(2, f(2)) = (2, 2)$. In the figure, f' (blue) has its minimum value $f'(2) = -3$ exactly where f'' (orange) crosses zero.

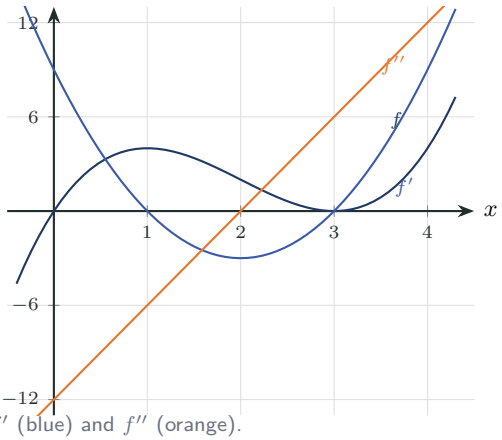


Figure 3.13. $f(x) = x^3 - 6x^2 + 9x$ (navy), f' (blue) and f'' (orange).

Example 2 The chain rule twice

Find $f''(x)$ for (a) $f(x) = e^{-x^2}$ and (b) $f(x) = \sin(2x)$.

(a) $f'(x) = -2x\,e^{-x^2}$. Now use the product rule: $f''(x) = -2e^{-x^2} + (-2x)(-2x)e^{-x^2} = (4x^2 - 2)\,e^{-x^2}$.
(b) $f'(x) = 2\cos 2x$, $f''(x) = -4\sin 2x$. Note that $f'' = -4f$.

Example 3 Second derivative of an implicit curve

For $x^2 + y^2 = 25$, find $\dfrac{d^2y}{dx^2}$ at $(3, 4)$.

Step 1. From Topic 3.2, $\dfrac{dy}{dx} = -\dfrac{x}{y}$.
Step 2. Differentiate with the quotient rule, treating y as a function of x:

$$\frac{d^2y}{dx^2} = -\frac{1\cdot y - x\dfrac{dy}{dx}}{y^2}.$$

Step 3. Substitute $\dfrac{dy}{dx} = -\dfrac{x}{y}$: $\dfrac{d^2y}{dx^2} = -\dfrac{y + \dfrac{x^2}{y}}{y^2} = -\dfrac{y^2 + x^2}{y^3} = -\dfrac{25}{y^3}$.

Step 4. At $(3, 4)$: $\dfrac{d^2y}{dx^2} = -\dfrac{25}{64} < 0$, so the upper semicircle is concave down there, as the figure of the circle suggests.

Second derivatives from graphs, in context, and nth derivatives

Example 4 f'' from the graph of f'

The graph of f', the derivative of f, consists of segments joining $(0, 3)$, $(2, -1)$, $(4, -1)$ and $(6, 2)$. Find $f''(1)$, $f''(3)$ and $f''(5)$, and state where the graph of f is concave up.

f'' is the slope of the graph of f'. On $(0,2)$ the slope is $\dfrac{-1-3}{2} = -2$; on $(2,4)$ it is 0; on $(4,6)$ it is $\dfrac{2-(-1)}{2} = \dfrac{3}{2}$.

So $f''(1) = -2$, $f''(3) = 0$ and $f''(5) = \dfrac{3}{2}$.

The graph of f is concave up where $f'' > 0$, that is on $4 < x < 6$ (and concave down on $0 < x < 2$; it has a straight-line piece on $2 < x < 4$).

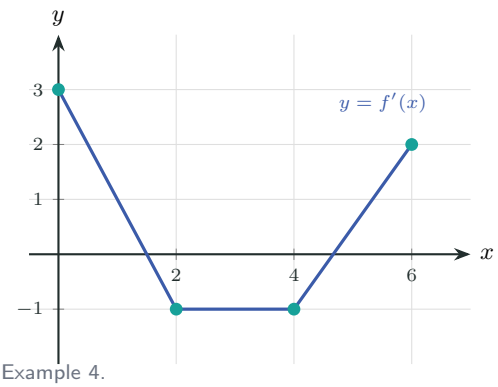


Figure 3.14. The graph of f' in Example 4.

Example 5 Velocity and acceleration

A particle moves along a line with position $s(t) = t^3 - 9t^2 + 24t$ metres, t in seconds. Find $v(t)$ and $a(t)$, and evaluate each at $t = 1$.

$v(t) = s'(t) = 3t^2 - 18t + 24 = 3(t-2)(t-4)$ and $a(t) = v'(t) = s''(t) = 6t - 18$. At $t = 1$: $v(1) = 9$ metres per second and $a(1) = -12$ metres per second per second. The velocity is positive but the acceleration is negative, so the particle is moving forward and slowing down.

Example 6 Patterns in nth derivatives

(a) Find $f^{(n)}(x)$ for $f(x) = e^{2x}$. (b) Find $\dfrac{d^{100}}{dx^{100}} \sin x$.

(a) $f' = 2e^{2x}$, $f'' = 4e^{2x}$, $f''' = 8e^{2x}$, so $f^{(n)}(x) = 2^n e^{2x}$.

(b) The derivatives of $\sin x$ repeat with period 4: $\cos x,\ -\sin x,\ -\cos x,\ \sin x$. Since $100 = 4 \cdot 25$, $\dfrac{d^{100}}{dx^{100}} \sin x = \sin x$.

Example 7 Concavity of $x \ln x$

Let $f(x) = x \ln x$ for $x > 0$. Find $f''(x)$ and decide where the graph is concave up.

Product rule: $f'(x) = \ln x + x \cdot \dfrac{1}{x} = \ln x + 1$. Then $f''(x) = \dfrac{1}{x}$. For every $x > 0$, $f''(x) > 0$, so the graph of f is concave up on $(0, \infty)$. (The $x \ln x$ curve never bends downward.)

Common errors

Stopping before substituting $\dfrac{dy}{dx}$. In an implicit second derivative, the answer must not contain $\dfrac{dy}{dx}$ unless the problem allows it.

Confusing the notations. $\dfrac{d^2y}{dx^2} \neq \left(\dfrac{dy}{dx}\right)^2$ and $f''(x) \neq \big(f'(x)\big)^2$.

Differentiating the unsimplified mess. Rewrite f' (factor, combine) before differentiating again; otherwise a sign slip in the quotient rule is likely.

AP Exam Tip

Many exam items ask for a second derivative at a point from a formula for f' (not f): differentiate f' once. If a table or graph of f' is given, $f''(a)$ is the slope of the graph of f' at a. Always state the units of f'' when it has a context, e.g. metres per second squared.

Check yourself 3.6

1. Find $f''(x)$ and $f'''(x)$ for $f(x) = x^5 - 3x^2$.
2. Find $f''(1)$ for $f(x) = \ln(x^2 + 1)$.
3. The curve $xy = 1$ defines y implicitly. Use implicit differentiation twice to find $\dfrac{d^2y}{dx^2}$ at $(1, 1)$.

Unit 3 summary

Formulas to know cold

$$\frac{d}{dx}f(g(x)) = f'(g(x))\,g'(x) \qquad \frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

$$\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}, \quad \frac{d}{dx}[h(y)] = h'(y)\frac{dy}{dx} \qquad (f^{-1})'(a) = \frac{1}{f'\big(f^{-1}(a)\big)}$$

$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}} \qquad \frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$$

$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}} \qquad \frac{d}{dx}\operatorname{arccot} u = -\frac{u'}{1+u^2}$$

$$y = u^v \;\Rightarrow\; \ln y = v\ln u, \quad \frac{dy}{dx} = y\cdot\frac{d}{dx}[v\ln u] \qquad f'' = \frac{d}{dx}f', \quad a(t) = v'(t) = s''(t)$$

How this unit is tested

Expect (i) a chain-rule evaluation from a table or graph, (ii) an implicit curve with a tangent line, a horizontal or vertical tangent, or $\frac{d^2y}{dx^2}$, (iii) an inverse derivative $g'(a)$ from a table, graph or formula, and (iv) an inverse trigonometric derivative inside a larger expression. These items appear in both multiple-choice sections and as parts of free-response questions, where the points go to the setup (the correct rule applied to the correct inner function), the correct value, and the units or justification.

Unit 3 Practice: Differentiation of Composite, Implicit, and Inverse Functions

Part A: Multiple Choice – No calculator

14 questions. Select the best answer.

1 If $f(x) = \sin(5x^2)$, then $f'(x) =$

(A) $\cos(5x^2)$
(B) $10x\cos(5x^2)$
(C) $-10x\cos(5x^2)$
(D) $\cos(10x)$

2 The table gives values of the differentiable functions f and g and their derivatives. If $h(x) = f(g(x))$, then $h'(2) =$

x	1	2	3
$f(x)$	3	1	2
$f'(x)$	-2	6	7
$g(x)$	2	3	1
$g'(x)$	5	-4	3

(A) 7
(B) -24
(C) 21
(D) -28

3 If $x^2 + xy = 10$, then $\dfrac{dy}{dx} =$

(A) $-\dfrac{2x+y}{x}$
(B) -2
(C) $-(2x+y)$
(D) $\dfrac{2x+y}{x}$

4 Let $f(x) = 2x^3 + 3x$ and let g be the inverse of f. What is the value of $g'(5)$?

(A) $\dfrac{1}{5}$
(B) 9
(C) $\dfrac{1}{9}$
(D) $\dfrac{1}{153}$

5 What is the value of $\dfrac{d}{dx}\left[\arctan(x^2)\right]$ at $x = 1$?

(A) $\dfrac{1}{2}$
(B) 1
(C) 2
(D) 4

6 If $f(x) = x^2e^x$, then $f''(1) =$

(A) $7e$
(B) $3e$
(C) $4e$
(D) 7

7 The function f is differentiable with $f'(2) = 3$ and $f'(4) = -2$. If $h(x) = f(x^2)$, then $h'(2) =$

(A) -2
(B) 12
(C) 1
(D) -8

8 Which of the following is an equation of the line tangent to the curve $x^3 + y^3 = 9$ at the point $(1, 2)$?

(A) $y - 2 = -4(x-1)$
(B) $y - 2 = \dfrac{1}{4}(x-1)$
(C) $y - 2 = -\dfrac{1}{4}(x-1)$
(D) $y - 2 = -\dfrac{1}{2}(x-1)$

9 If $f(x) = \arcsin\sqrt{x}$, then $f'\left(\frac{1}{2}\right) =$

(A) $\frac{1}{\sqrt{2}}$ **(B)** 1

(C) $\sqrt{2}$ **(D)** 2

10 What is the value of $\frac{d}{dx}\left[\frac{\cos x}{1+\sin x}\right]$ at $x = \frac{\pi}{2}$?

(A) $-\frac{1}{2}$ **(B)** -2

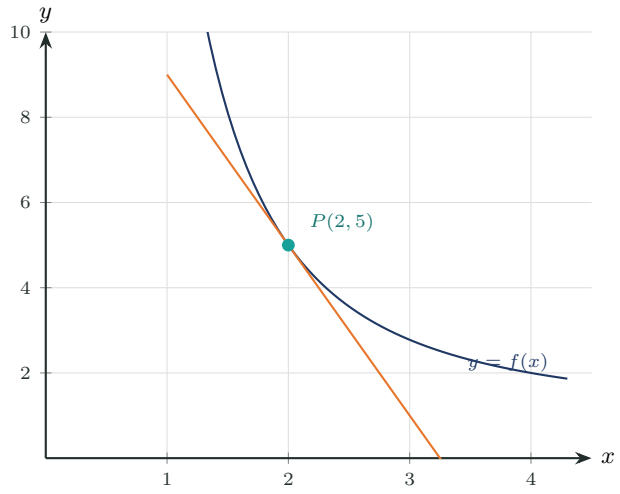
(C) $\frac{1}{2}$ **(D)** 0

11 If $x^2 + y^2 = 25$, then $\frac{d^2y}{dx^2}$ at the point $(3, 4)$ is

(A) $-\frac{3}{4}$ **(B)** $\frac{25}{64}$

(C) $-\frac{1}{4}$ **(D)** $-\frac{25}{64}$

12 The graph of the one-to-one differentiable function f is shown, together with its tangent line at $P(2, 5)$. The tangent line passes through the point $(3, 1)$. If g is the inverse of f, which of the following is an equation of the line tangent to the graph of g at $x = 5$?



(A) $y - 2 = -4(x - 5)$

(B) $y - 5 = -\frac{1}{4}(x - 2)$

(C) $y - 2 = -\frac{1}{4}(x - 5)$

(D) $y - 2 = \frac{1}{4}(x - 5)$

13 If $f(x) = e^{\arctan x}$, then $f'(1) =$

(A) $\frac{e^{\pi/4}}{2}$ **(B)** $e^{\pi/4}$

(C) $\frac{e^{\pi/4}}{\sqrt{2}}$ **(D)** $2e^{\pi/4}$

14 A curve is defined by $x^2 - 2xy + 2y^2 = 5$. At the point (a, b) with $a > 0$ the tangent line is horizontal. What is the value of a?

(A) 1 **(B)** $\sqrt{10}$

(C) $\sqrt{5}$ **(D)** 5

Part B: Multiple Choice – Graphing calculator allowed

7 questions. Use radian mode. Select the best answer.

15 If $f(x) = \sin(e^x)$, what is the value of $f'(1.2)$?

(A) -0.984 **(B)** 0.590
(C) 3.315 **(D)** -3.267

16 A curve is defined by $y + e^y = x^2 + 2$. At the point on the curve where $x = 2$ and $y > 0$, what is the value of $\dfrac{dy}{dx}$?

(A) 0.728 **(B)** 0.890
(C) 1.598 **(D)** 1.144

17 Let $f(x) = 2x + e^x$ and let g be the inverse of f. What is the value of $g'(6)$?

(A) 0.002 **(B)** 5.496
(C) 0.182 **(D)** 0.212

18 A drone rises vertically with height $h(t) = 3t^2$ metres at time t seconds. A camera on the ground 40 metres from the point below the drone measures the angle of elevation $\theta(t) = \arctan\left(\dfrac{h(t)}{40}\right)$. What is $\theta'(4)$, in radians per second?

(A) 0.600 **(B)** 0.246
(C) 0.273 **(D)** 9.836

19 A particle moves along a line with position $s(t) = t^2 e^{-t/2}$ for $t \geq 0$. What is the acceleration of the particle at time $t = 3$?

(A) 0.112 **(B)** 0.335
(C) -1.750 **(D)** -0.390

20 Let $f(x) = \ln(x^2 + 1) + \arctan x$. For $x > 0$, at what value of x is the slope of the graph of f equal to 0.8?

(A) 0.500 **(B)** 1.225
(C) 2.596 **(D)** 0.570

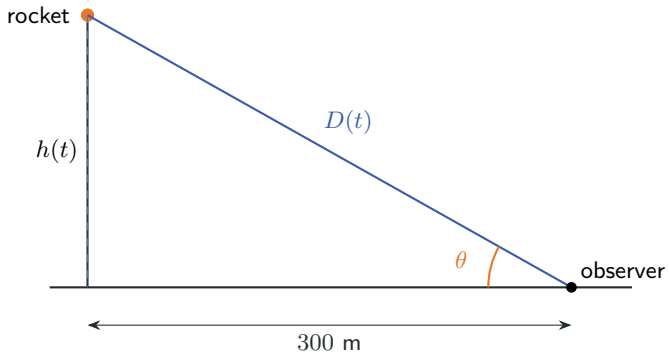
21 If $y = x^{\sin x}$ for $x > 0$, what is the value of $\dfrac{dy}{dx}$ at $x = 2$?

(A) 0.166 **(B)** 0.312
(C) 0.854 **(D)** 1.878

Part C: Free Response

2 questions, 9 points each. Show the work that leads to your answers.

22 [9 points] (Graphing calculator allowed.) A rocket is launched vertically from a pad. Its height above the ground is $h(t) = 10t^2$ metres, $0 \le t \le 20$, where t is in seconds. An observer stands on level ground 300 metres from the pad. Let $\theta(t)$ be the angle of elevation, in radians, from the observer to the rocket, and let $D(t)$ be the distance, in metres, from the observer to the rocket.



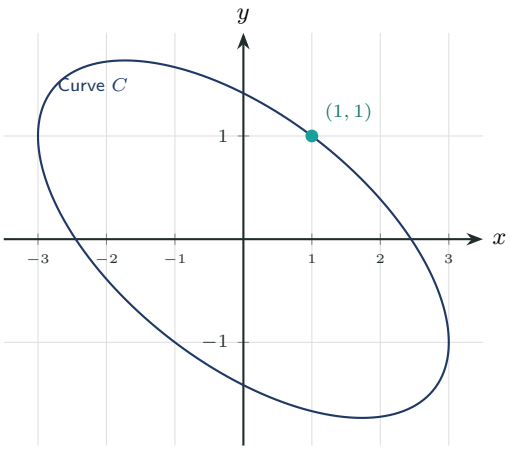
(a) Write $\theta(t)$ in terms of t. Find $\theta'(5)$. Indicate units of measure. *[2 points]*

(b) Let $T(h)$ be the time, in seconds, at which the rocket reaches height h metres. Use the derivative of an inverse function to find $T'(1000)$. Indicate units of measure. *[2 points]*

(c) Find the time t, for $0 < t < 20$, at which $\theta'(t)$ is greatest. Justify your answer. *[3 points]*

(d) Find the rate at which $D(t)$ is changing at $t = 5$. Indicate units of measure. *[2 points]*

23 [9 points] (No calculator.) Consider the curve C given by $x^2 + 2xy + 3y^2 = 6$. The point $(1, 1)$ lies on C.



(a) Show that $\dfrac{dy}{dx} = -\dfrac{x+y}{x+3y}$. *[2 points]*

(b) Write an equation of the line tangent to C at the point $(1, 1)$. *[2 points]*

(c) Find the coordinates of each point on C at which the tangent line is horizontal. *[3 points]*

(d) Find $\dfrac{d^2y}{dx^2}$ at the point $(1, 1)$. Is C concave up or concave down at $(1, 1)$? Give a reason for your answer. *[2 points]*

AP Calculus AB – Unit 4 Exam weighting: 10–15% of the AB exam

Contextual Applications of Differentiation

In this unit

In Unit 3 you learned the rules of differentiation. In this unit the derivative goes to work. A derivative is a *rate of change*, and nearly every quantity that matters in science, economics and engineering has a rate attached to it: the speed of a car, the rate at which a tank fills, the growth of a population, the marginal cost of one more item. The exam rarely asks "find the derivative" in isolation; it asks what the derivative *means* in a story, what its units are, and what it lets you conclude.

Topics.

4.1 Interpreting the meaning of the derivative in context
4.2 Straight-line motion: connecting position, velocity and acceleration
4.3 Rates of change in applied contexts other than motion
4.4 Introduction to related rates
4.5 Solving related rates problems
4.6 Approximating values of a function using local linearity and linearization
4.7 Using L'Hôpital's Rule for determining limits of indeterminate forms

Big ideas. (1) $f'(a)$ is an instantaneous rate with units *(output units) per (input unit)*. (2) The sign of v and a together decide whether a particle speeds up or slows down. (3) When two quantities are linked by an equation, their *rates* are linked by the chain rule. (4) Near a point, a differentiable function is almost its tangent line. (5) A $\frac{0}{0}$ or $\frac{\infty}{\infty}$ limit can be settled by comparing derivatives.

Topic 4.1 Interpreting the Meaning of the Derivative in Context

A derivative is a rate, and a rate has units

Formally, $f'(a) = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$. The quotient inside the limit is a change in output divided by a change in input, so the derivative inherits its units from that fraction:

$$\text{units of } f' \;=\; \frac{\text{units of the output } f}{\text{units of the input } x}.$$

If $V(t)$ is the volume of water in a tank in liters and t is measured in minutes, then $V'(t)$ is measured in *liters per minute*. This is the single most useful check you have: if your interpretation of $V'(5)$ does not end in "liters per minute", something is wrong.

The *sign* of $f'(a)$ tells you the direction of change. A positive value means the quantity is increasing at that instant; a negative value means it is decreasing; zero means it is momentarily not changing. The *size* $|f'(a)|$ tells you how fast. Notice that the derivative describes the situation *at one instant* (or at one input value). It does not tell you how much the quantity changed over a long period, and it is not the value of the quantity itself. Much of the work in this topic is keeping these three ideas apart: the value $f(a)$, the instantaneous rate $f'(a)$, and the average rate over an interval.

The exam asks for an interpretation in a precise way. A complete answer names the **quantity**, the **input value** (with units), whether it is **increasing or decreasing**, the **rate** (a positive number for the size), and the **units of the rate**. A useful template is:

> *"At $t = a$ [units], the [quantity] is [increasing/decreasing] at a rate of $|f'(a)|$ [output units] per [input unit]."*

Average rate versus instantaneous rate

The *average rate of change* of f on $[a,b]$ is $\dfrac{f(b)-f(a)}{b-a}$, the slope of the secant line. The *instantaneous rate* at $x=a$ is the slope of the tangent line. When a function is given only by a table of values, you cannot compute $f'(a)$ exactly; the best you can do is estimate it with the average rate over a small interval that contains a (the smaller and closer to a, the better). On the AP exam, use the *closest pair of table values that surround the input*, and show the quotient.

Key ideas: derivative in context

- Units of $f'(x)$: *(units of f) per (unit of x)*.
- $f'(a) > 0$: f is increasing at $x = a$; $f'(a) < 0$: f is decreasing at $x = a$.
- Average rate on $[a, b]$: $\dfrac{f(b) - f(a)}{b - a}$ (secant slope). Instantaneous rate at a: $f'(a)$ (tangent slope).
- Estimate from a table: $f'(c) \approx \dfrac{f(t_2) - f(t_1)}{t_2 - t_1}$ using the nearest values $t_1 \le c \le t_2$.
- **Marginal** cost, revenue or profit is the derivative: $C'(x)$ estimates the cost of producing one more unit, $C(x + 1) - C(x) \approx C'(x)$.

Example 1 — Interpreting $F'(5)$ with units

The volume of fuel in a tank is $F(t)$ liters, where t is in minutes. It is known that $F(5) = 84$ and $F'(5) = -3.2$. Interpret $F'(5)$, and use it to estimate $F(6)$.
Step 1 – units. F is in liters and t in minutes, so F' is in liters per minute.
Step 2 – sign. $F'(5) = -3.2 < 0$, so the volume is decreasing.
Step 3 – sentence. *At $t = 5$ minutes, the volume of fuel in the tank is decreasing at a rate of 3.2 liters per minute.*
Step 4 – estimate. In one more minute the volume changes by about $F'(5) \cdot 1 = -3.2$ liters, so $F(6) \approx 84 - 3.2 = 80.8$ liters.

Example 2 — Estimating a derivative from a table

The temperature $T(t)$ of a bowl of soup, in °C, is recorded t minutes after it is served.

t (min)	0	4	10	15
$T(t)$ (°C)	90	78	64	58

(a) Estimate $T'(7)$ and interpret it. (b) Estimate $T'(12)$. (c) Find the average rate of change of T on $[0, 15]$.
(a) The nearest table values around $t = 7$ are $t = 4$ and $t = 10$:

$$T'(7) \approx \frac{T(10) - T(4)}{10 - 4} = \frac{64 - 78}{6} = -\frac{7}{3} \approx -2.333.$$

At $t = 7$ minutes the soup is cooling at about 2.333°C per minute.
(b) Using $t = 10$ and $t = 15$: $T'(12) \approx \dfrac{58 - 64}{15 - 10} = -1.2$°C per minute.
(c) $\dfrac{T(15) - T(0)}{15 - 0} = \dfrac{58 - 90}{15} = -\dfrac{32}{15} \approx -2.133$°C per minute.
The cooling slows down ($|-2.333| > |-1.2|$), as Newton's law of cooling predicts: the hotter the soup, the faster it loses heat.

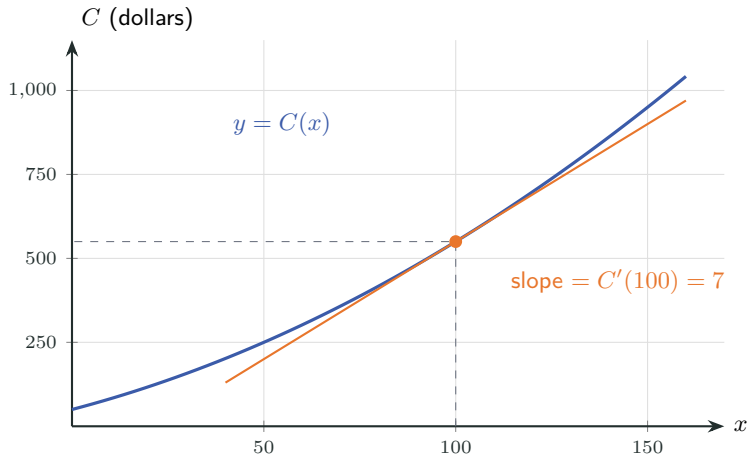


Figure 4.1: cost function $C(x) = 50 + 3x + 0.02x^2$ and its tangent line at $x = 100$; the slope 7 is the marginal cost.

Example 3 — Marginal cost

The cost, in dollars, of producing x chairs in one day is $C(x) = 50 + 3x + 0.02x^2$. (a) Find $C'(100)$ and interpret it. (b) Compare it with the actual cost of the 101st chair.
(a) $C'(x) = 3 + 0.04x$, so $C'(100) = 3 + 4 = 7$. *When 100 chairs are being produced per day, the cost is increasing at a rate of \$7 per chair.*
(b) $C(101) - C(100) = \big(50 + 303 + 204.02\big) - \big(50 + 300 + 200\big) = 557.02 - 550 = 7.02$. The derivative 7 is an excellent approximation of the true cost \$7.02 of one more chair: when the change in input is only 1 and the function is smooth, $C(x+1) - C(x) \approx C'(x)$. Figure 4.1 shows why – the tangent line hugs the curve near $x = 100$.

Common errors

- Confusing value and rate: $A'(3) = 12$ does *not* mean the area *is* 12, nor that it *grew by* 12 in three hours. It means that *at the instant* $t = 3$ the area is changing at 12 units of area per unit of time.
- Forgetting units or the word "per". "The rate is 12" earns no credit; "12 square meters per hour" does.
- Writing a negative rate of decrease: say "decreasing at a rate of 3.2", not "decreasing at -3.2".
- Using a table interval that does not surround the input (or is far wider than needed), or forgetting to divide by the change in t.

AP Exam Tip

Interpretations are graded on four ingredients: the **quantity in context**, the **input value with units**, the **direction** (increasing/decreasing), and the **rate with units**. If a part says "Using correct units, explain the meaning of…", a missing unit costs the point. Do not simplify an answer that is already numerical (such as $-\frac{7}{3}$ or $\frac{5}{6}$) unless asked; on calculator parts give three decimal places.

Check yourself

1. The volume of water in a reservoir is $W(t)$ million liters, t in days. Interpret $W'(10) = -1.8$.
2. An odometer records $s(t)$ miles at t hours: $s(0) = 0$, $s(1) = 55$, $s(3) = 170$, $s(4) = 212$. Estimate $s'(2)$ with units and interpret it.
3. The profit from selling x items is $P(x) = -0.5x^2 + 40x - 150$ dollars. Find $P'(20)$, interpret it, and compare it with $P(21) - P(20)$.

Topic 4.2 Straight-Line Motion: Connecting Position, Velocity, and Acceleration

Three functions, one story

When a particle moves on a line (the x-axis, or any number line), three functions describe the motion, and each is the derivative of the one before:

$$x(t) \text{ (position)} \xrightarrow{d/dt} v(t) = x'(t) \text{ (velocity)} \xrightarrow{d/dt} a(t) = v'(t) = x''(t) \text{ (acceleration)}.$$

Position says *where* the particle is. Velocity says how fast the position is changing *and in which direction*: $v > 0$ means the particle moves in the positive direction (to the right if the axis is horizontal), $v < 0$ means it moves in the negative direction, and $v = 0$ means it is momentarily at rest. Acceleration says how fast the *velocity* is changing. Note carefully that acceleration describes the change in velocity, not directly the change in speed.

Speed is the magnitude of velocity, $|v(t)|$. It is never negative, and it ignores direction. A particle with $v = -5$ is moving left at speed 5.

Speeding up and slowing down

Think of a car driving in reverse. If you press the accelerator while moving backward, v is negative and the velocity becomes *more* negative, so $a < 0$: the car goes faster. If you brake, the velocity moves toward zero, so $a > 0$ while $v < 0$: the car slows. This leads to the rule that every AP student must be able to state and justify:

> The speed is **increasing** when $v(t)$ and $a(t)$ have the **same sign**; the speed is **decreasing** when $v(t)$ and $a(t)$ have **opposite signs**.

A particle **changes direction** at a time when $v(t) = 0$ *and* v changes sign. A zero of v without a sign change (for example $v(t) = t^2$ at $t = 0$) is only a momentary stop.

Displacement versus total distance

The *displacement* from $t = t_1$ to $t = t_2$ is $x(t_2) - x(t_1)$: it can be positive, negative or zero. The *total distance* traveled is the sum of the lengths of all the legs between turning points. With a position function, find the times when the particle turns around, evaluate x at the endpoints and at every turning point, and add the absolute values of the changes along each leg. (Later, in Unit 6, you will compute distance as $\int_{t_1}^{t_2} |v(t)|\,dt$.)

Key ideas: rectilinear motion

- $v(t) = x'(t)$, $a(t) = v'(t) = x''(t)$, speed $= |v(t)|$.
- Moving right/up (positive direction): $v > 0$. Moving left/down: $v < 0$. At rest: $v = 0$.
- Speed increasing $\iff v\,a > 0$. Speed decreasing $\iff v\,a < 0$.
- Direction change: $v = 0$ and v changes sign.
- Displacement on $[t_1, t_2]$: $x(t_2) - x(t_1)$. Total distance: add the lengths of the legs between turning points.

Example 1 — A full motion analysis

A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $0 \le t \le 4$. (a) Find $v(t)$ and $a(t)$. (b) When is the particle at rest, and when does it change direction? (c) On which intervals is its speed increasing? (d) Find the total distance traveled on $[0, 4]$.

(a) $v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$ and $a(t) = 6t - 12 = 6(t-2)$.

(b) $v(t) = 0$ at $t = 1$ and $t = 3$. For $0 < t < 1$: $v > 0$; for $1 < t < 3$: $v < 0$; for $3 < t < 4$: $v > 0$. Since v changes sign at both times, the particle changes direction at $t = 1$ and $t = 3$.

(c) $a(t) = 0$ at $t = 2$; $a < 0$ for $t < 2$ and $a > 0$ for $t > 2$. Make a sign chart:

interval	$(0,1)$	$(1,2)$	$(2,3)$	$(3,4)$
sign of v	$+$	$-$	$-$	$+$
sign of a	$-$	$-$	$+$	$+$
speed	decreasing	**increasing**	decreasing	**increasing**

The speed is increasing on $(1,2)$ and $(3,4)$ because v and a have the same sign there.

(d) Evaluate position at the endpoints and turning points: $x(0) = 0$, $x(1) = 4$, $x(3) = 0$, $x(4) = 4$. The legs have lengths $|4-0| = 4$, $|0-4| = 4$, $|4-0| = 4$, so the total distance is 12. (The displacement is only $x(4) - x(0) = 4$.)

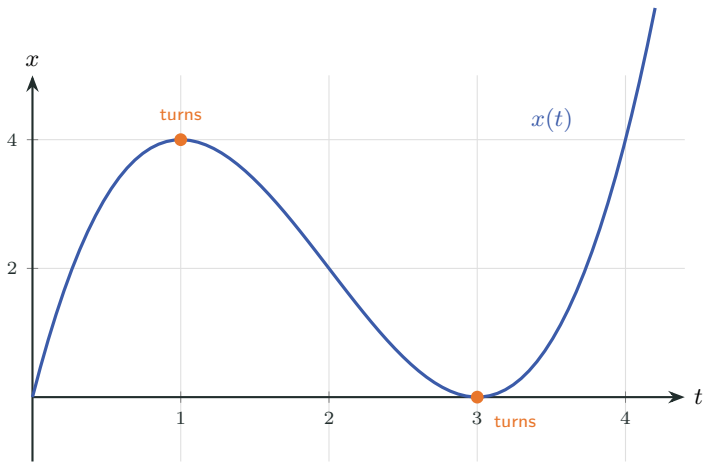


Figure 4.2: $x(t) = t^3 - 6t^2 + 9t$. The particle reverses at the local maximum ($t = 1$) and local minimum ($t = 3$).

Example 2 — Reading a velocity graph

The velocity $v(t)$ of a particle moving on the x-axis is shown; v is piecewise linear with $v(0) = 0$, $v(2) = 4$, $v(4) = 4$, $v(7) = -2$, $v(9) = -2$. (a) When is the particle at rest? (b) Find $a(t)$ on each piece. (c) When is the speed increasing?

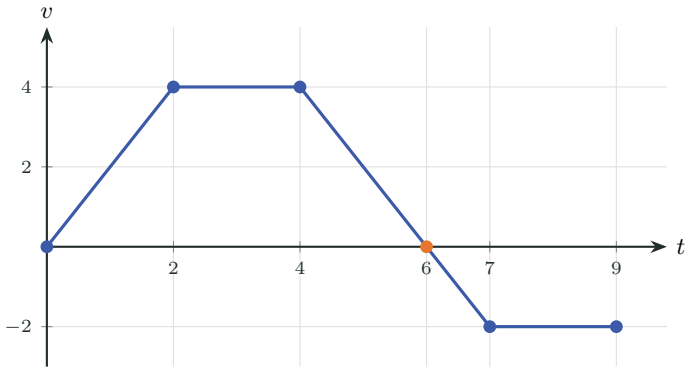


Figure 4.3: velocity graph for Example 2. The orange point is the only time (after $t = 0$) the particle is at rest.

(a) $v = 0$ at $t = 0$ and where the segment from $(4, 4)$ to $(7, -2)$ crosses the axis. That segment has slope $\dfrac{-2-4}{7-4} = -2$, so $v(t) = 4 - 2(t-4) = 12 - 2t$ and $v = 0$ at $t = 6$. The particle changes direction at $t = 6$ (positive to negative).
(b) Acceleration is the slope of the velocity graph: $a = 2$ on $(0, 2)$, $a = 0$ on $(2, 4)$, $a = -2$ on $(4, 7)$, $a = 0$ on $(7, 9)$.
(c) $(0, 2)$: $v > 0$, $a > 0$ – speeding up. $(2, 4)$: $a = 0$ – constant speed. $(4, 6)$: $v > 0$, $a < 0$ – slowing down. $(6, 7)$: $v < 0$, $a < 0$ – speeding up. $(7, 9)$: constant speed. So the speed is increasing on $(0, 2)$ and $(6, 7)$.

Example 3 — Justifying with signs

A particle has velocity $v(t) = 4t - t^2$ for $t \geq 0$. Is its speed increasing or decreasing at $t = 3$? At $t = 5$? Give reasons.
$a(t) = v'(t) = 4 - 2t$. At $t = 3$: $v(3) = 12 - 9 = 3 > 0$ and $a(3) = -2 < 0$. The signs are opposite, so the speed is *decreasing* at $t = 3$. At $t = 5$: $v(5) = 20 - 25 = -5 < 0$ and $a(5) = -6 < 0$. The signs are the same, so the speed is *increasing* at $t = 5$.

Common errors

- "The particle is slowing down because acceleration is negative." Wrong: you must compare the signs of v *and* a. A negative a with a negative v means speeding up.
- Treating $v(t) = 0$ as automatically a change of direction. You must see the sign of v change.
- Confusing velocity with speed: $v(t) = -5$ is a speed of 5, not -5.
- Computing total distance as $x(t_2) - x(t_1)$ (that is displacement) without checking for turnarounds.

AP Exam Tip

A justified conclusion cites *values*, not words: "The speed is increasing at $t = 5$ because $v(5) = -5 < 0$ and $a(5) = -6 < 0$." For a direction change write "$v(t)$ changes from positive to negative at $t = 6$". A graph of v answers questions about a through *slope*, and about speed through *distance from the axis*.

Check yourself

1. A particle has position $x(t) = t^2 - 6t + 5$. Find $v(1)$ and $a(1)$, state the direction of motion at $t = 1$, and decide whether the speed is increasing or decreasing at $t = 1$. When does the particle change direction?
2. A particle has velocity $v(t) = t^3 - 3t^2$ for $t > 0$. On which intervals is its speed increasing?
3. A particle has position $x(t) = t^2 - 4t$ for $0 \leq t \leq 5$. Find its displacement and the total distance traveled.

Topic 4.3 Rates of Change in Applied Contexts Other Than Motion

The same idea, in every subject

Topic 4.2 treated motion, where the derivative of position is velocity. The derivative is just as useful when the quantity is not a position: the number of bacteria in a culture, the volume of water in a tank, the cost of manufacturing, the concentration of a drug in the blood, the temperature of a cooling object. In every case the derivative with respect to time t (or with respect to some other input) is a *rate*, and its units are always the units of the quantity divided by the units of the input. Population in thousands and t in years gives *thousands of people per year*; cost in dollars and x the number of items gives *dollars per item*.

Many exam problems give you the *rate* directly, as a function or a graph of $P'(t)$, and expect you to reason about the original quantity $P(t)$. Two facts do almost all of the work:

- the **sign** of the rate decides whether the quantity is increasing ($P' > 0$) or decreasing ($P' < 0$);
- the **size** of the rate decides how fast, so the quantity grows fastest where P' is largest.

A second derivative adds one more layer. $P''(t)$ is the rate of change of the *rate*; its units are (units of P) per (unit of t) *per (unit of t)*, for example "thousand people per year per year". A positive P'' means the rate is increasing, which is not the same as the quantity increasing. A population can be decreasing ($P' < 0$) while $P'' > 0$: it is then still falling, but more and more slowly.

Rates that are percentages: the relative rate

Sometimes the interesting number is the rate *relative to the size* of the quantity, $\dfrac{P'(t)}{P(t)}$. For exponential models $P(t) = P_0 e^{kt}$ we get $P'(t) = kP(t)$, so the relative rate is the constant k: the quantity grows by the same percentage each unit of time, while the absolute rate $P'(t)$ keeps getting larger. (Differential equations of this form return in Unit 7.)

Key ideas: rates in applied contexts

- Units of $Q'(t)$: (units of Q) per (unit of t). Units of $Q''(t)$: (units of Q) per (unit of t) per (unit of t).
- $Q' > 0 \Rightarrow Q$ increasing; $Q' < 0 \Rightarrow Q$ decreasing. $Q'' > 0 \Rightarrow Q'$ increasing; $Q'' < 0 \Rightarrow Q'$ decreasing.
- If Q' is given by a graph, Q is increasing where the graph is above the t-axis, and Q increases *fastest* where the graph of Q' is highest.
- Marginal cost $C'(x)$, marginal revenue $R'(x)$, marginal profit $P'(x) = R'(x) - C'(x)$ approximate the change caused by one more unit.
- For $Q(t) = Q_0 e^{kt}$: $Q'(t) = kQ(t)$; the relative rate Q'/Q equals k.

Example 1 — A growing population (calculator)

The population of a town is $P(t) = 50\,000\,e^{0.02t}$, where t is the number of years since the start of 2020. Find $P'(10)$, interpret it with units, and find the relative rate of growth.

Step 1. $P'(t) = 50\,000(0.02)e^{0.02t} = 1000\,e^{0.02t}$.

Step 2. $P'(10) = 1000\,e^{0.2} \approx 1221.403$.

Step 3 – interpretation. *Ten years after the start of 2020, the population is increasing at a rate of about* 1221.403 *people per year.*

Step 4 – relative rate. $\dfrac{P'(t)}{P(t)} = \dfrac{1000e^{0.02t}}{50\,000e^{0.02t}} = 0.02$: the town grows by 2% per year at every moment, even though the number of people added each year grows.

Example 2 — A tank draining, and the rate of the rate

Water drains from a tank so that the volume at time t minutes is $V(t) = 400\left(1 - \dfrac{t}{20}\right)^2$ liters, for $0 \le t \le 20$. (a) Find $V'(5)$ and interpret it. (b) Find $V''(5)$ and interpret it. (c) Is the tank draining faster or more slowly as time goes on?

(a) $V'(t) = 800\left(1 - \dfrac{t}{20}\right)\left(-\dfrac{1}{20}\right) = -40 + 2t$, so $V'(5) = -30$. *At $t = 5$ minutes the volume of water is decreasing at a rate of* 30 *liters per minute.*

(b) $V''(t) = 2$, so $V''(5) = 2$. *At $t = 5$ minutes the rate of change of the volume is increasing at* 2 *liters per minute*

per minute. Because V' is negative and increasing toward 0, the water is still leaving, but the draining is slowing down.
(c) $V''(t) = 2 > 0$ for all t, so V' increases from $V'(0) = -40$ to $V'(20) = 0$. The tank drains *more slowly* as time goes on, and the flow stops at $t = 20$, when the tank is empty.

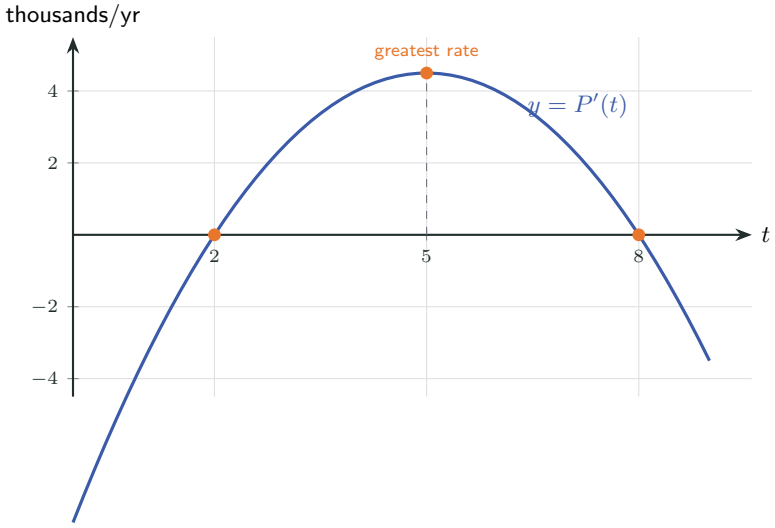


Figure 4.4: the *rate* $P'(t)$ for Example 3. P decreases on $(0, 2)$, increases on $(2, 8)$ and decreases again on $(8, 9)$; it increases fastest at $t = 5$.

Example 3 — Reading a graph of the rate

The number of fish in a lake is $P(t)$ thousand, t in years, $0 \le t \le 9$. Figure 4.4 shows the graph of $P'(t) = \dfrac{(t-2)(8-t)}{2}$. (a) When is the fish population increasing? (b) When is it increasing most rapidly? (c) Is the population increasing or decreasing at $t = 9$, and is that rate speeding up or slowing down?
(a) $P'(t) > 0$ exactly when the graph is above the axis, that is for $2 < t < 8$. The population is increasing on $(2, 8)$ (and decreasing on $(0, 2)$ and $(8, 9)$).
(b) The population grows fastest where P' is greatest. The parabola's vertex is at $t = 5$, where $P'(5) = \dfrac{3 \cdot 3}{2} = 4.5$. Because $P''(t) = \frac{1}{2}(10 - 2t) = 5 - t$ changes from positive to negative at $t = 5$, this is a maximum of P': the population is growing fastest at $t = 5$, by 4.5 thousand fish per year.
(c) $P'(9) = \dfrac{7 \cdot (-1)}{2} = -3.5 < 0$: the population is decreasing, at 3.5 thousand fish per year. $P''(9) = 5 - 9 = -4 < 0$, so P' is decreasing: the decline is getting steeper (the rate of decrease is increasing).

Common errors

- Mixing up "the quantity is increasing" ($Q' > 0$) with "the rate is increasing" ($Q'' > 0$). On a graph of Q': the first is about being *above the axis*, the second about the graph *rising*.
- Substituting into P when the question asks for P' (a value is not a rate), or reporting $P(5)$ as "the rate".
- Saying a quantity is "decreasing at an increasing rate" when $Q' < 0$ and $Q'' > 0$. The rate Q' is increasing, but it is negative and moving toward 0: the quantity decreases *more slowly*.
- Leaving off the second "per" in the units of a second derivative.

AP Exam Tip

Context answers need three things: the **units**, the **time or input value**, and the **direction**. Compare "$P'(5) = 4.5$" (no credit) with "At $t = 5$ years, the fish population is increasing at 4.5 thousand fish per year" (full credit). For a statement about *increasing fastest*, justify with the behaviour of P': "P' has its maximum at $t = 5$ because P'' changes from positive to negative there".

Check yourself

1. The number of people in a queue is $L(t)$, t in minutes. Given $L'(15) = -4$ and $L''(15) = 3$, interpret each statement. Is the queue getting shorter faster or more slowly at $t = 15$?
2. The temperature of a cup of coffee is $H(t) = 20 + 70e^{-0.1t}$ °C, t in minutes. Show that $H'(t) < 0$ and $H''(t) > 0$ for all t, and describe in words what this says about how the coffee cools.
3. The revenue from selling x items is $R(x) = 60x - 0.1x^2$ dollars. Find $R'(100)$, interpret it, and compare it with $R(101) - R(100)$. For what value of x is the marginal revenue zero?

Topic 4.4 Introduction to Related Rates

Quantities that move together

A circular ripple spreads across a pond. Its radius r grows, so its area $A = \pi r^2$ grows too. Neither can change without the other, and their *rates* are tied together in the same way. Whenever two or more quantities are linked by an equation and all of them change with time, the equation links their rates of change. Finding one unknown rate from the others is called a **related rates** problem.

The mechanism is nothing new: it is the chain rule. Every quantity in the problem is a function of time t, even if the problem never writes $r(t)$ or $A(t)$. So when we differentiate $A = \pi r^2$ *with respect to* t we must remember that r depends on t:

$$A = \pi r^2 \quad \Longrightarrow \quad \frac{dA}{dt} = \pi \cdot 2r \cdot \frac{dr}{dt} = 2\pi r \, \frac{dr}{dt}.$$

The factor $\dfrac{dr}{dt}$ is exactly what the chain rule produces, and it is the factor students forget most often. The result says: the rate at which the area grows equals the circumference $2\pi r$ times the rate at which the radius grows.

Where the equations come from

The equation relating the variables almost always comes from geometry: the area or volume formula of a circle, sphere, cone, cylinder or rectangle; the Pythagorean Theorem for a right triangle; similar triangles; or a trigonometric ratio. The skill is to choose a formula that contains exactly the quantities named in the problem. If a formula contains an extra variable, use the geometry (for example similar triangles or a given proportion) to eliminate it *before* differentiating.

Key ideas: setting up a related rates equation

- Every changing quantity is a function of t. Differentiate both sides of the relating equation **with respect to** t using the chain rule: $\dfrac{d}{dt}\big[g(u)\big] = g'(u)\dfrac{du}{dt}$.
- $A = \pi r^2 \Rightarrow \dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt}$. $\quad V = \tfrac{4}{3}\pi r^3 \Rightarrow \dfrac{dV}{dt} = 4\pi r^2\dfrac{dr}{dt}$. $\quad S = 4\pi r^2 \Rightarrow \dfrac{dS}{dt} = 8\pi r\dfrac{dr}{dt}$.
- $V = s^3 \Rightarrow \dfrac{dV}{dt} = 3s^2\dfrac{ds}{dt}$. $\quad x^2 + y^2 = z^2 \Rightarrow 2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 2z\dfrac{dz}{dt}$.
- Product of changing quantities: $A = \ell w \Rightarrow \dfrac{dA}{dt} = \dfrac{d\ell}{dt}w + \ell\dfrac{dw}{dt}$.
- **Sign convention:** a quantity that is increasing has a positive rate; one that is decreasing has a negative rate.

Example 1 — An expanding ripple

A pebble creates a circular ripple whose radius increases at a constant 3 cm/s. How fast is the area of the ripple increasing when the radius is 10 cm?

Step 1 – relation. $A = \pi r^2$.

Step 2 – differentiate with respect to t. $\dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt}$.

Step 3 – substitute $r = 10$, $\dfrac{dr}{dt} = 3$: $\dfrac{dA}{dt} = 2\pi(10)(3) = 60\pi$.

Answer. The area is increasing at $60\pi \approx 188.496$ square centimeters per second.

Note: the rate depends on r; when $r = 20$ the area grows twice as fast although $\frac{dr}{dt}$ is unchanged.

Example 2 — A balloon being inflated

Air is pumped into a spherical balloon at a constant rate of 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

Step 1. $V = \frac{4}{3}\pi r^3$.
Step 2. $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$.
Step 3. Substitute *after* differentiating: $100 = 4\pi(5)^2 \dfrac{dr}{dt} = 100\pi \dfrac{dr}{dt}$.
Step 4. $\dfrac{dr}{dt} = \dfrac{1}{\pi} \approx 0.318$ cm/s.
As the balloon gets bigger, the same inflow of air raises the radius more slowly, which is exactly what we expect from the r^2 in the denominator.

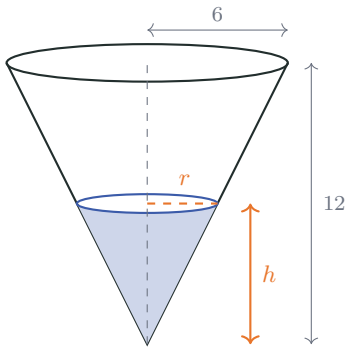


Figure 4.5: a conical tank, height 12, top radius 6, with water to depth h. Similar triangles give $\frac{r}{h} = \frac{6}{12}$, so $r = \frac{h}{2}$.

Example 3 — Eliminating a variable first

Water is poured at 10 cm^3/s into an inverted cone (point down) of height 12 cm and top radius 6 cm (Figure 4.5). How fast is the water level rising when the depth is $h = 4$ cm?

Step 1 – relation. The volume of water is $V = \frac{1}{3}\pi r^2 h$, which has two variables. By similar triangles $\dfrac{r}{h} = \dfrac{6}{12}$, so $r = \dfrac{h}{2}$ and

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi}{12} h^3.$$

Step 2. $\dfrac{dV}{dt} = \dfrac{\pi}{12} \cdot 3h^2 \dfrac{dh}{dt} = \dfrac{\pi h^2}{4} \dfrac{dh}{dt}$.
Step 3. Substitute $\dfrac{dV}{dt} = 10$ and $h = 4$: $10 = \dfrac{\pi \cdot 16}{4} \dfrac{dh}{dt} = 4\pi \dfrac{dh}{dt}$.
Step 4. $\dfrac{dh}{dt} = \dfrac{10}{4\pi} = \dfrac{5}{2\pi} \approx 0.796$ cm/s.
The water level is rising at about 0.796 cm per second.

Common errors

- **Substituting too early.** If you replace r by 10 in $A = \pi r^2$ you get $A = 100\pi$, a constant, and your derivative is 0. Only quantities that stay constant for the whole problem (a ladder's length, a cone's dimensions) may be substituted before differentiating; every quantity that varies must wait until *after* you differentiate.
- Forgetting the chain-rule factor: $\dfrac{d}{dt}(r^2) = 2r$ is wrong; it is $2r\dfrac{dr}{dt}$.
- Using a formula with an unnecessary variable and not eliminating it (as with r and h in the cone).
- Giving a positive rate for a quantity that is decreasing.

AP Exam Tip

Always write the relating equation *first* and the differentiated equation *second*, as separate lines. Scorers award separate points for (1) a correct relating equation, (2) a correct differentiated equation (including the $\frac{d}{dt}$ factors), (3) the substitution, and (4) the answer with units and direction. A correct derivative equation can earn credit even if the arithmetic afterwards is wrong.

Check yourself

1. Starting from $V = s^3$ for a cube of side s, write an equation relating $\dfrac{dV}{dt}$ and $\dfrac{ds}{dt}$. If s increases at 0.5 cm/s, find $\dfrac{dV}{dt}$ when $s = 4$ cm.
2. Starting from $x^2 + y^2 = 25$ with x and y functions of t, find $\dfrac{dy}{dt}$ in terms of x, y and $\dfrac{dx}{dt}$.
3. A spherical balloon is inflated so that $\dfrac{dr}{dt} = 2$ cm/s. How fast is the surface area $S = 4\pi r^2$ increasing when $r = 3$ cm?

Topic 4.5 Solving Related Rates Problems

A reliable method

Topic 4.4 gave the tools; this topic is about using them reliably on the whole range of problems the exam asks. Follow the same sequence every time.

1. **Sketch and label.** Draw the situation at a general moment, not a special one, and name every changing quantity with a letter. Write down what is given and what is wanted as derivatives, with signs (a ladder sliding down: $\frac{dy}{dt} < 0$).
2. **Write a relating equation** among the letters. Use geometry; eliminate extra variables with a constant proportion if you can.
3. **Differentiate with respect to t,** using the chain rule and, if two variables multiply, the product rule.
4. **Substitute** the instant's values of all variables and given rates. If a needed value (such as y in the ladder) is not given, find it from the relating equation at that moment.
5. **Solve** for the unknown rate and answer in a sentence: quantity, instant, direction, units.

How the exam asks

Related-rates items appear in both multiple-choice sections and as parts of free-response questions. The standard configurations are: two legs of a right triangle changing with a fixed hypotenuse (ladder) or a changing hypotenuse (two moving objects); similar triangles (lamp and shadow); volumes of tanks (cone, cylinder, sphere); and an angle of elevation or a camera angle, which uses a trigonometric ratio. In free-response parts you may also be asked "Is the quantity increasing or decreasing at that instant?", which is answered by the sign of your result.

Key ideas: strategy and configurations

- Pythagoras: $x^2 + y^2 = z^2 \;\Rightarrow\; x\,x' + y\,y' = z\,z'$ (divide the differentiated equation by 2).
- Similar triangles: equal ratios of sides, written *before* differentiating.
- Trigonometry: $\tan\theta = \dfrac{y}{x} \;\Rightarrow\; \sec^2\theta\,\dfrac{d\theta}{dt} = \dfrac{d}{dt}\left(\dfrac{y}{x}\right)$; find $\sec^2\theta = 1 + \tan^2\theta$ from the sides.
- Check the answer: its sign matches the story, its units are (unit of wanted quantity)/(unit of time).

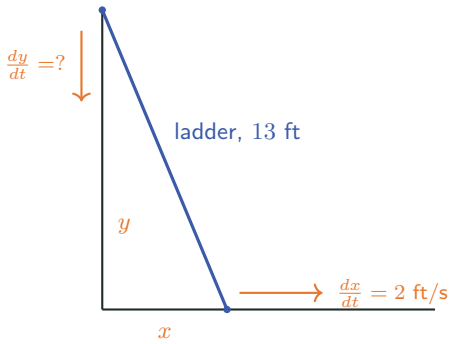


Figure 4.6: Example 1. The ladder has fixed length, so $x^2 + y^2 = 169$ throughout.

Example 1 — The sliding ladder

A 13-foot ladder leans against a vertical wall. The bottom is pulled away from the wall at 2 ft/s. How fast is the top sliding down the wall at the instant the bottom is 5 ft from the wall?

Step 1 – relation. $x^2 + y^2 = 13^2 = 169$ (the ladder's length is constant, so 169 may be used at once).

Step 2 – differentiate. $2x\dfrac{dx}{dt}+2y\dfrac{dy}{dt}=0 \;\Rightarrow\; \dfrac{dy}{dt}=-\dfrac{x}{y}\dfrac{dx}{dt}$.

Step 3 – missing value. When $x=5$: $y=\sqrt{169-25}=12$.

Step 4 – substitute. $\dfrac{dy}{dt}=-\dfrac{5}{12}(2)=-\dfrac{5}{6}$.

Answer. The top of the ladder is moving down the wall at $\dfrac{5}{6}$ foot per second (a negative rate: y is decreasing).

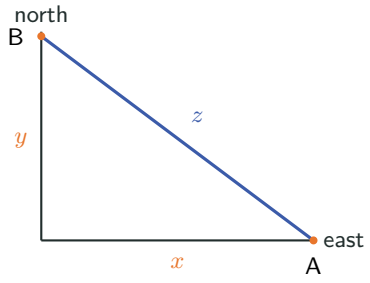


Figure 4.7: Example 2. Car A drives east, car B drives north; z is the distance between them.

Example 2 — Two vehicles

At noon, two cars leave the same junction. Car A travels east at 60 mph and car B travels north at 45 mph. How fast is the distance between them increasing at 2 p.m.?

Step 1 – relation. $z^2=x^2+y^2$.

Step 2. $2z\dfrac{dz}{dt}=2x\dfrac{dx}{dt}+2y\dfrac{dy}{dt}$, so $\dfrac{dz}{dt}=\dfrac{x\,x'+y\,y'}{z}$.

Step 3 – values at 2 p.m. After 2 hours: $x=120$, $y=90$, $z=\sqrt{120^2+90^2}=150$.

Step 4. $\dfrac{dz}{dt}=\dfrac{120(60)+90(45)}{150}=\dfrac{7200+4050}{150}=75$.

Answer. The cars are separating at 75 miles per hour. (Note 75 is less than $60+45=105$, since the cars are not moving directly away from each other.)

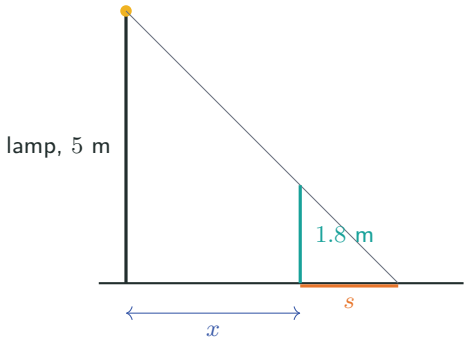


Figure 4.8: Example 3. The two triangles (lamp and tip of shadow; person and tip of shadow) are similar.

Example 3 — A shadow (similar triangles)

A 1.8 m tall person walks away from a 5 m lamp post at 1.2 m/s. Let x be the person's distance from the lamp post and s the length of the shadow. (a) How fast is the length of the shadow increasing? (b) How fast is the tip of the shadow moving?

(a) Similar triangles: $\dfrac{s}{1.8}=\dfrac{x+s}{5}\Rightarrow 5s=1.8x+1.8s\Rightarrow 3.2s=1.8x\Rightarrow s=\dfrac{9}{16}x$. Then $\dfrac{ds}{dt}=\dfrac{9}{16}\dfrac{dx}{dt}=\dfrac{9}{16}(1.2)=0.675$ m/s. The shadow lengthens at 0.675 m/s, *whatever the position*.

(b) The tip is at distance $x+s=\dfrac{25}{16}x$ from the lamp, so it moves at $\dfrac{25}{16}(1.2)=1.875$ m/s.

Example 4 — An angle of elevation

A balloon rises vertically at 3 m/s from a point on level ground. An observer stands 40 m from the launch point. How fast is the angle of elevation θ of the balloon changing when the balloon is 30 m high?

Step 1. With height h: $\tan\theta = \dfrac{h}{40}$.

Step 2. $\sec^2\theta\,\dfrac{d\theta}{dt} = \dfrac{1}{40}\dfrac{dh}{dt}$.

Step 3. At $h = 30$: $\tan\theta = \dfrac{3}{4}$, so $\sec^2\theta = 1 + \dfrac{9}{16} = \dfrac{25}{16}$.

Step 4. $\dfrac{25}{16}\dfrac{d\theta}{dt} = \dfrac{3}{40} \Rightarrow \dfrac{d\theta}{dt} = \dfrac{3}{40}\cdot\dfrac{16}{25} = \dfrac{6}{125} = 0.048$.

Answer. The angle of elevation is increasing at 0.048 radian per second.

Common errors

- Substituting a changing length (such as $y = 12$) into the relating equation *before* differentiating.
- Sign mistakes: when the ladder's top moves *down*, $\frac{dy}{dt}$ is negative. If you enter a positive value for a decreasing quantity your final answer has the wrong sign.
- Using degrees: in $\sec^2\theta\,\frac{d\theta}{dt}$ the angle is in radians, so $\frac{d\theta}{dt}$ is in radians per unit of time.
- Using the formula for a general cone (two variables) without relating r and h first.
- Omitting units, or the word "decreasing/increasing", in the final sentence.

AP Exam Tip

The scoring guide for a related-rates part typically awards: one point for the relating equation or the derivative equation, one point for the substitution, and one point for the answer *with units*. Do not skip the equation line to "go straight to the numbers". If the problem says "at the instant when…" your answer is for that instant only; state that.

Check yourself

1. The side s of a square increases at 3 cm/s. How fast is the area changing when $s = 8$ cm?
2. A spherical snowball melts so that its volume decreases at 12 cm^3/min. Find the rate of change of its radius when $r = 2$ cm.
3. A 10-foot ladder slides down a wall; at the instant the top is 8 ft above the ground it is falling at 1 ft/s. How fast is the bottom moving away from the wall at that instant?

Topic 4.6 Approximating Values of a Function Using Local Linearity and Linearization

Why a curve looks straight up close

Zoom in far enough on the graph of any differentiable function and the curve becomes indistinguishable from a straight line: its tangent line. This is **local linearity**, and it is the geometric meaning of differentiability. It gives us a practical tool. If we know $f(a)$ and $f'(a)$ exactly but $f(x)$ is hard to compute for some x close to a, we can walk along the tangent line instead of along the curve.

The tangent line at $x = a$ passes through $(a, f(a))$ with slope $f'(a)$, so its equation is $y - f(a) = f'(a)(x - a)$. The function

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of f at a, and the approximation $f(x) \approx L(x)$ is the **tangent line approximation**. Equivalently, if $\Delta x = x - a$ is small, then $\Delta f = f(x) - f(a) \approx f'(a)\,\Delta x$: the change in output is about the rate times the change in input. This is precisely the idea behind "$C(x+1) - C(x) \approx C'(x)$" from Topic 4.1.

Overestimate or underestimate?

The tangent line approximation is never exact (unless f is itself linear), and the *sign* of the error is decided by concavity. If $f'' > 0$ near a (concave up), the curve bends away *above* its tangent lines, so the tangent line lies below the curve and $L(x)$ is an **underestimate**. If $f'' < 0$ (concave down), the curve lies below its tangent lines and $L(x)$ is an **overestimate**. The error also grows as x moves away from a, so the approximation is only trustworthy close to a.

Key ideas: linearization

- Linearization of f at a: $L(x) = f(a) + f'(a)(x-a)$, and $f(x) \approx L(x)$ for x near a.
- Change form: $f(a+\Delta x) - f(a) \approx f'(a)\,\Delta x$.
- $f'' > 0$ on the interval: tangent line is *below* the graph, so the approximation is an **underestimate**. $f'' < 0$: tangent line *above* the graph, **overestimate**.
- Closer to a means better; $f(a)$ and $f'(a)$ must be known exactly.
- The slope of the tangent line at a is $f'(a)$, a *number*: do not leave x in it.

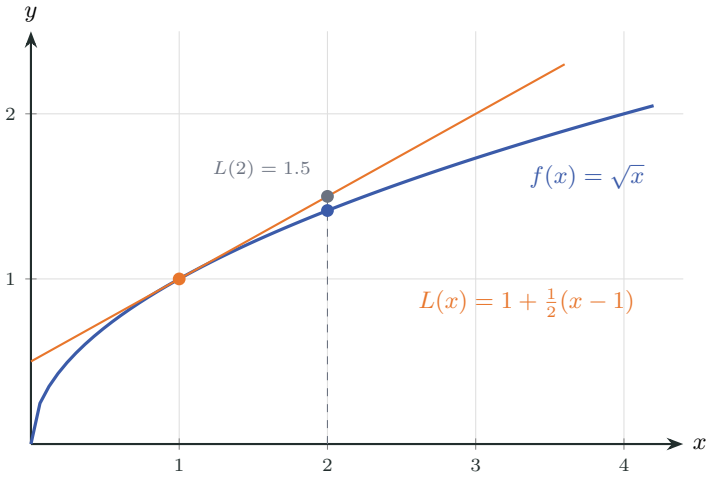


Figure 4.9: $f(x) = \sqrt{x}$ and its tangent line at $a = 1$. The curve is concave down, so the tangent line lies above it and overestimates; the error grows with distance from a.

Example 1 — Approximating a square root

Let $f(x) = \sqrt{x}$. (a) Write the linearization of f at $a = 1$. (b) Use it to approximate $\sqrt{1.1}$ and $\sqrt{2}$. (c) Is each an overestimate or underestimate? (d) Which is more accurate?

(a) $f(1) = 1$ and $f'(x) = \dfrac{1}{2\sqrt{x}}$, so $f'(1) = \dfrac{1}{2}$. Hence $L(x) = 1 + \dfrac{1}{2}(x-1)$.

(b) $\sqrt{1.1} \approx L(1.1) = 1 + \dfrac{1}{2}(0.1) = 1.05$ and $\sqrt{2} \approx L(2) = 1.5$.

(c) $f''(x) = -\dfrac{1}{4}x^{-3/2} < 0$ for $x > 0$, so f is concave down and the tangent line lies above the curve: both values are overestimates. (Check: $\sqrt{1.1} = 1.0488\ldots$, $\sqrt{2} = 1.4142\ldots$)

(d) The estimate at 1.1 (distance 0.1 from a) is off by about 0.0012; the estimate at 2 (distance 1) is off by about 0.086. The approximation is far better near the point of tangency.

Example 2 — Using only given information

The function g is twice differentiable with $g(3) = 8$, $g'(3) = -1.5$, and $g''(x) > 0$ for all x. (a) Write an equation for the line tangent to the graph of g at $x = 3$. (b) Use it to approximate $g(3.4)$. (c) Is your approximation less than or greater than $g(3.4)$? Give a reason.

(a) $y - 8 = -1.5(x-3)$, i.e. $y = 8 - 1.5(x-3)$.

(b) $g(3.4) \approx 8 - 1.5(0.4) = 8 - 0.6 = 7.4$.

(c) Since $g''(x) > 0$ for all x, the graph of g is concave up and lies above its tangent lines. Therefore the approximation 7.4 is *less than* $g(3.4)$.

Example 3 — Estimating an error in a measurement

The radius of a sphere is measured as 10 cm with a possible error of 0.2 cm. Use a linear approximation to estimate the resulting error in the computed volume $V = \frac{4}{3}\pi r^3$.

Step 1. $V'(r) = 4\pi r^2$, so $V'(10) = 400\pi$.

Step 2. $\Delta V \approx V'(10)\,\Delta r = 400\pi(0.2) = 80\pi \approx 251.327$.

Answer. The volume could be off by about 251.327 cm^3. Relative to $V(10) = \frac{4000}{3}\pi \approx 4188.790$, that is about 6%, three times the relative error of the radius (2%): relative errors multiply by the exponent 3 in a cubic formula.

Common errors

- Forgetting $f(a)$: writing $L(x) = f'(a)(x-a)$ describes a line through the origin of the change, not the tangent line.
- Leaving the slope as $f'(x)$ instead of the number $f'(a)$.
- Reversing the concavity rule. Memory aid: concave *up* like a cup holds the tangent *below* it, so L is too *low*.
- Using a linearization far from a, or writing "$=$" where only "$\approx$" is true.

AP Exam Tip

For over/underestimate questions write: "Since $f''(x) < 0$ on the interval, the graph of f is concave down and lies below its tangent line, so the tangent line approximation is greater than $f(\ldots)$." State the concavity *and* the relationship of the line to the graph. If f'' is only known through a table or the graph of f', use whether f' is decreasing or increasing.

Check yourself

1. Write the linearization of $f(x) = e^{2x}$ at $x = 0$ and use it to approximate $e^{0.06}$. Is it an overestimate or underestimate?
2. The function g satisfies $g(5) = 10$, $g'(5) = 0.4$ and $g''(x) < 0$ for all x. Approximate $g(4.5)$ with the tangent line and say whether it is an overestimate or underestimate.
3. Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$ and use it to approximate $\sqrt[3]{28}$.

Topic 4.7 Using L'Hôpital's Rule for Determining Limits of Indeterminate Forms

When substitution gives no answer

In Unit 1 you evaluated limits by substitution, by factoring and cancelling, or by multiplying by a conjugate. Some limits resist all of these. If $\lim\limits_{x\to a}\dfrac{f(x)}{g(x)}$ gives $\dfrac{0}{0}$ (both numerator and denominator tend to 0) or $\dfrac{\pm\infty}{\pm\infty}$ (both tend to infinity in size), the form is **indeterminate**: the limit could be 0, a finite number, or infinite, depending on *how fast* each part approaches its limit. The classic example is $\lim\limits_{x\to 0}\dfrac{\sin x}{x}$, where both sides tend to 0 at the same speed and the quotient tends to 1 (Figure 4.10).

L'Hôpital's Rule says that, in these two forms, we can compare the *rates* at which numerator and denominator move, that is, their derivatives. Near $x = a$ the numerator behaves like $f'(a)(x-a)$ and the denominator like $g'(a)(x-a)$ (linearization!), so the quotient behaves like $\dfrac{f'(a)}{g'(a)}$.

L'Hôpital's Rule

Suppose f and g are differentiable near a (except possibly at a) and $g'(x) \neq 0$ there. If $\lim\limits_{x\to a}\dfrac{f(x)}{g(x)}$ has the indeterminate form $\dfrac{0}{0}$ or $\dfrac{\pm\infty}{\pm\infty}$, then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)},$$

provided the limit on the right exists or is $\pm\infty$. The rule also holds for $x \to \infty$, $x \to -\infty$ and one-sided limits.

- **Check the form first.** If substitution gives a normal number (for example $\frac{2}{5}$), the rule does not apply.
- Differentiate the numerator and the denominator **separately**; do *not* use the quotient rule.
- If the new quotient is still indeterminate, apply the rule again.
- Growth rates as $x \to \infty$: exponentials (e^x) grow faster than any power x^n, which grow faster than any logarithm $\ln x$.

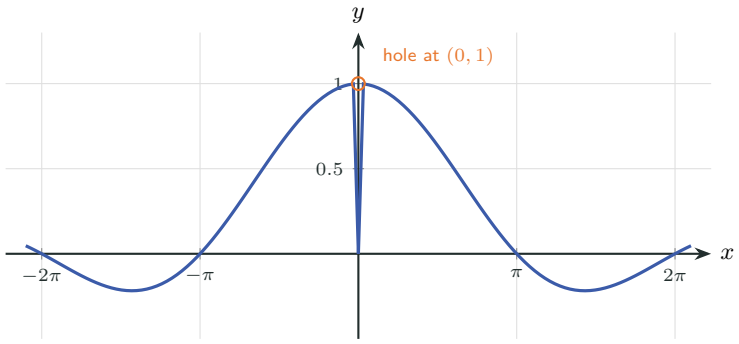


Figure 4.10: $y = \dfrac{\sin x}{x}$ is undefined at $x = 0$ ($\frac{0}{0}$ form) but its limit there is 1.

Example 1 — Zero over zero

Evaluate (a) $\lim\limits_{x\to 3} \dfrac{x^2-9}{x-3}$ (b) $\lim\limits_{x\to 0} \dfrac{\sin x}{x}$ (c) $\lim\limits_{x\to 0} \dfrac{e^{3x}-1-3x}{x^2}$.

(a) Substitution gives $\frac{0}{0}$, so L'Hôpital applies: $\lim\limits_{x\to 3} \dfrac{2x}{1} = 6$ (the same value we get by factoring $\frac{(x-3)(x+3)}{x-3}$).

(b) $\frac{0}{0}$; $\lim\limits_{x\to 0} \dfrac{\cos x}{1} = 1$.

(c) At $x = 0$: $e^0 - 1 - 0 = 0$ and the denominator is 0, so the form is $\frac{0}{0}$. Apply the rule: $\lim\limits_{x\to 0} \dfrac{3e^{3x}-3}{2x}$. This is again $\frac{0}{0}$, so apply it once more: $\lim\limits_{x\to 0} \dfrac{9e^{3x}}{2} = \dfrac{9}{2}$.

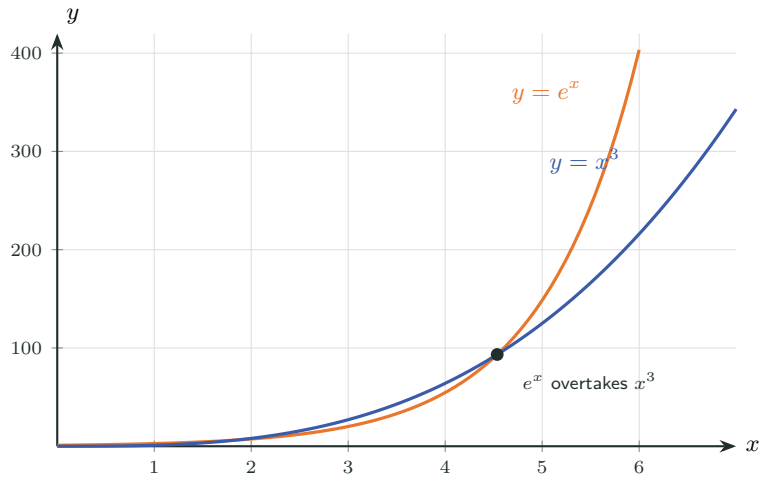


Figure 4.11: although $e^x < x^3$ for $1.86 < x < 4.54$, the exponential eventually wins; that is why $\lim\limits_{x\to\infty} \dfrac{x^3}{e^x} = 0$.

Example 2 — Infinity over infinity, and growth rates

Evaluate (a) $\lim\limits_{x\to\infty} \dfrac{x^3}{e^x}$ (b) $\lim\limits_{x\to\infty} \dfrac{\ln x}{\sqrt{x}}$ (c) $\lim\limits_{x\to\infty} \dfrac{e^x}{x^2+1}$.

(a) Form $\frac{\infty}{\infty}$. Apply the rule three times: $\dfrac{x^3}{e^x} \to \dfrac{3x^2}{e^x} \to \dfrac{6x}{e^x} \to \dfrac{6}{e^x} \to 0$. So the limit is 0 (Figure 4.11).

(b) Form $\frac{\infty}{\infty}$. $\lim\limits_{x\to\infty} \dfrac{1/x}{\frac{1}{2\sqrt{x}}} = \lim\limits_{x\to\infty} \dfrac{2\sqrt{x}}{x} = \lim\limits_{x\to\infty} \dfrac{2}{\sqrt{x}} = 0$.

(c) Form $\frac{\infty}{\infty}$. $\lim\limits_{x\to\infty} \dfrac{e^x}{2x}$ is still $\frac{\infty}{\infty}$; one more step gives $\lim\limits_{x\to\infty} \dfrac{e^x}{2} = \infty$. The limit does not exist (it is $+\infty$).

Example 3 — Limits from given values

Functions f and g are differentiable with the values in the table. Find $\lim_{x\to 2}\dfrac{f(x)}{g(x)}$ and $\lim_{x\to 2}\dfrac{g(x)}{f(x)}$.

	$f(2)$	$g(2)$	$f'(2)$	$g'(2)$
value	0	0	6	-3

Because f and g are differentiable they are continuous, so $\lim_{x\to 2} f(x) = f(2) = 0$ and $\lim_{x\to 2} g(x) = 0$: the quotient has the form $\frac{0}{0}$. By L'Hôpital's Rule, $\lim_{x\to 2}\dfrac{f(x)}{g(x)} = \lim_{x\to 2}\dfrac{f'(x)}{g'(x)} = \dfrac{6}{-3} = -2$, where the last step requires f' and g' to be continuous at 2 (assumed in this context). The reciprocal quotient has limit $\dfrac{-3}{6} = -\dfrac{1}{2}$.

Common errors

- **Using the rule when the form is not indeterminate.** $\lim_{x\to 1}\dfrac{x^2+1}{x+1} = \dfrac{2}{2} = 1$. Applying the rule gives $\dfrac{2x}{1} \to 2$, which is wrong.
- Applying the quotient rule to $\dfrac{f}{g}$ instead of differentiating numerator and denominator separately.
- Stopping after one step while the new limit is still $\frac{0}{0}$, or continuing after it is no longer indeterminate.
- Believing that $\frac{\infty}{\infty} = 1$ or $\frac{0}{0} = 0$; these are exactly the situations that are *not* determined by their form.
- Writing "$=$" between a limit and its $\frac{0}{0}$ form: write the form as a remark, for example "(form $\frac{0}{0}$)".

AP Exam Tip

Scorers want to see that you recognised the hypothesis. Write "Since $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, L'Hôpital's Rule applies" (or the $\frac{\infty}{\infty}$ version) *before* the derivative step. An answer of ∞ or $-\infty$ is acceptable for an unbounded limit, but "the limit does not exist" is also correct if the context asks about existence.

Check yourself

1. Evaluate $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$.
2. Evaluate $\lim_{x\to 2}\dfrac{x^3-8}{x^2-4}$ and check it by factoring.
3. Evaluate $\lim_{x\to\infty}\dfrac{x^2}{e^x}$ and explain what it says about growth.

Unit 4 Practice: Official Exam Style

Part A – Multiple choice, NO CALCULATOR

14 questions · choose the best answer

1 The temperature of a cup of tea is $H(t)$ degrees Celsius, where t is the time in minutes after the tea is poured. It is known that $H'(10) = -2.5$. Which of the following is the best interpretation of this statement?

(A) After 10 minutes the tea has cooled by a total of $2.5°$C.

(B) At $t = 10$ minutes the temperature of the tea is $2.5°$C.

(C) At $t = 10$ minutes the temperature of the tea is decreasing at a rate of $2.5°$C per minute.

(D) During the first 10 minutes the temperature of the tea decreased at an average rate of $2.5°$C per minute.

2 The number of subscribers to a channel, $N(t)$ thousand, is recorded t days after launch.

t (days)	0	3	7	10
$N(t)$	20	38	66	74

Using the data in the table, what is the best estimate of $N'(5)$, in thousand subscribers per day?

(A) 7 **(B)** 5.4

(C) 2.8 **(D)** 28

3 A particle moves along the x-axis with position $x(t) = \dfrac{t^3}{3} - 2t^2 + 3t$ for $t \geq 0$. For which values of t is the particle moving to the left?

(A) $0 < t < 1$ **(B)** $1 < t < 3$

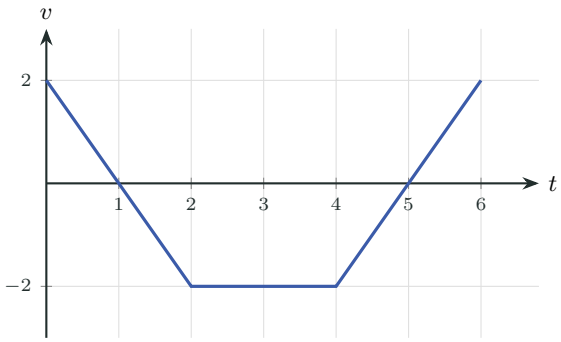
(C) $t > 3$ **(D)** $0 < t < 1$ and $t > 3$

4 A particle moves along the x-axis with velocity $v(t) = t^2 - 6t$. Which of the following describes the motion at $t = 2$?

(A) Moving right and speeding up **(B)** Moving right and slowing down

(C) Moving left and slowing down **(D)** Moving left and speeding up

5 The graph of the velocity $v(t)$ of a particle moving along the x-axis is shown for $0 \leq t \leq 6$; the graph consists of three line segments. On which of the following intervals is the speed of the particle increasing?



(A) $0 < t < 1$ **(B)** $1 < t < 2$

(C) $2 < t < 4$ **(D)** $4 < t < 5$

6 The area of an oil spill is $A(t)$ square meters at time t hours. For $1 < t < 5$, $A'(t) > 0$ and $A''(t) < 0$. Which statement is true for $1 < t < 5$?

(A) The area is decreasing and the rate of decrease is increasing.

(B) The area is decreasing and the rate of decrease is slowing.

(C) The area is increasing at an increasing rate.

(D) The area is increasing at a decreasing rate.

7 The volume of a spherical balloon, $V = \frac{4}{3}\pi r^3$, is increasing at the rate of 72π cm^3/s. At the instant when $r = 3$ cm, how fast is the radius increasing, in cm/s?

(A) 2 **(B)** $\frac{1}{2}$
(C) 6 **(D)** 8

8 A 10-meter ladder leans against a vertical wall. The bottom of the ladder slides away from the wall at 0.5 m/s. At the instant the bottom is 6 m from the wall, what is $\frac{dy}{dt}$, the rate of change of the height y of the top of the ladder, in m/s?

(A) $\frac{3}{8}$ **(B)** $-\frac{2}{3}$
(C) $-\frac{3}{8}$ **(D)** -3

9 The length ℓ of a rectangle is increasing at 3 cm/s while its width w is decreasing at 2 cm/s. At the instant when $\ell = 10$ cm and $w = 6$ cm, what is the rate of change of the area of the rectangle, in cm^2/s?

(A) 2 **(B)** -2
(C) 38 **(D)** 60

10 Let $f(x) = \sqrt{x}$. The linearization of f at $x = 16$ is used to approximate $\sqrt{17}$. What is the approximation?

(A) $\frac{17}{4}$ **(B)** $\frac{65}{16}$
(C) $\frac{129}{32}$ **(D)** $\frac{33}{8}$

11 The twice-differentiable function f satisfies $f(3) = 5$, $f'(3) = -2$, and $f''(x) < 0$ for all x. The line tangent to the graph of f at $x = 3$ is used to approximate $f(3.5)$. Which statement is true?

(A) The approximation is 4, which is greater than $f(3.5)$.
(B) The approximation is 4, which is less than $f(3.5)$.
(C) The approximation is 6, which is greater than $f(3.5)$.
(D) The approximation is 6, which is less than $f(3.5)$.

12 $\lim\limits_{x\to 0} \dfrac{\sin 3x}{5x} =$

(A) 0 **(B)** $\frac{5}{3}$
(C) $\frac{3}{5}$ **(D)** 15

13 $\lim\limits_{x\to \infty} \dfrac{e^x}{x^2+1} =$

(A) 0 **(B)** $\frac{1}{2}$
(C) 1 **(D)** ∞

14 $\lim\limits_{x\to 4} \dfrac{\sqrt{x}-2}{x-4} =$

(A) 0 **(B)** $\frac{1}{4}$
(C) $\frac{1}{2}$ **(D)** 4

Part B — Multiple choice, CALCULATOR ALLOWED

7 questions · a graphing calculator is required for some

15 The population of a colony of bacteria is $P(t) = 1200\,e^{0.04t}$, where t is in hours. At what rate is the population changing at $t = 5$, in bacteria per hour?

(A) 48.000 **(B)** 59.816

(C) 58.627 **(D)** 1465.683

16 A particle moves along the x-axis with velocity $v(t) = t\cos t$ for $0 \le t \le 4$. Which statement is true at $t = 2$?

(A) The speed is increasing, because $v(2)$ and $a(2)$ are both negative.

(B) The speed is decreasing, because $a(2)$ is negative.

(C) The speed is decreasing, because $v(2)$ is negative.

(D) The speed is increasing, because $|a(2)| > |v(2)|$.

17 The cost, in dollars, of producing x pounds of a chemical is $C(x) = 500 + 20x^{0.8}$. Which of the following is closest to the rate of change of cost with respect to production when $x = 50$, in dollars per pound?

(A) 22.865 **(B)** 365.844

(C) 957.305 **(D)** 7.317

18 Air is pumped into a spherical balloon at a constant rate of 25 cm^3/s. At the instant when the radius is 6 cm, how fast is the surface area $S = 4\pi r^2$ increasing, in cm^2/s?

(A) 0.055 **(B)** 8.333

(C) 25.000 **(D)** 150.796

19 A rocket rises vertically at 80 m/s from a launch pad. A camera on the ground 500 m from the pad tracks it. At the instant the rocket is 1200 m high, how fast is the camera's angle of elevation θ changing, in radians per second?

(A) 0.024 **(B)** 0.136

(C) 0.160 **(D)** 1.082

20 Let $f(x) = \ln(x^2 + 3)$. The linearization of f at $x = 1$ is used to approximate $f(1.2)$. Which of the following is the approximation?

(A) 1.386 **(B)** 1.491

(C) 1.486 **(D)** 1.586

21 $\displaystyle\lim_{x\to 0} \frac{\ln(1+3x)}{\sin 2x} =$

(A) 0 **(B)** $\dfrac{2}{3}$

(C) 3 **(D)** $\dfrac{3}{2}$

Part C — Free response

2 questions · 9 points each · show the work that leads to your answers

22 **[9 points]** *(Calculator allowed.)* A particle moves along the x-axis. For $0 \le t \le 8$, its position is $x(t) = 4t\,e^{-t/2}$, where x is measured in centimeters and t in seconds.

(a) Find $v(3)$ and $a(3)$. Is the speed of the particle increasing or decreasing at $t = 3$? Give a reason for your answer.

(b) Find all times t in the interval $0 < t < 8$ at which the particle changes direction. Justify your answer.

(c) Find the first time t at which $x(t) = 1$. Find the velocity of the particle at that time and, using correct units, interpret your answer in the context of the problem.

(d) Is the particle moving toward the origin or away from the origin at $t = 7$? Give a reason for your answer.

Write all decimal answers correct to three places.

23 **[9 points]** *(No calculator.)* Water drains from an inverted right circular conical tank (vertex down) whose height is 12 ft and whose top radius is 6 ft. Let h be the depth of the water in feet and V the volume of the water in cubic feet, both functions of time t in minutes. The water drains at the constant rate $\dfrac{dV}{dt} = -3\pi$ cubic feet per minute. (The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.)

(a) Show that $V = \dfrac{\pi}{12}h^3$.

(b) Find $\dfrac{dh}{dt}$ at the instant when $h = 6$ ft. Indicate units of measure.

(c) At time $t = t_0$ the depth is $h(t_0) = 6$ ft. Write an equation for the line tangent to the graph of h at $t = t_0$ and use it to approximate the depth of the water one minute later.

(d) Is your approximation in part (c) greater than or less than the actual depth at $t_0 + 1$? Give a reason for your answer.

AP Calculus AB – Unit 5 Exam weighting: 15–18% of the AB exam

Analytical Applications of Differentiation

In this unit

Unit 2 and Unit 3 taught you how to *compute* derivatives. This unit teaches you what a derivative *tells you about a function*. The sign of f' says where f rises and falls; the places where f' changes sign are the peaks and valleys; the sign of f'' says whether the graph bends upward or downward; and a handful of theorems (the Mean Value Theorem, the Extreme Value Theorem and their consequences) guarantee that the things you are looking for exist and tell you where to find them. This is the largest unit of the exam, and it is where *justification* is rewarded most: on the free-response section you will be asked not just for an answer but for a reason, written in the language of the theorems below.

Topics.

5.1 Using the Mean Value Theorem 5.2 Extreme Value Theorem, Global Versus Local Extrema, and Critical Points
5.3 Determining Intervals on Which a Function Is Increasing or Decreasing
5.4 Using the First Derivative Test to Determine Relative (Local) Extrema
5.5 Using the Candidates Test to Determine Absolute (Global) Extrema
5.6 Determining Concavity of Functions over Their Domains
5.7 Using the Second Derivative Test to Determine Extrema
5.8 Sketching Graphs of Functions and Their Derivatives
5.9 Connecting a Function, Its First Derivative, and Its Second Derivative
5.10 Introduction to Optimization Problems 5.11 Solving Optimization Problems
5.12 Exploring Behaviors of Implicit Relations

Big ideas. (1) $f' > 0$ means f increases, $f' < 0$ means f decreases, and a sign change of f' marks a relative extremum. (2) $f'' > 0$ means the graph is concave up (f' is increasing), $f'' < 0$ means concave down, and a sign change of f'' marks an inflection point. (3) On a closed interval a continuous function *must* have an absolute maximum and minimum; they occur at critical points or endpoints. (4) Optimization is the same idea applied to a model you build yourself. (5) Every conclusion needs a *reason* that names a theorem, a sign change, or a value.

Topic 5.1 Using the Mean Value Theorem

From averages to instants

Suppose you drive 130 miles in exactly two hours. Your *average* speed is 65 miles per hour. Does the speedometer ever read exactly 65? Common sense says yes: if you were always below 65 you could not have averaged 65, and if you were always above 65 you could not have either, so at some moment you must have passed through 65 exactly. The Mean Value Theorem (MVT) is the precise mathematical form of that argument. It connects an *average rate of change* over an interval, which needs only the two end values, with an *instantaneous rate of change* at a point inside the interval, which needs the derivative.

Geometrically, the average rate of change of f on $[a, b]$ is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$. The theorem says that, provided the graph is a smooth unbroken curve over $[a, b]$, somewhere strictly between a and b there is a point where the tangent line is *parallel* to that secant line (Figure 5.1). Slide the secant line up or down, keeping its slope, and the last place it touches the curve is such a point.

The theorems

Mean Value Theorem (MVT) and Rolle's Theorem

MVT. If f is *continuous* on the closed interval $[a,b]$ and *differentiable* on the open interval (a,b), then there is at least one number c in (a,b) such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Rolle's Theorem (the special case $f(a)=f(b)$): then $f'(c)=0$ for some c in (a,b).

Hypotheses to state on the exam: (1) f is continuous on $[a,b]$; (2) f is differentiable on (a,b). If either fails, the theorem gives *no* guarantee.

Notice the shape of the hypotheses: continuity is required on the *closed* interval (the graph has no break, and it meets the end points), while differentiability is required only on the *open* interval (a one-sided corner at an endpoint is allowed). A polynomial, e^x, $\sin x$ and any rational function on an interval that avoids its zeros of the denominator satisfy both hypotheses automatically; on the exam it is enough to say so ("f is a polynomial, so it is continuous and differentiable"). The functions that fail are the usual suspects: a function with a jump or a vertical asymptote in $[a,b]$ (not continuous), and a function with a corner or cusp in (a,b) (not differentiable).

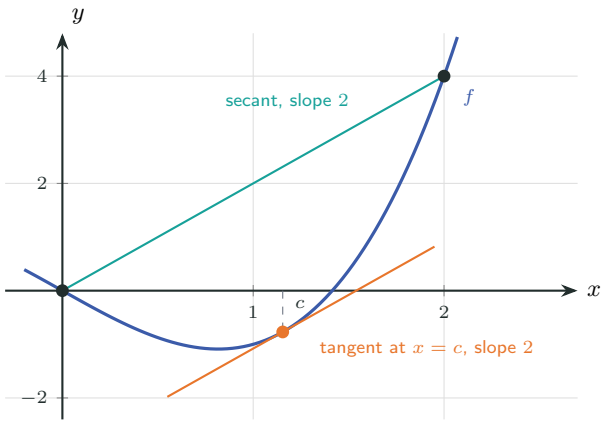


Figure 5.1. For $f(x)=x^3-2x$ on $[0,2]$ the secant line has slope 2; at $x=c=\frac{2}{\sqrt{3}}\approx 1.155$ the tangent line has the same slope.

How the exam uses the MVT

There are three common tasks. (i) *Find the value(s) of c*: compute the average rate, set $f'(x)$ equal to it, solve, and keep only solutions inside the open interval. (ii) *Existence from data*: a table or a story gives two values of a differentiable function; you conclude that the derivative (velocity, growth rate, …) equals the average rate at some time in between. (iii) *Bounding*: if you know f' is always at most M, then no average rate can exceed M, so $f(b)-f(a)\le M(b-a)$.

Example 1 — Finding c

Let $f(x)=x^3-2x$ on $[0,2]$. Verify the hypotheses of the MVT and find all values of c that the theorem guarantees.

Step 1 (hypotheses). f is a polynomial, so it is continuous on $[0,2]$ and differentiable on $(0,2)$.

Step 2 (average rate). $\dfrac{f(2)-f(0)}{2-0}=\dfrac{4-0}{2}=2.$

Step 3 (solve $f'(c)=2$). $f'(x)=3x^2-2$, so $3c^2-2=2$, $c^2=\dfrac{4}{3}$, $c=\pm\dfrac{2}{\sqrt{3}}$.

Step 4 (check the interval). Only $c=\dfrac{2}{\sqrt{3}}\approx 1.155$ lies in $(0,2)$; $c=-\dfrac{2}{\sqrt{3}}$ is rejected. See Figure 5.1.

Example 2 — Existence and bounding

The function f is differentiable for all real numbers with $f(1)=3$ and $f(5)=11$.

(a) Explain why there is a number c with $1<c<5$ and $f'(c)=2$.
(b) Is it possible that $f'(x)\le 1.5$ for every x in $[1,5]$? Justify.

(a) Since f is differentiable on $\mathbb{R}$, it is continuous on $[1,5]$ and differentiable on $(1,5)$. By the MVT there is c in

$(1,5)$ with $f'(c) = \dfrac{f(5)-f(1)}{5-1} = \dfrac{11-3}{4} = 2$.
(b) No. By part (a), $f'(c) = 2 > 1.5$ for some c in $(1,5)$, which contradicts $f'(x) \le 1.5$ for every x.

Example 3 — When the MVT does not apply

Let $f(x) = |x-1|$ on $[-1,3]$. Show that no value of c satisfies $f'(c) = \dfrac{f(3)-f(-1)}{3-(-1)}$, and explain why this does not contradict the MVT.

Step 1. The average rate is $\dfrac{f(3)-f(-1)}{4} = \dfrac{2-2}{4} = 0$.
Step 2. For $x < 1$, $f(x) = 1-x$ and $f'(x) = -1$; for $x > 1$, $f'(x) = 1$. So $f'(x)$ is never 0.
Step 3. f is continuous on $[-1,3]$, but it is **not differentiable at** $x = 1$, a point inside $(-1,3)$ (the graph has a corner). The second hypothesis fails, so the MVT promises nothing.

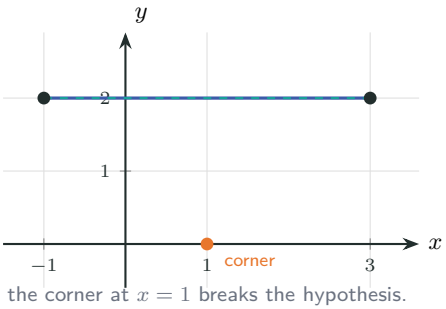


Figure 5.2. The secant line is horizontal, but no tangent line is: the corner at $x = 1$ breaks the hypothesis.

Common errors

- **Forgetting to check the interval.** Solving $f'(x) = m$ often gives several roots. The MVT gives values of c in the *open* interval (a,b) only; reject the others.
- **Reading the converse.** "The hypotheses fail" does *not* prove that no c exists; it only means the theorem is silent.
- **Averaging the wrong thing.** The slope to match is $\dfrac{f(b)-f(a)}{b-a}$, not $f(b)-f(a)$ alone and not the midpoint $\dfrac{a+b}{2}$ (the midpoint is almost never the value of c).

AP Exam Tip

Always *name both hypotheses* before you invoke the theorem: "Since f is continuous on $[a,b]$ and differentiable on (a,b), the Mean Value Theorem guarantees a c in (a,b) such that $f'(c) = \dots$." A conclusion stated without the hypotheses typically loses the justification point. When f is given only by a table, say that f is differentiable (so continuous) as stated in the problem.

Check yourself

1. Find all c guaranteed by the MVT for $f(x) = \sqrt{x}$ on $[1,9]$.
2. The function g is differentiable with $g(1) = 4$ and $g(4) = -5$. Show that $g'(c) = -3$ for some c in $(1,4)$.
3. Explain why the MVT cannot be applied to $h(x) = \dfrac{1}{x-2}$ on $[0,4]$, and show that no c in $(0,4)$ satisfies the conclusion.

Topic 5.2 Extreme Value Theorem, Global Versus Local Extrema, and Critical Points

Highest, lowest, and merely locally highest

When you look at a graph the eye is drawn to peaks and valleys, but there are two different questions you can ask about them. A *relative* (or *local*) maximum is a point that is higher than all the points *near* it: it is the top of a hill, even if a taller mountain stands elsewhere. An *absolute* (or *global*) maximum is higher than every other point of the function on the whole interval (or domain) under discussion. Every absolute extremum that is not at an endpoint is also a relative extremum, but a relative extremum need not be absolute. Minima are the mirror image.

Two details of the definitions matter on the exam. First, a relative extremum is defined using an *open* interval around c on which f is defined, so an endpoint of the interval can never be a relative extremum, although it can be an absolute one. Second, we speak of the *value* of the extremum, $f(c)$, and the *location* $x = c$; a question that asks "for what value of x" wants c, and one that asks for "the maximum value of f" wants $f(c)$.

Definitions and the Extreme Value Theorem

- f has an **absolute maximum** at $x = c$ on an interval I if $f(c) \geq f(x)$ for all x in I. Absolute minimum: $f(c) \leq f(x)$ for all x in I.
- f has a **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for all x in some open interval containing c. Relative minimum: $f(c) \leq f(x)$ nearby.
- A **critical point** of f is a number c in the domain of f at which $f'(c) = 0$ or $f'(c)$ does not exist.
- **Extreme Value Theorem (EVT).** If f is continuous on the closed interval $[a, b]$, then f attains both an absolute maximum and an absolute minimum on $[a, b]$.
- **Relative extrema occur at critical points.** If f has a relative extremum at $x = c$ (with c inside the domain, not an endpoint), then c is a critical point. The converse is false.

Critical points: where to look

Why must a relative extremum occur at a critical point? Suppose f is differentiable at a peak. If the tangent line there had a positive slope, the curve would be still rising to the right of the point, so the point could not be a maximum; a negative slope leads to the same contradiction on the left. The only slope that survives is $f'(c) = 0$ (a horizontal tangent). If f is *not* differentiable at the peak, then the argument does not apply and the peak can be a sharp corner or a cusp. Hence the candidates for relative extrema are the solutions of $f'(x) = 0$ together with the x-values where f' fails to exist but f is defined.

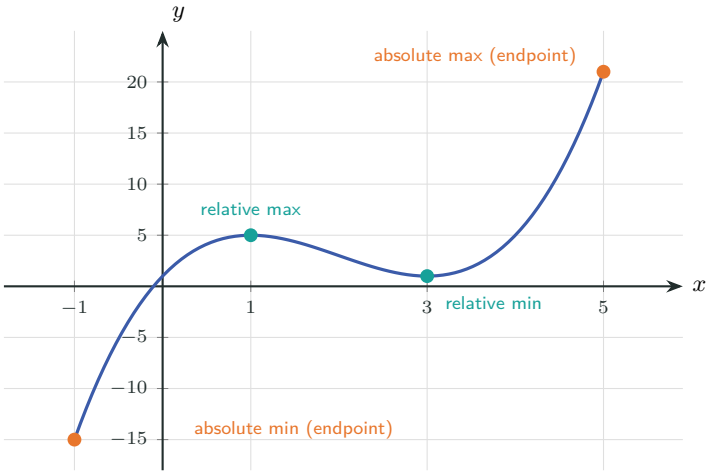


Figure 5.3. $f(x) = x^3 - 6x^2 + 9x + 1$ on $[-1, 5]$. The critical points $x = 1$ and $x = 3$ give a relative maximum and a relative minimum; the absolute extrema occur at the endpoints.

Example 1 — Critical points of a polynomial

Find the critical points of $f(x) = x^3 - 6x^2 + 9x + 1$.

Step 1. f is a polynomial, so f' exists everywhere; only $f'(x) = 0$ matters.
Step 2. $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$.
Step 3. $f'(x) = 0$ at $x = 1$ and $x = 3$. These are the critical points. (Figure 5.3 shows a peak at $x = 1$ with $f(1) = 5$ and a valley at $x = 3$ with $f(3) = 1$.)

Example 2 — A critical point where f' does not exist

Find the critical points of $f(x) = x^{2/3}(x-5)$.

Step 1. Rewrite $f(x) = x^{5/3} - 5x^{2/3}$, so

$$f'(x) = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3} = \frac{5(x-2)}{3x^{1/3}}.$$

Step 2. $f'(x) = 0$ when $x = 2$.
Step 3. $f'(x)$ does not exist at $x = 0$ (zero denominator), but $f(0) = 0$ is defined, so $x = 0$ is a critical point.
The critical points are $x = 0$ and $x = 2$. The graph has a sharp *cusp* at the origin.

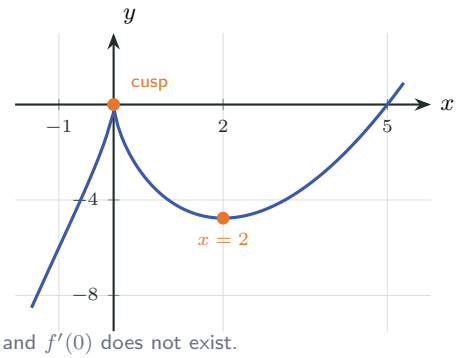


Figure 5.4. $f(x) = x^{2/3}(x-5)$: $f'(2) = 0$ and $f'(0)$ does not exist.

Example 3 — A trigonometric function on $[0, 2\pi]$

Find the critical points of $h(x) = 2\sin x + \cos 2x$ for $0 < x < 2\pi$.

Step 1. $h'(x) = 2\cos x - 2\sin 2x = 2\cos x - 4\sin x\cos x = 2\cos x\,(1 - 2\sin x)$.

Step 2. $h'(x) = 0$ when $\cos x = 0$ or $\sin x = \dfrac{1}{2}$.

Step 3. In $(0, 2\pi)$: $\cos x = 0$ gives $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$; $\sin x = \dfrac{1}{2}$ gives $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$.

The four critical points are $x = \dfrac{\pi}{6}, \dfrac{\pi}{2}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$. (Factoring first matters: dividing both sides of $\cos x = 2\sin x\cos x$ by $\cos x$ would lose the solutions of $\cos x = 0$.)

Example 4 — When does the EVT apply?

For each function, decide whether the EVT guarantees an absolute maximum and an absolute minimum.

Function and interval	Guaranteed?	Reason
$f(x) = x^2 - 4x$ on $[0, 5]$	Yes	continuous on a closed interval
$g(x) = \dfrac{1}{x}$ on $[-1, 1]$	No	not continuous at $x = 0$ (and g has no max or min)
$h(x) = x^2$ on $(0, 2)$	No	interval is not closed (neither a max nor a min exists)
$k(x) = \begin{cases} x & x < 1 \\ 0 & x \geq 1 \end{cases}$ on $[0, 2]$	No	jump at $x = 1$; here no maximum exists (values approach 1 but never reach it) although a minimum, 0, does

A "No" means only that the theorem gives no guarantee; some such functions do happen to have extrema (as k has a minimum).

Common errors

- **"$f'(c) = 0$, so f has an extremum at c."** Not necessarily: $f(x) = x^3$ has $f'(0) = 0$ but is increasing on both sides of 0. A critical point is only a *candidate*.
- **Calling an endpoint a critical point or a relative extremum.** Endpoints are not critical points and cannot be relative extrema, but they *must* be checked when you look for absolute extrema on $[a, b]$ (Topic 5.5).
- **Dropping points where f' does not exist.** Cusps, corners and vertical tangents are critical points when f is defined there, but a point outside the domain (such as a vertical asymptote) is not.

AP Exam Tip

Write "$f'(x) = 0$ *or* $f'(x)$ is undefined" every time you list critical points. On the free-response section, a table of candidates that mentions only the zeros of f' is often missing a point.

Check yourself

1. Find the critical points of $f(x) = x^4 - 4x^3$.
2. Find the critical points of $g(x) = (x-1)^{2/3}$ and describe the behaviour of the graph there.
3. True or false: if f is continuous on $(0, 5)$, then f must have an absolute maximum on $(0, 5)$. Explain.

Topic 5.3 Determining Intervals on Which a Function Is Increasing or Decreasing

The sign of f' controls the direction of f

Think of walking left to right along the graph of f. If the tangent slope is positive you are climbing; if it is negative you are descending. A function is *increasing* on an interval if larger inputs always give larger outputs there, and *decreasing* if larger inputs give smaller outputs. The derivative gives a test that needs no picture. The reason is the Mean Value Theorem: if $x_1 < x_2$ lie in an interval where $f' > 0$, then $f(x_2) - f(x_1) = f'(c)\,(x_2 - x_1)$ for some c between them, and both factors on the right are positive, so $f(x_2) > f(x_1)$.

The sign of f' can change only at a point where $f' = 0$ or f' is undefined (a continuous function cannot change sign without passing through 0 or breaking). This is why the critical points from Topic 5.2 are the natural dividing points. We must also split at every x-value where f itself is not defined, such as a vertical asymptote or a hole, because the function is not increasing or decreasing *across* a break in its domain.

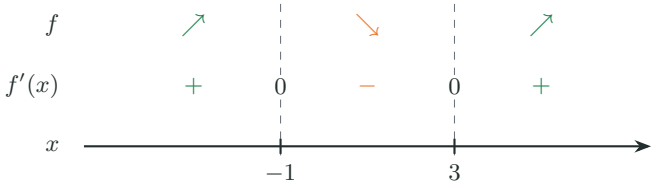
Increasing/decreasing test and the sign-chart procedure

- If $f'(x) > 0$ on an interval, then f is **increasing** on that interval. If $f'(x) < 0$ on an interval, then f is **decreasing** on that interval.
- **Procedure.** (1) Find f'. (2) Find every x with $f'(x) = 0$, every x where f' is undefined, and every x where f is undefined. (3) Mark these on a number line; they cut it into intervals. (4) Pick one test value in each interval (or factor the sign) and record the sign of f'. (5) State the conclusion with a reason.

Example 1 — A polynomial

Find the intervals on which $f(x) = x^3 - 3x^2 - 9x + 5$ is increasing and decreasing.

Step 1. $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$.
Step 2. $f'(x) = 0$ at $x = -1$ and $x = 3$; f' exists everywhere.
Step 3. Test values: $f'(-2) = 3(-5)(-1) = 15 > 0$; $f'(0) = 3(-3)(1) = -9 < 0$; $f'(4) = 3(1)(5) = 15 > 0$.

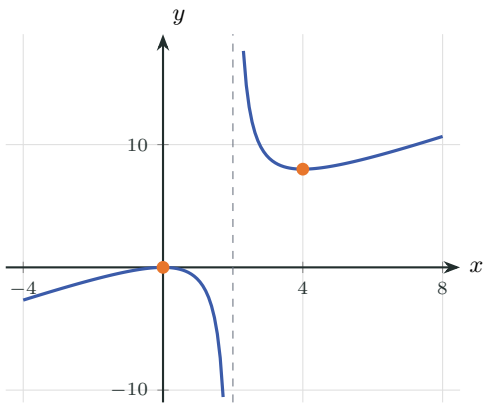


Step 4. f is increasing on $(-\infty, -1)$ and $(3, \infty)$, and decreasing on $(-1, 3)$, since $f'(x) > 0$ on the first two intervals and $f'(x) < 0$ on the third.

Example 2 — A function with a vertical asymptote

Find the intervals on which $f(x) = \dfrac{x^2}{x-2}$ is increasing or decreasing.

Step 1. $f'(x) = \dfrac{2x(x-2) - x^2}{(x-2)^2} = \dfrac{x^2 - 4x}{(x-2)^2} = \dfrac{x(x-4)}{(x-2)^2}$.
Step 2. $f' = 0$ at $x = 0$ and $x = 4$. Neither f nor f' is defined at $x = 2$ (vertical asymptote). The denominator $(x-2)^2$ is positive, so the sign of f' is the sign of $x(x-4)$.
Step 3. See the chart. f is increasing on $(-\infty, 0)$ and on $(4, \infty)$; decreasing on $(0, 2)$ and on $(2, 4)$.
Note the intervals $(0, 2)$ and $(2, 4)$ are *not* merged: $x = 2$ is not in the domain.



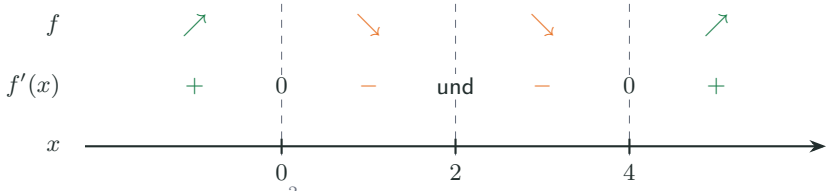


Figure 5.5. $f(x) = \dfrac{x^2}{x-2}$ (dashed line: $x = 2$) and the sign chart of f'.

Example 3 — Reading a graph of f'

The graph of f', the derivative of a function f, is shown on $[-2.2, 4]$. On which intervals is f increasing? Decreasing?

The graph of f' is *below* the x-axis for $-2.2 < x < -1$, so $f' < 0$ and f is decreasing on $[-2.2, -1]$. It is above the axis on $(-1, 2)$ and again on $(2, 4)$, so f is increasing on $[-1, 2]$ and on $[2, 4]$.
At $x = 2$ the graph of f' only *touches* the axis: $f'(2) = 0$ but f' stays positive on both sides, so f does not turn around there. Because the two increasing intervals share the endpoint 2, we may write: **f is increasing on $[-1, 4]$.**

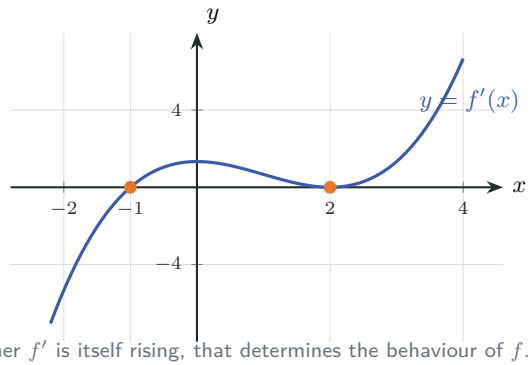


Figure 5.6. The graph of f'. It is the *sign* of f', not whether f' is itself rising, that determines the behaviour of f.

Common errors

- **Confusing f with f' on a graph of f'.** If the graph of f' is above the x-axis, f is increasing, even if the graph of f' is itself falling.
- **Merging intervals across a break.** Never write "decreasing on $(0, 4)$" when $x = 2$ is not in the domain. Split at every point where f is undefined.
- **Treating a zero of f' as a turning point.** A double root such as $(x-2)^2$ does not change the sign of f', so f keeps increasing (or decreasing) through that point.

AP Exam Tip

Justify with the sign of the derivative, in a sentence: "$f'(x) > 0$ on $(3, \infty)$, so f is increasing on $(3, \infty)$." The scoring guide gives the point for the *interval and the reason together*; an interval alone, or a table of signs with no sentence, is usually not enough. Intervals may be written open or closed at the endpoints; the exam accepts either as long as the interval is correct.

Check yourself

1. On what intervals is $f(x) = x^3 - 12x$ increasing?
2. $f'(x) = (x-1)^2(x+3)$. On what interval is f increasing? Does the zero of f' at $x = 1$ matter?
3. On what interval is $g(x) = e^{-x^2}$ increasing? Decreasing?

Topic 5.4 Using the First Derivative Test to Determine Relative (Local) Extrema

A peak is where rising turns into falling

Topic 5.2 told us where relative extrema can be: at critical points. Topic 5.3 told us how to read the direction of f from the sign of f'. Put them together and you get a complete test. Walk along the graph from left to right. If you are climbing before the critical point and descending after it, you have crossed a peak: a relative maximum. If you are descending and then climbing, you have crossed a valley: a relative minimum. If the direction does not change, the critical point is neither (think of the flat spot of $y = x^3$). In derivative language, a peak is a point at which f' changes from positive to negative, and a valley is a point at which f' changes from negative to positive.

The test needs f to be *continuous* at c, but it does not need $f'(c)$ to exist: a cusp or corner is classified exactly the same way, by looking at the sign of f' on either side. This makes the First Derivative Test more widely applicable than the Second Derivative Test of Topic 5.7.

The First Derivative Test

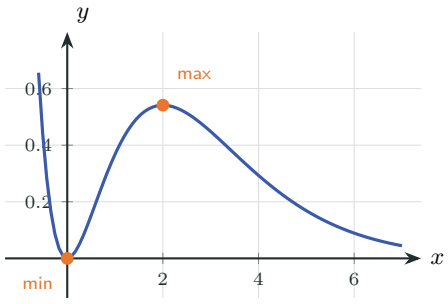
Let f be continuous at the critical point $x = c$, and let f' exist on an open interval around c except possibly at c.

- If f' changes from **positive to negative** at c, then f has a **relative maximum** at $x = c$.
- If f' changes from **negative to positive** at c, then f has a **relative minimum** at $x = c$.
- If f' does **not change sign** at c, then f has **no relative extremum** at c.

Example 1 — Classifying with a sign chart

Find the relative extrema of $f(x) = x^2 e^{-x}$.

Step 1. Product rule: $f'(x) = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2-x)$.
Step 2. Since $e^{-x} > 0$, $f'(x) = 0$ only at $x = 0$ and $x = 2$.
Step 3. The sign of f' is the sign of $x(2-x)$: negative for $x < 0$, positive on $(0, 2)$, negative for $x > 2$.
Step 4. At $x = 0$, f' changes from negative to positive: **relative minimum**, $f(0) = 0$. At $x = 2$, f' changes from positive to negative: **relative maximum**, $f(2) = \dfrac{4}{e^2} \approx 0.541$.



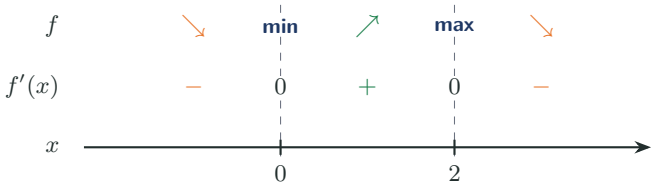
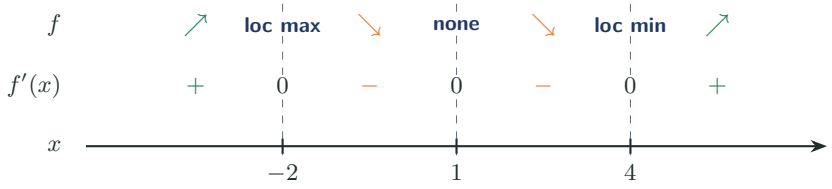


Figure 5.7. $f(x) = x^2e^{-x}$ and the sign chart of f'.

Example 2 — A zero of f' with no sign change

The function f is continuous for all x, and $f'(x) = (x-1)^2(x+2)(x-4)$. At which x does f have a relative maximum? A relative minimum?

Step 1. The critical points are $x = -2$, $x = 1$ and $x = 4$.
Step 2. The factor $(x-1)^2$ is never negative, so the sign of f' is the sign of $(x+2)(x-4)$ everywhere except at $x = 1$.

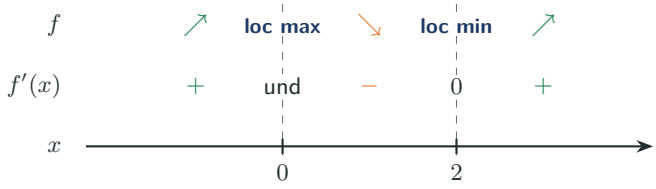


Step 3. f' changes from positive to negative at $x = -2$: relative maximum. f' changes from negative to positive at $x = 4$: relative minimum. At $x = 1$, f' is negative on both sides, so there is no relative extremum.

Example 3 — A critical point where f' is undefined

Classify the critical points of $f(x) = x^{2/3}(x-5)$ found in Example 2 of Topic 5.2.

Step 1. $f'(x) = \dfrac{5(x-2)}{3x^{1/3}}$, with critical points $x = 0$ (f' undefined) and $x = 2$.
Step 2. For $x < 0$: $x^{1/3} < 0$ and $x - 2 < 0$, so $f' > 0$. For $0 < x < 2$: $x^{1/3} > 0$, $x - 2 < 0$, so $f' < 0$. For $x > 2$: $f' > 0$.



Step 3. f' changes from positive to negative at $x = 0$, so f has a relative maximum at $x = 0$ (the cusp), with $f(0) = 0$. f' changes from negative to positive at $x = 2$, so f has a relative minimum at $x = 2$, with $f(2) = -3 \cdot 2^{2/3} \approx -4.762$ (Figure 5.4).

Common errors

- **Saying "$f'(c) = 0$, so c is a maximum or minimum."** You must show the sign change. Without it the point may be neither.
- **Mixing up f and f'.** The correct statement is "f' changes from positive to negative", not "f changes from positive to negative"; the value of f at the extremum can be negative, positive, or zero.
- **Giving the y-value when the question asks for x.** A relative maximum *occurs at* $x = c$; the maximum *value* is $f(c)$.
- **Reversing the signs.** Positive-to-negative is a *maximum*: climbing, then descending.

AP Exam Tip

The model sentence for a free-response answer is: *"Since $f'(x)$ changes from positive to negative at $x = c$, f has a relative maximum at $x = c$."* It contains the three things graders look for: the function (f'), the direction of the sign change, and the conclusion at a named value of x. Do not write "f' is zero" as the only reason, and do not use the Second Derivative Test unless it is asked for, since it can fail.

Check yourself

1. Find the relative extrema of $f(x) = x^4 - 8x^2$ and justify.
2. $f'(x) = (x+1)(x-2)^2$ for all x. Where does f have a relative extremum? Justify.
3. Show that $f(x) = \dfrac{x}{x^2+1}$ has a relative maximum at $x = 1$ and a relative minimum at $x = -1$.

Topic 5.5 Using the Candidates Test to Determine Absolute (Global) Extrema

Absolute extrema on a closed interval

The First Derivative Test finds hills and valleys, but the question "what is the greatest value of f on this interval?" is a different question. The highest point on an interval might be the top of a hill, but it might equally be an end of the interval where the graph is still climbing. The Extreme Value Theorem guarantees that for a continuous function on $[a, b]$ the absolute maximum and minimum *exist*; the **Candidates Test** says where they must be. Any absolute extremum is either at an endpoint or at an interior point, where it is also a relative extremum and therefore a critical point. So the list of candidates is finite: the critical points in (a, b) and the two endpoints. Compute f at each candidate; the largest value is the absolute maximum and the smallest is the absolute minimum.

The Candidates Test (closed interval method)

Let f be continuous on $[a, b]$.

1. Find all critical points of f in (a, b).
2. Evaluate f at each critical point *and* at a and b.
3. The largest of these values is the **absolute maximum value**; the smallest is the **absolute minimum value**. State where each occurs.

On an interval that is not closed the EVT does not apply. If f is continuous on an interval I with exactly one critical point c in I, and f' changes from negative to positive (positive to negative) at c, then $f(c)$ is the absolute minimum (maximum) of f on I.

Example 1 — A candidates table

Find the absolute maximum and minimum values of $h(x) = 2\sin x + \cos 2x$ on $[0, 2\pi]$.

Step 1. From Topic 5.2, the critical points in $(0, 2\pi)$ are $\dfrac{\pi}{6}, \dfrac{\pi}{2}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$.

Step 2. Evaluate h at these and at the endpoints:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	$\frac{3\pi}{2}$	2π
$h(x)$	1	$\frac{3}{2}$	1	$\frac{3}{2}$	-3	1

Step 3. The absolute maximum value is $\dfrac{3}{2}$ (at $x = \dfrac{\pi}{6}$ and $x = \dfrac{5\pi}{6}$); the absolute minimum value is -3 (at $x = \dfrac{3\pi}{2}$).

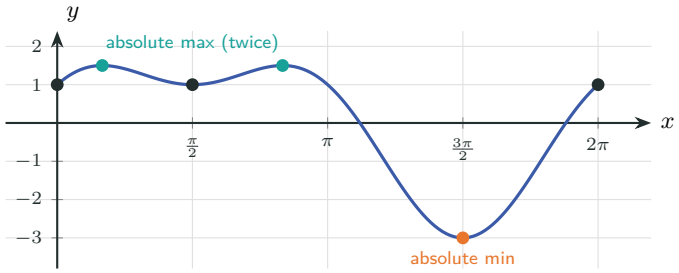
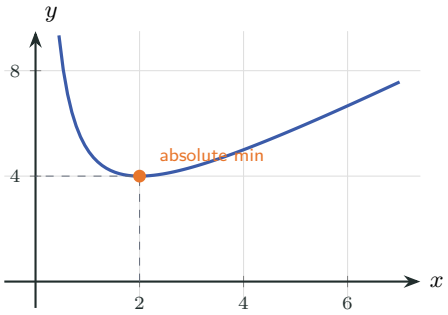


Figure 5.8. $h(x) = 2\sin x + \cos 2x$ on $[0, 2\pi]$.

Example 2 — An interval that is not closed

Find the absolute extrema, if any, of $f(x) = x + \dfrac{4}{x}$ on $(0, \infty)$.

Step 1. The EVT does not apply: the interval is open and unbounded.
Step 2. $f'(x) = 1 - \dfrac{4}{x^2} = \dfrac{(x-2)(x+2)}{x^2}$. On $(0, \infty)$ the only critical point is $x = 2$.
Step 3. $f' < 0$ on $(0, 2)$ and $f' > 0$ on $(2, \infty)$. Because f' changes from negative to positive at the *only* critical point, $f(2) = 4$ is the absolute minimum on $(0, \infty)$.
Step 4. There is no absolute maximum: $f(x) \to \infty$ as $x \to 0^+$ and as $x \to \infty$.



Example 3 — Candidates where f' does not exist

Find the absolute extrema of $f(x) = |x^2 - 4|$ on $[-3, 3]$.

Step 1. For $|x| < 2$, $f(x) = 4 - x^2$ and $f'(x) = -2x$, which is 0 at $x = 0$. For $|x| > 2$, $f(x) = x^2 - 4$ and $f'(x) = 2x \neq 0$. At $x = \pm 2$ the graph has corners, so f' does not exist: these are critical points as well.
Step 2. Candidates: $x = -3, -2, 0, 2, 3$, with $f = 5, 0, 4, 0, 5$.
Step 3. Absolute maximum value 5 (at $x = \pm 3$); absolute minimum value 0 (at $x = \pm 2$).

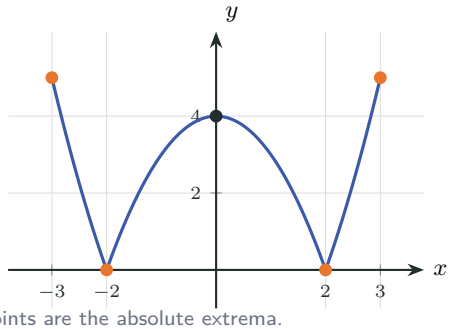


Figure 5.9. $f(x) = |x^2 - 4|$ on $[-3, 3]$: orange points are the absolute extrema.

Common errors

- **Forgetting the endpoints.** On $[a, b]$ the absolute extremum is very often at an endpoint (Figure 5.3).
- **Reporting the location when the value is requested**, or the reverse. Answer both: "the absolute maximum value is . . . , and it occurs at $x = \ldots$."
- **Using a relative extremum as an absolute one** without comparing it with the endpoints and the other candidates.
- **Using the test on a non-closed interval.** For (a, b) or $(0, \infty)$ you need a sign-change argument (and limits at the ends), and the extremum may not exist.

AP Exam Tip

For absolute extrema, a justification that earns full credit shows the *candidates and their function values* (a short table), and states the hypothesis: "f is continuous on $[a, b]$, so by the Extreme Value Theorem the absolute extrema occur at critical points or endpoints." On a calculator-active part, you may find critical points with the calculator's equation solver, but you must still list the values of f at all candidates.

Check yourself

1. Find the absolute extrema of $f(x) = x^3 - 3x$ on $[-2, 3]$.
2. Find the absolute maximum and minimum values of $g(x) = xe^{-x}$ on $[0, 3]$.
3. Why can the Candidates Test not be used for $f(x) = x^2$ on $(0, 2)$? Does f have an absolute maximum there?

Topic 5.6 Determining Concavity of Functions over Their Domains

How the graph bends

Two functions can both be increasing and yet look completely different: one climbs ever more steeply, like a ski jump, and the other climbs ever more slowly, flattening out. The difference is *concavity*. A graph is **concave up** on an interval if it bends upward, like a cup that holds water, and **concave down** if it bends downward, like a cap. The tangent lines tell the story: on a concave-up graph every tangent line lies *below* the curve, and the slopes of those tangent lines are increasing from left to right. On a concave-down graph the tangent lines lie *above* the curve and their slopes decrease.

Since the slope of the tangent line is f', "the slopes are increasing" says that f' *is an increasing function*. And we already know how to decide when a function is increasing: look at the sign of its derivative. The derivative of f' is f''. So $f'' > 0$ means f' is increasing, i.e. f is concave up; $f'' < 0$ means f is concave down. A point where the concavity changes is called a **point of inflection**.

Concavity and inflection points

- If $f''(x) > 0$ on an interval, f is **concave up** there (f' is increasing). If $f''(x) < 0$ on an interval, f is **concave down** there (f' is decreasing).
- A **point of inflection** is a point $(c, f(c))$ at which the graph has a tangent line (possibly vertical) and the concavity changes, i.e. f'' changes sign at $x = c$ (equivalently, f' changes from increasing to decreasing or vice versa).
- Candidates for inflection points: the x-values where $f''(x) = 0$ or $f''(x)$ does not exist. **They are only candidates:** the sign of f'' must actually change.
- Procedure: find f''; mark its zeros, the points where it is undefined, and the points where f is undefined; test the sign of f'' in each interval.

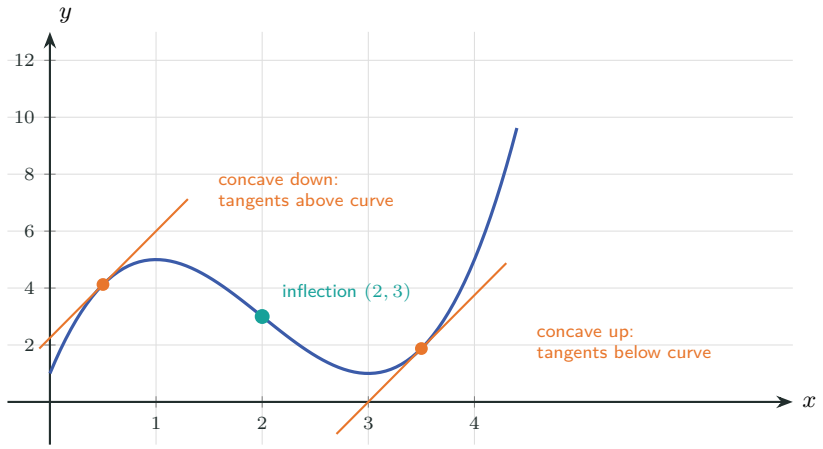
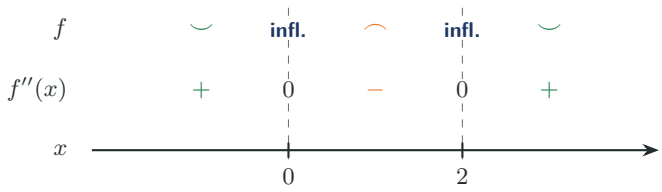


Figure 5.10. $f(x) = x^3 - 6x^2 + 9x + 1$: concave down for $x < 2$ (tangents above the curve), concave up for $x > 2$ (tangents below). Here $f''(x) = 6x - 12$.

Example 1 — Concavity and inflection points of a polynomial

Find the intervals of concavity and the inflection points of $f(x) = x^4 - 4x^3$.

Step 1. $f'(x) = 4x^3 - 12x^2$ and $f''(x) = 12x^2 - 24x = 12x(x - 2)$.
Step 2. $f''(x) = 0$ at $x = 0$ and $x = 2$. Test: $f''(-1) = 36 > 0$, $f''(1) = -12 < 0$, $f''(3) = 36 > 0$.



Step 3. f is concave up on $(-\infty, 0)$ and $(2, \infty)$, and concave down on $(0, 2)$. Since f'' changes sign at both $x = 0$ and $x = 2$, the inflection points are $(0, 0)$ and $(2, -16)$.

Example 2 — Candidates that are not inflection points, and one that is

(a) Does $f(x) = x^4$ have an inflection point at $x = 0$? (b) Does $g(x) = x^{1/3}$ have one at $x = 0$?

(a) $f''(x) = 12x^2$, which is 0 at $x = 0$ but is positive on both sides. No sign change, so **no** inflection point: the graph is concave up everywhere.

(b) $g'(x) = \frac{1}{3}x^{-2/3}$ and $g''(x) = -\frac{2}{9}x^{-5/3}$, which does not exist at 0. For $x < 0$, $g'' > 0$ (concave up); for $x > 0$, $g'' < 0$ (concave down). The concavity changes at $x = 0$ and the graph has a (vertical) tangent line there, so $(0, 0)$ **is** an inflection point even though $g''(0)$ does not exist.

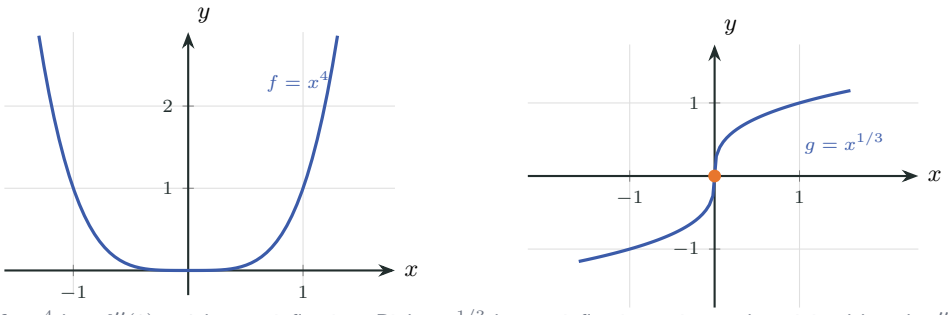


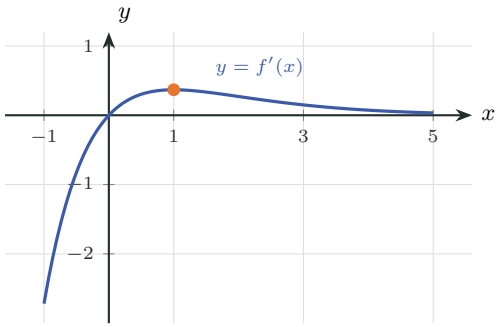
Figure 5.11. Left: x^4 has $f''(0) = 0$ but no inflection. Right: $x^{1/3}$ has an inflection point at the origin although $g''(0)$ is undefined.

Example 3 — Concavity from the graph of f'

The graph of f' is shown for $-1 \le x \le 5$. On which intervals is f concave up? Where does f have a point of inflection?

Concavity of f is read from the *slope* of the graph of f'. The graph of f' is increasing on $(-1, 1)$, so f' is increasing and f is concave up on $(-1, 1)$. It is decreasing on $(1, 5)$, so f is concave down on $(1, 5)$.

At $x = 1$ the graph of f' has a maximum: f' changes from increasing to decreasing, so f has a point of inflection at $x = 1$. (For this graph $f'(x) = xe^{-x}$, so $f''(x) = (1 - x)e^{-x}$, which agrees.)



Common errors

- **"$f''(c) = 0$, so there is an inflection point."** Not unless f'' changes sign (compare x^4).
- **Concave up does not mean increasing.** A function can be decreasing and concave up (the right half of a parabola opening upward, to the left of the vertex).
- **Reading f'' from a graph of f' by height.** On a graph of f', concavity of f is the *slope* (increasing or decreasing) of the graph, not whether it is above the axis.
- **Describing a point of inflection by its x-value only when the question wants the point.** Check whether the answer should be $x = c$ or $(c, f(c))$.

AP Exam Tip

A complete justification of an inflection point has two parts: "$f''(x)$ changes sign at $x = c$" (or "f' changes from increasing to decreasing at $x = c$") *and* a conclusion naming $x = c$. For concave up or down, write "$f''(x) > 0$ on (a, b), so the graph of f is concave up on (a, b)."

Check yourself

1. Find the intervals of concavity and the x-coordinate of every inflection point of $f(x) = x^3 - 6x^2 + 5$.
2. Does $h(x) = x^6$ have a point of inflection at $x = 0$? Justify.
3. The graph of f' is a parabola opening upward with vertex at $x = 3$. On what intervals is f concave down? Concave up?

Topic 5.7 Using the Second Derivative Test to Determine Extrema

Let the concavity decide

Suppose $f'(c) = 0$, so the tangent line at $x = c$ is horizontal. If the graph is concave up near c it looks like the bottom of a bowl, so $f(c)$ is a relative minimum; if it is concave down it looks like the top of a hill, so $f(c)$ is a relative maximum. The sign of $f''(c)$ tells you which. This is the **Second Derivative Test**. It is usually quicker than the First Derivative Test because you only evaluate f'' at a point, rather than testing the sign of f' on several intervals.

The test has a limitation that the exam loves to exploit: if $f''(c) = 0$ (or does not exist) it gives *no information*. The three functions $y = x^4$, $y = -x^4$ and $y = x^3$ all have $f'(0) = 0$ and $f''(0) = 0$, yet they have a minimum, a maximum and neither, respectively. When the Second Derivative Test is inconclusive you return to the First Derivative Test.

The Second Derivative Test

Suppose $f'(c) = 0$ and f'' exists near c.

- If $f''(c) < 0$, then f has a **relative maximum** at $x = c$.
- If $f''(c) > 0$, then f has a **relative minimum** at $x = c$.
- If $f''(c) = 0$, the test is **inconclusive**: use the First Derivative Test.

The test applies only at points where $f'(c) = 0$. It says nothing about a critical point where f' is undefined.

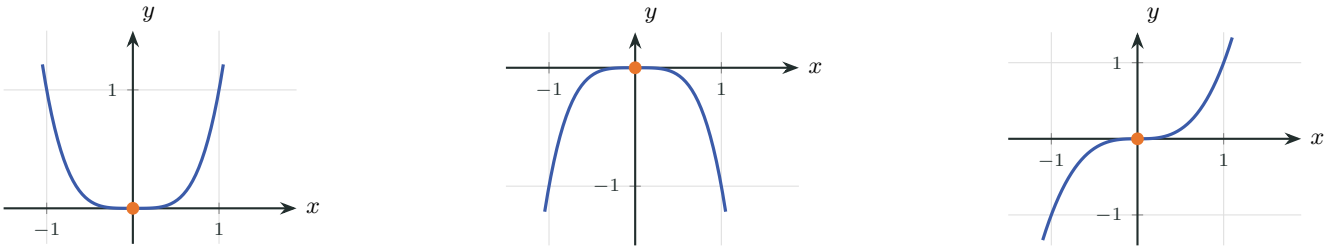


Figure 5.12. Left: $y = x^4$ (minimum). Middle: $y = -x^4$ (maximum). Right: $y = x^3$ (neither). In all three cases $f'(0) = 0$ and $f''(0) = 0$, so the Second Derivative Test cannot decide.

Example 1 — A clean application

Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 6x^2 + 9x + 1$.

Step 1. $f'(x) = 3(x-1)(x-3)$, so $f' = 0$ at $x = 1$ and $x = 3$.
Step 2. $f''(x) = 6x - 12$.
Step 3. $f''(1) = -6 < 0$: relative **maximum** at $x = 1$, with value $f(1) = 5$. $f''(3) = 6 > 0$: relative **minimum** at $x = 3$, with value $f(3) = 1$. (Compare Figure 5.3.)

Example 2 — An inconclusive test

Classify the critical points of $f(x) = x^4 - 4x^3$.

Step 1. $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$, so $f' = 0$ at $x = 0$ and $x = 3$.
Step 2. $f''(x) = 12x^2 - 24x$. At $x = 3$: $f''(3) = 108 - 72 = 36 > 0$, so f has a relative minimum at $x = 3$, with $f(3) = -27$.
Step 3. At $x = 0$: $f''(0) = 0$. The test is inconclusive. Use the First Derivative Test: $f'(x) = 4x^2(x-3)$ is negative for $x < 0$ and negative for $0 < x < 3$. There is no sign change at $x = 0$, so f has **no** relative extremum at $x = 0$. (Note that $x = 0$ is an inflection point: see Example 1 of Topic 5.6.)

Example 3 — Values from a table

The function f is twice differentiable. Values of f' and f'' are given. What can be concluded about f at each x?

x	1	3	5	7
$f'(x)$	0	0	0	3
$f''(x)$	5	−2	0	−4

$x = 1$: $f'(1) = 0$ and $f''(1) = 5 > 0$, so f has a relative minimum. $x = 3$: $f'(3) = 0$ and $f''(3) = -2 < 0$: relative maximum. $x = 5$: $f'(5) = 0$ but $f''(5) = 0$: no conclusion from this test. $x = 7$: $f'(7) = 3 \neq 0$, so $x = 7$ is not a critical point; there is no extremum conclusion (the graph is rising and concave down there).

Common errors

- **Concluding "inflection point" (or "no extremum") when $f''(c) = 0$.** The test is silent; check the sign of f'.
- **Using the test where $f'(c) \neq 0$.** The sign of f'' alone never makes c an extremum.
- **Reversing the signs:** *negative* second derivative means *maximum* (concave down, a cap).
- **Using it for cusps and corners.** At a critical point where f' does not exist, only the First Derivative Test applies.

AP Exam Tip

Both tests earn full credit if they are stated correctly. For the Second Derivative Test write all of it: "$f'(c) = 0$ and $f''(c) < 0$, so f has a relative maximum at $x = c$." If a problem says "justify using the first derivative", the Second Derivative Test will not be accepted.

Check yourself

1. Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 12x$.
2. Show that $f(x) = xe^x$ has a relative minimum at $x = -1$.
3. Show that the Second Derivative Test fails for $g(x) = (x-2)^4$ at $x = 2$, and decide by another method what happens there.

Topic 5.8 Sketching Graphs of Functions and Their Derivatives

Three graphs, one story

The functions f, f' and f'' are three views of the same curve, and the exam constantly moves between them: you are given the graph of one and asked about another. Everything you have learned in Topics 5.2–5.7 can be summarised as a dictionary between features of f and features of f'.

Dictionary: features of f and of f'

If f ...	**then the graph of f' ...**
is increasing	is above the x-axis ($f' > 0$)
is decreasing	is below the x-axis ($f' < 0$)
has a horizontal tangent	touches or crosses the x-axis ($f' = 0$)
has a relative extremum	crosses the x-axis (sign change)
is concave up	is increasing (rising)
is concave down	is decreasing (falling)
has an inflection point	has a relative maximum or minimum (a turning point)
has a corner or cusp	has a jump, or is undefined, at that x

The same dictionary applies one level up: replace f by f' and f' by f''.

Notice that a graph of f' never tells you the *height* of f: if F and G differ by a constant, they have the same derivative, so they share every feature in the dictionary. Without a given value such as $f(0) = 1$ you may only sketch the *shape*.

Example 1 — From f to f'

The graph of $f(x) = xe^{-x}$ for $x \geq 0$ is shown. Describe f' and sketch it.

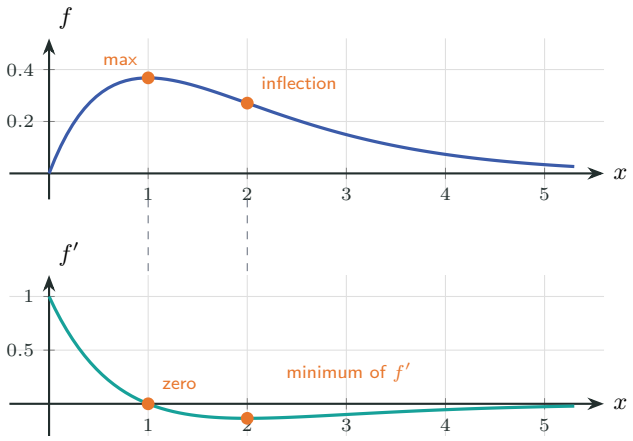


Figure 5.13. $f(x) = xe^{-x}$ (top) and $f'(x) = (1-x)e^{-x}$ (bottom). Dashed lines join the corresponding features.

Step 1. f rises on $(0,1)$ and falls afterwards, so $f' > 0$ on $(0,1)$ and $f' < 0$ for $x > 1$, with $f'(1) = 0$ where f has its maximum.
Step 2. f is concave down on $(0,2)$ and concave up for $x > 2$ (inflection at $x = 2$): thus f' is decreasing on $(0,2)$ and increasing for $x > 2$, with a minimum of f' at $x = 2$.
Step 3. For large x, f levels off, so $f' \to 0$ from below. Check: $f'(x) = (1-x)e^{-x}$ has $f'(0) = 1$, $f'(1) = 0$, $f'(2) = -e^{-2} \approx -0.135$.

Example 2 — A piecewise linear f

The graph of the continuous function f consists of line segments joining $(0,0)$, $(2,4)$, $(4,4)$, $(6,0)$ and $(8,2)$. Sketch f' and state where f is not differentiable.

Step 1. The slope on each piece: $\dfrac{4-0}{2-0} = 2$ on $(0,2)$; 0 on $(2,4)$; $\dfrac{0-4}{6-4} = -2$ on $(4,6)$; $\dfrac{2-0}{8-6} = 1$ on $(6,8)$.
Step 2. So f' is a step function. At $x = 2, 4, 6$ the slope jumps, so f has corners there and f' does not exist; f is not differentiable at $x = 2$, $x = 4$ and $x = 6$.

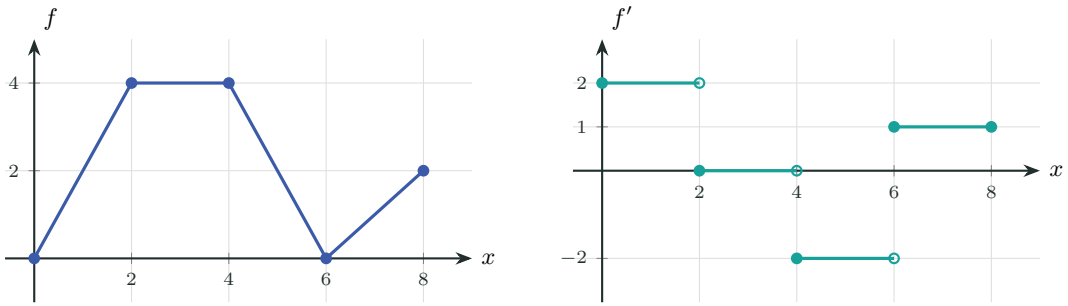


Figure 5.14. A piecewise linear f and its derivative. Open circles show that f' is undefined at the corners.

Example 3 — From f' to the shape of f

The derivative of f is $f'(x) = x^2 - 4$ and $f(0) = 0$. Describe the behaviour of f and sketch it.

Step 1 (increasing/decreasing). $f'(x) = (x-2)(x+2)$ is positive for $x < -2$, negative on $(-2, 2)$, positive for $x > 2$. So f increases, decreases, then increases: relative maximum at $x = -2$ and relative minimum at $x = 2$.
Step 2 (concavity). $f''(x) = 2x$: concave down for $x < 0$, concave up for $x > 0$; inflection point at $x = 0$ (where f' has its minimum, -4).

Step 3 (the sketch). An antiderivative with $f(0) = 0$ is $f(x) = \dfrac{x^3}{3} - 4x$. Then $f(-2) = \dfrac{16}{3} \approx 5.33$ and $f(2) = -\dfrac{16}{3}$. The graph through $(0, 0)$ rises to a peak at $(-2, \frac{16}{3})$, falls through the origin to a valley at $(2, -\frac{16}{3})$, then rises.

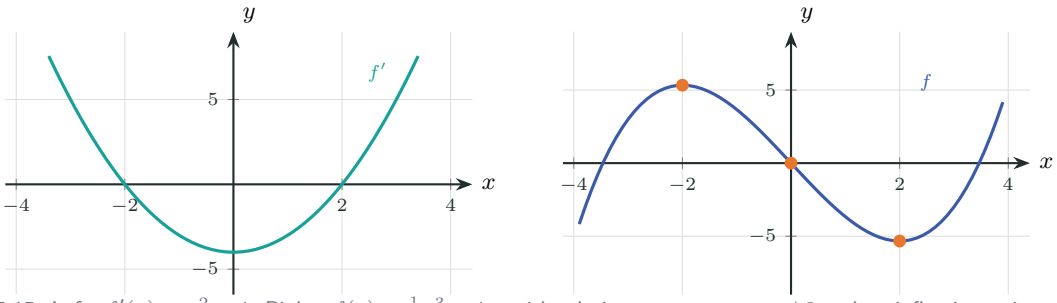


Figure 5.15. Left: $f'(x) = x^2 - 4$. Right: $f(x) = \frac{1}{3}x^3 - 4x$, with relative extrema at $x = \pm 2$ and an inflection point at $x = 0$.

Common errors

- **Using the height of f' as the height of f.** A graph of f' tells you about slopes of f only.
- **Putting extrema of f where f' has extrema.** Extrema of f are at the *zeros* of f'; extrema of f' are the *inflection points* of f.
- **Smoothing a corner.** At a corner of f the derivative has a jump; do not draw f' as a continuous curve there.

AP Exam Tip

When you sketch, mark the *key x-values* first (zeros of f', turning points of f') and draw vertical dashed guide lines, as in Figure 5.13. The exam rewards correct signs and shape, not artistic precision; label your features.

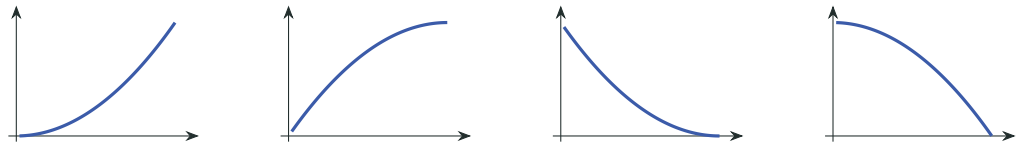
Check yourself

1. The graph of f is a downward-opening parabola with vertex at $(2, 5)$. Describe the graph of f'.
2. f is continuous with $f'(x) = -1$ for $x < 3$ and $f'(x) = 2$ for $x > 3$. Describe f and say whether f is differentiable at $x = 3$.
3. If f' has a relative minimum at $x = 4$, what can you say about f at $x = 4$?

Topic 5.9 Connecting a Function, Its First Derivative, and Its Second Derivative

Four shapes and one rule

Every smooth piece of a graph that is not a straight line and does not turn around has one of exactly four shapes, determined by the two signs f' (is the function rising or falling?) and f'' (is it bending up or down?). These four shapes are the building blocks of every graph you will meet. A relative maximum looks like the second and third shapes joined at a flat top; an inflection point is where one shape turns into its neighbour.



$f' > 0,\ f'' > 0$	$f' > 0,\ f'' < 0$	$f' < 0,\ f'' > 0$	$f' < 0,\ f'' < 0$
increasing, concave up	increasing, concave down	decreasing, concave up	decreasing, concave down

Figure 5.16. The four combinations of signs of f' and f''.

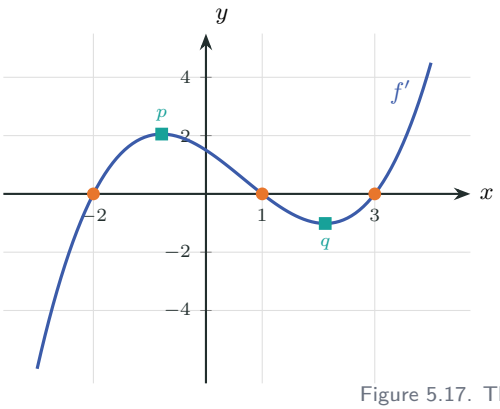
Reading f from f' and f''

Evidence	Conclusion about f	Reason to write
$f' > 0$	increasing	sign of f'
$f' < 0$	decreasing	sign of f'
f' changes $+ \to -$ at c	relative maximum at c	First Derivative Test
f' changes $- \to +$ at c	relative minimum at c	First Derivative Test
$f'' > 0$ (or f' increasing)	concave up	sign of f'' / slope of f'
$f'' < 0$ (or f' decreasing)	concave down	sign of f'' / slope of f'
f'' changes sign at c	point of inflection at c	concavity changes

The exam will usually give you one of the three functions as a graph or formula (most often f') and ask for conclusions about another. The method is always the same: translate the evidence into signs, build a sign chart, and write one sentence for each conclusion that cites the evidence.

Example 1 — Everything from the graph of f'

The graph of f', the derivative of f, is shown on $[-3, 4]$: it crosses the x-axis at $x = -2$, $x = 1$, $x = 3$, and has a relative maximum at $x = p \approx -0.79$ and a relative minimum at $x = q \approx 2.12$. Describe f.



Increasing/decreasing. $f' < 0$ on $(-3, -2)$ and $(1, 3)$: f is decreasing there. $f' > 0$ on $(-2, 1)$ and $(3, 4)$: f is increasing there.
Extrema. f' changes $- \to +$ at $x = -2$ and $x = 3$: relative minima. f' changes $+ \to -$ at $x = 1$: relative maximum.
Concavity. f' is increasing on $(-3, p)$ and $(q, 4)$: f is concave up there. f' is decreasing on (p, q): f is concave down on (p, q).
Inflection. f' has a relative maximum at p and a relative minimum at q, so the concavity of f changes at $x = p$ and $x = q$.

Figure 5.17. The graph of $f'(x) = \frac{1}{4}(x + 2)(x - 1)(x - 3)$.

Example 2 — A piecewise linear f'

The graph of f' is made of segments joining $(0, -2)$, $(2, 2)$, $(6, -2)$ and $(8, 0)$. Find the x-values of the relative extrema and the inflection points of f on $(0, 8)$.

Step 1 (extrema). On $(0, 2)$, f' rises from -2 to 2, crossing 0 at $x = 1$. On $(2, 6)$ it falls from 2 to -2, crossing 0 at $x = 4$. So f' changes $- \to +$ at $x = 1$ (relative minimum of f) and $+ \to -$ at $x = 4$ (relative maximum).
Step 2 (inflection). f' is increasing on $(0, 2)$, decreasing on $(2, 6)$, increasing on $(6, 8)$. The graph of f' turns at $x = 2$ and $x = 6$, so the concavity of f changes at $x = 2$ and $x = 6$: these are the inflection points.
Step 3. On $(6, 8)$, $f' < 0$ increasing to 0 at $x = 8$ (the endpoint), so f is still decreasing there; there is no extremum inside $(0, 8)$ other than $x = 1, 4$.

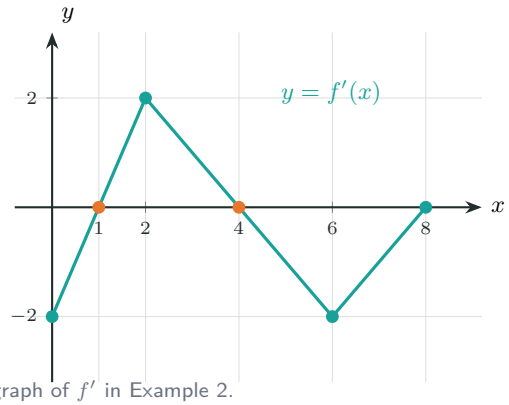


Figure 5.18. The piecewise linear graph of f' in Example 2.

Example 3 — Which graph is which?

The three graphs below are f, f' and f'' in some order. Identify them.

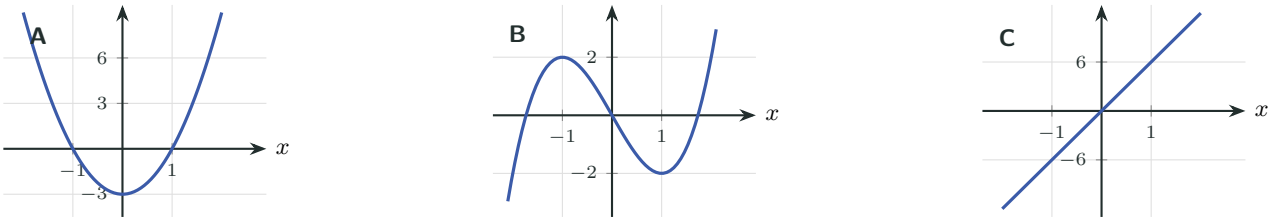


Figure 5.19. Graphs A, B and C.

Graph C is a straight line, and a line can only be the derivative of a parabola-shaped graph, so C is the *derivative of A*, and A is the derivative of B. Check: A is zero at $x = \pm 1$, exactly where B has its relative extrema; A has its minimum at $x = 0$, where B has its inflection point and where C crosses the axis. Therefore **B is f, A is f', and C is f''.** (They are $f(x) = x^3 - 3x$, $f'(x) = 3x^2 - 3$, $f''(x) = 6x$.)

Common errors

- **Mixing the two derivatives:** "f' is decreasing, so f is decreasing" is wrong; f' decreasing means f is *concave down*. f decreases where f' is *negative*.
- **Using f to answer a concavity question on a graph of f'.** Use the slope of the graph of f'.
- **Missing the endpoint.** Relative extrema and inflection points are interior points; do not list an endpoint of the interval as one.
- **Giving a conclusion with no evidence.** Every statement about f needs its reason from f' or f''.

AP Exam Tip

Write each conclusion as *[statement about f] because [statement about f' or f'']*: "f is concave down on (p, q) because f' is decreasing on (p, q)." If you are given the graph of f', you may point to it ("the graph of f' is below the x-axis on $(1, 3)$"); you do not need to find f' as a formula.

Check yourself

1. The function f satisfies $f'(x) < 0$ and $f''(x) > 0$ for all x in $[0, 3]$. Which table could give values of f at $x = 0, 1, 2, 3$: (a) $10, 7, 5, 4$ (b) $10, 8, 5, 1$? Explain.
2. $f'(x) = x^2 - 6x + 5$. On which intervals is f increasing? On which is it concave up?
3. The graph of f' is a horizontal line at height 3. What can you say about f and f''?

Topic 5.10 Introduction to Optimization Problems

Building the function

Optimization means finding the *best* value of something: the greatest area for a given amount of fencing, the least material for a given volume, the shortest distance, the lowest cost. You already own the calculus needed to find an extremum (Topics 5.2–5.7). What is new is that the function is not handed to you: you must *build* it from a word problem. This topic is about building it correctly, and it is where most of the errors in optimization problems happen.

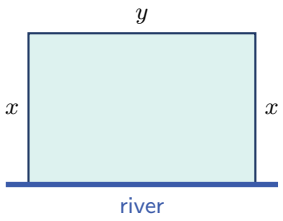
A four-step plan for optimization

1. **Draw and name.** Sketch the situation; give a letter to every changing quantity and write down the fixed numbers.
2. **Objective and constraint.** The *objective* is the quantity you want to maximize or minimize (area, cost, distance, …). The *constraint* is the equation that links the variables (fixed perimeter, fixed volume, a point lying on a curve).
3. **One variable.** Use the constraint to eliminate all but one variable so the objective becomes a function of a single variable, e.g. $A(x)$.
4. **Domain.** State the values of the variable that make physical sense (lengths are nonnegative, a point on a curve has its x-values restricted, …). The domain is an interval; it tells you whether to use the Candidates Test.

Then (Topic 5.11) find the critical points, decide whether each gives the maximum or minimum, and *answer the question that was asked*, with units.

Example 1 — Fencing a field beside a river

A farmer has 200 metres of fencing and wants to enclose a rectangular field. One side of the field is a straight river and needs no fence. Write the area as a function of one variable, state its domain, and find the largest possible area.

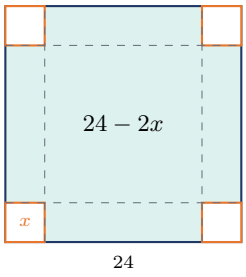


Step 1. Let x be the length (m) of each of the two sides perpendicular to the river and y the length of the side parallel to it.
Step 2. Objective: $A = xy$. Constraint: $2x + y = 200$, so $y = 200 - 2x$.
Step 3. $A(x) = x(200 - 2x) = 200x - 2x^2$.
Step 4. Lengths cannot be negative: $x \geq 0$ and $y = 200 - 2x \geq 0$, so the domain is $0 \leq x \leq 100$.

Step 5 (solve). $A'(x) = 200 - 4x = 0$ at $x = 50$. Candidates: $A(0) = 0$, $A(50) = 5000$, $A(100) = 0$. The maximum area is 5000 m^2, with $x = 50$ m and $y = 100$ m.

Example 2 — An open box from a square sheet

A square sheet of cardboard, 24 cm on each side, has a square of side x cm cut from each corner. The sides are folded up to form a box with no lid. Write the volume $V(x)$ and its domain, and estimate the best value of x from a table.



The base is a square of side $24 - 2x$ and the height is x, so

$$V(x) = x(24 - 2x)^2.$$

The cut cannot be negative, and the two cuts along one side use $2x \leq 24$: domain $0 \leq x \leq 12$.

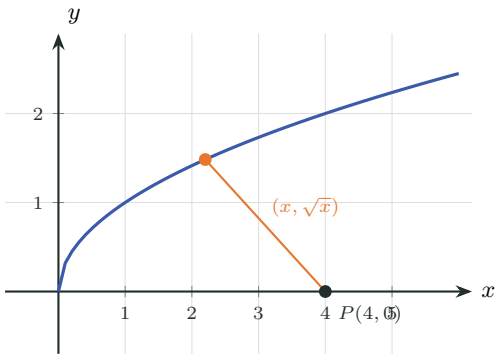
x	2	3	4	5	6
$V(x)$	800	972	1024	980	864

The table suggests a maximum near $x = 4$; we confirm it with calculus in Topic 5.11 (Example 1).

Example 3 — The point of a curve closest to a given point

Write the *square* of the distance from the point $P(4, 0)$ to a point (x, y) on the curve $y = \sqrt{x}$ as a function of x, and state its domain.

Step 1. A point on the curve is $(x, \sqrt{x})$ with $x \geq 0$.
Step 2. Distance^2: $D(x) = (x - 4)^2 + (\sqrt{x} - 0)^2 = (x - 4)^2 + x$.
Step 3. Domain: $x \geq 0$ (the curve exists only there).
Why the square? $D = \sqrt{(x - 4)^2 + x}$ is smallest exactly where the expression under the root is smallest, and the square has a far simpler derivative.



Common errors

- **Leaving two variables in the objective.** Differentiate only after the constraint has been used to eliminate one variable.
- **Forgetting the domain.** The domain decides whether endpoints are candidates and rules out solutions such as a negative length.
- **Optimizing the wrong quantity.** If a problem gives a fixed *volume* and asks for least *surface area*, the volume is the constraint, not the objective.
- **Using the whole fence for one equation.** Count only the sides that are actually fenced (Example 1 has three

sides).

AP Exam Tip

Free-response optimization problems score points for the *model*: "$A(x) = x(200 - 2x)$" with its *domain* stated, then for the derivative, the critical point, the justification that it is a maximum, and the answer *with units*. Do not skip writing the objective function even if you can do the algebra in your head.

Check yourself

1. A rectangle has perimeter 40 cm. Write its area as a function of its width w and give the domain.
2. Two nonnegative numbers x and y satisfy $x + y = 12$. Write $P = xy^2$ as a function of y and give the domain.
3. Write the square of the distance from $(3, 0)$ to a point on $y = x^2$ as a function of x.

Topic 5.11 Solving Optimization Problems

From the model to the answer

With the model $f(x)$ and its domain in hand, the solution is the Candidates Test or a sign-change argument. If the domain is a *closed* interval $[a, b]$ (the usual case when the variable has natural bounds), evaluate the objective at the critical points and at the endpoints, and choose the largest or smallest value. If the domain is open or unbounded, show that the critical point is an absolute extremum by one of two arguments: the First Derivative Test (f' changes sign at the *only* critical point), or concavity ($f'' > 0$ everywhere on the domain means the critical point is an absolute minimum, and $f'' < 0$ everywhere an absolute maximum). The exam expects one of these justifications; the answer alone earns only a fraction of the points.

Solving and justifying an optimization problem

1. Build $f(x)$ and the domain (Topic 5.10).
2. Solve $f'(x) = 0$ (and find where f' is undefined) inside the domain.
3. **Justify** the extremum: Candidates Test on a closed interval; or sign of f' around the only critical point; or sign of f''.
4. Answer in words, with units, and give *every* quantity the problem asks for (dimensions, maximum value, the point).

Example 1 — Finishing the box

In Example 2 of Topic 5.10, find the value of x that maximizes $V(x) = x(24 - 2x)^2$ on $[0, 12]$, and the maximum volume.

Step 1. $V'(x) = (24 - 2x)^2 + x \cdot 2(24 - 2x)(-2) = (24 - 2x)(24 - 2x - 4x) = (24 - 2x)(24 - 6x)$.
Step 2. $V'(x) = 0$ at $x = 12$ and $x = 4$.
Step 3 (Candidates Test). $V(0) = 0$, $V(4) = 4 \cdot 16^2 = 1024$, $V(12) = 0$.
The volume is largest when $x = 4$ cm: the box is $16 \times 16 \times 4$ cm and its volume is 1024 cm^3.

Example 2 — A can of minimum surface area

A closed cylindrical can must hold 1000 cm^3. Find the radius and height that minimize the surface area of the can.

Step 1 (model). Let r be the radius and h the height, in cm. Constraint: $\pi r^2 h = 1000$, so $h = \dfrac{1000}{\pi r^2}$. Surface area: $S = 2\pi r^2 + 2\pi r h$, so

$$S(r) = 2\pi r^2 + \frac{2000}{r}, \qquad r > 0.$$

Step 2 (critical point). $S'(r) = 4\pi r - \dfrac{2000}{r^2} = 0 \Rightarrow r^3 = \dfrac{500}{\pi}$, so $r = \sqrt[3]{500/\pi} \approx 5.419$ cm. The only critical point in $(0, \infty)$.

Step 3 (justify). $S''(r) = 4\pi + \dfrac{4000}{r^3} > 0$ for every $r > 0$, so S is concave up on its whole domain, and the critical point is the absolute minimum.

Step 4 (answer). $r \approx 5.419$ cm and $h = \dfrac{1000}{\pi r^2} \approx 10.839$ cm, which is $h = 2r$ (the can is as tall as it is wide). The

minimum surface area is $S \approx 553.6$ cm^2.

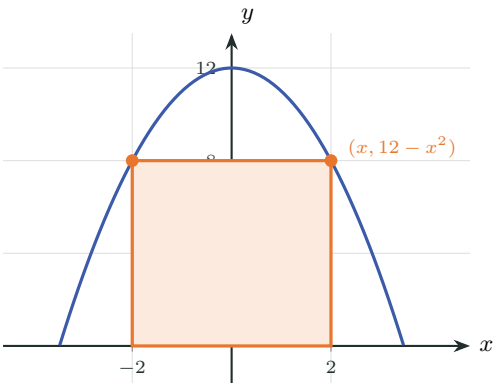
Example 3 — A rectangle under a parabola

A rectangle has its base on the x-axis and its upper two vertices on the parabola $y = 12 - x^2$. Find the largest possible area.

Step 1 (model). Let the vertices on the parabola be $(x, 12 - x^2)$ and $(-x, 12 - x^2)$ with $0 \le x \le \sqrt{12}$. Width $2x$, height $12 - x^2$:

$$A(x) = 2x(12 - x^2) = 24x - 2x^3.$$

Step 2. $A'(x) = 24 - 6x^2 = 0 \Rightarrow x = 2$ (since $x \ge 0$).
Step 3. $A(0) = 0$, $A(2) = 32$, $A(\sqrt{12}) = 0$. The maximum area is 32 square units, for a rectangle 4 wide and 8 high.



Example 4 — Minimum cost, calculator allowed

A cable must run from point A on a straight shore to a point B on an island 3 km from the shore; the nearest shore point to B is C, and A is 8 km from C along the shore. Laying cable under water costs \$5000 per km and along the shore \$3000 per km. The cable runs along the shore from A to a point D at distance x km from C, then straight to B under water. Find x for minimum cost.

Step 1 (model). Shore length $AD = 8 - x$, water length $DB = \sqrt{9 + x^2}$, so

$$C(x) = 3000(8 - x) + 5000\sqrt{9 + x^2}, \qquad 0 \le x \le 8.$$

Step 2. $C'(x) = -3000 + \dfrac{5000x}{\sqrt{9 + x^2}} = 0$ gives $5x = 3\sqrt{9 + x^2}$, so $25x^2 = 81 + 9x^2$, $x = \dfrac{9}{4} = 2.25$ km. (A graphing calculator's solver for $C'(x) = 0$ gives the same value.)
Step 3 (Candidates Test).

x	0	2.25	8
$C(x)$	39000	36000	≈ 42720

The minimum cost is \$36000, when the cable leaves the shore 2.25 km from C (that is, 5.75 km along the shore from A).

Common errors

- **Solving $f' = 0$ and stopping.** Without a justification (candidates, sign change or concavity) the answer is incomplete.
- **Answering with the wrong quantity**: the problem may ask for the *dimensions*, the *maximum area*, or the *value of* x. Read the last sentence again.
- **Ignoring the endpoints**, where the extremum sometimes lies (a degenerate rectangle gives area 0, but a cost or distance can be extreme at an end).
- **Dropping units**, or rounding too early; on calculator parts, keep three decimal places unless told otherwise.

AP Exam Tip

A global argument is the heart of the justification. The three accepted forms are: (i) "f is continuous on $[a, b]$, so the absolute maximum is the largest of the values at the critical point and endpoints"; (ii) "f' changes from positive to negative at $x = c$, the only critical point, so $f(c)$ is the absolute maximum on the interval"; (iii) "$f''(x) < 0$ for all x in the domain, so f is concave down and the critical point is an absolute maximum."

Check yourself

1. Two positive numbers have product 36. Find the least possible value of their sum, and justify.
2. A square sheet 12 cm on a side is used to make an open box as in Topic 5.10. Find the maximum volume.
3. Find the point of the parabola $y = x^2$ that is closest to $(0, 3)$, and the distance.

Topic 5.12 Exploring Behaviors of Implicit Relations

Tangent lines and concavity without solving for y

In Unit 3 you learned implicit differentiation: when x and y are tied by an equation such as $x^2+y^2 = 25$, you differentiate both sides with respect to x and solve for $\dfrac{dy}{dx}$. The result is a formula that usually contains *both* x and y. This topic puts that formula to work, to ask the same analytical questions that we asked of $f(x)$.

The key points of an implicit curve are where the tangent is horizontal (numerator of $\dfrac{dy}{dx}$ zero) or vertical (denominator zero). Solve the numerator or denominator *together with the original equation*, because the point must lie on the curve. For concavity, differentiate $\dfrac{dy}{dx}$ again, substitute the formula for $\dfrac{dy}{dx}$ wherever it appears, and simplify with the original equation if you can.

Analyzing an implicit relation

- Differentiate implicitly to get $\dfrac{dy}{dx} = \dfrac{N(x,y)}{D(x,y)}$.
- **Horizontal tangent**: $N = 0$ and $D \neq 0$. **Vertical tangent**: $D = 0$ and $N \neq 0$. Always combine with the equation of the curve to get the points.
- **Second derivative**: differentiate $\dfrac{dy}{dx}$ again with respect to x (remember y' appears again), then replace y' by $\frac{N}{D}$.
- At a point where $\dfrac{dy}{dx} = 0$: $\dfrac{d^2y}{dx^2} < 0$ gives a relative maximum of y (concave down) and $\dfrac{d^2y}{dx^2} > 0$ a relative minimum.

Example 1 — Horizontal and vertical tangents of a slanted ellipse

The curve $x^2 + xy + y^2 = 12$ is shown. Find all points with a horizontal tangent and all points with a vertical tangent, and the tangent line at $(2, 2)$.

Step 1. Differentiate: $2x + y + x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} = 0$, so

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y}.$$

Step 2 (horizontal). $2x + y = 0 \Rightarrow y = -2x$. Substitute: $x^2 - 2x^2 + 4x^2 = 3x^2 = 12$, $x = \pm 2$. Points $(2, -4)$ and $(-2, 4)$.
Step 3 (vertical). $x+2y = 0 \Rightarrow x = -2y$: $4y^2-2y^2+y^2 = 3y^2 = 12$, $y = \pm 2$. Points $(-4, 2)$ and $(4, -2)$.
Step 4. At $(2, 2)$: $\dfrac{dy}{dx} = -\dfrac{6}{6} = -1$, so $y - 2 = -(x - 2)$, i.e. $y = -x + 4$.

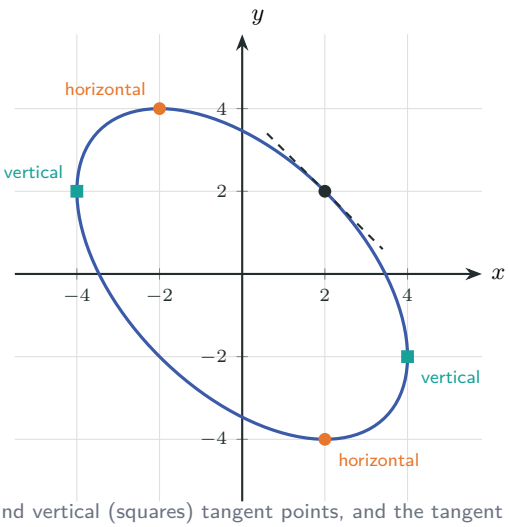


Figure 5.20. The curve $x^2 + xy + y^2 = 12$ with its horizontal (circles) and vertical (squares) tangent points, and the tangent line at $(2, 2)$.

Example 2 — Concavity from implicit differentiation

For the circle $x^2 + y^2 = 25$, find $\dfrac{d^2y}{dx^2}$ and use it to decide where the curve is concave up or down, and to classify the point $(0, 5)$.

Step 1. $2x + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

Step 2. Differentiate again (quotient rule, with $\dfrac{dy}{dx} = -\dfrac{x}{y}$):

$$\frac{d^2y}{dx^2} = -\frac{y - x\dfrac{dy}{dx}}{y^2} = -\frac{y + \dfrac{x^2}{y}}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{25}{y^3}.$$

Step 3. If $y > 0$ (upper half) then $\dfrac{d^2y}{dx^2} < 0$: concave down; if $y < 0$ (lower half), $\dfrac{d^2y}{dx^2} > 0$: concave up. For example, at $(3, 4)$ it is $-\dfrac{25}{64}$, and at $(3, -4)$ it is $\dfrac{25}{64}$.

Step 4. At $(0, 5)$: $\dfrac{dy}{dx} = 0$ and $\dfrac{d^2y}{dx^2} = -\dfrac{25}{125} = -\dfrac{1}{5} < 0$, so $(0, 5)$ is a relative maximum of y.

Example 3 — An elliptic curve, calculator allowed

Consider the curve $y^2 = x^3 - 3x + 3$. Find the points with a horizontal tangent, and the point with a vertical tangent.

Step 1. $2y\dfrac{dy}{dx} = 3x^2 - 3$, so $\dfrac{dy}{dx} = \dfrac{3(x^2 - 1)}{2y}$.

Step 2 (horizontal). Numerator 0: $x = \pm 1$. $x = 1$: $y^2 = 1 - 3 + 3 = 1$, $y = \pm 1$. $x = -1$: $y^2 = -1 + 3 + 3 = 5$, $y = \pm\sqrt{5} \approx \pm 2.236$. Four points: $(1, \pm 1)$, $(-1, \pm\sqrt{5})$.

Step 3 (vertical). Denominator 0: $y = 0$, so $x^3 - 3x + 3 = 0$. A calculator gives the only real root $x \approx -2.104$. The point $(-2.104, 0)$ has a vertical tangent (numerator $3(x^2 - 1) \neq 0$ there).

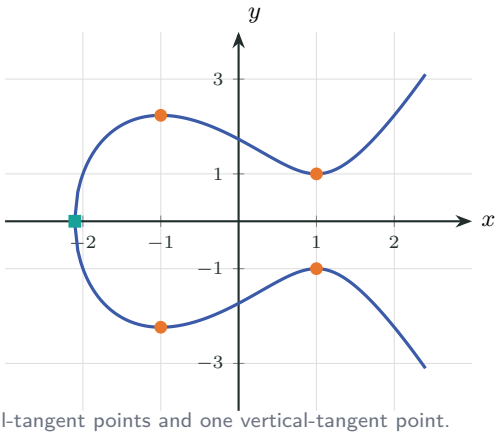


Figure 5.21. The curve $y^2 = x^3 - 3x + 3$: four horizontal-tangent points and one vertical-tangent point.

Common errors

- **Not checking that the point lies on the curve.** Setting the numerator to zero gives a condition on x and y; you must substitute into the original equation to find the actual points.
- **Second derivative slips.** Differentiating $\dfrac{dy}{dx}$ makes every y produce another factor $\dfrac{dy}{dx}$; substitute the first-derivative formula before you decide a sign.

AP Exam Tip

An implicit curve often has *several* y-values for one x, so evaluate $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ with both coordinates of the point. "At what points is the tangent horizontal?" wants *ordered pairs*, found from $N = 0$ *and* the equation.

Check yourself

1. For the circle $x^2 + y^2 - 4x + 2y = 11$, find the points with horizontal and vertical tangents.
2. For $xy = 4$, find $\dfrac{d^2y}{dx^2}$ and evaluate it at $(2, 2)$.
3. For $x^2 - xy + y^2 = 3$, find $\dfrac{dy}{dx}$ and the points with a horizontal tangent.

Unit 5 Practice: Analytical Applications of Differentiation

Part A Multiple choice: no calculator

14 questions

1 Let $f(x) = 2x^3 - 9x^2 + 12x$. On which of the following intervals is f increasing?

(A) $(-\infty, 1) \cup (2, \infty)$ **(B)** $(1, 2)$

(C) $(-\infty, 2)$ **(D)** $(1, \infty)$

2 The Mean Value Theorem is applied to $f(x) = x^3 - x$ on $[0, 2]$. Which of the following is the value of c guaranteed by the theorem?

(A) $\frac{4}{3}$ **(B)** $\frac{2}{\sqrt{3}}$

(C) 1 **(D)** $\sqrt{\frac{7}{3}}$

3 The function f is continuous for all x, and $f'(x) = (x+3)(x-1)^2(x-2)$. At which value(s) of x does f have a relative minimum?

(A) $x = -3$ only **(B)** $x = 1$ only

(C) $x = 2$ only **(D)** $x = 1$ and $x = 2$

4 The function f is twice differentiable with $f'(3) = 0$ and $f''(3) = -4$. Which of the following must be true?

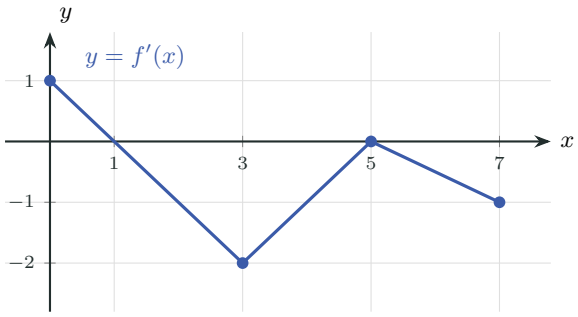
(A) f has a relative minimum at $x = 3$. **(B)** f has a point of inflection at $x = 3$.

(C) The Second Derivative Test is inconclusive at $x = 3$. **(D)** f has a relative maximum at $x = 3$.

5 The second derivative of a function f is $f''(x) = x(x-2)^2(x+1)$. At which values of x does the graph of f have a point of inflection?

(A) $x = 0$ only **(B)** $x = 2$ only

(C) $x = -1$ and $x = 0$ **(D)** $x = -1$, $x = 0$ and $x = 2$



Graph for Questions 6 and 7: f', the derivative of f, on $0 \le x \le 7$ (line segments).

6 On which of the following intervals is the graph of f concave up?

(A) $(3, 5)$ **(B)** $(1, 5)$

(C) $(0, 3)$ **(D)** $(0, 1)$

7 At $x = 5$, the graph of f has

(A) a relative maximum **(B)** a relative minimum

(C) neither a relative extremum nor a point of inflection **(D)** a point of inflection, but no relative extremum

8 What is the absolute minimum value of $f(x) = x^3 - 3x^2$ on the closed interval $[-1, 4]$?

(A) -2 **(B)** -4
(C) 0 **(D)** 16

9 If $x \geq 0$, $y \geq 0$ and $x + y = 12$, what is the maximum value of xy^2?

(A) 216 **(B)** 128
(C) 256 **(D)** 0

10 The curve $x^2 + xy + y^2 = 12$ contains the point $(2, -4)$. At which of the following points does the curve have a horizontal tangent?

(A) $(2, -4)$ **(B)** $(4, -2)$
(C) $(2, 2)$ **(D)** $(0, 2\sqrt{3})$

11 The function f is differentiable with $f(1) = -2$ and $f(5) = 10$. Which of the following must be true?

(A) $f'(3) = 3$ **(B)** $f'(x) = 3$ for all x in $(1, 5)$
(C) $f'(c) = 12$ for some c in $(1, 5)$ **(D)** $f'(c) = 3$ for some c in $(1, 5)$

12 For a function f, $f'(x) > 0$ and $f''(x) < 0$ for all x in $[0, 3]$. Which of the following could be a table of values of f?

(A)

x	0	1	2	3
f	2	5	9	14

(B)

x	0	1	2	3
f	2	6	9	11

(C)

x	0	1	2	3
f	11	9	6	2

(D)

x	0	1	2	3
f	11	5	2	1

13 The graph of $f(x) = ax^3 + bx^2$ has a point of inflection at $(1, 4)$. What is the value of b?

(A) 6 **(B)** -2
(C) 4 **(D)** 12

14 Which of the following guarantees that a function f has a relative maximum at $x = c$?

(A) $f'(c) = 0$
(B) $f'(c) = 0$ and $f''(c) > 0$
(C) f is continuous at c, $f'(x) > 0$ for x slightly less than c, and $f'(x) < 0$ for x slightly greater than c
(D) $f''(c) = 0$ and f'' changes sign at c

Part B Multiple choice: graphing calculator allowed

7 questions

15 What is the x-coordinate of the relative maximum of $f(x) = x^4 - 3x^2 + x$?

(A) -1.301 **(B)** 0.170
(C) 1.131 **(D)** 0.084

16 The Mean Value Theorem is applied to $f(x) = e^{x/2}$ on $[0, 4]$. What is the value of c guaranteed by the theorem?

(A) 1.597 **(B)** 2.000
(C) 3.195 **(D)** 2.323

17 The derivative of g is $g'(x) = \ln(x^2 + 1) - 1$. On which of the following is g increasing?

(A) $x > 1.311$ only **(B)** $-1.311 < x < 1.311$
(C) $x < -1.311$ or $x > 1.311$ **(D)** $x > 1.718$

18 What is the maximum value of $f(x) = x \sin x$ on the closed interval $[0, \pi]$?

(A) 1.820 **(B)** 2.029
(C) 1.571 **(D)** 0

19 What is the x-coordinate of the point on the graph of $y = e^x$ that is closest to the origin?

(A) 0.653 **(B)** -0.426
(C) 0.780 **(D)** 0

20 What is the absolute minimum value of $f(x) = e^x - x^3$ on the closed interval $[-1, 4]$?

(A) -9.402 **(B)** 0.729
(C) 1.731 **(D)** -10.216

21 A temperature $T(t)$, in °C, is a differentiable function of time t in minutes. Values are given in the table. Which of the following must be true?

t	0	3	7	12
$T(t)$	20	41	62	80

(A) $T'(c) = 5$ for some c in $(0, 12)$
(B) $T'(5) = 5.25$
(C) $T'(t) > 0$ for all t in $(0, 12)$
(D) $T'(c) = 60$ for some c in $(0, 12)$

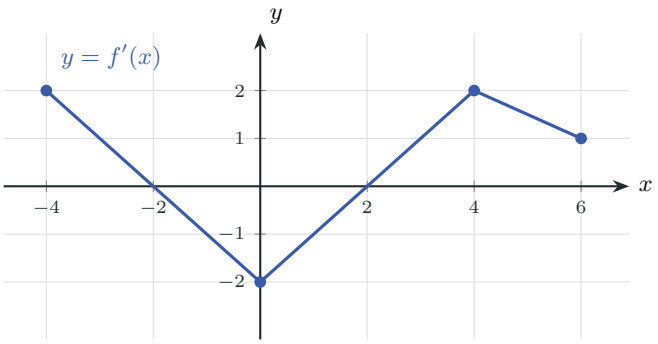
Part C Free response: two questions, 9 points each

Show the work that leads to your answers

22 **[9 points]** *(A graphing calculator is allowed.)* After an injection, the concentration of a medicine in a patient's blood is modelled by $C(t) = 6te^{-t/3}$ milligrams per litre, for $0 \le t \le 10$, where t is measured in hours.

(a) Find $C'(5)$. Using correct units, explain the meaning of $C'(5)$ in the context of the problem. Is the concentration increasing or decreasing at $t = 5$? Give a reason for your answer.
(b) Find the time t at which the concentration is greatest. Justify your answer.
(c) Find $C''(t)$. On what interval is the graph of C concave down? Give a reason for your answer.
(d) At what time t, for $0 \le t \le 10$, is the concentration decreasing most rapidly? Justify your answer.

23 **[9 points]** *(No calculator is allowed for this question.)* The function f is continuous on $[-4, 6]$, and $f(-4) = 1$. The graph of f', the derivative of f, consists of line segments and is shown below. It is known that $f(-2) = 3$, $f(2) = -1$ and $f(6) = 4$.



(a) Find the x-coordinate of each relative extremum of f on $(-4, 6)$. Classify each as a relative maximum or a relative minimum, and justify your answers.
(b) Find the x-coordinate of each point of inflection of the graph of f. Give a reason for your answer.
(c) Find the absolute minimum value and the absolute maximum value of f on $[-4, 6]$. Justify your answers.
(d) Write an equation of the line tangent to the graph of f at $x = -4$. Use this line to approximate $f(-3)$. Is this approximation greater than or less than $f(-3)$? Give a reason for your answer.

AP Calculus AB – Unit 6 Exam weighting: 17–20% of the AB exam score

Integration and Accumulation of Change

In this unit

Unit 5 used derivatives to analyse a function. This unit runs the process backwards and answers a different kind of question: if you know *how fast* a quantity is changing, how much has it changed? The answer is an *accumulation*, and the mathematics of accumulation is the definite integral. You will approximate accumulations with Riemann sums, define the integral as their limit, and then meet the Fundamental Theorem of Calculus, which connects integration to differentiation and lets you evaluate integrals exactly with antiderivatives. Accumulation functions, whose derivative is the integrand, appear in almost every Free Response section, and the substitution method extends your antidifferentiation skills to composite integrands.

Topics.

6.1 Exploring Accumulations of Change 6.2 Approximating Areas with Riemann Sums
6.3 Riemann Sums, Summation Notation, and Definite Integral Notation
6.4 The Fundamental Theorem of Calculus and Accumulation Functions
6.5 Interpreting the Behavior of Accumulation Functions Involving Area
6.6 Applying Properties of Definite Integrals 6.7 The Fundamental Theorem of Calculus and Definite Integrals
6.8 Finding Antiderivatives and Indefinite Integrals: Basic Rules and Notation
6.9 Integrating Using Substitution 6.10 Integrating Functions Using Long Division and Completing the Square
6.14 Selecting Techniques for Antidifferentiation

Topics 6.11 (integration by parts), 6.12 (partial fractions) and 6.13 (improper integrals) are *BC only* and are not covered.

Big ideas. (1) A rate of change, accumulated over an interval, gives a net change; geometrically it is a signed area. (2) The definite integral is the limit of Riemann sums. (3) If $g(x) = \int_a^x f(t)\,dt$, then $g'(x) = f(x)$: differentiation undoes accumulation. (4) $\int_a^b f(x)\,dx = F(b) - F(a)$ whenever $F' = f$, so antiderivatives evaluate integrals. (5) Substitution is the chain rule run backwards.

Topic 6.1 Exploring Accumulations of Change

From a rate to a total

If a car travels at a constant 60 km/h for 3 hours, it covers $60 \times 3 = 180$ km. The product of a rate and a time is a *total change*, and on a graph of speed against time it is the area of a rectangle. Calculus asks what to do when the rate is *not* constant. The answer you will build in this unit is that the same idea survives: the total change is still the area between the graph of the rate and the horizontal axis, but now the region may be a triangle, a trapezoid, a curved shape or a combination of these. A total obtained in this way is called an *accumulation of change*.

Three words need to be kept apart. A *rate of change* $r(t)$ tells you how fast a quantity $Q(t)$ is changing at the instant t; it is the derivative $Q'(t)$. The *net change* of Q from $t = a$ to $t = b$ is $Q(b) - Q(a)$. The *accumulation* of the rate over $[a, b]$ is the signed area between the graph of r and the t-axis, and the central fact of this unit is that the accumulation *equals* the net change. When $r(t) > 0$ the quantity is growing and the area counts as positive; when $r(t) < 0$ the quantity is shrinking and the area counts as negative. For this reason the answer is called a *signed* or *net* area.

Accumulation of change

If $r(t) = Q'(t)$ is the rate of change of Q, then for $a \le b$

$$Q(b) = Q(a) + \underbrace{\int_a^b r(t)\,dt}_{\text{net change of } Q \text{ on } [a,b]} \qquad \text{where } \int_a^b r(t)\,dt = (\text{area above the axis}) - (\text{area below the axis}).$$

Units. The units of the accumulation are (units of r)$\times$(units of t): litres per minute $\times$ minutes $=$ litres. **Net vs total.** Net change adds signed areas; the *total amount* that has entered (or the distance travelled) adds the *absolute* areas.

Areas you need without a calculator: rectangle bh; triangle $\frac{1}{2}bh$; trapezoid $\frac{1}{2}(b_1 + b_2)h$; circle of radius ρ: $\pi\rho^2$,

so a semicircle is $\frac{1}{2}\pi\rho^2$.

Reading a rate graph

On the exam, accumulation questions are almost always posed with a graph or a table of the rate. The routine is the same each time: (i) split the interval where the graph changes shape, (ii) find each area with a geometry formula, (iii) attach the sign, (iv) add, and (v) *add the starting value* if the question asks for the amount rather than the change.

Example 1 Water flowing into a tank

Water flows into a tank at a rate $R(t)$ litres per minute, shown in Figure 6.1, for $0 \le t \le 10$ minutes. The tank holds 20 litres at $t = 0$. Find the number of litres in the tank at $t = 10$.

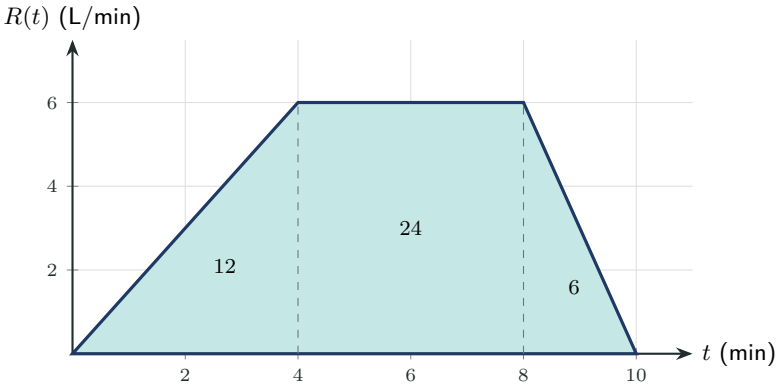


Figure 6.1. The rate $R(t)$; the three regions have areas 12, 24 and 6.

Step 1. The accumulated amount is the area under the graph: triangle $\frac{1}{2}(4)(6) = 12$, rectangle $(8-4)(6) = 24$, triangle $\frac{1}{2}(2)(6) = 6$. Total $= 42$ litres.
Step 2. $V(10) = V(0) + \int_0^{10} R(t)\,dt = 20 + 42 = 62$ litres.

Example 2 Net change versus total distance

A particle moves along a line with velocity $v(t) = 4 - 1.5t$ metres per second for $0 \le t \le 4$. Find (a) the particle's displacement over $[0, 4]$ and (b) the total distance travelled.

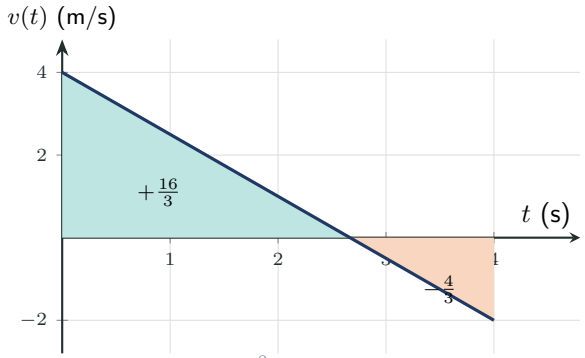


Figure 6.2. The velocity is zero at $t = \frac{8}{3}$; the region below the axis has negative area.

(a) The graph crosses the axis where $4 - 1.5t = 0$, i.e. $t = \frac{8}{3}$. Above the axis: $\frac{1}{2} \cdot \frac{8}{3} \cdot 4 = \frac{16}{3}$. Below the axis: $\frac{1}{2}\left(4 - \frac{8}{3}\right)(2) = \frac{4}{3}$, since $v(4) = -2$. Displacement $= \frac{16}{3} - \frac{4}{3} = 4$ m.
(b) The distance adds absolute areas: $\frac{16}{3} + \frac{4}{3} = \frac{20}{3}$ m.

Example 3 Interpreting an accumulation

People enter a stadium at a rate $E(t)$ people per minute, where t is the number of minutes after the gates open. Explain the meaning of $\displaystyle\int_{10}^{30} E(t)\,dt = 3600$.

The integrand has units of people per minute and dt has units of minutes, so the integral is a number of people. It

says: *between* 10 *and* 30 *minutes after the gates open,* 3600 *people entered the stadium.* Because E is a rate of entering, the number is the net change in the count of people inside over those 20 minutes (people leaving would have to be subtracted).

Common errors

Counting negative area as positive when asked for net change. A region below the axis *subtracts*. The correct move for displacement is above − below; use the sum of the absolute areas only when asked for the *total distance*.
Forgetting the initial value. The integral gives the *change*. The amount at time b is $Q(a) + \int_a^b r(t)\,dt$.
Dropping units. An interpretation on the exam must include units: "3600 people", not "3600".

AP Exam Tip

For "explain the meaning of $\int_a^b r(t)\,dt$" write a full sentence that names (1) the quantity, (2) the *interval* with units, and (3) the units of the answer, for example: "the number of litres that entered the tank from $t = 0$ to $t = 10$ minutes."

Check yourself 6.1

1. Water enters a tank at $r(t) = 2t$ litres per minute for $0 \le t \le 5$. Use a triangle to find the amount that enters.
2. A particle has velocity $v(t) = 3 - t$ m/s for $0 \le t \le 5$. Find the displacement and the total distance.
3. The rate at which a bakery sells loaves is $S(t)$ loaves per hour, t hours after opening. What does $\int_2^6 S(t)\,dt$ represent?

Topic 6.2 Approximating Areas with Riemann Sums

When the graph is not a polygon

Most rate functions are curved, and then no geometry formula gives the area. The remedy, due in essence to Archimedes, is to cut the region into thin vertical strips, replace each strip by a rectangle (or trapezoid) whose area you *can* compute, and add. A sum of this kind is a *Riemann sum*. The approximation improves as the strips narrow, a fact that Topic 6.3 turns into a definition.

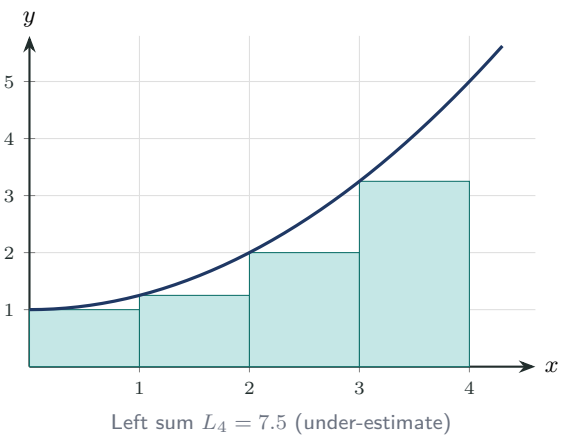
Partition $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ with widths $\Delta x_i = x_i - x_{i-1}$. On each subinterval choose a *sample point*, and build a rectangle whose height is the value of f there. The choice defines the type of sum: the left endpoint (LRAM), the right endpoint (RRAM), or the midpoint (MRAM). A trapezoidal sum instead joins the two endpoint heights with a slanted top.

Riemann sums on $[a, b]$ with subinterval widths Δx_i

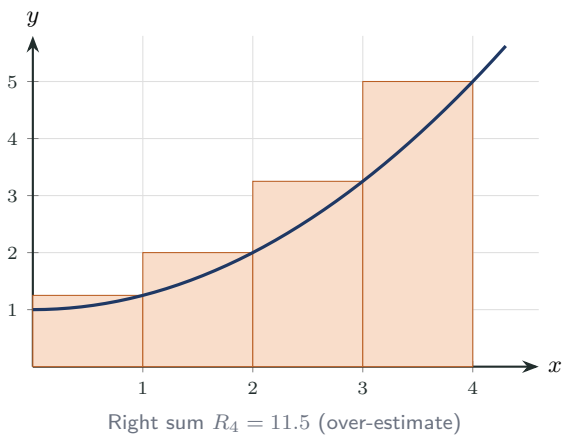
$$\text{Left: } L_n = \sum_{i=1}^{n} f(x_{i-1})\,\Delta x_i \qquad \text{Right: } R_n = \sum_{i=1}^{n} f(x_i)\,\Delta x_i$$

$$\text{Midpoint: } M_n = \sum_{i=1}^{n} f\!\left(\tfrac{x_{i-1}+x_i}{2}\right)\Delta x_i \qquad \text{Trapezoidal: } T_n = \sum_{i=1}^{n} \frac{f(x_{i-1}) + f(x_i)}{2}\,\Delta x_i$$

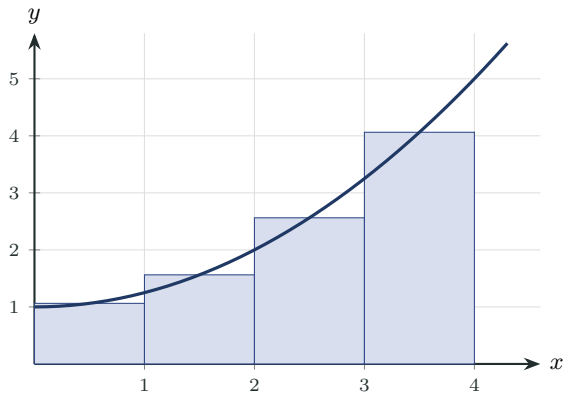
In every case $T_n = \frac{1}{2}(L_n + R_n)$. For **unequal** widths (tables), each term uses its own Δx_i; equal widths are $\Delta x = \frac{b-a}{n}$.



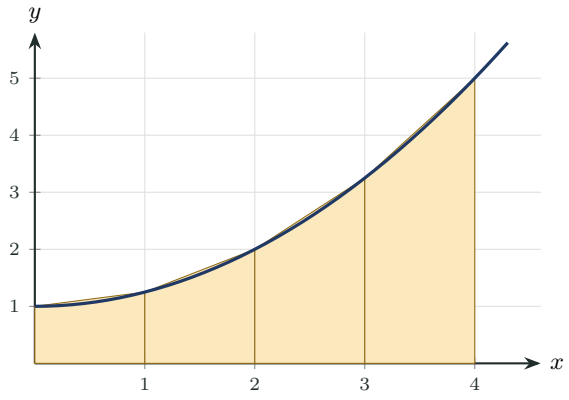
Left sum $L_4 = 7.5$ (under-estimate)



Right sum $R_4 = 11.5$ (over-estimate)



Midpoint sum $M_4 = 9.25$



Trapezoidal sum $T_4 = 9.5$ (over-estimate)

Figure 6.3. Four approximations to the area under $f(x) = 1 + \frac{1}{4}x^2$ on $[0, 4]$ with $n = 4$. The exact area is $\frac{28}{3} \approx 9.333$.

Over- or under-estimate?

Because the true value is rarely known, the exam asks you to decide whether an approximation is too large or too small. The decision depends only on whether f is increasing or decreasing (for left and right sums) and on the *concavity* (for trapezoids and midpoints). Draw a quick sketch each time rather than trusting memory.

Over-/under-estimate rules for $\int_a^b f(x)\,dx$

	Left	Right	Trapezoidal	Midpoint
f increasing	under	over	—	—
f decreasing	over	under	—	—
f concave up	—	—	over	under
f concave down	—	—	under	over

(Left and right rules need f monotonic on $[a, b]$; trapezoid and midpoint rules need a fixed concavity.)

Example 1 Four sums for the same curve

For $f(x) = 1 + \frac{1}{4}x^2$ on $[0, 4]$ with $n = 4$ (Figure 6.3), the values are $f(0) = 1,\ f(1) = 1.25,\ f(2) = 2,\ f(3) = 3.25,\ f(4) = 5$ and $\Delta x = 1$.
$L_4 = 1 + 1.25 + 2 + 3.25 = 7.5$. $R_4 = 1.25 + 2 + 3.25 + 5 = 11.5$. $T_4 = \frac{1}{2}(7.5 + 11.5) = 9.5$.
Midpoints $0.5, 1.5, 2.5, 3.5$ give $f = 1.0625,\ 1.5625,\ 2.5625,\ 4.0625$, so $M_4 = 9.25$.
Since f is increasing, $L_4 < I < R_4$. Since f is concave up, $M_4 < I < T_4$. All of these agree with $I = \frac{28}{3} \approx 9.33$.

Example 2 A table with unequal subintervals

The rate $r(t)$, in litres per hour, at which a pump delivers fuel is measured at the times below. Approximate $\int_0^{10} r(t)\,dt$ with a left sum, a right sum and a trapezoidal sum using the four subintervals.

t (h)	0	2	5	9	10
$r(t)$ (L/h)	3	5	8	6	4

The widths are $2,\ 3,\ 4,\ 1$.
Left: $3(2) + 5(3) + 8(4) + 6(1) = 6 + 15 + 32 + 6 = 59$.
Right: $5(2) + 8(3) + 6(4) + 4(1) = 10 + 24 + 24 + 4 = 62$.
Trapezoidal: $\frac{3+5}{2}(2) + \frac{5+8}{2}(3) + \frac{8+6}{2}(4) + \frac{6+4}{2}(1) = 8 + 19.5 + 28 + 5 = 60.5$.
The results are in litres. Note $T = \frac{1}{2}(L + R) = \frac{1}{2}(59 + 62) = 60.5$, a useful check.

Example 3 Deciding over or under

The function f is positive, decreasing and concave up on $[1,5]$. Let L_4, R_4, M_4, T_4 be the four-subinterval sums and $I = \int_1^5 f(x)\,dx$. Order them.

Decreasing: the left rectangles stick out above the curve, so $L_4 > I$; the right rectangles lie below it, so $R_4 < I$. Concave up: chords lie above the curve, so $T_4 > I$; the midpoint rectangle's flat top lies below the curve, so $M_4 < I$. Also $T_4 = \frac{1}{2}(L_4 + R_4) < L_4$. Combining: $R_4 < I < T_4 < L_4$ and $M_4 < I$.

Common errors

Using one Δx for a table with unequal spacing. Each term needs the width of *its own* subinterval, $x_i - x_{i-1}$.
Mixing up which endpoint a "left" sum uses. A left sum uses $f(x_0), \ldots, f(x_{n-1})$ and never the last value; a right sum never uses $f(x_0)$.
Averaging the heights for a trapezoid but forgetting the factor $\frac{1}{2}$. The trapezoid height term is $\frac{f(x_{i-1})+f(x_i)}{2}$.

AP Exam Tip

The exam rewards the *reason*: "the left Riemann sum is an under-estimate because f is increasing on $[a,b]$." If f is given only by a table, you may justify monotonicity from f' or from the context (a positive rate means the amount is increasing); you may not assume concavity from a table.

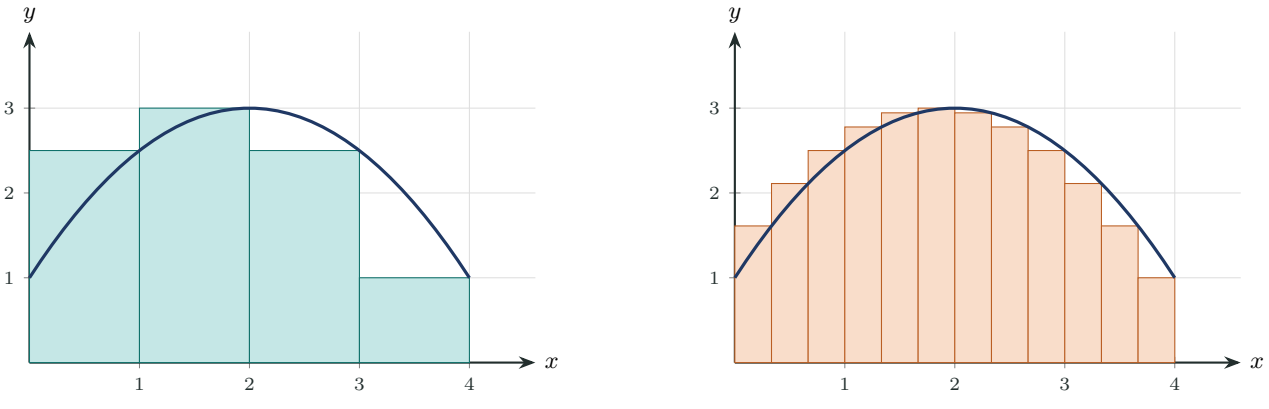
Check yourself 6.2

1. For $f(x) = \sqrt{x}$ on $[0,4]$ with four equal subintervals, find L_4 and R_4 (three decimals). State which is an under-estimate and why.
2. f takes the values $f(1) = 2,\ f(3) = 5,\ f(4) = 6,\ f(8) = 3$. Find the trapezoidal approximation to $\int_1^8 f(x)\,dx$.
3. If f is decreasing on $[a,b]$, is L_n an over- or an under-estimate of $\int_a^b f$? Is R_n?

Topic 6.3 Riemann Sums, Summation Notation, and Definite Integral Notation

From sums to the integral

Figure 6.3 showed that the sums differ by quite a lot when only four strips are used. Figure 6.4 repeats a right-hand sum for the function $f(x) = 1 + \frac{1}{2}x(4-x)$ with 4 strips and then 12. The rectangles hug the curve far more closely when they are thin, and the sum steadies down towards a single number. That number is defined to be the *definite integral*.



$n = 4$: $R_4 = 9$ $\qquad$ $n = 12$: $R_{12} \approx 9.296$

Figure 6.4. Right Riemann sums for $f(x) = 1 + \frac{1}{2}x(4-x)$ on $[0,4]$. As n grows the sums approach $\frac{28}{3} \approx 9.333$.

With n equal subintervals, $\Delta x = \dfrac{b-a}{n}$ and the endpoints are $x_i = a + i\,\Delta x$. The right sum is $\displaystyle\sum_{i=1}^{n} f(x_i)\Delta x$. The symbol $\sum$ is a capital Greek sigma; the index i runs from the lower number to the upper number and each value of i produces one term. The integral symbol $\int$ is an elongated S, for "sum", and dx remembers the width Δx of a strip as it shrinks.

Definite integral as a limit of Riemann sums

If f is continuous on $[a,b]$,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x, \qquad \Delta x = \frac{b-a}{n},$$

for any choice of sample points x_i^* in the i-th subinterval (left, right, midpoint …). The number $\int_a^b f(x)\,dx$ is the *signed area* between the graph and the x-axis. The symbols a and b are the *limits of integration*, $f(x)$ the *integrand*.

Reading a limit of sums as an integral. In $\lim_{n\to\infty}\sum_{i=1}^{n} g\big(a+i\,\frac{b-a}{n}\big)\,\frac{b-a}{n}$: the factor $\frac{b-a}{n}$ fixes $b-a$; the value added to a shows where the sample point begins, so a is read off the constant term, and x replaces $a+i\frac{b-a}{n}$ in g.

$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$: the variable of integration is a *dummy* variable.

Example 1 A limit of sums as an integral

Write $\lim_{n\to\infty}\sum_{i=1}^{n}\left(2+\frac{3i}{n}\right)^2\frac{3}{n}$ as a definite integral and evaluate it.

Step 1. $\Delta x = \frac{3}{n}$, so $b-a=3$. The sample points are $x_i = 2+i\cdot\frac{3}{n}$, so $a=2$ and $b=5$.

Step 2. The function is $f(x)=x^2$. The limit is $\displaystyle\int_2^5 x^2\,dx = \left[\frac{x^3}{3}\right]_2^5 = \frac{125-8}{3} = 39.$

Example 2 Writing a Riemann sum in sigma notation

Write the right Riemann sum with $n=4$ equal subintervals for $\displaystyle\int_1^3 \ln x\,dx$ and evaluate it.

$\Delta x = \frac{3-1}{4} = \frac{1}{2}$ and $x_i = 1+\frac{i}{2}$. So

$$R_4 = \sum_{i=1}^{4}\ln\big(1+\tfrac{i}{2}\big)\cdot\tfrac{1}{2} = \tfrac{1}{2}\big[\ln 1.5+\ln 2+\ln 2.5+\ln 3\big] \approx 1.557.$$

Since $\ln x$ is increasing, R_4 is an over-estimate; the exact value is $3\ln 3-2\approx 1.296$.

Example 3 Evaluating by geometry

Evaluate (a) $\displaystyle\int_{-3}^{3}\sqrt{9-x^2}\,dx$ (b) $\displaystyle\int_0^6 |x-3|\,dx.$

(a) The graph of $y=\sqrt{9-x^2}$ is the upper half of the circle $x^2+y^2=9$, so the integral is the area of a semicircle of radius 3: $\frac{1}{2}\pi(3)^2 = \frac{9\pi}{2}$.
(b) The graph of $y=|x-3|$ is a V with vertex $(3,0)$. Two triangles, each with base 3 and height 3, give $2\cdot\frac{1}{2}(3)(3)=9$.

Common errors

Reading Δx from the wrong place. In $\sum\left(2+\frac{3i}{n}\right)^2\frac{3}{n}$ the factor $\frac{3}{n}$ is Δx; the 3 in the *argument* is the same 3, which is $b-a$, and the lower limit is the constant 2—not 0 and not 5.
Evaluating the sum for a specific n and calling it the integral. A Riemann sum with finite n is only an approximation.

AP Exam Tip

On the exam you are asked to "write" an expression ("Write a left Riemann sum …"). Provide the sum with all of its terms or in sigma notation, and then give the numerical value if asked. Do not stop at the sum if the question says "approximate".

Check yourself 6.3

1. Express $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{\frac{i}{n}}\cdot\frac{1}{n}$ as a definite integral and evaluate it.
2. Use geometry to evaluate $\int_{-2}^{2}\sqrt{4-x^2}\,dx$.
3. Write the right Riemann sum with $n=4$ for $\int_{0}^{2}e^{x}\,dx$ and evaluate it to three decimals.

Topic 6.4 The Fundamental Theorem of Calculus and Accumulation Functions

A function defined by an integral

So far $\int_a^b f$ has been a number. Now let the upper limit *vary*. For a fixed number a define

$$g(x)=\int_a^x f(t)\,dt.$$

This is an *accumulation function*: $g(x)$ is the signed area between the graph of f and the t-axis from $t=a$ up to $t=x$. The letter t is a dummy variable, used so that it is not confused with the upper limit x. Two facts are immediate: $g(a)=0$ (no accumulation over an interval of zero width) and the *sign of* f controls whether g grows or shrinks.

Why should $g'(x)$ equal $f(x)$? Increase x by a small h. The accumulated area grows by a thin strip of width h and height almost $f(x)$, so $g(x+h)-g(x)\approx f(x)\,h$ and $\dfrac{g(x+h)-g(x)}{h}\approx f(x)$. In the limit, $g'(x)=f(x)$. This is the first form of the **Fundamental Theorem of Calculus**: the rate of change of an accumulated amount is the rate that is being accumulated, just as in Topic 6.1.

Fundamental Theorem of Calculus (accumulation form)

If f is continuous on an interval containing a, then $g(x)=\int_a^x f(t)\,dt$ is differentiable and

$$\frac{d}{dx}\int_a^x f(t)\,dt=f(x).$$

With the chain rule: if $u(x)$ is differentiable, $\dfrac{d}{dx}\int_a^{u(x)} f(t)\,dt=f\big(u(x)\big)\cdot u'(x)$.

Variable lower limit: $\dfrac{d}{dx}\int_x^b f(t)\,dt=-f(x)$ (swapping the limits changes the sign).

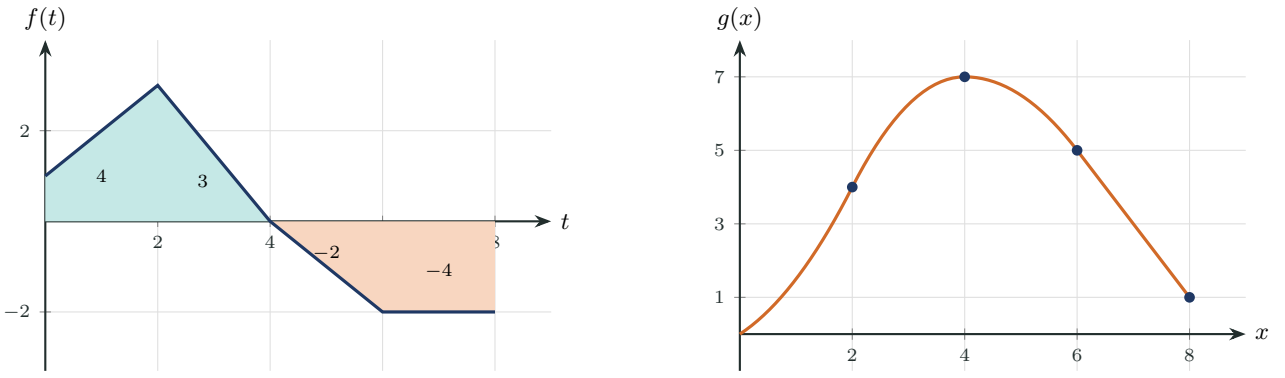


Figure 6.5. Left: the graph of f (line segments through $(0,1),(2,3),(4,0),(6,-2),(8,-2)$) with signed areas. Right: $g(x)=\int_0^x f(t)\,dt$, built from those areas: $g(2)=4,\ g(4)=7,\ g(6)=5,\ g(8)=1$.

Example 1 Differentiating an accumulation function

Find the derivative. (a) $g(x)=\int_1^x\sqrt{t^2+3}\,dt$ (b) $h(x)=\int_2^{3x}\dfrac{dt}{t^2+1}$ (c) $k(x)=\int_x^5 e^{t^2}\,dt$ (d) $m(x)=$

$$\int_0^{x^2} \cos t\, dt$$

(a) $g'(x) = \sqrt{x^2+3}$. **(b)** $h'(x) = \dfrac{1}{(3x)^2+1}\cdot 3 = \dfrac{3}{9x^2+1}$.

(c) $k(x) = -\displaystyle\int_5^x e^{t^2}\,dt$, so $k'(x) = -e^{x^2}$. **(d)** $m'(x) = \cos(x^2)\cdot 2x$.

No integration was done: the theorem gives the derivative directly.

Example 2 Using the graph of f

Let $g(x) = \displaystyle\int_0^x f(t)\,dt$ where f is the graph in Figure 6.5. Find $g(2)$, $g(4)$, $g(6)$, $g'(3)$ and $g''(3)$.

Step 1: values of g are signed areas. $g(2) = \frac{1+3}{2}(2) = 4$ (trapezoid). $g(4) = g(2) + \frac{1}{2}(2)(3) = 7$ (triangle). $g(6) = g(4) - \frac{1}{2}(2)(2) = 5$ (triangle below the axis).
Step 2: derivatives. $g'(x) = f(x)$, so $g'(3) = f(3) = 1.5$ (the segment from $(2,3)$ to $(4,0)$ has slope -1.5, so $f(3) = 3 - 1.5 = 1.5$).
Step 3: $g''(x) = f'(x)$, so $g''(3) = -1.5$.

Example 3 An accumulation with a variable upper limit

With f as in Figure 6.5, let $F(x) = \displaystyle\int_0^{x^2} f(t)\,dt$. Find $F'(1)$ and $F'(2)$.

By the chain rule form, $F'(x) = f(x^2)\cdot 2x$.
$F'(1) = f(1)\cdot 2 = 2\cdot 2 = 4$ since $f(1) = 2$. $F'(2) = f(4)\cdot 4 = 0\cdot 4 = 0$.

Common errors

Forgetting the chain-rule factor. For $\frac{d}{dx}\int_0^{x^2}\cos t\,dt$ the answer is $2x\cos(x^2)$, not $\cos(x^2)$.
Putting x inside the integrand. Write $f(u(x))$, not an expression that still contains t. After differentiating, no t remains.
Ignoring the sign when x is the lower limit. $\frac{d}{dx}\int_x^b f = -f(x)$.

AP Exam Tip

Always cite the theorem: "By the Fundamental Theorem of Calculus, $g'(x) = f(x)$." A correct value with no reference to the theorem is often not awarded the justification point.

Check yourself 6.4

1. If $F(x) = \displaystyle\int_1^x \ln(t^2+1)\,dt$, find $F'(2)$.
2. Find $\dfrac{d}{dx}\displaystyle\int_0^{x^3}\sin(t^2)\,dt$.
3. With g as in Figure 6.5, find $g'(6)$ and $g''(3)$.

Topic 6.5 Interpreting the Behavior of Accumulation Functions Involving Area

Reading g from the graph of f

Because $g' = f$, everything you learned in Unit 5 about the first and second derivative of g transfers to the graph of f, which is the thing the exam gives you. Where f is positive, g is increasing; where f is negative, g is decreasing; g has a relative maximum where f changes from positive to negative; and because $g'' = f'$, the graph of g is concave up where f is *increasing* and concave down where f is *decreasing*. Points of inflection of g are therefore the places where f has a local extreme value (where f' changes sign), not where f is zero.

Behaviour of $g(x) = \int_a^x f(t)\,dt$

Facts about f	Conclusion about g
$f > 0$ $(f < 0)$ on an interval	g increasing (decreasing)
f changes $+$ to $-$ at c	g has a relative maximum at c
f changes $-$ to $+$ at c	g has a relative minimum at c
f increasing (decreasing)	g concave up (down)
f' changes sign at c	g has an inflection point at c

Absolute extremes of g on $[p, q]$: compute g at the critical points $(f = 0)$ and the endpoints, using signed areas, and compare. Changing the lower limit a to a' shifts g by the constant $\int_a^{a'} f$ and does not change where it increases or is concave.

Example 1 Extreme values and inflection points

Let $g(x) = \int_0^x f(t)\,dt$ for $0 \le x \le 8$, with f as in Figure 6.5. (a) On what interval is g increasing? (b) Where does g have a relative maximum? (c) Where is g concave down, and where does it have a point of inflection? (d) Find the absolute minimum and maximum of g on $[0, 8]$.

(a) $g'(x) = f(x) > 0$ for $0 \le x < 4$, so g is increasing on $[0, 4]$.
(b) $g' = f$ changes from positive to negative at $x = 4$, so g has a relative maximum at $x = 4$.
(c) $g''(x) = f'(x)$. f is increasing on $(0, 2)$ and decreasing on $(2, 6)$. Thus g is concave down on $(2, 6)$, and $g'' = f'$ changes from positive to negative at $x = 2$, so $(2, g(2))$ is a point of inflection. (At $x = 6$, f' goes from -1 to 0, with no sign change.)
(d) Candidates $x = 0, 4, 8$ (critical point $x = 4$ and endpoints): $g(0) = 0$, $g(4) = 7$, $g(8) = 1$. The absolute maximum is 7 at $x = 4$ and the absolute minimum is 0 at $x = 0$.

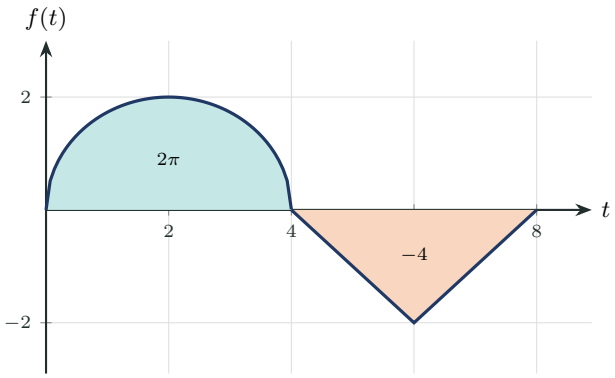


Figure 6.6. The graph of f: a semicircle of radius 2 on $[0, 4]$ and two line segments.

Example 2 A semicircle and line segments

Let $g(x) = \int_0^x f(t)\,dt$ with f as in Figure 6.6. (a) Find $g(4)$ and $g(8)$. (b) On which intervals is g decreasing? (c) Find the x-coordinates of the inflection points of g. (d) Let $h(x) = \int_4^x f(t)\,dt$. How does h compare with g?

(a) $g(4) = \frac{1}{2}\pi(2)^2 = 2\pi$. The triangle below the axis has area $\frac{1}{2}(4)(2) = 4$, so $g(8) = 2\pi - 4 \approx 2.283$.
(b) $f < 0$ on $(4, 8)$, so g is decreasing on $[4, 8]$.
(c) f increases on $(0, 2)$, decreases on $(2, 6)$ and increases on $(6, 8)$. So f' changes sign at $x = 2$ and $x = 6$: those are the inflection points.
(d) $h(x) = g(x) - g(4) = g(x) - 2\pi$ for all x: h is g shifted down by 2π. Same critical points and the same inflection points, but $h(4) = 0$.

Common errors

Confusing f with g. "g is increasing because f is increasing" is wrong: g increases where $f > 0$; f increasing makes g *concave up*.
Calling a zero of f an inflection point of g. Zeros of f give critical points of g. Inflection points occur where f has a local maximum or minimum.
Skipping the endpoints when asked for an *absolute* extreme on a closed interval.

AP Exam Tip

Model sentences: "g has a relative maximum at $x = 4$ because $g'(x) = f(x)$ changes from positive to negative at $x = 4$." "The graph of g has a point of inflection at $x = 2$ because $g''(x) = f'(x)$ changes from positive to negative there."

Check yourself 6.5

1. Let $g(x) = \displaystyle\int_1^x (t^2 - 4)\,dt$ for $x > 0$. Find the critical point of g, classify it, and find where g is concave up.
2. f is continuous with $f > 0$ on $(0, 3)$, $f < 0$ on $(3, 7)$, $f > 0$ on $(7, 9)$. For $g(x) = \int_0^x f(t)\,dt$, where does g have a relative maximum and a relative minimum?
3. In Figure 6.6, is g concave up or down at $x = 1$? Give a reason.

Topic 6.6 Applying Properties of Definite Integrals

Rules that save work

Definite integrals behave like signed areas, and several rules follow directly from that picture. Reversing the direction of integration reverses the sign. Integrating over zero width gives zero. An interval can be split at any point, so a complicated region can be cut into pieces whose areas you know. Constants factor out and sums split. If one function is never smaller than another on an interval, its integral is not smaller either. Finally, symmetry can make part of the work disappear: an even function has equal signed areas on each side of the y-axis, and an odd function has opposite ones.

Properties of definite integrals (f, g continuous)

$$\int_a^a f = 0, \qquad \int_b^a f = -\int_a^b f, \qquad \int_a^b f = \int_a^c f + \int_c^b f \quad \text{for any } c,$$

$$\int_a^b \big[k\,f(x) \pm m\,g(x)\big]dx = k\int_a^b f \pm m\int_a^b g, \qquad \text{if } f \ge g \text{ on } [a,b] \text{ then } \int_a^b f \ge \int_a^b g.$$

Symmetry on $[-a, a]$: if f is even, $\displaystyle\int_{-a}^a f = 2\int_0^a f$; if f is odd, $\displaystyle\int_{-a}^a f = 0$.

Constants: $\displaystyle\int_a^b k\,dx = k(b-a)$. **Not a rule:** $\int fg \neq \int f \cdot \int g$ and $\int \frac{f}{g} \neq \frac{\int f}{\int g}$.

Example 1 Using given values

Suppose $\displaystyle\int_0^3 f(x)\,dx = 5$, $\displaystyle\int_3^7 f(x)\,dx = -2$ and $\displaystyle\int_0^7 g(x)\,dx = 6$. Find (a) $\int_0^7 f$ (b) $\int_7^0 f$ (c) $\int_0^7 \big[2f(x) - g(x) + 1\big]dx$ (d) $\int_3^3 g$.

(a) $\int_0^7 f = \int_0^3 f + \int_3^7 f = 5 + (-2) = 3$. **(b)** $\int_7^0 f = -3$.
(c) $2(3) - 6 + \int_0^7 1\,dx = 6 - 6 + 7 = 7$. **(d)** 0.

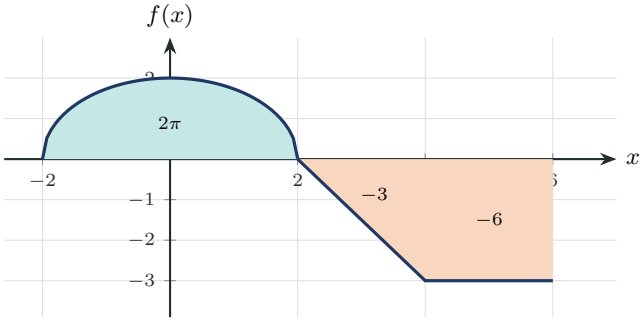


Figure 6.7. A semicircle of radius 2 and line segments through $(2,0),(4,-3),(6,-3)$.

Example 2 Using a graph

For f in Figure 6.7 find (a) $\int_{-2}^{6} f$ (b) $\int_{-2}^{6} |f(x)|\,dx$ (c) $\int_{-2}^{6} [f(x)+2]dx$ (d) $\int_{6}^{-2} f$.

Signed areas: semicircle $\frac{1}{2}\pi(2)^2 = 2\pi$; triangle $\frac{1}{2}(2)(3) = 3$ below the axis, so -3; rectangle $(2)(3) = 6$ below the axis, so -6.

(a) $2\pi - 3 - 6 = 2\pi - 9 \approx -2.717$. **(b)** Absolute values: $2\pi + 3 + 6 = 2\pi + 9$.

(c) $\int f + \int_{-2}^{6} 2\,dx = (2\pi - 9) + 2(8) = 2\pi + 7$. **(d)** $-(2\pi - 9) = 9 - 2\pi$.

Example 3 Symmetry

(a) f is even and $\int_0^3 f = 4$. Find $\int_{-3}^{3} (f(x)+2)dx$. (b) Find $\int_{-2}^{2} (x^3+5)\,dx$.

(a) $\int_{-3}^{3} f = 2(4) = 8$ and $\int_{-3}^{3} 2\,dx = 12$, so the answer is 20.

(b) x^3 is odd, so $\int_{-2}^{2} x^3 dx = 0$; $\int_{-2}^{2} 5\,dx = 20$. The answer is 20.

Common errors

Splitting a product. $\int_0^3 fg \neq \int_0^3 f \cdot \int_0^3 g$.

Forgetting that the property for sums requires the *same* limits. $\int_0^3 f + \int_0^5 g$ cannot be combined.

Misreading the sign after swapping the limits. $\int_b^a f = -\int_a^b f$, so a positive area read "backwards" is negative.

Applying the odd/even rule to a sum. $\int_{-a}^{a}(x^3+5)\,dx$ uses the odd rule only on the x^3 part.

AP Exam Tip

When a question gives $\int_a^b f$ for several intervals, write each interval on a number line first; the answer is then always a sum or difference of the given values, never a new integration.

Check yourself 6.6

1. If $\int_1^5 f = 8$ and $\int_1^3 f = 3$, find $\int_3^5 f$ and $\int_5^1 3f(x)\,dx$.
2. Evaluate $\displaystyle\int_{-3}^{3} \left(x^3 + \sqrt{9-x^2}\right) dx$.
3. Is $\displaystyle\int_0^1 x^2 dx \geq \int_0^1 x^3 dx$? Justify with the comparison property.

Topic 6.7 The Fundamental Theorem of Calculus and Definite Integrals

Evaluating integrals exactly

Topic 6.4 showed that differentiation undoes accumulation. The second form of the theorem turns that insight into a calculating tool. Suppose F is any antiderivative of f, that is $F' = f$. The accumulation function $g(x) = \int_a^x f(t)\,dt$ is also an antiderivative of f, and two antiderivatives of the same function differ by a constant. Since $g(a) = 0$, we get $g(x) = F(x) - F(a)$. Setting $x = b$ gives the evaluation theorem below. In words: to find the net change of a quantity over $[a,b]$, you only need the quantity itself at the two ends.

Fundamental Theorem of Calculus (evaluation form)

If f is continuous on $[a,b]$ and $F'(x)=f(x)$ on $[a,b]$, then

$$\int_a^b f(x)\,dx = F(b)-F(a) = \Big[F(x)\Big]_a^b.$$

Net change form: $\displaystyle\int_a^b Q'(t)\,dt = Q(b)-Q(a)$, i.e. $Q(b)=Q(a)+\int_a^b Q'(t)\,dt$.
The constant C is not needed: it cancels in $F(b)-F(a)$. **Continuity matters:** the theorem cannot be applied across a point where f is not defined (see the common errors).

Example 1 A definite integral by antiderivatives

Evaluate $\displaystyle\int_1^4 \left(3\sqrt{x}-\frac{2}{x^2}\right)dx.$

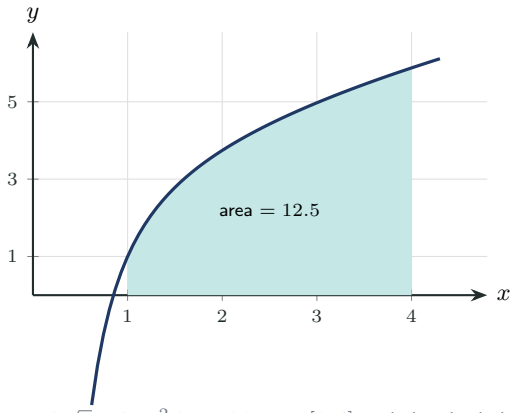


Figure 6.8. $y=3\sqrt{x}-2x^{-2}$ is positive on $[1,4]$ and the shaded area is 12.5.

Rewrite $3\sqrt{x}-\frac{2}{x^2}=3x^{1/2}-2x^{-2}$, so $F(x)=2x^{3/2}+\frac{2}{x}$. Then

$$\Big[2x^{3/2}+\tfrac{2}{x}\Big]_1^4 = \big(2\cdot 8+\tfrac{1}{2}\big)-(2+2) = 16.5-4 = 12.5.$$

Example 2 Trigonometric integrand

Evaluate $\displaystyle\int_0^{\pi/2}(3\cos x+2\sin x)\,dx.$

$F(x)=3\sin x-2\cos x$. $F\big(\tfrac{\pi}{2}\big)=3-0=3$ and $F(0)=0-2=-2$, so the value is $3-(-2)=5$.

Example 3 Net change

(a) f is differentiable with $f(1)=3$ and $f'(x)=2x-\dfrac{1}{x}$. Find $f(4)$. (b) A tank holds 50 litres at $t=0$. Water flows in at $R(t)=6e^{-t/2}$ litres per minute. Write an expression for the volume at $t=4$ and evaluate it exactly.

(a) $f(4)=f(1)+\displaystyle\int_1^4 \big(2x-\tfrac{1}{x}\big)\,dx = 3+\Big[x^2-\ln x\Big]_1^4 = 3+(16-\ln 4)-1 = 18-2\ln 2.$

(b) $V(4)=50+\displaystyle\int_0^4 6e^{-t/2}dt = 50+\Big[-12e^{-t/2}\Big]_0^4 = 50+12-12e^{-2} = 62-12e^{-2}$ litres (≈ 60.38).

Common errors

Using the theorem when the integrand is discontinuous. $\displaystyle\int_{-1}^{1}\frac{1}{x^2}dx=\Big[-\tfrac{1}{x}\Big]_{-1}^{1}=-2$ is *wrong*: $\frac{1}{x^2}$ is positive, so no integral over an interval can be negative, and the integrand is unbounded at $x=0$. The theorem needs f continuous on $[a,b]$.

Reversing the subtraction: $F(a) - F(b)$.
Adding $+C$ and then carrying it: it cancels; do not write "$+C$" in a definite integral.
Substituting into f, not F. You evaluate the *antiderivative* at the limits.

AP Exam Tip

Show the antiderivative and the substitution of limits, e.g. $\left[2x^{3/2} + \frac{2}{x}\right]_1^4 = (\cdots) - (\cdots)$. If a question says "Show the work that leads to your answer", a bare calculator value earns no credit on a no-calculator part.

Check yourself 6.7

1. Evaluate $\displaystyle\int_{-1}^{2} (x^2 - 2x)\,dx$.
2. Evaluate $\displaystyle\int_{0}^{\pi/3} \sec^2 x\,dx$.
3. A tank holds 10 litres at $t = 1$ and the water flows in at $3t^2$ litres per hour. How much is in the tank at $t = 3$?

Topic 6.8 Finding Antiderivatives and Indefinite Integrals: Basic Rules and Notation

Running differentiation backwards

An *antiderivative* of f is a function F with $F' = f$. The antiderivative is not unique: if F works then so does $F+C$ for any constant C, and the Mean Value Theorem shows that these are all the antiderivatives. The family of all antiderivatives is the *indefinite integral*, written $\displaystyle\int f(x)\,dx = F(x) + C$. The constant C must not be forgotten: in Figure 6.9 each value of C produces a vertical translate of the same curve, and an extra condition (an *initial value*) is needed to pick out one of them.

Basic antiderivatives (check each by differentiating)

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1) \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$
$$\int a^x\,dx = \frac{a^x}{\ln a} + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sin x\,dx = -\cos x + C$$
$$\int \sec^2 x\,dx = \tan x + C \qquad \int \csc^2 x\,dx = -\cot x + C \qquad \int \sec x \tan x\,dx = \sec x + C$$
$$\int \csc x \cot x\,dx = -\csc x + C \qquad \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \qquad \int \frac{dx}{1+x^2} = \arctan x + C$$

Linearity: $\displaystyle\int [kf \pm mg]\,dx = k\int f\,dx \pm m\int g\,dx$. **There is no product or quotient rule for integrals:** first *rewrite* the integrand as a sum of powers.

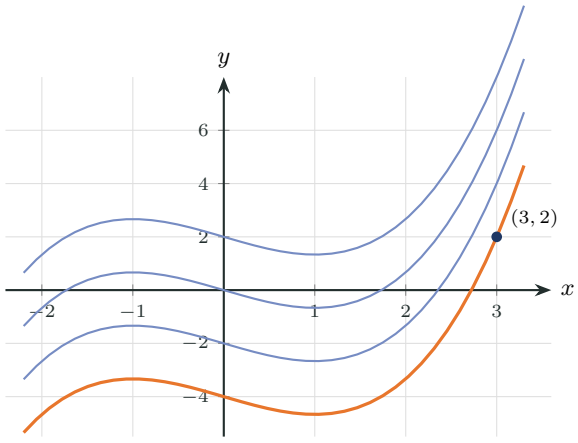


Figure 6.9. Antiderivatives of $f(x) = x^2 - 1$: $F(x) = \frac{x^3}{3} - x + C$. The condition $F(3) = 2$ selects $C = -4$ (orange).

Example 1 **Rewrite first**

Find (a) $\displaystyle\int \frac{x^2+3}{\sqrt{x}}\,dx$ (b) $\displaystyle\int \left(\frac{5}{x}+2e^x-3\sin x\right)dx$ (c) $\displaystyle\int \left(\frac{1}{1+x^2}+\frac{3}{\sqrt{1-x^2}}\right)dx.$

(a) $\dfrac{x^2+3}{x^{1/2}} = x^{3/2}+3x^{-1/2}$, so the integral is $\dfrac{2}{5}x^{5/2}+6x^{1/2}+C.$
(b) $5\ln|x|+2e^x+3\cos x+C.$ **(c)** $\arctan x+3\arcsin x+C.$

Example 2 **Initial value problems**

(a) Find F with $F'(x)=x^2-1$ and $F(3)=2.$ (b) Find f if $f''(x)=12x$, $f'(0)=-1$ and $f(1)=4.$

(a) $F(x)=\frac{x^3}{3}-x+C$; $F(3)=9-3+C=2$ gives $C=-4$. So $F(x)=\frac{x^3}{3}-x-4$ (the orange curve in Figure 6.9).
(b) $f'(x)=6x^2+C_1$ and $f'(0)=C_1=-1$, so $f'(x)=6x^2-1$. Then $f(x)=2x^3-x+C_2$ and $f(1)=2-1+C_2=4$ gives $C_2=3$. So $f(x)=2x^3-x+3$.

Common errors

Omitting $+C$ in an indefinite integral (a lost point on the exam), or adding it too early in an initial value problem.
Integrating the product termwise: $\int x\cos x\,dx \neq \frac{x^2}{2}\sin x.$
The power rule for $n=-1$**:** $\int x^{-1}dx=\ln|x|+C$, not $\frac{x^0}{0}$.
Forgetting to divide by the new exponent (or the chain factor): $\int 4x^3dx=x^4$, not $4x^4$.

AP Exam Tip

Use the initial condition *immediately after* finding the general antiderivative. For a second-order problem, use each condition after each integration (as in Example 2b) so that no constants pile up.

Check yourself 6.8

1. Find $\displaystyle\int \left(4x^3-\frac{2}{x^2}+3\cos x\right)dx.$
2. Find $\displaystyle\int \frac{(x+1)^2}{x}\,dx.$
3. Find f if $f''(x)=6x$, $f'(0)=2$ and $f(0)=-1.$

Topic 6.9 Integrating Using Substitution

The chain rule in reverse

The chain rule says $\frac{d}{dx}F(g(x)) = F'(g(x))\,g'(x)$. Read backwards, an integrand of the form $f(g(x))\,g'(x)$ has the antiderivative $F(g(x))$ where $F'=f$. The method of *substitution* (or *u-substitution*) makes this recognition systematic: let $u=g(x)$, so that $du=g'(x)\,dx$, and the integral becomes the simpler $\int f(u)\,du$. The art lies in choosing u: it is usually the inner function of a composite (inside a root, a power, an exponent, a trigonometric function or a denominator) *whose derivative, up to a constant multiple, is also present* in the integrand.

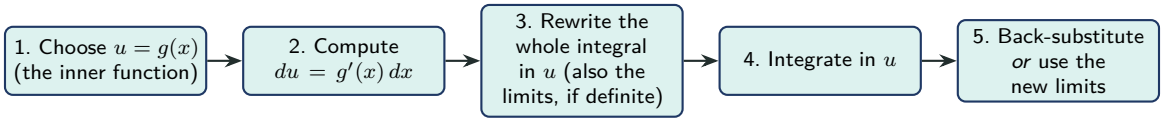


Figure 6.10. The five steps of u-substitution.

Substitution rules

Indefinite: $\displaystyle\int f(g(x))\,g'(x)\,dx=\int f(u)\,du=F(u)+C=F(g(x))+C.$

Definite (change the limits): $\displaystyle\int_a^b f(g(x))\,g'(x)\,dx=\int_{g(a)}^{g(b)} f(u)\,du.$

Linear inner function: $\displaystyle\int f(ax+b)\,dx=\frac{1}{a}F(ax+b)+C.$ **Useful forms:** $\displaystyle\int \frac{g'(x)}{g(x)}dx=\ln|g(x)|+C,$

$$\int g'(x)e^{g(x)}dx = e^{g(x)} + C.$$

Example 1 **Indefinite integral**

Find $\int 6x\,(x^2+1)^4\,dx$.

Let $u = x^2+1$. Then $du = 2x\,dx$, so $6x\,dx = 3\,du$.

$$\int 6x(x^2+1)^4dx = 3\int u^4\,du = \tfrac{3}{5}u^5 + C = \tfrac{3}{5}(x^2+1)^5 + C.$$

Example 2 **Definite integral, new limits**

Evaluate $\displaystyle\int_0^1 x\sqrt{1+3x^2}\,dx$.

Let $u = 1+3x^2$, so $du = 6x\,dx$ and $x\,dx = \tfrac{1}{6}du$. When $x=0$, $u=1$; when $x=1$, $u=4$.

$$\int_0^1 x\sqrt{1+3x^2}\,dx = \tfrac{1}{6}\int_1^4 u^{1/2}\,du = \tfrac{1}{6}\left[\tfrac{2}{3}u^{3/2}\right]_1^4 = \tfrac{1}{9}(8-1) = \tfrac{7}{9}.$$

Example 3 **A quotient**

Evaluate $\displaystyle\int_0^{\pi/4} \tan x\,dx$.

$\tan x = \dfrac{\sin x}{\cos x}$. Let $u = \cos x$, $du = -\sin x\,dx$; $x=0 \Rightarrow u=1$ and $x=\tfrac{\pi}{4} \Rightarrow u = \tfrac{\sqrt{2}}{2}$.

$$\int_0^{\pi/4} \tan x\,dx = -\int_1^{\sqrt{2}/2} \frac{du}{u} = -\Big[\ln u\Big]_1^{\sqrt{2}/2} = -\ln\tfrac{\sqrt{2}}{2} = \ln\sqrt{2} = \tfrac{1}{2}\ln 2.$$

Example 4 **Substitution with a table**

The function f is differentiable and some values are given.

x	1	3	9
$f(x)$	4	2	-1

Evaluate $\displaystyle\int_1^3 2x\,f'(x^2)\,dx$.

Let $u = x^2$, $du = 2x\,dx$; the limits become 1 and 9. The integral is $\displaystyle\int_1^9 f'(u)\,du = f(9) - f(1) = -1 - 4 = -5$.

Common errors

Leaving x in the integrand after substituting: every x must be converted to u.
Mixing limits. Either change the limits to u-values and never go back to x, or back-substitute and use the original x-limits. Using x-limits on a u-integral is wrong.
Dropping the constant factor. If $du = 6x\,dx$ and the integrand has $x\,dx$, you must multiply by $\tfrac{1}{6}$.
Substituting when no derivative factor is present: $\int e^{x^2}dx$ cannot be done by $u = x^2$ (the factor $2x$ is missing).

AP Exam Tip

Write "Let $u = \ldots$, $du = \ldots$" as the first line of the work. On definite integrals the scoring guides award a point for the limits $g(a)$ and $g(b)$ in the u-integral.

Check yourself 6.9

1. Find $\displaystyle\int x^2\cos(x^3)\,dx$.

2. Evaluate $\displaystyle\int_1^e \frac{(\ln x)^2}{x}\,dx$.

3. Evaluate $\displaystyle\int_0^2 \frac{2x}{x^2+4}\,dx$.

Topic 6.10 Integrating Functions Using Long Division and Completing the Square

Two algebraic preparations

Some integrals cannot be done by a table or by substitution as written but become routine after an algebraic rewrite. When the integrand is a *rational function* whose numerator has degree at least that of the denominator, divide first: the quotient is a polynomial (easy to integrate) and the remainder over the divisor is of the form $\dfrac{k}{x-r}$, whose antiderivative is $k\ln|x-r|$. When the denominator is an irreducible quadratic, or a quadratic stands under a square root, complete the square to match $\dfrac{1}{u^2+a^2}$ or $\dfrac{1}{\sqrt{a^2-u^2}}$.

Patterns

Long division: $\dfrac{x^2+3x+5}{x+1} = x+2+\dfrac{3}{x+1}$ because $(x+1)(x+2)+3 = x^2+3x+5$.

Completing the square: $x^2+bx+c = \left(x+\frac{b}{2}\right)^2 + \left(c-\frac{b^2}{4}\right)$.

$$\int \frac{du}{u^2+a^2} = \frac{1}{a}\arctan\frac{u}{a}+C, \qquad \int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\frac{u}{a}+C \qquad (a>0).$$

Example 1 Long division

Evaluate $\displaystyle\int_0^1 \frac{x^2+3x+5}{x+1}\,dx$. *Divide the leading terms:* $x^2 \div x = x$; subtract $x(x+1) = x^2+x$, leaving $2x+5$. *Again:* $2x \div x = 2$; subtract $2(x+1) = 2x+2$, leaving the remainder 3. Hence quotient $x+2$, remainder 3. So the integrand is $x+2+\dfrac{3}{x+1}$ and

$$\int_0^1 \left(x+2+\frac{3}{x+1}\right)dx = \left[\tfrac{x^2}{2}+2x+3\ln|x+1|\right]_0^1 = \tfrac{5}{2}+3\ln 2.$$

Example 2 Completing the square

(a) Find $\displaystyle\int \frac{dx}{x^2+4x+13}$. (b) Find $\displaystyle\int \frac{dx}{\sqrt{-x^2+6x-5}}$.

(a) $x^2+4x+13 = (x+2)^2+9$. With $u=x+2$, $a=3$: $\dfrac{1}{3}\arctan\dfrac{x+2}{3}+C$.

(b) $-x^2+6x-5 = -(x^2-6x+5) = -\big[(x-3)^2-4\big] = 4-(x-3)^2$. With $u=x-3$, $a=2$: $\arcsin\dfrac{x-3}{2}+C$.

Common errors

Splitting a denominator. You may split a numerator over one denominator, but never the reverse: $\dfrac{1}{x^2+4} \neq \dfrac{1}{x^2}+\dfrac{1}{4}$, and $\displaystyle\int \frac{dx}{x^2+4} \neq \ln|x^2+4|$.
Forgetting the factor $\frac{1}{a}$ in the arctangent formula.
Completing the square with a leading coefficient $\neq 1$ without first factoring it out.

Check yourself 6.10

1. Find $\displaystyle\int \frac{x+5}{x-2}\,dx$.

2. Find $\displaystyle\int \frac{dx}{x^2-6x+10}$.

Topic 6.14 Selecting Techniques for Antidifferentiation

Choosing a method

You now have a toolkit: basic forms (6.8), algebraic rewriting, substitution (6.9), long division and completing the square (6.10). The skill tested on the exam is choosing the right tool in a few seconds. A good habit is to ask the same questions in the same order, from the simplest tool to the most elaborate, as in Figure 6.11. When none of them works (for example $\int e^{-x^2}\,dx$, which has no elementary antiderivative), the question will be a calculator-active one that asks for a numerical value of a *definite* integral.

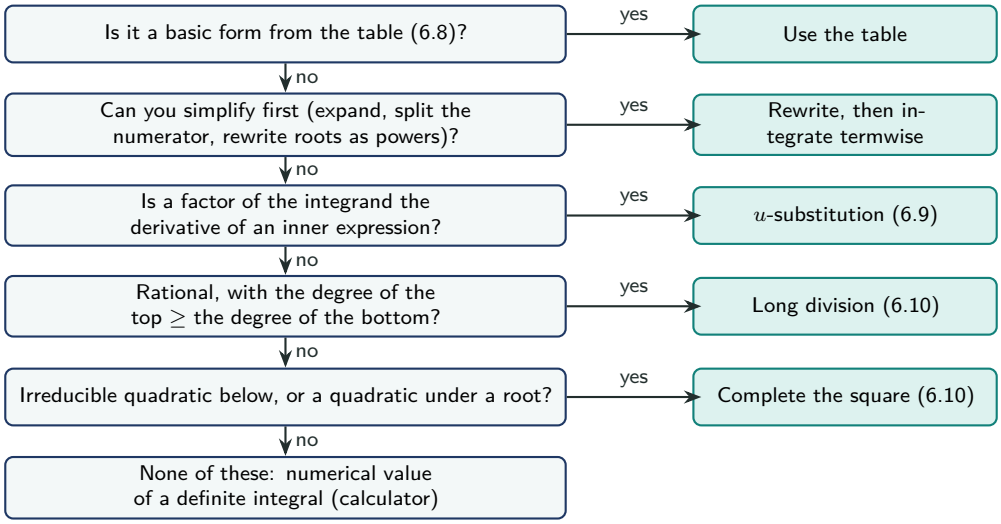


Figure 6.11. Which technique? Work down the list; the first "yes" is the tool to use.

Example 1 Match the method

Name a method and find an antiderivative.

Integral	Method	Result
(a) $\displaystyle\int \frac{x^3-1}{x^2}\,dx$	rewrite as $x - x^{-2}$	$\dfrac{x^2}{2} + \dfrac{1}{x} + C$
(b) $\displaystyle\int \frac{x}{x^2+1}\,dx$	$u = x^2+1$	$\frac{1}{2}\ln(x^2+1) + C$
(c) $\displaystyle\int \cos x\, e^{\sin x}\,dx$	$u = \sin x$	$e^{\sin x} + C$
(d) $\displaystyle\int \frac{x+5}{x-2}\,dx$	long division: $1 + \dfrac{7}{x-2}$	$x + 7\ln\lvert x-2\rvert + C$
(e) $\displaystyle\int \frac{dx}{x^2-6x+10}$	$(x-3)^2+1$	$\arctan(x-3) + C$
(f) $\displaystyle\int \frac{3}{\sqrt{1-x^2}}\,dx$	table	$3\arcsin x + C$

Example 2 Combining methods

Evaluate $\displaystyle\int_0^{\ln 2} e^x\sqrt{1+e^x}\,dx$.

Let $u = 1+e^x$, $du = e^x\,dx$; $x = 0 \Rightarrow u = 2$ and $x = \ln 2 \Rightarrow u = 3$.

$$\int_2^3 u^{1/2}du = \left[\tfrac{2}{3}u^{3/2}\right]_2^3 = \tfrac{2}{3}\left(3\sqrt{3} - 2\sqrt{2}\right) \approx 1.578.$$

Common errors

Using a substitution that does not cancel. If after $u = \ldots$ some x remain, the choice (or the integrand) is wrong; try another u or another method.

Splitting a denominator (see 6.10).
Expecting every integral to have a closed form: some do not, and the exam will then give you a calculator.

AP Exam Tip

Differentiate your answer to *check* it: if the derivative is the integrand, you are done. This takes ten seconds and catches missing constant factors.

Check yourself 6.14

1. Name the best method for each and integrate: (a) $\displaystyle\int \frac{e^x}{1+e^x}\,dx$ (b) $\displaystyle\int (x^2+2)^2 dx$ (c) $\displaystyle\int \frac{dx}{x^2+4x+8}$.
2. Which of these has no elementary antiderivative: $\int xe^{x^2}dx$, $\int e^{x^2}dx$, $\int \frac{dx}{1+x^2}$?

Unit 6 Practice: Integration and Accumulation of Change

Part A Multiple choice: NO CALCULATOR

14 questions. Choose the best answer for each.

1 $\displaystyle\int_1^3 (2x+1)\,dx =$

(A) 8 **(B)** 10

(C) 12 **(D)** 7

2 Let $G(x) = \displaystyle\int_0^{x^2} \sqrt{t+5}\,dt$. What is $G'(2)$?

(A) 3 **(B)** 6

(C) $4\sqrt{7}$ **(D)** 12

3 Water is pumped into a tank at a rate of $r(t)$ litres per minute. Values of r are shown in the table. A trapezoidal sum with three subintervals approximates $\displaystyle\int_0^6 r(t)\,dt$ as

t (min)	0	2	5	6
$r(t)$ (L/min)	4	7	9	11

(A) 45 **(B)** 38

(C) 52 **(D)** 90

4 If $f'(x) = 6x^2 - 4x$ and $f(1) = 2$, then $f(2) =$

(A) 8 **(B)** 4

(C) 10 **(D)** 16

5 $\displaystyle\int_0^1 x\sqrt{1+3x^2}\,dx =$

(A) $\dfrac{14}{3}$ **(B)** $\dfrac{14}{9}$

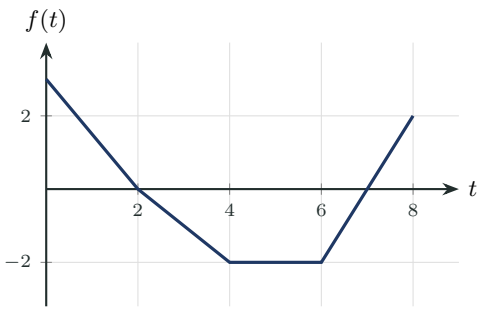
(C) $\dfrac{1}{9}$ **(D)** $\dfrac{7}{9}$

6 The function f is even and $\displaystyle\int_0^3 f(x)\,dx = 4$. What is $\displaystyle\int_{-3}^3 \big(f(x)+2\big)\,dx$?

(A) 20 **(B)** 16

(C) 8 **(D)** 32

7 The graph of f is shown, and $g(x) = \displaystyle\int_0^x f(t)\,dt$ for $0 \le x \le 8$. On which interval is g both increasing and concave up?



(A) $(0,2)$ **(B)** $(2,4)$

(C) $(7,8)$ **(D)** $(6,7)$

8 A particle moves on a line with velocity $v(t) = 3t^2 - 12t + 9$ for $0 \le t \le 4$. What is the total distance travelled?

(A) 4 **(B)** 12
(C) 8 **(D)** 0

9 $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \left(1 + \dfrac{3k}{n}\right)^2 \cdot \dfrac{3}{n}$ equals

(A) 9 **(B)** 63
(C) $\dfrac{64}{3}$ **(D)** 21

10 $\displaystyle\int x e^{x^2}\, dx =$

(A) $\dfrac{1}{2}e^{x^2} + C$ **(B)** $e^{x^2} + C$
(C) $\dfrac{x^2}{2}e^{x^2} + C$ **(D)** $\dfrac{e^{x^2}}{2x} + C$

11 The function f is positive, decreasing and concave up on $[1, 5]$. Let L_4, R_4, T_4 be the left, right and trapezoidal sums with four equal subintervals, and $I = \int_1^5 f(x)\, dx$. Which statement is true?

(A) $R_4 < T_4 < I < L_4$
(B) $L_4 < I < T_4 < R_4$
(C) $R_4 < I < T_4 < L_4$
(D) $I < R_4 < T_4 < L_4$

12 Let $g(x) = \displaystyle\int_0^x (t^2 - 4t)\, dt$. On the interval $-1 < x < 6$, at what value of x does g have a relative minimum?

(A) 0 **(B)** 4
(C) 2 **(D)** 6

13 $\displaystyle\int_0^{\ln 3} \frac{e^x}{1 + e^x}\, dx =$

(A) $\ln 2$ **(B)** $\ln 4$
(C) $\dfrac{1}{4}$ **(D)** $\ln 3$

14 If $\displaystyle\int_1^5 f(x)\, dx = 12$, then $\displaystyle\int_0^2 f(2x + 1)\, dx =$

(A) 12 **(B)** 24
(C) 4 **(D)** 6

Part B Multiple choice: CALCULATOR ALLOWED

7 questions. Give decimals to three places where applicable.

15 $\displaystyle\int_0^2 e^{-x^2}\, dx \approx$

(A) 0.018 **(B)** 0.865
(C) 0.882 **(D)** 1.764

16 Cars enter a car park at a rate $R(t)$ cars per minute. Values are given below. A trapezoidal sum with the four subintervals approximates $\displaystyle\int_0^{12} R(t)\,dt$ as

t (min)	0	3	5	8	12
$R(t)$	10.2	14.6	13.1	9.4	6.3

(A) 130.05 **(B)** 136.7
(C) 123.4 **(D)** 260.1

17 The volume of water in a pond is $V(t)$ litres, where $V'(t) = e^{-0.2t} - 0.4$ for $0 \le t \le 10$ hours. At what time t is V greatest?

(A) 0.916 **(B)** 2.291
(C) 3.000 **(D)** 4.581

18 The midpoint Riemann sum with three equal subintervals for $\displaystyle\int_0^3 \sqrt{x^3+1}\,dx$ is

(A) 5.414 **(B)** 7.230
(C) 9.706 **(D)** 7.560

19 The function f satisfies $f(0) = 5$ and $f'(x) = e^{\sin x}$. What is $f(2)$?

(A) 4.237 **(B)** 2.483
(C) 9.237 **(D)** 7.483

20 Let $H(x) = \displaystyle\int_0^{x^2} e^{-t^2}\,dt$. What is $H'(1.2)$?

(A) 0.302 **(B)** 0.126
(C) 0.569 **(D)** 0.251

21 A tank holds 40 litres of water at $t = 0$. Water flows in at a rate $R(t) = 8 + 3\sin(0.5t)$ litres per minute. How many litres are in the tank at $t = 8$ minutes?

(A) 73.92 **(B)** 45.73
(C) 104.01 **(D)** 113.92

Part C Free response

2 questions, 9 points each. Show the work that leads to your answers.

22 **[9 points] (Calculator active.)** Water flows into a storage tank at a rate modelled by $E(t) = 4 + 3\sin\left(\dfrac{t}{2}\right)$ and flows out at a rate $L(t)$, where both rates are in thousands of litres per hour and t is the number of hours after the tank begins operating, $0 \le t \le 8$. The tank holds 25 thousand litres at $t = 0$. Selected values of $L(t)$ are given.

t (hours)	0	2	5	7	8
$L(t)$ (thousand L/h)	3.0	5.0	6.4	5.6	4.8

(a) Use a trapezoidal sum with the four subintervals indicated by the table to approximate $\displaystyle\int_0^8 L(t)\,dt$. Using correct units, explain the meaning of $\displaystyle\int_0^8 L(t)\,dt$ in the context of the problem.

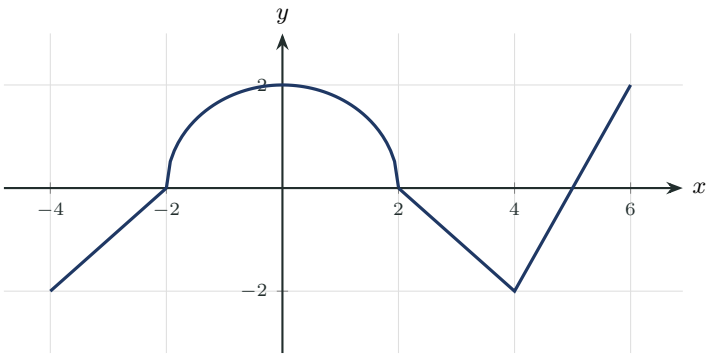
(b) Find $\displaystyle\int_0^8 E(t)\,dt$. Using your answer from part (a), approximate the amount of water in the tank, in thousands of litres, at $t = 8$.

(c) Is the amount of water in the tank increasing or decreasing at $t = 5$? Give a reason for your answer.

(d) Assume L is continuous. Is there a time t, $2 < t < 5$, at which the amount of water in the tank is neither increasing

nor decreasing? Justify your answer.

23 **[9 points] (No calculator.)** The graph of the continuous function f, shown below, consists of two line segments, a semicircle centred at the origin and a third line segment. Let $g(x) = \int_0^x f(t)\,dt$ for $-4 \leq x \leq 6$.



The graph of f for Question 23.

(a) Find $g(-2)$ and $g(4)$.

(b) Find the slope of the line tangent to the graph of g at $x = 4$ and write an equation of this tangent line.

(c) Find the x-coordinate of each point of inflection of the graph of g on the interval $-4 < x < 6$. Give a reason for your answer.

(d) Find the absolute minimum value of g on the interval $[-4, 6]$. Justify your answer.

AP Calculus AB – Unit 7 Exam weighting: 6–12% of the AB exam

Differential Equations

In this unit

A *differential equation* (DE) is an equation that contains a function together with one or more of its derivatives. Calculus was invented to describe how things change, and a differential equation is the most direct way to state a law of change: "the rate at which the temperature falls is proportional to how much hotter the cup is than the room", or "salt enters a tank at one rate and leaves at another". Unit 7 teaches you to (i) translate such statements into equations, (ii) decide whether a proposed function really is a solution, (iii) understand solutions *geometrically* through slope fields, and (iv) solve the most important family of DEs, the separable ones, which produces the exponential models used throughout science.

Topics (AB scope).

7.1	Modeling situations with differential equations	7.6	General solutions by separation of variables
7.2	Verifying solutions	7.7	Particular solutions (initial conditions)
7.3	Sketching slope fields	7.8	Exponential models
7.4	Reasoning using slope fields		

Topics 7.5 (Euler's method) and 7.9 (logistic models) are Calculus BC only and are not part of the AB course.

Big ideas. *Change:* a DE describes how a quantity changes in terms of its current state. *Analysis of functions:* the sign of $\frac{dy}{dx}$ and of $\frac{d^2y}{dx^2}$, found from the DE itself, tells you where solutions rise, fall and bend. *Fundamental theorem and antiderivatives:* separating variables turns a DE into two antiderivative problems. *Exam skill:* on the free-response section a DE question usually carries 4–5 points for a clean separation-of-variables solution, so the *format* of the work matters as much as the answer.

Topic 7.1 Modeling Situations with Differential Equations

What a differential equation says

A differential equation relates an unknown function to its own derivative. Compare $y = 3x^2$, which *tells you* y, with $\frac{dy}{dx} = 6x$, which tells you only how y changes; many different functions ($3x^2$, $3x^2 + 5$, $3x^2 - 1$, …) have this derivative. In a DE, the derivative may also depend on the function itself, as in $\frac{dy}{dx} = 2y$, and that is what makes DEs so effective at modelling: in nature the speed at which a quantity changes very often depends on how much of it is already present. A DE is *first order* when the highest derivative is $\frac{dy}{dx}$ (or $\frac{dP}{dt}$ etc.). Almost everything on the AB exam is first order; a second-order equation such as $y'' + y = 0$ appears only when you are asked to *verify* a given solution (Topic 7.2).

Translating words into a derivative equation

The key is to identify (1) the quantity that changes, (2) the independent variable, usually time t, and (3) what the sentence says the *rate* $\frac{d}{dt}$ is equal to. The phrase "is proportional to" always means "equals a constant times"; the constant is the *constant of proportionality* and its sign carries information. If you are told the rate of change of P is proportional to P, write $\frac{dP}{dt} = kP$. Whether k is positive or negative is decided by the story: growth gives $k > 0$, decay gives $k < 0$ (or, equivalently, you can write $-kP$ and say $k > 0$). Be careful about *what* is proportional to what: "proportional to the *difference* between T and 20" gives $k(T - 20)$, not kT.

A second great family of models is *rate in minus rate out*. If $A(t)$ is the amount of a substance in a container, then

$$\frac{dA}{dt} = (\text{rate at which the substance enters}) - (\text{rate at which it leaves}),$$

and each rate is (flow rate)×(concentration), with units checked at every step.

Key ideas: modelling with differential equations

- "The rate of change of y is proportional to y" $\Longrightarrow \dfrac{dy}{dt} = ky$.
- "…proportional to y^2 / $\sqrt{y}$ / the difference $y - b$" $\Longrightarrow \dfrac{dy}{dt} = ky^2,\ k\sqrt{y},\ k(y-b)$.
- "…inversely proportional to y" $\Longrightarrow \dfrac{dy}{dt} = \dfrac{k}{y}$.
- "The slope of the curve at each point (x, y) equals …" $\Longrightarrow \dfrac{dy}{dx} = \ldots$
- Mixing/flow: $\dfrac{dA}{dt}$ = (rate in) − (rate out); rate = flow rate × concentration.
- The sign of $\dfrac{dy}{dt}$ tells you whether y is increasing (> 0) or decreasing (< 0); a value of y with $\dfrac{dy}{dt} = 0$ is an *equilibrium* value.

Example 1 Translating statements

Write a differential equation for each statement.

Statement	Differential equation
(a) The mass M of a bacterial culture grows at a rate equal to 0.15 times the mass present.	$\dfrac{dM}{dt} = 0.15M$
(b) The temperature T of a cup of coffee changes at a rate proportional to the difference between T and the room temperature $22°$C.	$\dfrac{dT}{dt} = k(T - 22)$
(c) At every point (x, y) on a curve the slope is the product of the coordinates.	$\dfrac{dy}{dx} = xy$
(d) The rate of change of y with respect to x is inversely proportional to y.	$\dfrac{dy}{dx} = \dfrac{k}{y}$
(e) The volume V of water in a tank decreases at a rate proportional to $\sqrt{V}$.	$\dfrac{dV}{dt} = -k\sqrt{V},\ k > 0$

Note on (b). A cooling cup has $T > 22$, so $T - 22 > 0$ and $\dfrac{dT}{dt} < 0$; therefore $k < 0$ in the form $k(T - 22)$. Writing $-k(T - 22)$ with $k > 0$ is equally correct.

Example 2 A mixing problem

A tank contains 200 litres of well-stirred brine. Brine with salt concentration 0.4 kg per litre enters at 5 L/min, and the stirred mixture leaves at 5 L/min, so the volume stays 200 L. Let $A(t)$ be the kilograms of salt in the tank at time t minutes.

1. Salt in: $5\ \frac{\text{L}}{\text{min}} \cdot 0.4\ \frac{\text{kg}}{\text{L}} = 2$ kg/min.
2. Concentration in the tank: $\dfrac{A}{200}$ kg/L, so salt out: $5 \cdot \dfrac{A}{200} = \dfrac{A}{40}$ kg/min.
3. Hence $\dfrac{dA}{dt} = 2 - \dfrac{A}{40}$.
4. Equilibrium: $2 - \dfrac{A}{40} = 0 \Rightarrow A = 80$ kg.
5. If $A = 100$: $\dfrac{dA}{dt} = 2 - 2.5 = -0.5 < 0$, so the amount of salt is decreasing toward 80.

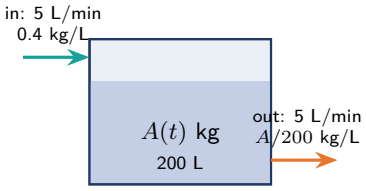


Figure 7.1: rate in − rate out

Example 3 Reading a model

The number of fish $P(t)$, in thousands, in a managed lake satisfies $\dfrac{dP}{dt} = 0.05P - 30$, where t is in years. The term

$0.05P$ is natural growth (5% per year) and 30 is the number of thousands harvested per year.

P (thousands)	400	500	600	700	800
$\frac{dP}{dt}$ (thousands/yr)	-10	-5	0	5	10

1. $\frac{dP}{dt} = 0$ when $0.05P = 30$, i.e. $P = 600$: the harvest exactly balances growth, so the population is constant at 600 thousand.
2. For $P > 600$ the rate is positive and the population grows; for $P < 600$ it is negative and the population falls. The equilibrium at 600 is therefore *unstable*: a small error in either direction grows. A sustainable harvest policy must keep P above 600.

Common errors

Sign and meaning of k. Writing $\frac{dV}{dt} = k\sqrt{V}$ for a *decreasing* volume and then treating k as positive contradicts the story. Say "$k > 0$" only after you have fixed the sign in the equation. **Mixing up the variable:** "proportional to the difference" is $k(T - 22)$, not $kT - 22$.

AP Exam Tip

On the exam a modelling sentence is usually followed by "Write a differential equation…" or by a given DE that you must *interpret*. When you interpret, link the sign to the context: "$\frac{dA}{dt} < 0$, so the amount of salt in the tank is decreasing at that time". Always include units.

Check yourself 7.1

1. The number $N(t)$ of fish in a lake changes at a rate proportional to the difference between 5000 and N. Write a differential equation for N.
2. A 200-litre tank holds a solution. A solution containing 0.3 kg of sugar per litre enters at 4 L/min, and the well-mixed solution leaves at 4 L/min. If $S(t)$ is the kilograms of sugar in the tank, write a differential equation for S.
3. A population P (thousands) satisfies $\frac{dP}{dt} = 0.04P - 20$. Find the value of P for which P is constant, and decide whether P is increasing or decreasing when $P = 400$.

Topic 7.2 Verifying Solutions for Differential Equations

What "solution" means

A function $y = f(x)$ is a *solution* of a differential equation if, when you substitute $y = f(x)$ and its derivatives into the equation, you obtain a true statement for every x in an interval. Verifying a solution requires no integration at all: it is a *substitution check*. Differentiate the proposed function, compute both sides of the DE, and show they are equal. The check is mechanical, but exam scoring rewards *showing both sides*: write $\frac{dy}{dx} = \ldots$ from differentiating, then the right-hand side computed from y, and conclude they match.

Most differential equations have infinitely many solutions, usually a whole family that differs by a constant. The family $y = Ce^{x^2}$ solves $\frac{dy}{dx} = 2xy$ for every constant C; this is the *general solution*. When an *initial condition* such as $y(0) = 4$ is also given, one member of the family is singled out: the *particular solution*. To verify a particular solution you therefore perform *two* checks: the DE and the initial condition.

Procedure: verify that $y = f(x)$ solves a DE with initial condition $y(a) = b$

1. Differentiate f (use the chain rule!), giving $f'(x)$ (and $f''(x)$ if needed).
2. Substitute $y = f(x)$ into the *right-hand side* of the DE and simplify.
3. Show the left side (the derivative from Step 1) equals the right side for all x.
4. If an initial condition is given, show $f(a) = b$.

To *find* an unknown constant (for example k in $y = e^{kx}$), substitute and require the identity to hold for all x.

Example 1 Verifying a family member

Show that $y = 3e^{2x} + x$ is a solution of $\dfrac{dy}{dx} = 2y - 2x + 1$.

1. $\dfrac{dy}{dx} = 6e^{2x} + 1$.
2. Right side: $2y - 2x + 1 = 2(3e^{2x} + x) - 2x + 1 = 6e^{2x} + 2x - 2x + 1 = 6e^{2x} + 1$.
3. The two sides agree for every x, so $y = 3e^{2x} + x$ is a solution.

Example 2 A particular solution

Show that $y = 4e^{-x^2}$ is the solution of $\dfrac{dy}{dx} = -2xy$ with $y(0) = 4$.

1. By the chain rule, $\dfrac{dy}{dx} = 4e^{-x^2} \cdot (-2x) = -8xe^{-x^2}$.
2. Right side: $-2xy = -2x \cdot 4e^{-x^2} = -8xe^{-x^2}$. They are equal.
3. Initial condition: $y(0) = 4e^0 = 4$. ✓

Example 3 Testing several candidates (table)

Which of the following functions satisfy $x\dfrac{dy}{dx} = 3y$ for $x > 0$?

Candidate y	$x\dfrac{dy}{dx}$	$3y$	Solution?
$y = x^3$	$x \cdot 3x^2 = 3x^3$	$3x^3$	**Yes**
$y = 3x$	$x \cdot 3 = 3x$	$9x$	No
$y = x^3 + 5$	$x \cdot 3x^2 = 3x^3$	$3x^3 + 15$	No (constant breaks it)
$y = -2x^3$	$x \cdot (-6x^2) = -6x^3$	$-6x^3$	**Yes**

The solutions are exactly $y = Cx^3$; adding a constant is *not* allowed here because the DE contains y on the right, unlike $\dfrac{dy}{dx} = 3x^2$, where $y = x^3 + C$ is fine.

Example 4 Finding an unknown constant, second derivative

For which values of k is $y = e^{kx}$ a solution of $y'' - y' - 6y = 0$?

1. $y' = ke^{kx}$ and $y'' = k^2e^{kx}$.
2. Substituting: $k^2e^{kx} - ke^{kx} - 6e^{kx} = e^{kx}(k^2 - k - 6) = 0$.
3. Since $e^{kx} \neq 0$, we need $k^2 - k - 6 = (k - 3)(k + 2) = 0$, so $k = 3$ or $k = -2$.

Common errors

Forgetting the chain rule: for $y = 4e^{-x^2}$, $y' \neq -4e^{-x^2}$; the factor $-2x$ is required. **Checking only the DE:** a "particular solution" must also satisfy the initial condition. **Adding $+C$ everywhere:** $y = Cx^3$ and $y = x^3 + C$ are different families; only substitution can tell you which is valid.

AP Exam Tip

Write the verification in three visible lines: "$\dfrac{dy}{dx} = \ldots$", "right side $= \ldots$", "these are equal, so y is a solution". You do not earn the point for a bare "it works".

Check yourself 7.2

1. Show that $y = e^{3x} + x$ is a solution of $\dfrac{dy}{dx} = 3y - 3x + 1$.
2. For what values of k is $y = e^{kx}$ a solution of $y'' - 5y' + 6y = 0$?
3. Show that $y = \dfrac{1}{1 - x}$ satisfies $\dfrac{dy}{dx} = y^2$ and $y(0) = 1$.

Topic 7.3 Sketching Slope Fields

Seeing a differential equation

Suppose $\dfrac{dy}{dx} = x - y$. You cannot yet write down y, but the equation still tells you something concrete: *at any point* (x, y)*, a solution curve passing through that point must have slope* $x - y$. At $(1, 3)$ the slope is -2; at $(2, 2)$ it is 0; at $(0, -1)$ it is 1. A *slope field* (or direction field) records this information: choose a grid of points, compute $\dfrac{dy}{dx}$ at each, and draw at that point a short line segment having that slope. A solution curve is then any curve that is tangent to the segments wherever it goes, like a leaf carried along a stream whose current direction is drawn at every point.

Because the field is computed from the differential equation alone, slope fields work for *every* first-order equation, including ones you cannot solve by hand. The exam asks for two skills: *drawing* the segments at specified points (this topic), and *using* a given field to sketch solution curves and reason about behaviour (Topic 7.4). Slope-field questions are almost always worth 2 of the 9 points on a free-response question, and they are the easiest two points on the page if you are systematic: build a table of slopes first, then draw.

How to sketch a slope field

1. Make a table. For each grid point (x, y) substitute *both* coordinates into the right-hand side to get $m = \dfrac{dy}{dx}$.
2. At each point draw a short segment with slope m: $m = 0$ horizontal; $m > 0$ rising; $m < 0$ falling; $m = 1$ at $45°$; $|m|$ large nearly vertical; m undefined vertical.
3. Look for patterns that save work. If the right side contains only x, all segments in a *column* are parallel. If it contains only y (an *autonomous* equation), all segments in a *row* are parallel. Wherever the right side is 0 the segments are horizontal.
4. To sketch the solution through (a, b), start at (a, b) and follow the segments to the right and to the left; the curve must be tangent to the field everywhere and can never cross a segment's direction.

A curve on which $\dfrac{dy}{dx}$ has the same value c is called an *isocline*; every segment on it is parallel.

Example 1 A table of slopes, then the field

For $\dfrac{dy}{dx} = x - y$, compute the slope at the integer points with $-2 \le x, y \le 2$, then sketch the solution through $(0, 1)$.

$y \backslash x$	-2	-1	0	1	2
2	-4	-3	-2	-1	0
1	-3	-2	-1	0	1
0	-2	-1	0	1	2
-1	-1	0	1	2	3
-2	0	1	2	3	4

Entry $= x - y$. The zeros lie on the line $y = x$.

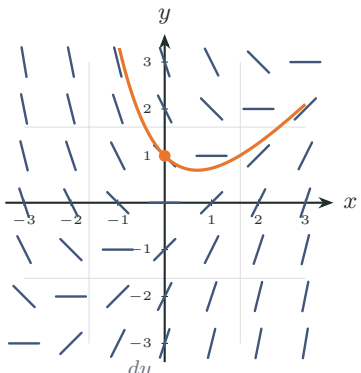


Figure 7.2: slope field of $\dfrac{dy}{dx} = x - y$ with the solution through $(0, 1)$.

1. The table shows that segments are horizontal exactly on the line $y = x$ (where $x - y = 0$), rise to the right below that line, and fall above it.
2. Starting at $(0, 1)$ the slope is -1, so the solution falls; it reaches the line $y = x$ at $x = \ln 2 \approx 0.69$, where the slope is 0 (a minimum), and then rises, approaching the line $y = x - 1$ as x grows.

Example 2 Slope depends on both variables

Describe the slope field of $\dfrac{dy}{dx} = xy$ on $-2 \le x, y \le 2$.

1. If $x = 0$ or $y = 0$ the slope is 0: the segments along *both axes* are horizontal.
2. The slope has the sign of xy: positive in Quadrants I and III, negative in II and IV.
3. $|\text{slope}| = |x||y|$ is largest at the corners, e.g. at $(2, 2)$ the slope is 4 (steep), at $(1, 1)$ it is 1 ($45°$).
4. The solution through $(0, 1)$ has slope 0 there, then rises on both sides: it has a minimum at $x = 0$.

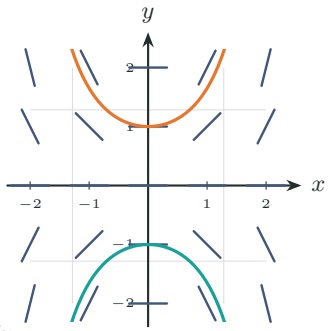


Figure 7.3: $\dfrac{dy}{dx} = xy$ with solutions through $(0, 1)$ and $(0, -1)$.

Example 3 An equation with only y (autonomous)

Sketch the slope field of $\dfrac{dy}{dx} = \dfrac{y^2 - 4}{4}$ and identify its horizontal solutions.

y	-3	-2	-1	0	1	2	3
$\dfrac{dy}{dx}$	$\frac{5}{4}$	0	$-\frac{3}{4}$	-1	$-\frac{3}{4}$	0	$\frac{5}{4}$

1. The right side contains no x, so every row has identical segments; one calculation per row suffices.
2. The slope is 0 at $y = \pm 2$, so $y = -2$ and $y = 2$ are constant (*equilibrium*) solutions.
3. For $-2 < y < 2$ the slope is negative; for $|y| > 2$ it is positive.

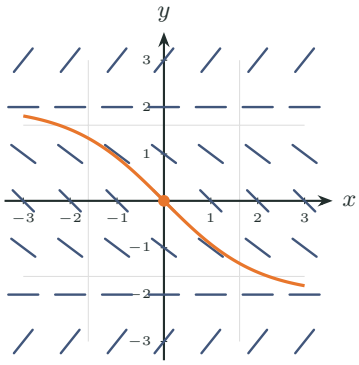


Figure 7.4: rows are constant; the curve through $(0, 0)$ falls toward $y = -2$.

Common errors

Substituting only one variable: for $\dfrac{dy}{dx} = x - y$ the slope at $(2, 5)$ is $2 - 5 = -3$, not 2. **Confusing zero with undefined:** slope 0 is horizontal, an undefined slope (denominator zero) is vertical. **Drawing every segment at $45°$:** a slope of 4 must look much steeper than a slope of 1. **Letting a solution curve cross the segments:** the curve must be tangent to the segment at every point it passes.

AP Exam Tip

When asked to "sketch the slope field at the points shown", show the computed slope for each point beside the grid (the scoring guide awards the point for correct segments, but your numbers protect you if a segment is ambiguous). Make horizontal segments clearly horizontal and vertical ones clearly vertical.

Check yourself 7.3

1. For $\dfrac{dy}{dx} = x + y$ find the slope at $(0, 0)$, $(1, -1)$, $(-2, 1)$ and $(2, 3)$.
2. Without computing every point, describe the slope field of $\dfrac{dy}{dx} = 2 - y$: where are the segments horizontal, and where do they rise?
3. The slope field of $\dfrac{dy}{dx} = \dfrac{x}{2}$ has identical segments in each vertical column. Explain why, and state the slope at $x = -2$.

Topic 7.4 Reasoning Using Slope Fields

Reading a field

A slope field is a picture of the differential equation, and exam questions in this topic give you the picture (or the equation) and ask you to *reason* from it. The same four questions keep recurring. (1) *Sketch the solution through a given point*, by following the segments. (2) *Match* a field to its differential equation, by testing a few features. (3) *Describe behaviour*: where is y increasing, where does it level off, what is $\lim_{x\to\infty} y$? (4) *Concavity and extrema*, which require the *second derivative of a solution computed from the differential equation itself*.

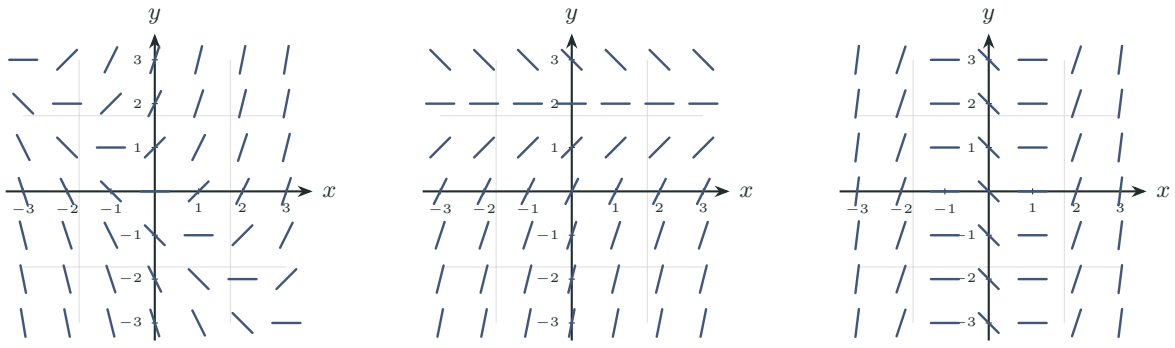
For (4), differentiate both sides of the DE with respect to x, remembering that y is a function of x so the chain rule produces factors of $\frac{dy}{dx}$. For $\frac{dy}{dx} = x - y$ you get $\frac{d^2y}{dx^2} = 1 - \frac{dy}{dx}$, and then you may substitute $\frac{dy}{dx} = x - y$ again to write the answer in terms of x and y alone. At any point (a, b) you can then find the sign of $\frac{d^2y}{dx^2}$ without ever knowing the solution formula.

Reasoning with a slope field

- **Increasing / decreasing:** the solution rises where segments rise ($\frac{dy}{dx} > 0$) and falls where they fall.
- **Equilibrium solutions:** a horizontal line $y = c$ is a solution if $\frac{dy}{dx} = 0$ at every point of it, i.e. the right side vanishes when $y = c$ for all x.
- **Matching:** test (i) where the segments are horizontal ($\frac{dy}{dx} = 0$), (ii) the sign of the slope in each region, (iii) whether columns or rows are identical, (iv) the steepness at one or two points.
- **Local extrema:** at a point where a solution has $\frac{dy}{dx} = 0$, use $\frac{d^2y}{dx^2} > 0$ (minimum) or < 0 (maximum), or the sign change of $\frac{dy}{dx}$.
- **Concavity:** $\frac{d^2y}{dx^2} > 0$ means concave up (segments get steeper to the right), < 0 concave down.
- **Limits:** follow the segments as $x \to \infty$; a solution cannot cross an equilibrium solution of a well-behaved DE, so it stays on one side of it.

Example 1 Matching fields to equations

Three slope fields are shown. Match each to one of $\frac{dy}{dx} = x + y$, $\frac{dy}{dx} = 2 - y$, $\frac{dy}{dx} = x^2 - 1$.



Field I Field II Field III

1. Field II has identical segments along each *row* and is horizontal along $y = 2$: it depends only on y and vanishes at $y = 2$, so it is $\frac{dy}{dx} = 2 - y$.
2. Field III has identical segments in each *column*, horizontal at $x = \pm 1$ (where $x^2 - 1 = 0$) and steep for $|x| = 3$ (slope 8): $\frac{dy}{dx} = x^2 - 1$.
3. Field I is horizontal along the line $y = -x$ (where $x + y = 0$): $\frac{dy}{dx} = x + y$.

Example 2 Behaviour of solutions of $\frac{dy}{dx} = 2 - y$

Let $y = f(x)$ be the solution of $\frac{dy}{dx} = 2 - y$ with $f(0) = -1$. Describe f using the field.

1. At $(0, -1)$ the slope is $2 - (-1) = 3 > 0$, so f is increasing there. Since $y < 2$ the slope stays positive, so f is increasing for all x.
2. $y = 2$ is an equilibrium solution; f cannot cross it, so $f(x) < 2$.
3. Increasing and bounded above by 2, with segments flattening near $y = 2$: $\lim_{x \to \infty} f(x) = 2$.
4. Concavity: $\frac{d^2y}{dx^2} = -\frac{dy}{dx} = -(2 - y) = y - 2 < 0$ when $y < 2$: f is concave down.

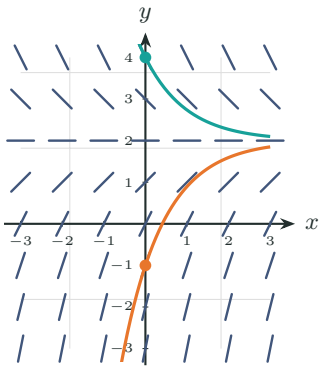


Figure 7.5: solutions through $(0, -1)$ (orange) and $(0, 4)$ (teal).

Example 3 Extremum and concavity from the equation

Let $y = f(x)$ solve $\frac{dy}{dx} = x + y$ with $f(0) = 1$. (a) Find $\frac{d^2y}{dx^2}$ in terms of x and y. (b) Find the concavity of f at $x = 0$. (c) Does f have a critical point where $x + y = 0$ and what type is it?

1. (a) $\frac{d^2y}{dx^2} = 1 + \frac{dy}{dx} = 1 + (x + y) = 1 + x + y$.
2. (b) At $(0, 1)$: $\frac{dy}{dx} = 1$ and $\frac{d^2y}{dx^2} = 1 + 0 + 1 = 2 > 0$, so f is concave up at $x = 0$.
3. (c) Critical points of f occur where $\frac{dy}{dx} = 0$, i.e. $x + y = 0$. There $\frac{d^2y}{dx^2} = 1 + 0 = 1 > 0$, so any such point is a *relative minimum*: $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$.
4. (Check with Topic 7.2: $f(x) = 2e^x - x - 1$ satisfies the DE and $f(0) = 1$; its minimum is at $x = -\ln 2$, where $f = \ln 2$.)

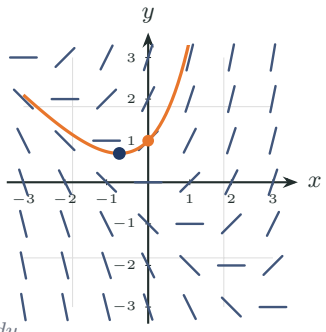


Figure 7.6: $\frac{dy}{dx} = x + y$, $f(0) = 1$; minimum at the navy dot.

Common errors

Forgetting the chain rule when differentiating the DE: $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$, not $2y$. **Using the slope field to decide concavity from a single point:** you need $\frac{d^2y}{dx^2}$ (or several nearby segments). **Claiming a solution crosses an equilibrium line:** a solution that starts above $y = 2$ stays above it. **Giving a limit without a reason:** "the segments level off" is not enough; say why (increasing, bounded above by the equilibrium).

AP Exam Tip

Justification language: "Since $\frac{dy}{dx} = x - y = 0$ at (a, b) and $\frac{d^2y}{dx^2} = 1 - (x - y) > 0$ there, f has a relative minimum at $x = a$." A conclusion without the numerical evidence from the DE earns no credit. A sketch of a solution curve is scored for passing through the given point, being tangent to the field, and having the correct increasing/decreasing and concavity behaviour.

Check yourself 7.4

1. Which of $\frac{dy}{dx} = x - 1$, $\frac{dy}{dx} = y - 1$ has horizontal segments along the line $y = 1$? Give the second derivative of a solution of that equation in terms of y.

2. Let $\dfrac{dy}{dx} = x - y$ and $f(1) = 1$. Is f concave up or concave down at $x = 1$? Give a reason.
3. For $\dfrac{dy}{dx} = 2 - y$ and $f(0) = 5$, find $\lim_{x \to \infty} f(x)$ and say whether f is increasing or decreasing. Justify.

Topic 7.6 General Solutions Using Separation of Variables

Solving a differential equation

Slope fields show the *shape* of solutions; to get a formula you must solve the equation. The technique on the AB exam is *separation of variables*. It applies when the right side can be written as a product (or quotient) of a function of x alone and a function of y alone:

$$\frac{dy}{dx} = g(x)\,h(y).$$

Treat $\dfrac{dy}{dx}$ as a ratio of differentials, move every y to the left with dy and every x to the right with dx, and integrate each side:

$$\int \frac{1}{h(y)}\,dy = \int g(x)\,dx.$$

Each side is an ordinary antiderivative problem from Unit 6. The result is the *general solution*: a whole family of curves, one for each value of the constant C. It is often left as an *implicit* relation between x and y; solve for y explicitly when the algebra allows it.

Why does this work? If $y = f(x)$ is a solution, then $\dfrac{1}{h(f(x))}f'(x) = g(x)$, and the left side is the derivative of $H(f(x))$ where $H' = 1/h$ (chain rule). So $H(f(x))$ and $G(x)$ have the same derivative, and differ only by a constant, which is exactly the equation $H(y) = G(x) + C$.

Separation of variables

1. **Separate:** rewrite as $\dfrac{1}{h(y)}\,dy = g(x)\,dx$ (all y with dy, all x with dx).
2. **Integrate both sides,** and put a *single* constant C on the x-side *at this step*.
3. **Solve for** y if possible. Remember $\displaystyle\int \frac{dy}{y} = \ln|y|$.

Absolute values. From $\ln|y| = g_1(x) + C$ we get $|y| = e^C e^{g_1(x)}$, hence $y = Ae^{g_1(x)}$ where $A = \pm e^C$ is an arbitrary nonzero constant. The constant solution $y = 0$ (lost when you divide by y) is the case $A = 0$.
Check: differentiate your answer and compare with the original equation.

Example 1 A general solution that is a family of curves

Find the general solution of $\dfrac{dy}{dx} = xy$.

1. Separate: $\dfrac{1}{y}\,dy = x\,dx$ (for $y \neq 0$).
2. Integrate: $\ln|y| = \dfrac{x^2}{2} + C$.
3. Exponentiate: $|y| = e^C e^{x^2/2}$, so $y = Ae^{x^2/2}$ with $A = \pm e^C$.
4. $y = 0$ also satisfies the equation ($A = 0$), so the general solution is $\boxed{y = Ae^{x^2/2}}$, any real A.

Figure 7.7 compares this family with the slope field of Figure 7.3: each member follows the segments.

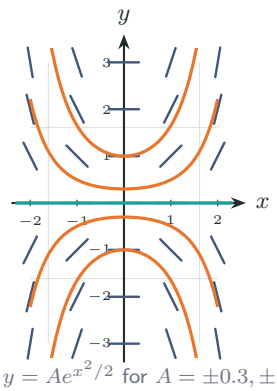


Figure 7.7: $y = Ae^{x^2/2}$ for $A = \pm 0.3, \pm 1$ and $A = 0$.

Example 2 An implicit general solution

Find the general solution of $\dfrac{dy}{dx} = \dfrac{x}{y^2}$.

1. Separate: $y^2\,dy = x\,dx$.

2. Integrate: $\dfrac{y^3}{3} = \dfrac{x^2}{2} + C$.
3. Solve: $y^3 = \dfrac{3x^2}{2} + 3C$, so $y = \sqrt[3]{\dfrac{3x^2}{2} + K}$ where $K = 3C$ is an arbitrary constant.

Check: $y^3 = \frac{3}{2}x^2 + K \Rightarrow 3y^2\dfrac{dy}{dx} = 3x \Rightarrow \dfrac{dy}{dx} = \dfrac{x}{y^2}$. ✓

Example 3 Absolute values and a power function

Find the general solution of $\dfrac{dy}{dx} = \dfrac{2y}{x}$ for $x > 0$.

1. Separate: $\dfrac{dy}{y} = \dfrac{2\,dx}{x}$.
2. Integrate: $\ln|y| = 2\ln x + C = \ln x^2 + C$ (since $x > 0$).
3. Exponentiate: $|y| = e^C x^2$, hence $y = Ax^2$, A any real number (including $A = 0$).

The solutions are parabolas through the origin. Compare with Topic 7.2, Example 3: $x\dfrac{dy}{dx} = 3y$ gave Cx^3.

Common errors

Not separating: writing $y = \int xy\,dx$ or integrating xy with respect to x treating y as constant earns no credit. **Constant of integration omitted** (or added after solving for y): $y = e^{x^2/2} + C$ is wrong; the constant belongs before exponentiating. **Losing the dy:** $\displaystyle\int y^2\,dy = \frac{y^3}{3}$, not $\frac{y^3}{3}$ times x. **Splitting a sum:** $\ln|y+1| \neq \ln|y| + \ln 1$.

AP Exam Tip

On the free-response section, a DE solved by separation is scored with a typical rubric: **1 point** for separating the variables, **1 point** for the antiderivatives (including $+C$), **1 point** for using the initial condition, **1 point** for the solution $y = f(x)$. If you do not separate the variables, you earn *none* of these points, so show the line $\dfrac{1}{y}\,dy = x\,dx$ explicitly.

Check yourself 7.6

1. Find the general solution of $\dfrac{dy}{dx} = 3x^2y$.
2. Find the general solution of $\dfrac{dy}{dx} = \dfrac{e^x}{y^2}$.
3. For $x > 0$, find the general solution of $\dfrac{dy}{dx} = \dfrac{y+1}{x}$.

Topic 7.7 Particular Solutions Using Initial Conditions and Separation of Variables

Choosing one curve from the family

An *initial condition* such as $y(1) = -3$ (or $f(1) = -3$) says that the solution curve passes through the point $(1, -3)$. Substituting that point into the general solution fixes the constant and picks out the one *particular solution*. The exam has a strict order for this work, and following it also prevents most errors: separate, integrate (with $+C$), *substitute the initial condition to find C*, and only then solve for y. Finding C before you solve for y keeps the numbers simple, and it also tells you which sign of a square root to take.

You must also state the *domain*. A particular solution is only valid on the interval containing the initial x-value on which the formula is defined and the derivative exists. So the domain is not decided by the formula alone: it is the largest interval around the initial x-value on which the formula stays a solution (no division by zero, no negative number under a square root).

Particular solution procedure

1. Separate variables and integrate: obtain an equation with $+C$.
2. Substitute $x = a$, $y = b$ from the initial condition $y(a) = b$ to find C.
3. Solve for y, using the initial condition to select the correct root or sign.

4. State the interval of x on which the formula is a solution (containing a).

Example 1 Choosing the sign of a root

Find the particular solution of $\dfrac{dy}{dx} = \dfrac{4x}{y}$ with $y(1) = -3$.

1. Separate: $y\,dy = 4x\,dx$.
2. Integrate: $\dfrac{y^2}{2} = 2x^2 + C$.
3. Use $(1, -3)$: $\dfrac{9}{2} = 2 + C$, so $C = \dfrac{5}{2}$. Hence $y^2 = 4x^2 + 5$.
4. Since $y(1) = -3 < 0$ we take the negative root: $\boxed{y = -\sqrt{4x^2 + 5}}$.
5. $4x^2 + 5 > 0$ for all x, so the solution is defined for all real x.

Example 2 A solution that blows up

Find the particular solution of $\dfrac{dy}{dx} = y^2$ with $y(0) = 1$, and state its domain.

1. Separate: $\dfrac{1}{y^2}\,dy = dx$.
2. Integrate: $-\dfrac{1}{y} = x + C$.
3. Use $(0, 1)$: $-1 = 0 + C$, so $C = -1$.
4. Solve: $-\dfrac{1}{y} = x - 1 \Rightarrow y = \dfrac{1}{1 - x}$.
5. The formula is undefined at $x = 1$. The solution must be defined at $x = 0$ and be continuous, so its domain is $x < 1$: as $x \to 1^-$, $y \to \infty$.

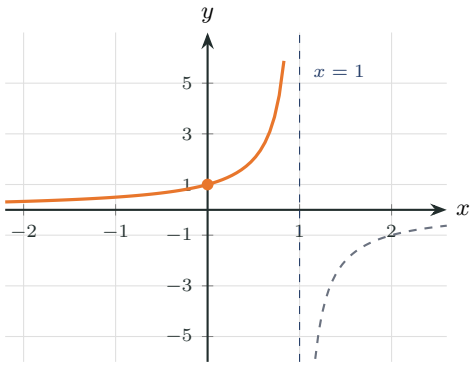


Figure 7.8: only the orange branch (through $(0, 1)$) is the solution.

Example 3 A periodic right side

Find the particular solution of $\dfrac{dy}{dx} = y\cos x$ with $y(0) = 2$, then find $y\left(\frac{\pi}{2}\right)$.

1. Separate: $\dfrac{dy}{y} = \cos x\,dx$ (note $y \neq 0$ since $y(0) = 2$).
2. Integrate: $\ln|y| = \sin x + C$.
3. Use $(0, 2)$: $\ln 2 = 0 + C$, so $C = \ln 2$.
4. Since $y(0) > 0$, $|y| = y$: $y = e^{\sin x + \ln 2} = 2e^{\sin x}$.
5. $y\left(\frac{\pi}{2}\right) = 2e^1 = 2e$.

Common errors

Using the initial condition too early or too late: substituting before integrating loses the equation; solving for y before finding C and then forgetting $\pm$ also loses points. **Dropping the sign:** $y^2 = 4x^2 + 5$ gives two functions; the initial condition $y(1) = -3$ selects the negative one. **Ignoring the domain:** $y = \dfrac{1}{1 - x}$ solves the problem only for $x < 1$. **Reading the condition backwards:** $y(2) = 5$ means $x = 2$, $y = 5$.

AP Exam Tip

A typical full-credit response is five lines: separated equation, antiderivatives with C, the initial-condition substitution, then "$y = \ldots$". Do *not* write $y = \pm\ldots$ in the final answer; use the initial condition to choose one. If the question asks for $f(x)$ "for x in some interval", state the interval; the answer $y = 1/(1 - x)$ is incomplete without "$x < 1$".

Check yourself 7.7

1. Find the particular solution of $\dfrac{dy}{dx} = -\dfrac{x}{y}$ with $y(0) = 2$, and state its domain.

2. Find the particular solution of $\dfrac{dy}{dx} = e^x y^2$ with $y(0) = 1$, and state its domain.
3. Find y if $\dfrac{dy}{dx} = 3y$ and $y(1) = 5$.

Topic 7.8 Exponential Models with Differential Equations

Growth, decay and Newton's law of cooling

The simplest and most important differential equation says that a quantity changes at a rate proportional to its current size:

$$\frac{dy}{dt} = ky.$$

Separating gives $\dfrac{dy}{y} = k\,dt$, then $\ln|y| = kt + C$, then $y = Ae^{kt}$. The initial condition $y(0) = y_0$ gives $A = y_0$, so $y = y_0 e^{kt}$. When $k > 0$ this is exponential *growth*; when $k < 0$ it is exponential *decay*. The constant k is the *relative rate of change* $\dfrac{1}{y}\dfrac{dy}{dt}$: a population growing at "3% per year" has $k = 0.03$.

Two quantities describe the speed of growth or decay without mentioning k: the *doubling time* $t_d = \dfrac{\ln 2}{k}$ (for $k > 0$) and the *half-life* $t_h = \dfrac{\ln 2}{|k|}$ (for $k < 0$). Both come from solving $e^{kt} = 2$ or $e^{kt} = \frac{1}{2}$.

A second model is *Newton's law of cooling*: the temperature T of an object changes at a rate proportional to the difference between its temperature and the temperature T_s of the surroundings,

$$\frac{dT}{dt} = k\,(T - T_s).$$

The substitution $u = T - T_s$ (so $\dfrac{du}{dt} = \dfrac{dT}{dt} = ku$) turns this into the previous model and gives $T = T_s + (T_0 - T_s)e^{kt}$. For a cooling object $k < 0$ and $T \to T_s$ as $t \to \infty$. The same form, $\dfrac{dy}{dt} = k(y - b)$, describes the tank problem of Topic 7.1: it has the equilibrium value $y = b$ and every solution approaches it (if $k < 0$).

Exponential models

- $\dfrac{dy}{dt} = ky,\ y(0) = y_0 \implies y = y_0 e^{kt}$. Growth if $k > 0$, decay if $k < 0$.
- $\dfrac{dy}{dt} = k(y - b),\ y(0) = y_0 \implies y = b + (y_0 - b)e^{kt}$. Equilibrium $y = b$.
- Doubling time $\dfrac{\ln 2}{k}$; half-life $\dfrac{\ln 2}{|k|}$. Neither depends on the amount present.
- To find k: use a second data point, e.g. $y(t_1) = y_1 \Rightarrow k = \dfrac{1}{t_1}\ln\dfrac{y_1}{y_0}$.
- The rate $\dfrac{dy}{dt} = ky$ is proportional to y: it is *largest* when y is largest.

Example 1 Population growth

A bacterial culture has 500 cells at $t = 0$ (hours) and its size $B(t)$ satisfies $\dfrac{dB}{dt} = kB$. After 3 hours there are 2000 cells. (a) Find $B(t)$. (b) When will there be 10,000 cells? (c) How fast is the culture growing at $t = 3$?

1. (a) $B = 500e^{kt}$. From $B(3) = 2000$: $e^{3k} = 4$, $k = \dfrac{\ln 4}{3} \approx 0.462$. So $B(t) = 500\,e^{(\ln 4/3)t} = 500 \cdot 4^{t/3}$.
2. (b) $500{\cdot}4^{t/3} = 10{,}000 \Rightarrow 4^{t/3} = 20 \Rightarrow t = \dfrac{3\ln 20}{\ln 4} \approx 6.483$ hours.
3. (c) $\dfrac{dB}{dt} = kB = 0.4621 \times 2000 \approx 924.2$ cells per hour.

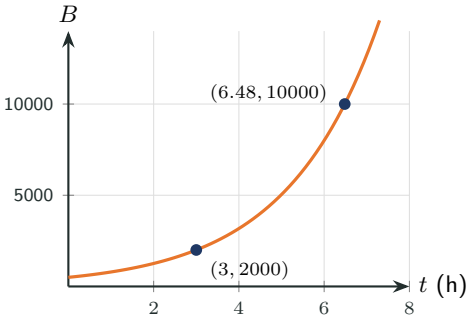


Figure 7.9: $B(t) = 500 \cdot 4^{t/3}$.

Example 2 Radioactive decay and half-life

Carbon-14 has half-life 5730 years. The mass $M(t)$ satisfies $\dfrac{dM}{dt} = kM$. (a) Find k. (b) What fraction remains after 2000 years? (c) A sample has 30% of its original carbon-14. How old is it?

1. (a) $M = M_0 e^{kt}$ and $e^{5730k} = \frac{1}{2}$, so $k = -\dfrac{\ln 2}{5730} \approx -1.210 \times 10^{-4}$ per year.
2. (b) $\dfrac{M(2000)}{M_0} = e^{-2000 \ln 2/5730} = 2^{-2000/5730} \approx 0.785$.
3. (c) $e^{kt} = 0.30 \Rightarrow t = \dfrac{\ln 0.30}{k} = \dfrac{5730 \ln(10/3)}{\ln 2} \approx 9953$ years.

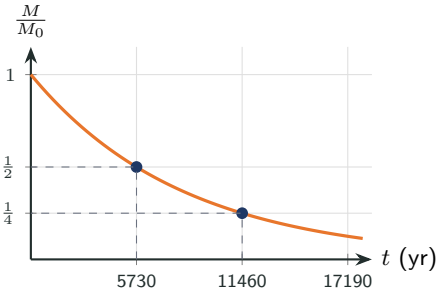


Figure 7.10: each half-life halves the mass.

Example 3 Newton's law of cooling

A cup of coffee at 90°C is placed in a room at 20°C. Its temperature satisfies $\dfrac{dT}{dt} = -0.1\,(T - 20)$, t in minutes. Find $T(t)$ and the time when the coffee reaches 50°C.

1. Separate: $\dfrac{dT}{T - 20} = -0.1\,dt$ (note $T > 20$).
2. Integrate: $\ln(T - 20) = -0.1t + C$, so $T - 20 = Ae^{-0.1t}$.
3. $T(0) = 90$: $A = 70$. So $T(t) = 20 + 70e^{-0.1t}$.
4. $T = 50$: $70e^{-0.1t} = 30 \Rightarrow e^{-0.1t} = \frac{3}{7} \Rightarrow t = 10 \ln \dfrac{7}{3} \approx 8.473$ min.
5. $T \to 20$ as $t \to \infty$: the coffee approaches room temperature but never goes below it.

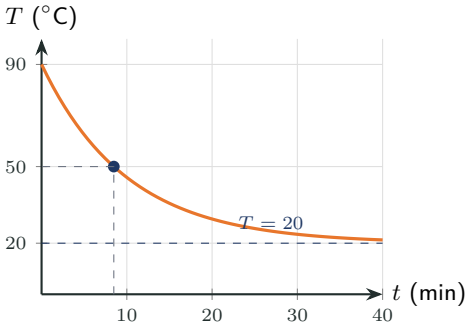


Figure 7.11: cooling toward the room temperature.

Example 4 Back to the tank

In Topic 7.1 the amount of salt satisfied $\dfrac{dA}{dt} = 2 - \dfrac{A}{40} = -\dfrac{1}{40}(A - 80)$. If the tank starts with fresh water, $A(0) = 0$, find $A(t)$ and $A(40)$.

1. This is $\dfrac{dA}{dt} = k(A - b)$ with $k = -\frac{1}{40}$, $b = 80$: $\dfrac{dA}{A - 80} = -\dfrac{1}{40}\,dt$.
2. $\ln|A - 80| = -\dfrac{t}{40} + C$, so $A - 80 = Be^{-t/40}$. With $A(0) = 0$: $B = -80$.
3. $A(t) = 80\left(1 - e^{-t/40}\right)$, and $A(40) = 80(1 - e^{-1}) \approx 50.57$ kg.

Common errors

Using $T = T_0 e^{kt}$ for cooling: the decay is toward the surroundings, not toward 0; use $T = T_s + (T_0 - T_s)e^{kt}$. **Forgetting the sign of k for decay:** half-life 5730 gives $k < 0$. **Using $t_{1/2} = \dfrac{1}{k}$:** the half-life is $\dfrac{\ln 2}{|k|}$. **Rounding k too early:** keep $k = \dfrac{\ln 4}{3}$ exactly until the last step, or three decimal places are not guaranteed.

AP Exam Tip

For every exponential-model question write the differential equation first, then the general solution, then use two conditions (the initial value and one more data point) to find the two constants. If the question says "find y" or "solve", show the separation of variables; the shortcut $y = y_0 e^{kt}$ is acceptable only when you are told the quantity changes at a rate proportional to itself and the problem does not ask you to solve the DE. Always state units and give the three-decimal answer last.

Check yourself 7.8

1. A population satisfies $\dfrac{dP}{dt} = 0.03P$ with $P(0) = 250$. Find $P(10)$.
2. An isotope has half-life 12 days. How long does it take for a sample to fall to $\frac{1}{8}$ of its original mass?
3. An object satisfies $\dfrac{dT}{dt} = -0.2\,(T - 10)$ with $T(0) = 30$. Find $T(5)$.

Unit 7 Practice: Differential Equations

Part A – Multiple Choice, No Calculator

15 questions

Directions: Solve each problem, using the available space for scratch work. Then decide which is the best of the choices given.

1 The rate of change of a population P with respect to time t is proportional to the square root of P. Which of the following is a differential equation for P?

(A) $\dfrac{dP}{dt} = \dfrac{k}{\sqrt{P}}$ **(B)** $\dfrac{dP}{dt} = kP^2$

(C) $\dfrac{dP}{dt} = k\sqrt{P}$ **(D)** $\dfrac{dP}{dt} = \sqrt{P} + k$

2 Which of the following is the solution of the differential equation $\dfrac{dy}{dx} = y - x$ with the initial condition $y(0) = 3$?

(A) $y = e^x + 2$

(B) $y = x + 1 + 2e^x$

(C) $y = x + 3e^x$

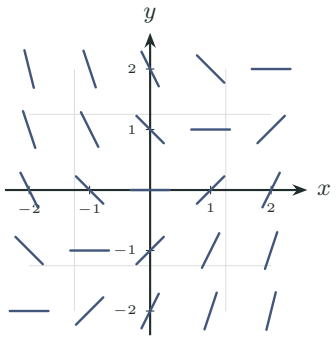
(D) $y = 3e^x$

3 If $\dfrac{dy}{dx} = x^2 - 2y$, what is the slope of the line tangent to the solution curve at the point $(-1, 3)$?

(A) -5 **(B)** -7

(C) 7 **(D)** 11

4 The slope field shown is for which of the following differential equations?



(A) $\dfrac{dy}{dx} = x + y$ **(B)** $\dfrac{dy}{dx} = y - x$

(C) $\dfrac{dy}{dx} = xy$ **(D)** $\dfrac{dy}{dx} = x - y$

5 If $\dfrac{dy}{dx} = y - x$, what is $\dfrac{d^2y}{dx^2}$ at the point $(1, 3)$?

(A) 2 **(B)** 3

(C) 1 **(D)** -1

6 The general solution of $\dfrac{dy}{dx} = x^2 y$ is

(A) $y = Ce^{x^3/3}$ **(B)** $y = \dfrac{x^3}{3} + C$

(C) $y = e^{x^3/3} + C$ **(D)** $y = Ce^{x^3}$

7 If $\dfrac{dy}{dx} = \dfrac{4x}{y}$ and $y(1) = -3$, then $y =$

(A) $\sqrt{4x^2+5}$ **(B)** $-\sqrt{4x^2+9}$

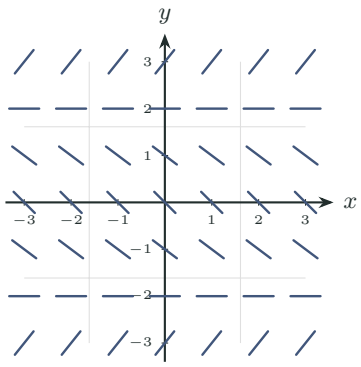
(C) $-\sqrt{2x^2+7}$ **(D)** $-\sqrt{4x^2+5}$

8 If $\dfrac{dy}{dt} = \dfrac{1}{2}y$ and $y(0) = 6$, then $y(2) =$

(A) $6e^2$ **(B)** $6e$

(C) 12 **(D)** $6+e$

9 The slope field for the differential equation $\dfrac{dy}{dx} = \dfrac{y^2-4}{4}$ is shown. If $y = f(x)$ is the solution with $f(0) = 1$, what is $\lim\limits_{x\to\infty} f(x)$?



(A) -2 **(B)** 0 **(C)** 2 **(D)** $-\infty$

10 Let $y = f(x)$ be the solution of $\dfrac{dy}{dx} = xy$ with $f(0) = 1$. Which of the following is true?

(A) f has a relative maximum at $x = 0$.

(B) f has a relative minimum at $x = 0$.

(C) f has neither a relative extremum nor an inflection point at $x = 0$.

(D) f has a point of inflection at $x = 0$.

11 If $\dfrac{dT}{dt} = -\dfrac{1}{2}(T-10)$ and $T(0) = 30$, then $T(2) =$

(A) $30e^{-1}$ **(B)** $20e^{-1}$

(C) $10+20e^{-1}$ **(D)** $10+20e^{-1/2}$

12 The amount $A(t)$ of salt, in kilograms, in a tank satisfies $\dfrac{dA}{dt} = 3 - \dfrac{A}{50}$, with $A(0) = 0$. What is $\lim\limits_{t\to\infty} A(t)$?

(A) 3 **(B)** 50

(C) 0 **(D)** 150

13 If $\dfrac{dy}{dx} = y^2\cos x$ and $y(0) = 1$, then $y\left(\dfrac{\pi}{6}\right) =$

(A) $\dfrac{2}{3}$ **(B)** $\sqrt{e}$

(C) 2 **(D)** $-\dfrac{2}{3}$

14 Which of the following differential equations has $y = e^{-3x}$ as a solution?

(A) $y'' - y' - 6y = 0$ **(B)** $y'' + y' - 6y = 0$

(C) $y'' + y' + 6y = 0$ **(D)** $y'' - 6y = 0$

15 The general solution of $\dfrac{dy}{dx} = \dfrac{y}{x^2}$, for $x > 0$, is

(A) $y = Ce^{-1/x}$ **(B)** $y = Ce^{1/x}$

(C) $y = -\dfrac{1}{x} + C$ **(D)** $y = Ce^{-2/x^3}$

Part B – Multiple Choice, Graphing Calculator Required

7 questions

Directions: A graphing calculator is required for some problems or parts of problems. Unless stated otherwise, give decimal answers to three places.

16 A cup of tea at $95°$C is placed in a room at $20°$C. Its temperature $T(t)$, in °C, satisfies $\dfrac{dT}{dt} = -0.08\,(T - 20)$, where t is in minutes. At what time t is $T = 50$?

(A) 11.454 **(B)** 8.023

(C) 5.068 **(D)** 5.000

17 A colony of bacteria grows at a rate proportional to its size. If the colony doubles every 8 hours, how many hours after the start will it be 5 times its initial size?

(A) 12.876 **(B)** 18.575

(C) 40.000 **(D)** 2.322

18 If $\dfrac{dy}{dx} = xe^{-y}$ and $y(0) = 0$, then $y(2) =$

(A) 0.693 **(B)** 1.609

(C) 2.000 **(D)** 1.099

19 A drug is infused into a patient at 5 mg per hour, and the body removes it at a rate of 10% of the amount present per hour. The amount $Q(t)$, in mg, satisfies $\dfrac{dQ}{dt} = 5 - 0.1Q$ with $Q(0) = 0$. What is $Q(8)$?

(A) 40.000 **(B)** 22.466

(C) 27.534 **(D)** 50.000

20 If $\dfrac{dy}{dx} = 1 + y^2$ and $y(0) = 0$, then $y(1) =$

(A) 0.785 **(B)** 1.000

(C) 0.818 **(D)** 1.557

21 The population $P(t)$ of a town satisfies $\dfrac{dP}{dt} = 0.03P$, where t is in years and $P(0) = 250$ (thousand people). At what rate, in thousands of people per year, is P changing at $t = 10$?

(A) 337.465 **(B)** 10.124

(C) 7.500 **(D)** 75.000

22 A thermometer reading $80°$C is placed in a room at $20°$C. Its temperature satisfies $\dfrac{dT}{dt} = k\,(T - 20)$, where t is in minutes. After 5 minutes the thermometer reads $50°$C. What is the value of k?

(A) 0.139 **(B)** -0.094

(C) -0.100 **(D)** -0.139

Part C – Free Response

2 questions, 9 points each

Directions: Show all of your work, even though a question may not explicitly remind you to do so. Clearly label any functions, graphs, tables or other objects that you use. Your work must be expressed in standard mathematical notation. Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If you use decimal approximations, give three places.

23 **[9 points]** *(Graphing calculator allowed.)* A cup of tea at $90°$C is placed in a room whose temperature is $22°$C. The temperature $T(t)$ of the tea, in degrees Celsius, t minutes after it is placed in the room, satisfies the differential equation

$$\frac{dT}{dt} = -\frac{1}{5}(T - 22), \qquad T(0) = 90.$$

(a) Write an equation for the line tangent to the graph of T at $t = 0$. Use this line to approximate $T(1)$. **[2 points]**

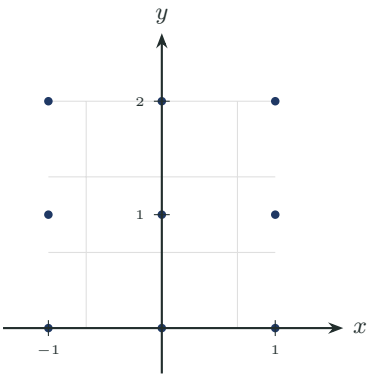
(b) Find $\dfrac{d^2T}{dt^2}$ in terms of T. Is the approximation from part (a) an overestimate or an underestimate of $T(1)$? Give a reason for your answer. **[2 points]**

(c) Use separation of variables to find $T(t)$, the particular solution of the differential equation with $T(0) = 90$. **[4 points]**

(d) Using your solution from part (c), find the time t at which the tea reaches $40°$C. **[1 point]**

24 **[9 points]** *(No calculator.)* Consider the differential equation $\dfrac{dy}{dx} = x\,(y - 1)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated. **[2 points]**



(b) Let $y = f(x)$ be the particular solution of the differential equation with $f(0) = 3$. Find $\dfrac{d^2y}{dx^2}$ in terms of x and y. Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer. **[2 points]**

(c) Find $y = f(x)$, the particular solution of the differential equation with $f(0) = 3$. **[4 points]**

(d) Find $\lim\limits_{x \to \infty} f(x)$. **[1 point]**

AP Calculus AB – Unit 8 | Exam weighting: 10–15% of the AB exam score

Applications of Integration

In this unit

Unit 8 is where the definite integral earns its keep. Having learned in Unit 6 that $\int_a^b f(x)\,dx$ is a limit of Riemann sums and the net change of an antiderivative, you now use it to *measure* things: the average value of a function, how far a particle has moved, how much water is in a tank, the area trapped between two curves, and the volume of a solid. Every application here follows the same recipe—**slice, approximate each slice by a simple piece, add the pieces, and take a limit**—and the sum of slices becomes an integral.

Topics. 8.1 Average value of a function · 8.2 Position, velocity and acceleration · 8.3 Accumulation in applied contexts · 8.4 Area between curves (x) · 8.5 Area between curves (y) · 8.6 Curves that intersect more than twice · 8.7 Cross sections: squares and rectangles · 8.8 Cross sections: triangles and semicircles · 8.9–8.10 Disc method · 8.11–8.12 Washer method. (Topic 8.13, arc length, is BC only and is not covered.)

Big ideas. *Change* and *accumulation*: a rate integrated over an interval gives net change, and a cross-sectional area integrated over a length gives volume. On the exam, expect a rate or area/volume free-response question each year, plus several multiple-choice items.

Topic 8.1 Finding the Average Value of a Function on an Interval

From averaging numbers to averaging a function

The average of n numbers is their sum divided by n. A continuous function takes infinitely many values on $[a,b]$, so we cannot add them one by one—but we can sample. Split $[a,b]$ into n equal pieces of width $\Delta x = \frac{b-a}{n}$ and average the sampled heights $f(x_1),\dots,f(x_n)$:

$$\frac{f(x_1)+\cdots+f(x_n)}{n} = \frac{1}{b-a}\sum_{i=1}^{n} f(x_i)\,\Delta x \;\xrightarrow[n\to\infty]{}\; \frac{1}{b-a}\int_a^b f(x)\,dx.$$

The right-hand side is a Riemann sum divided by the length of the interval, so in the limit the average of *all* the values of f is the integral divided by $b-a$. Geometrically, the average value is the height of the rectangle on $[a,b]$ whose area equals the area under the curve: the part of the graph above that height exactly fills in the part below it.

The *Mean Value Theorem for Integrals* adds that a continuous function actually *attains* its average value somewhere on the interval. Do not confuse the average *value* of f with the average *rate of change* of f. The latter is $\frac{f(b)-f(a)}{b-a}$, which (by the Fundamental Theorem of Calculus) is the average value of the derivative f'. Which of the two a question wants is decided entirely by the wording: "average value of f" needs an integral, "average rate of change of f" needs only two function values.

Average value and the Mean Value Theorem for Integrals

If f is continuous on $[a,b]$, the **average value** of f on $[a,b]$ is

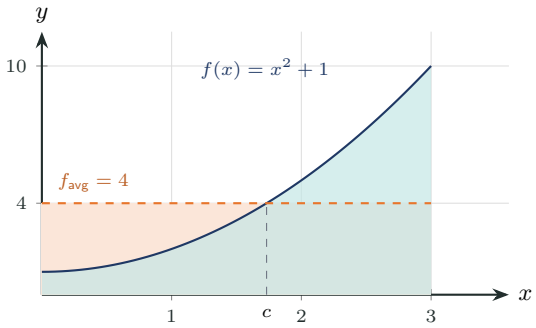
$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

MVT for integrals. There exists c in $[a,b]$ with $f(c) = f_{\text{avg}}$, i.e. $\displaystyle\int_a^b f(x)\,dx = f(c)\,(b-a)$.

Link to rates. Average rate of change of f on $[a,b] = \dfrac{f(b)-f(a)}{b-a} = \dfrac{1}{b-a}\displaystyle\int_a^b f'(x)\,dx$. Units: those of f (not of $\int f$).

Example 1 Average value and the point where it is attained

Let $f(x) = x^2 + 1$ on $[0,3]$. (a) Find the average value of f. (b) Find c with $f(c) = f_{\text{avg}}$.



The orange rectangle (height 4) has the same area, 12, as the region under the curve.

1. $f_{\text{avg}} = \dfrac{1}{3-0}\displaystyle\int_0^3 (x^2+1)\,dx = \frac{1}{3}\left[\frac{x^3}{3} + x\right]_0^3 = \frac{1}{3}(9+3) = 4.$
2. Solve $c^2 + 1 = 4$: $c^2 = 3$, so $c = \sqrt{3} \approx 1.732$ (the root $-\sqrt{3}$ is outside $[0,3]$).

Example 2 Average value from a table (trapezoids)

The temperature $T(t)$, in °C, of a greenhouse is recorded at unequal times. Estimate the average temperature over $0 \le t \le 9$ hours using a trapezoidal sum with the three subintervals shown.

t (hours)	0	3	5	9
$T(t)$ (°C)	12	18	20	14

1. $\displaystyle\int_0^9 T(t)\,dt \approx \tfrac{1}{2}(3)(12+18) + \tfrac{1}{2}(2)(18+20) + \tfrac{1}{2}(4)(20+14) = 45 + 38 + 68 = 151.$
2. $T_{\text{avg}} \approx \dfrac{151}{9} \approx 16.778$ °C. (Divide by the *length of the interval*, 9, not by the number of subintervals.)

Example 3 A substitution inside the average

Find the average value of $f(x) = xe^{x^2}$ on $[0,2]$.

1. $f_{\text{avg}} = \dfrac{1}{2}\displaystyle\int_0^2 xe^{x^2}\,dx$. Let $u = x^2$, $du = 2x\,dx$; limits $0 \to 0$, $2 \to 4$.
2. $\displaystyle\int_0^2 xe^{x^2}\,dx = \frac{1}{2}\int_0^4 e^u\,du = \frac{1}{2}(e^4 - 1)$, so $f_{\text{avg}} = \dfrac{e^4-1}{4} \approx 13.400.$

Common errors

(1) Forgetting the factor $\frac{1}{b-a}$ and reporting the integral itself (this is the area, not the average). **(2)** "Averaging the endpoints," $\frac{f(a)+f(b)}{2}$; this is correct only when f is linear. **(3)** Using $\frac{f(b)-f(a)}{b-a}$ when the question says average *value* of f.

AP Exam Tip

Always write the setup $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$ before evaluating, with the limits and the correct $b-a$. When a calculator question says "find the average value," that setup earns the first point; the decimal answer (three places) earns the second.

Check yourself 8.1.

1. Find the average value of $f(x) = x^3$ on $[0,2]$.
2. f is continuous and $\int_1^5 f(x)\,dx = 14$. Find the average value of f on $[1,5]$.
3. (Calculator) Find every c in $[0,\pi]$ with $\sin c = f_{\text{avg}}$, where f_{avg} is the average value of $\sin x$ on $[0,\pi]$.

Topic 8.2 Connecting Position, Velocity, and Acceleration Using Integrals

Motion along a line

For a particle on a number line with position $x(t)$, velocity is $v(t) = x'(t)$ and acceleration is $a(t) = v'(t)$. Integration runs these relationships backwards: since x is an antiderivative of v, the Fundamental Theorem gives $\int_{t_1}^{t_2} v(t)\,dt = x(t_2) - x(t_1)$, the *net* change in position, called **displacement**. Likewise $\int_{t_1}^{t_2} a(t)\,dt = v(t_2) - v(t_1)$. To recover a *position* (not just a change) you need an initial condition: $x(t) = x(t_1) + \int_{t_1}^{t} v(s)\,ds$.

Displacement and distance are different. If the particle turns around, the part of the motion to the left cancels part of the motion to the right in $\int v\,dt$. **Total distance** is the sum of all the lengths travelled, so it integrates the *speed* $|v(t)|$: find where $v = 0$, split the interval there, and add the absolute values of the pieces. The *average velocity* on $[a, b]$ is displacement$/(b-a)$, i.e. the average value of v; the *average speed* is distance$/(b-a)$.

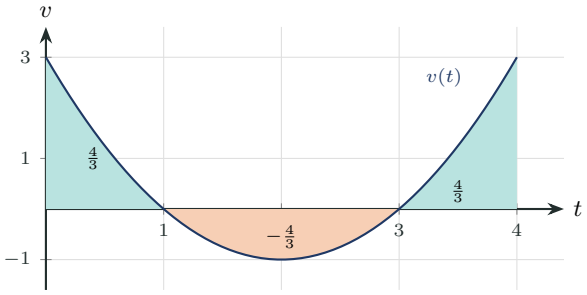
Finally, the particle is *speeding up* when v and a have the same sign and *slowing down* when they have opposite signs. Speed is a magnitude, so a negative velocity with a negative acceleration means the particle is moving left ever faster.

Motion with integrals

Displacement on $[a, b]$	$\displaystyle\int_a^b v(t)\,dt = x(b) - x(a)$
Position at time t	$\displaystyle x(t) = x(a) + \int_a^t v(s)\,ds$
Total distance on $[a, b]$	$\displaystyle\int_a^b \lvert v(t)\rvert\,dt$ (split where $v = 0$)
Change in velocity	$\displaystyle\int_a^b a(t)\,dt = v(b) - v(a)$
Speed	$\lvert v(t)\rvert$; speeding up $\iff v(t)\,a(t) > 0$

Example 1 Displacement versus total distance

A particle moves on a line with velocity $v(t) = t^2 - 4t + 3 = (t-1)(t-3)$ for $0 \le t \le 4$. Find the displacement and the total distance.



Signed areas of v: teal is moving right, orange is moving left.

1. An antiderivative is $V(t) = \frac{t^3}{3} - 2t^2 + 3t$. Displacement $= V(4) - V(0) = \frac{64}{3} - 32 + 12 = \frac{4}{3}$.
2. $v = 0$ at $t = 1, 3$. $\int_0^1 v = V(1) - V(0) = \frac{4}{3}$; $\int_1^3 v = V(3) - V(1) = 0 - \frac{4}{3} = -\frac{4}{3}$; $\int_3^4 v = V(4) - V(3) = \frac{4}{3}$.
3. Total distance $= \left|\frac{4}{3}\right| + \left|-\frac{4}{3}\right| + \left|\frac{4}{3}\right| = 4$.

Example 2 From acceleration to position

A particle has acceleration $a(t) = 2t - 4$, velocity $v(1) = 3$, and position $x(0) = 5$. Find $x(3)$, and decide whether the speed is increasing or decreasing at $t = 1$ and at $t = 3$.

1. $v(t) = \int (2t-4)\,dt = t^2 - 4t + C$; $v(1) = 1 - 4 + C = 3$ gives $C = 6$, so $v(t) = t^2 - 4t + 6$.
2. $x(3) = x(0) + \int_0^3 (t^2 - 4t + 6)\,dt = 5 + \left[\frac{t^3}{3} - 2t^2 + 6t\right]_0^3 = 5 + (9 - 18 + 18) = 14$.
3. $t = 1$: $v(1) = 3 > 0$, $a(1) = -2 < 0$: opposite signs, so the speed is **decreasing**. $t = 3$: $v(3) = 3 > 0$, $a(3) = 2 > 0$: same sign, so the speed is **increasing**.

Example 3 Average velocity and average speed

For the motion in Example 1, find the average velocity and the average speed on $[0, 4]$.

Average velocity $= \frac{1}{4}\int_0^4 v\,dt = \frac{4/3}{4} = \frac{1}{3}$. Average speed $= \frac{1}{4}\int_0^4 |v|\,dt = \frac{4}{4} = 1$. They differ because the particle changes direction.

Common errors

(1) Reporting $\int v\,dt$ when asked for *distance*; split at the zeros of v. **(2)** Reporting displacement when asked for *position*: add $x(a)$. **(3)** Deciding speeding up/slowing down from the sign of a alone: you must compare the signs of v *and* a.

AP Exam Tip

Justify speed with a sentence: "The speed is increasing at $t = 3$ because $v(3)$ and $a(3)$ have the same sign." Write the displacement integral, then the initial position, then the sum.

Check yourself 8.2.

1. $v(t) = 3t^2 - 12$. Find the displacement and the total distance on $[0, 3]$.
2. $a(t) = 4t$, $v(0) = -2$, $x(0) = 3$. Find $x(3)$.
3. $v(t) = t^2 - 4t$. At $t = 1$, is the speed increasing or decreasing? Give a reason.

Topic 8.3 Using Accumulation Functions and Definite Integrals in Applied Contexts

Rates in, rates out, and what accumulates

The idea of Topic 8.2 is not special to motion. Whenever a quantity $Q(t)$ changes at a rate $Q'(t)$ (people per minute, litres per hour, dollars per year), the **net change** in the quantity over $[a, b]$ is $\int_a^b Q'(t)\,dt$ and the amount at time t is

$$Q(t) = Q(a) + \int_a^t Q'(s)\,ds.$$

A definite integral of a rate answers "how much *in total*?"; its units are (units of the rate)×(units of t). On the exam you will be asked to interpret integrals in this way ("$\int_0^{30} R(t)\,dt$ is the total number of cars that entered the garage during the first 30 minutes") and to compute with them.

When a quantity has both an inflow $R_{\text{in}}(t)$ and an outflow $R_{\text{out}}(t)$, the **net rate** is $R_{\text{in}} - R_{\text{out}}$, and the amount present is $Q(0) + \int_0^t (R_{\text{in}} - R_{\text{out}})\,ds$. The amount is *increasing* when the net rate is positive and *decreasing* when it is negative, so a maximum or minimum amount occurs where the net rate changes sign. To find an *absolute* extremum on a closed interval, compare the amounts at each such time *and* at the endpoints (the candidates test of Unit 5).

Accumulation checklist

- Amount at time t = initial amount + $\int$ (rate in − rate out). Do not forget the initial amount.
- $\int_a^b r(t)\,dt$ means *total change* in the quantity from $t = a$ to $t = b$; give units.
- The quantity is increasing where its rate is > 0, decreasing where it is < 0; extrema occur where the rate changes sign (or at endpoints).
- Tables: use left/right/trapezoidal sums to approximate $\int r$; the sums approximate *change*, not amount.

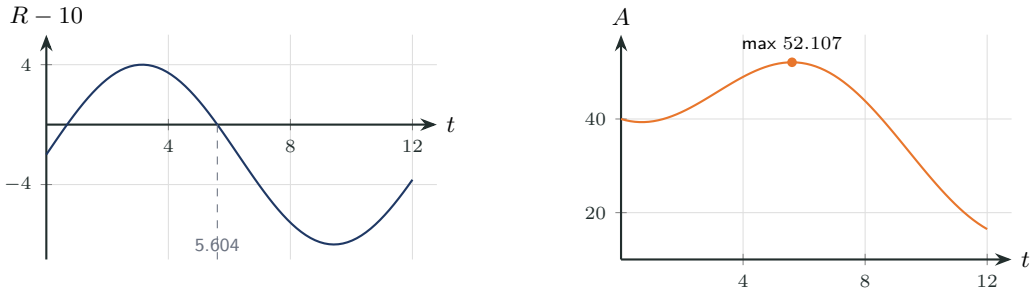
Example 1 A draining tank (no calculator)

The net rate at which the amount of water in a tank changes is $r(t) = 12 - 3t$ gallons per minute for $0 \le t \le 8$. The tank holds 100 gallons at $t = 0$. At what time is the amount of water greatest, and how much is there then?

1. $r(t) = 0$ at $t = 4$; $r > 0$ for $t < 4$ and $r < 0$ for $t > 4$. So the amount is increasing and then decreasing, and has an absolute maximum at $t = 4$ (the amount at $t = 0$ and $t = 8$ is no larger, as the amount at $t = 8$ is $100 + \int_0^8 r = 100 + (96 - 96) = 100$).
2. $A(4) = 100 + \int_0^4 (12 - 3t)\,dt = 100 + \left[12t - \frac{3}{2}t^2\right]_0^4 = 100 + (48 - 24) = 124$ gallons.

Example 2 Inflow and outflow (calculator)

Water flows into a tank at rate $R(t) = 8 + 6\sin\left(\frac{t}{2}\right)$ and leaves at a constant 10 litres per minute, $0 \le t \le 12$. The tank contains 40 litres at $t = 0$. (a) How much water is in the tank at $t = 12$? (b) At what time is the amount greatest?



Left: net rate $R(t) - 10$. Right: amount $A(t)$; it peaks where the net rate changes from $+$ to $-$.

1. Net rate $N(t) = R(t) - 10 = 6\sin\frac{t}{2} - 2$. $A(12) = 40 + \int_0^{12}\left(6\sin\frac{t}{2} - 2\right)dt = 40 + \left[-12\cos\frac{t}{2} - 2t\right]_0^{12} = 28 - 12\cos 6 \approx 16.478$ litres.
2. $N(t) = 0$ when $\sin\frac{t}{2} = \frac{1}{3}$: $t = 2\arcsin\frac{1}{3} \approx 0.680$ and $t = 2\left(\pi - \arcsin\frac{1}{3}\right) \approx 5.604$. N changes from $-$ to $+$ at 0.680 (a local minimum of A) and from $+$ to $-$ at 5.604 (a local maximum).
3. Candidates: $A(0) = 40$, $A(5.604) \approx 52.107$, $A(12) \approx 16.478$. The amount is greatest at $t \approx 5.604$ minutes.

Example 3 Interpreting an integral

Water leaks from a reservoir at rate $r(t)$ million litres per hour. Given $\int_2^8 r(t)\,dt = 35$ and a volume of 90 million litres at $t = 2$, interpret the integral and find the volume at $t = 8$.
The integral is the *total volume that leaked out* between hours 2 and 8 (35 million litres). Because r is a loss rate, the volume at $t = 8$ is $90 - 35 = 55$ million litres.

Common errors

(1) Treating $\int_0^t r(s)\,ds$ as the *amount* instead of the *change* in amount. **(2)** Adding an outflow rate instead of subtracting it. **(3)** Claiming a maximum amount "where the rate is largest"; the amount is largest where the *rate* changes from positive to negative.

AP Exam Tip

Justify an extremum of an accumulated amount with: "$A'(t) = R(t) - 10$ changes from positive to negative at $t = 5.604$, and $A(5.604) > A(0)$ and $A(5.604) > A(12)$, so the amount is greatest at $t = 5.604$." Include units in every interpretation.

Check yourself 8.3.

1. Water leaks from a tank at $L(t) = 12 - 2t$ litres per hour for $0 \le t \le 6$. The tank holds 80 litres at $t = 0$. How much water is in it at $t = 6$?
2. $R(t)$ is the rate, in people per minute, at which people enter a stadium. Explain the meaning and units of $\int_0^{10} R(t)\,dt$.
3. The net rate of change of the amount in a tank is $R(t) = t^2 - 6t + 8$ for $0 < t < 5$. At what time does the amount have a local maximum? Justify.

Topic 8.4 Finding the Area Between Curves Expressed as Functions of x

Top minus bottom

The area under a single curve above the x-axis is $\int_a^b f(x)\,dx$. For a region trapped between two curves, slice it into thin vertical strips of width Δx. A strip at x runs from the lower curve $y = g(x)$ up to the upper curve $y = f(x)$, so its height is $f(x) - g(x)$ and its area is approximately $\big[f(x) - g(x)\big]\Delta x$. Adding the strips and letting $\Delta x \to 0$ gives an integral.

The recipe is: **(1)** sketch (or at least understand) the curves; **(2)** find the x-values that bound the region, which are intersection points (solve $f(x) = g(x)$) or given vertical lines; **(3)** decide which curve is on top *throughout* the interval, by testing a point; **(4)** integrate (top − bottom). This formula works even when part of the region lies below the x-axis: subtracting a negative height *adds* to the strip's length, so no special treatment is needed as long as you take (top − bottom). If the curves cross inside the interval, the roles swap and you must split the integral (Topic 8.6).

Area between curves, vertical strips

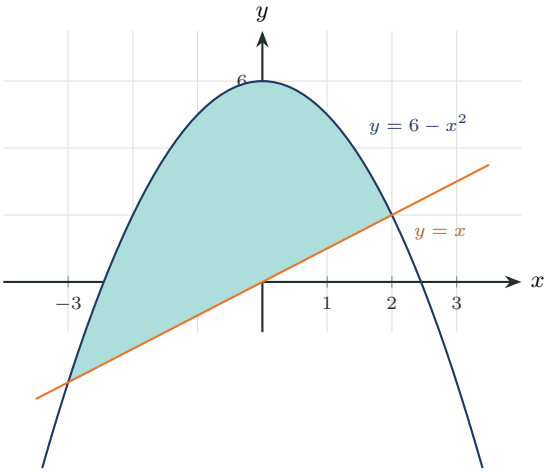
If $f(x) \geq g(x)$ on $[a, b]$, the area of the region bounded by $y = f(x)$, $y = g(x)$, $x = a$, $x = b$ is

$$A = \int_a^b \big[f(x) - g(x)\big]\,dx = \int_a^b (\text{top} - \text{bottom})\,dx.$$

Area is always positive. If the curves cross, use $\displaystyle\int_a^b |f(x) - g(x)|\,dx$ (split at the crossings).

Example 1 A parabola and a line

Find the area of the region enclosed by $y = 6 - x^2$ and $y = x$.



1. Intersections: $6 - x^2 = x \Rightarrow x^2 + x - 6 = 0 \Rightarrow (x+3)(x-2) = 0$, so $x = -3$ and $x = 2$.
2. At $x = 0$: $6 - 0^2 = 6 > 0$, so the parabola is on top.
3. $A = \displaystyle\int_{-3}^{2} \big[(6 - x^2) - x\big]\,dx = \left[6x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-3}^{2} = \frac{22}{3} - \left(-\frac{27}{2}\right) = \frac{125}{6} \approx 20.833.$

Example 2 Bounded by vertical lines

Find the area bounded by $y = e^x$, $y = x$, $x = 0$ and $x = 1$.

On $[0, 1]$, $e^x \geq 1 > x$, so e^x is on top. $A = \displaystyle\int_0^1 (e^x - x)\,dx = \left[e^x - \tfrac{x^2}{2}\right]_0^1 = \left(e - \tfrac{1}{2}\right) - 1 = e - \tfrac{3}{2} \approx 1.218.$

Example 3 The curves cross inside the interval

Find the area of the region bounded by $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$.

1. $\sin x = \cos x$ at $x = \frac{\pi}{4}$. For $0 \leq x < \frac{\pi}{4}$, $\cos x > \sin x$; for $\frac{\pi}{4} < x \leq \frac{\pi}{2}$, $\sin x > \cos x$.

2. $A = \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx$. Each integral equals $\sqrt{2}-1$ (for example $[\sin x + \cos x]_0^{\pi/4} = \sqrt{2}-1$), so $A = 2(\sqrt{2}-1) \approx 0.828$.

Common errors

(1) Writing (bottom − top) and getting a negative "area." **(2)** Integrating $f - g$ over an interval on which the curves cross, so positive and negative parts cancel. **(3)** Taking the limits from the picture's axes instead of the intersection points.

AP Exam Tip

On calculator parts, store the intersection points in A and B (to full calculator precision) and use them as limits; then write $\int_A^B (f(x) - g(x))dx$ with the integrand *identified* ("$f(x) = \ldots$ is above $g(x)$ on $[A, B]$") before the number.

Check yourself 8.4.

1. Find the area enclosed by $y = x^2$ and $y = 8 - x^2$.
2. Find the area enclosed by $y = x + 1$ and $y = x^2 - 1$.

Topic 8.5 Finding the Area Between Curves Expressed as Functions of y

Horizontal strips

Sometimes it is easier—or necessary—to slice the region with *horizontal* strips. A strip at height y runs from the left curve $x = g(y)$ to the right curve $x = f(y)$, has length $f(y) - g(y)$ and thickness Δy, so the area is $\int_c^d [f(y) - g(y)]\,dy$ with limits that are y-values. The rule is the same as before with the roles of the variables swapped: **(right − left)** instead of (top − bottom), and the limits are y-**coordinates**.

Choose horizontal strips when the boundary curves are given as $x = h(y)$ (such as $x = y^2$), or when vertical strips would force you to split the region because the *upper* or *lower* boundary changes formula partway across (while the left and right boundaries do not). To use y as the variable, solve each curve for x; for example $y = \sqrt{x}$ becomes $x = y^2$ (for $y \geq 0$) and $y = \ln x$ becomes $x = e^y$.

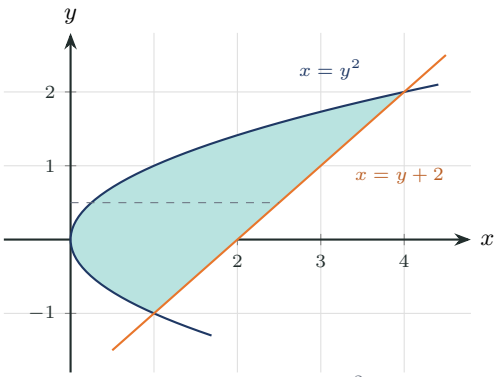
Area between curves, horizontal strips

If $f(y) \geq g(y)$ for $c \leq y \leq d$, the area between $x = f(y)$ (right) and $x = g(y)$ (left) is

$$A = \int_c^d [f(y) - g(y)]\,dy = \int_c^d (\text{right} - \text{left})\,dy.$$

Example 1 Where y-slicing saves work

Find the area of the region bounded by $x = y^2$ and $x = y + 2$.



Right curve $x = y + 2$, left curve $x = y^2$, for $-1 \leq y \leq 2$.

1. Intersections: $y^2 = y + 2 \Rightarrow (y - 2)(y + 1) = 0$, so $y = -1$ and $y = 2$ (points $(1, -1)$ and $(4, 2)$).
2. For $-1 < y < 2$, e.g. $y = 0$: $x = y + 2 = 2$ is to the right of $x = y^2 = 0$.

3. $A = \displaystyle\int_{-1}^{2} \left[(y+2) - y^2\right] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2} = \frac{10}{3} - \left(-\frac{11}{6}\right) = \frac{9}{2}.$
With vertical strips you would need two integrals (x from 0 to 1 with $\pm\sqrt{x}$, then 1 to 4).

Example 2 A logarithmic boundary

Find the area of the region bounded by $y = \ln x$, the y-axis, $y = 0$ and $y = 2$.
Solve $y = \ln x$ for x: $x = e^y$. The right boundary is $x = e^y$ and the left boundary is $x = 0$ (the y-axis), for $0 \le y \le 2$. $A = \displaystyle\int_0^2 e^y\, dy = e^2 - 1 \approx 6.389$. (Slicing vertically would need $\int \ln x$, which also works but is longer.)

Common errors

(1) Using x-values as limits for a dy integral. **(2)** Integrating (top − bottom) with the y-variable still unsolved; the integrand must be a function of y. **(3)** Subtracting in the wrong order: in dy integrals it is right − left.

Check yourself 8.5.

1. Find the area enclosed by $x = y^2$ and $x = 4$.
2. Use horizontal strips to find the area of the region bounded by $y = \sqrt{x}$, $y = x - 2$ and the x-axis.

Topic 8.6 Finding the Area Between Curves That Intersect at More Than Two Points

Where the curves trade places

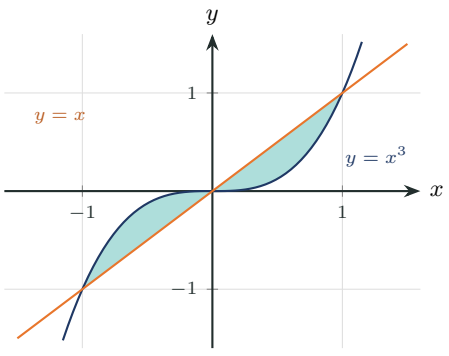
When two curves intersect at three or more points, the curve on top changes at the intermediate crossings, so a single integral of $f - g$ would let positive and negative contributions cancel and understate the area. The cure is mechanical: find *all* intersection points $x_1 < x_2 < \cdots < x_n$ and write one integral per piece, each time subtracting the lower curve from the upper curve *on that piece*:

$$A = \int_{x_1}^{x_2} |f - g|\, dx + \int_{x_2}^{x_3} |f - g|\, dx + \cdots .$$

If the curves are symmetric you may double a half, but only after checking the symmetry. On the calculator section, find the intersections with the graphing calculator and keep full precision.

Example 1 A cubic and its tangent-like line (no calculator)

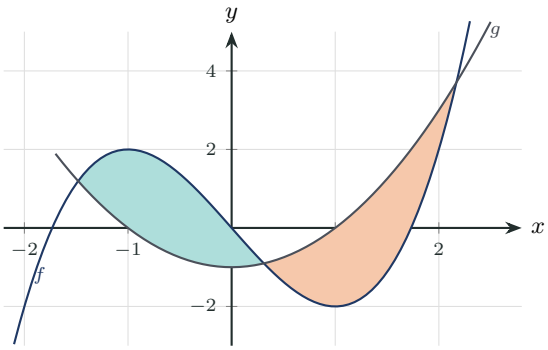
Find the area of the region(s) enclosed by $y = x^3$ and $y = x$.



1. $x^3 = x \Rightarrow x(x-1)(x+1) = 0$: $x = -1, 0, 1$.
2. On $[-1, 0]$, $x^3 \ge x$; on $[0, 1]$, $x \ge x^3$.
3. $A = \displaystyle\int_{-1}^{0} (x^3 - x)\, dx + \int_{0}^{1} (x - x^3)\, dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. (The integral $\int_{-1}^{1}(x - x^3)\, dx$ alone would give 0, which is wrong.)

Example 2 Three crossings found with a calculator

Find the area enclosed by $f(x) = x^3 - 3x$ and $g(x) = x^2 - 1$.



1. Solve $x^3 - 3x = x^2 - 1$, i.e. $x^3 - x^2 - 3x + 1 = 0$, with the calculator: $x_1 \approx -1.481$, $x_2 \approx 0.311$, $x_3 \approx 2.170$.
2. Test points: at $x = 0$, $f = 0 > g = -1$, so f is on top on $[x_1, x_2]$; at $x = 1$, $f = -2 < g = 0$, so g is on top on $[x_2, x_3]$.
3. $A = \int_{x_1}^{x_2} (f - g)\,dx + \int_{x_2}^{x_3} (g - f)\,dx \approx 2.644 + 2.914 = 5.558.$

Common errors

(1) Using the outermost intersections as limits for a single integral of $f - g$. **(2)** Missing a crossing (always check how many real solutions $f = g$ has). **(3)** Using $|f - g|$ with only the *outer* limits and no splitting by hand: on the exam either split, or write $\int_{x_1}^{x_3} |f - g|\,dx$ and evaluate with a calculator.

Check yourself 8.6.

1. Find the total area enclosed by $y = x^3 - 4x$ and the x-axis.
2. Find the total area enclosed by $y = x^3$ and $y = 2x$.

Topic 8.7 Volumes with Cross Sections: Squares and Rectangles

Slice the solid, add up the slices

Area is the integral of lengths; *volume* is the integral of areas. Suppose a solid lies between $x = a$ and $x = b$ and every plane perpendicular to the x-axis cuts it in a region of known area $A(x)$. A thin slab at position x with thickness Δx has volume about $A(x)\,\Delta x$. Adding the slabs and letting $\Delta x \to 0$ gives

$$V = \int_a^b A(x)\,dx.$$

In an exam question the solid is described, not drawn: a *base* is a region in the xy-plane bounded by given curves, and the *cross sections* (perpendicular to the x-axis, or to the y-axis) are squares, rectangles, triangles or semicircles standing up on the base. Your job is to turn the words into a formula for $A(x)$. The key move is always the same: **measure the length of the base segment cut off by the slicing plane**—it is top − bottom for slices perpendicular to the x-axis and right − left for slices perpendicular to the y-axis—call it s, and then express the area of the cross section in terms of s. For a square cross section with a side in the base the area is s^2; for a rectangle with base s and height h it is hs, where h may be a constant or another expression in x (for instance "height three times the base" means $h = 3s$, so $A = 3s^2$).

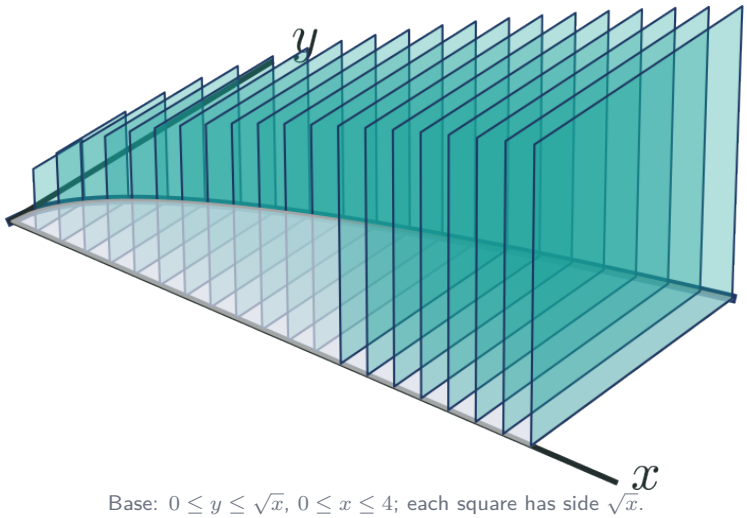
Volume by cross sections

Perpendicular to the x-axis, $a \le x \le b$: $V = \int_a^b A(x)\,dx$. Perpendicular to the y-axis, $c \le y \le d$: $V = \int_c^d A(y)\,dy$.

Let s be the length of the base segment (top − bottom, or right − left). Square: $A = s^2$. Rectangle of height h: $A = hs$. Units of volume are cubic units.

Example 1 Square cross sections

The base of a solid is the region in the first quadrant bounded by $y = \sqrt{x}$, the x-axis and $x = 4$. Each cross section perpendicular to the x-axis is a square with one side in the base. Find the volume.



Base: $0 \le y \le \sqrt{x}$, $0 \le x \le 4$; each square has side $\sqrt{x}$.

1. The side of the square at x is the base segment from $y = 0$ to $y = \sqrt{x}$, so $s = \sqrt{x}$.
2. $A(x) = s^2 = x$.
3. $V = \displaystyle\int_0^4 x\,dx = \left[\frac{x^2}{2}\right]_0^4 = 8.$

Example 2 Rectangles whose height depends on the base

The base is the region enclosed by $y = x$ and $y = x^2$. Cross sections perpendicular to the x-axis are rectangles whose height is 3 times the length of the base. Find the volume.

1. The curves meet where $x = x^2$, i.e. $x = 0, 1$; on $[0, 1]$, $x \ge x^2$, so $s = x - x^2$.
2. Height $= 3s$, so $A(x) = s \cdot 3s = 3(x - x^2)^2$.
3. $V = 3\displaystyle\int_0^1 (x^2 - 2x^3 + x^4)\,dx = 3\left[\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5}\right]_0^1 = 3 \cdot \frac{1}{30} = \frac{1}{10}.$

Example 3 Cross sections perpendicular to the y-axis

The base is the region bounded by $x = y^2$ and $x = 4$. Cross sections perpendicular to the y-axis are squares. Find the volume.

1. The parabola meets $x = 4$ at $y = \pm 2$. A horizontal slice at height y runs from the left curve $x = y^2$ to the right line $x = 4$, so $s = 4 - y^2$.
2. $V = \displaystyle\int_{-2}^{2} (4 - y^2)^2\,dy = 2\int_0^2 (16 - 8y^2 + y^4)\,dy = 2\left[16y - \frac{8y^3}{3} + \frac{y^5}{5}\right]_0^2 = 2 \cdot \frac{256}{15} = \frac{512}{15}.$

Example 4 Estimating a volume from a table

A solid lies between $x = 0$ and $x = 8$ cm. Its cross sections perpendicular to the x-axis are squares whose side $w(x)$ is measured at selected positions:

x (cm)	0	2	5	8
$w(x)$ (cm)	2	4	5	3

Use a trapezoidal sum with the three subintervals to approximate the volume.
The cross-sectional area is $A = w^2$: $4,\ 16,\ 25,\ 9$. Then $V \approx \frac{1}{2}(2)(4 + 16) + \frac{1}{2}(3)(16 + 25) + \frac{1}{2}(3)(25 + 9) = 20 + 61.5 + 51 = 132.5$ cm^3. (Square the widths *first*; trapezoids of w would give the wrong integrand.)

Common errors

(1) Forgetting to square the side: $\int s\,dx$ is an area, not a volume. **(2)** Using the curve $y = f(x)$ as the side when the base is bounded by *two* curves; the side is top − bottom. **(3)** For slices perpendicular to the y-axis, writing the integrand in x or using x-limits.

AP Exam Tip

Write "$A(x) = \ldots$" explicitly, then "$V = \int_a^b A(x)\,dx$" with limits. Graders award one point for the correct integrand and one for the limits and answer, so a missing $A(x)$ or an unsquared side costs the first.

Check yourself 8.7.

1. The base is the triangle bounded by $y = x$, the x-axis and $x = 3$. Cross sections perpendicular to the x-axis are squares. Find the volume.
2. The base is the region enclosed by $y = x$ and $y = x^2$. Cross sections perpendicular to the x-axis are rectangles of height 5. Find the volume.
3. $w(x)$ is the side of a square cross section: $w(0) = 1,\ w(1) = 3,\ w(3) = 4,\ w(4) = 2$. Use a trapezoidal sum with three subintervals to approximate the volume for $0 \le x \le 4$.

Topic 8.8 Volumes with Cross Sections: Triangles and Semicircles

Other shapes, same method

The only new work is the area formula. When the cross section is a triangle or semicircle standing on a base segment of length s, express its area in terms of s using geometry: the height of an equilateral triangle on side s is $\frac{\sqrt{3}}{2}s$, so its area is $\frac{1}{2}\cdot s\cdot\frac{\sqrt{3}}{2}s = \frac{\sqrt{3}}{4}s^2$; a semicircle whose *diameter* lies in the base has radius $\frac{s}{2}$ and area $\frac{1}{2}\pi\left(\frac{s}{2}\right)^2 = \frac{\pi}{8}s^2$. For an isosceles right triangle you must be told whether the *hypotenuse* or a *leg* lies in the base: with hypotenuse s the legs are $\frac{s}{\sqrt{2}}$ and the area is $\frac{s^2}{4}$, with a leg s the area is $\frac{s^2}{2}$.

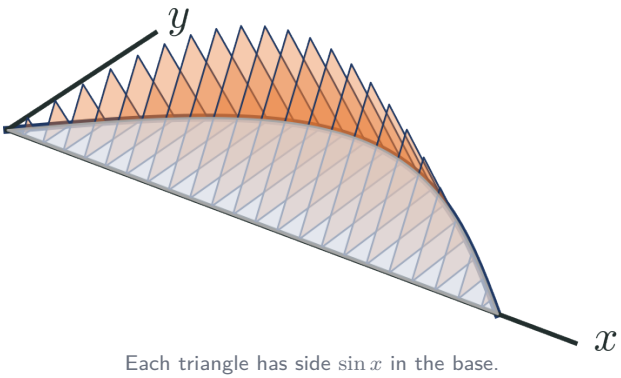
Reading the wording carefully is half of the problem. "A semicircle whose diameter lies in the base" uses s as the diameter, not the radius. "Whose height is half the base" makes the height $\frac{s}{2}$. Once A is a function of s, and s a function of the variable of integration, the integral is routine.

Areas of common cross sections in terms of the base length s

Square (side s)	s^2
Equilateral triangle (side s)	$\dfrac{\sqrt{3}}{4}s^2$
Isosceles right triangle, hypotenuse s in base	$\dfrac{s^2}{4}$
Isosceles right triangle, leg s in base	$\dfrac{s^2}{2}$
Semicircle with diameter s in base	$\dfrac{\pi}{8}s^2$

Example 1 Equilateral triangles

The base is the region bounded by $y = \sin x$ and the x-axis for $0 \le x \le \pi$. Cross sections perpendicular to the x-axis are equilateral triangles with one side in the base. Find the volume.

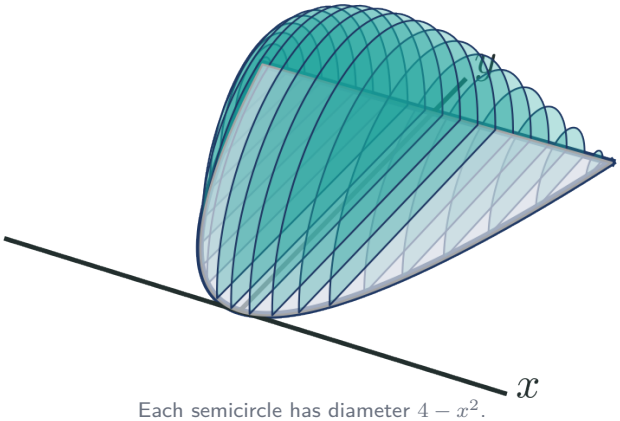


Each triangle has side $\sin x$ in the base.

1. $s = \sin x - 0 = \sin x$ and $A(x) = \frac{\sqrt{3}}{4}\sin^2 x$.
2. $V = \dfrac{\sqrt{3}}{4}\displaystyle\int_0^{\pi}\sin^2 x\,dx = \frac{\sqrt{3}}{4}\int_0^{\pi}\frac{1-\cos 2x}{2}\,dx = \frac{\sqrt{3}}{4}\left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\pi} = \frac{\sqrt{3}}{4}\cdot\frac{\pi}{2} = \frac{\sqrt{3}\,\pi}{8} \approx 0.680.$

Example 2 Semicircles

The base is the region enclosed by $y = x^2$ and $y = 4$. Cross sections perpendicular to the x-axis are semicircles whose diameters lie in the base. Find the volume.



Each semicircle has diameter $4 - x^2$.

1. The curves meet at $x = \pm 2$. The diameter is $d = 4 - x^2$, so the radius is $\frac{4-x^2}{2}$ and $A(x) = \frac{1}{2}\pi\left(\frac{4-x^2}{2}\right)^2 = \frac{\pi}{8}(4-x^2)^2$.
2. $V = \dfrac{\pi}{8}\displaystyle\int_{-2}^{2}(4-x^2)^2\,dx = \frac{\pi}{8}\cdot\frac{512}{15} = \frac{64\pi}{15} \approx 13.404.$

Example 3 Isosceles right triangles with the hypotenuse in the base

The base is the disc $x^2 + y^2 \le 9$. Cross sections perpendicular to the x-axis are isosceles right triangles whose hypotenuse lies in the base. Find the volume.

1. At x the base chord runs from $y = -\sqrt{9-x^2}$ to $y = \sqrt{9-x^2}$, so $s = 2\sqrt{9-x^2}$.
2. $A = \dfrac{s^2}{4} = \dfrac{4(9-x^2)}{4} = 9 - x^2.$
3. $V = \displaystyle\int_{-3}^{3}(9-x^2)\,dx = 2\left[9x - \frac{x^3}{3}\right]_0^3 = 2(27-9) = 36.$

Common errors

(1) Using s as the *radius* of a semicircle when it is the diameter (a factor of 4 error). **(2)** Using $\frac{1}{2}s^2$ for a right triangle with the hypotenuse in the base. **(3)** Taking the triangle's height to be s for an equilateral triangle; the height is $\frac{\sqrt{3}}{2}s$.

AP Exam Tip

Draw one typical cross section and *label its dimensions in terms of* x before writing the area. The setup $V = \int_a^b A(x)\,dx$ with $A(x)$ shown is where the points are.

Check yourself 8.8.

1. The base is bounded by $y = \sqrt{x}$, the x-axis and $x = 4$. Cross sections perpendicular to the x-axis are semicircles whose diameters lie in the base. Find the volume.
2. The base is enclosed by $y = x$ and $y = x^2$. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume.
3. The base is enclosed by $x = y^2$ and $x = y + 2$. Cross sections perpendicular to the y-axis are semicircles with diameters in the base. Write and evaluate the volume.

Topic 8.9 Volume with the Disc Method: Revolving Around the x- or y-Axis

A solid of revolution is a stack of discs

Spin a plane region about a line that borders it and you sweep out a *solid of revolution*. Slice that solid with a plane perpendicular to the axis of revolution: every slice is a circular **disc**. If the slice at x has radius $R(x)$, its area is $\pi R(x)^2$ and the volume is $\int \pi R(x)^2\,dx$. The radius is the distance from the axis to the curve.

When the region lies between the curve $y = f(x)$ and the x-axis, $a \le x \le b$, and is revolved about the x-axis, the radius is $R = f(x)$. When the region lies between the curve $x = g(y)$ and the y-axis, $c \le y \le d$, and is revolved about the y-axis, the slices are horizontal, the radius is $R = g(y)$ and the variable of integration is y. In both cases the slicing is *perpendicular to the axis*, so the integration variable is the coordinate that runs along the axis—x for the x-axis, y for the y-axis. If a curve is given as $y = h(x)$ and the axis is the y-axis, first solve for x in terms of y.

Disc method

Revolving about the x-axis the region between $y = f(x)$ and the axis, $a \le x \le b$:

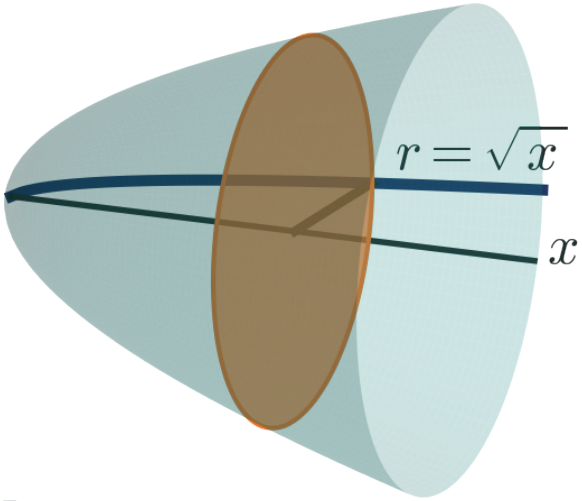
$$V = \int_a^b \pi\big[f(x)\big]^2\,dx.$$

Revolving about the y-axis the region between $x = g(y)$ and the axis, $c \le y \le d$: $\quad V = \displaystyle\int_c^d \pi\big[g(y)\big]^2\,dy.$

The square is on the *radius*: $\pi \int f^2$, never $\pi\big(\int f\big)^2$ or $\pi \int f$.

Example 1 Revolving about the x-axis

The region bounded by $y = \sqrt{x}$, the x-axis and $x = 4$ is revolved about the x-axis. Find the volume.

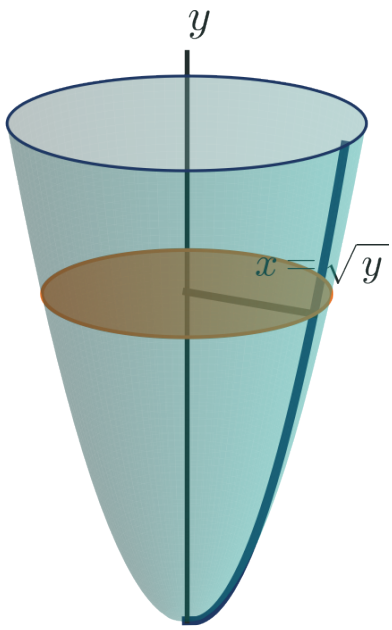


A disc of radius $\sqrt{x}$ at position x (orange); the solid is a paraboloid-shaped bowl lying on its side.

1. The radius of the disc at x is $R = \sqrt{x}$ and its area is $\pi(\sqrt{x})^2 = \pi x$.
2. $V = \pi \displaystyle\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi \approx 25.133.$

Example 2 Revolving about the y-axis

The region in the first quadrant bounded by $y = x^2$, the y-axis and $y = 9$ is revolved about the y-axis. Find the volume.



Horizontal disc of radius $\sqrt{y}$ at height y.

1. Solve for x: $x = \sqrt{y}$ (first quadrant). The radius at height y is $R = \sqrt{y}$, $0 \le y \le 9$.
2. $V = \pi \displaystyle\int_0^9 y\,dy = \pi \cdot \frac{81}{2} = \frac{81\pi}{2} \approx 127.235.$

Example 3 A calculator volume

The region bounded by $y = e^{-x^2}$, the x-axis, $x = 0$ and $x = 1$ is revolved about the x-axis. Find the volume to three decimal places.

$V = \pi \displaystyle\int_0^1 \left(e^{-x^2}\right)^2 dx = \pi \int_0^1 e^{-2x^2} dx \approx \pi(0.5981) \approx 1.879.$ (There is no elementary antiderivative, so the integral is evaluated numerically; the setup is what earns the first point.)

Common errors

(1) Omitting the π, or computing $\pi \int R\,dx$ with the radius not squared. **(2)** Squaring *after* integrating. **(3)** Revolving about the y-axis but integrating dx with x-limits.

AP Exam Tip

Write the three ingredients explicitly: "radius $R = \ldots$," the integral $\pi \int_a^b R^2\,dx$ (or dy) with limits, then the value. Make the limits the *values of the variable that runs along the axis*.

Check yourself 8.9.

1. The region bounded by $y = x$, the x-axis and $x = 3$ is revolved about the x-axis. Find the volume.
2. The region bounded by $y = x^3$, the y-axis and $y = 8$ is revolved about the y-axis. Find the volume.
3. The region bounded by $y = \sin x$ and the x-axis, $0 \le x \le \pi$, is revolved about the x-axis. Find the volume.

Topic 8.10 Volume with the Disc Method: Revolving Around Other Axes

When the axis is a line like $y = 4$ or $x = 2$

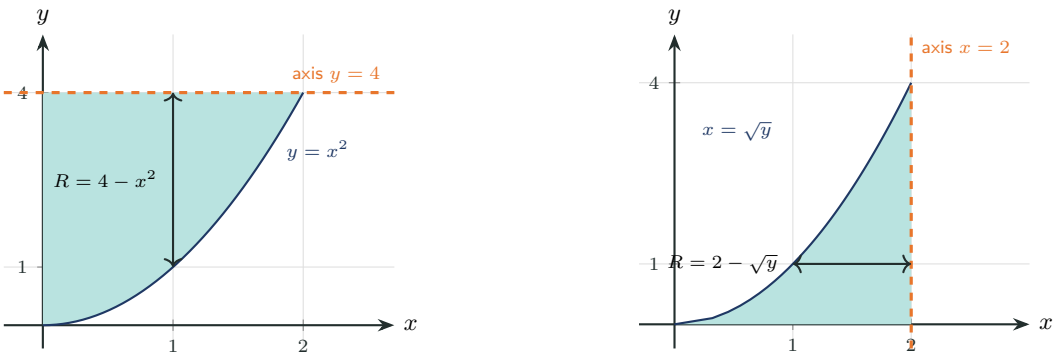
The disc method works around any horizontal or vertical line, provided the region *touches* that line (so there is no hole). Only the radius changes: it is the **distance from the axis of revolution to the curve**, measured along the slice. For an axis $y = k$ the radius is $|k - f(x)|$ and you integrate dx; for an axis $x = k$ the slices are horizontal, the radius is $|k - g(y)|$ and you integrate dy. Because the radius is squared the absolute value is unnecessary in the integrand, but you must still get the *distance* right: for the region under $y = 4$ revolved about $y = 4$, the radius is $4 - f(x)$, not $f(x)$.

Disc method about $y=k$ or $x=k$ (region touching the axis)

Axis $y=k$: $V=\displaystyle\int_a^b \pi\big[k-f(x)\big]^2\,dx$. Axis $x=k$: $V=\displaystyle\int_c^d \pi\big[k-g(y)\big]^2\,dy$.
Radius = distance from the axis to the boundary curve; the limits are those of the variable along the axis direction.

Example 1 Revolving about $y=4$

The region bounded by $y=x^2$, $y=4$ and the y-axis ($x\ge 0$) is revolved about the line $y=4$. Find the volume.



Left: Example 1 (axis $y=4$). Right: Example 2 (axis $x=2$). The radius is the length of the double arrow.

1. The curve $y=x^2$ meets $y=4$ at $x=2$. The region touches the axis along $y=4$, so the slices are discs with radius $R=4-x^2$.
2. $V=\pi\displaystyle\int_0^2 (4-x^2)^2dx=\pi\int_0^2 (16-8x^2+x^4)\,dx=\pi\left[16x-\frac{8x^3}{3}+\frac{x^5}{5}\right]_0^2=\frac{256\pi}{15}\approx 53.617.$

Example 2 Revolving about a vertical line $x=2$

The region in the first quadrant bounded by $y=x^2$, the x-axis and $x=2$ is revolved about the line $x=2$. Find the volume.

1. The axis is vertical, so slice horizontally and integrate in y. Solve $y=x^2$ for $x=\sqrt{y}$; the region runs from $y=0$ to $y=4$ and touches $x=2$.
2. The radius is the distance from the axis to the curve: $R=2-\sqrt{y}$.
3. $V=\pi\displaystyle\int_0^4 (2-\sqrt{y})^2dy=\pi\int_0^4 (4-4\sqrt{y}+y)dy=\pi\left[4y-\frac{8}{3}y^{3/2}+\frac{y^2}{2}\right]_0^4=\pi\left(16-\frac{64}{3}+8\right)=\frac{8\pi}{3}\approx 8.378.$

Common errors

(1) Using $f(x)$ as the radius when the axis is not the x-axis. **(2)** Expanding $(4-x^2)^2$ as $16-x^4$ (the middle term is $-8x^2$). **(3)** Using x-limits with a vertical axis of revolution after slicing horizontally.

AP Exam Tip

A sketch with the axis drawn and one radius labelled prevents almost every error here. Write the radius as "*farther boundary − nearer boundary*" measured from the axis, e.g. $R=2-\sqrt{y}$.

Check yourself 8.10.

1. The region bounded by $y=x^2$ and $y=1$ ($x\ge 0$) is revolved about the line $y=1$. Find the volume.
2. The region bounded by $y=x^2$, the x-axis and $x=1$ is revolved about the line $x=1$. Find the volume.

Topic 8.11 Volume with the Washer Method: Revolving Around the x- or y-Axis

Discs with a hole

If the region is separated from the axis by a gap—it lies between two curves, neither of which is the axis—the slice is a **washer**: a disc with a circular hole. A washer with outer radius R and inner radius r has area $\pi R^2-\pi r^2=\pi\big(R^2-r^2\big)$. The outer radius is the distance from the axis to the *farther* boundary and the inner radius the distance to the *nearer* boundary.

Washer method

Revolving about the x-axis the region between $y = f(x)$ (outer) and $y = g(x)$ (inner), $0 \le g \le f$ on $[a, b]$:

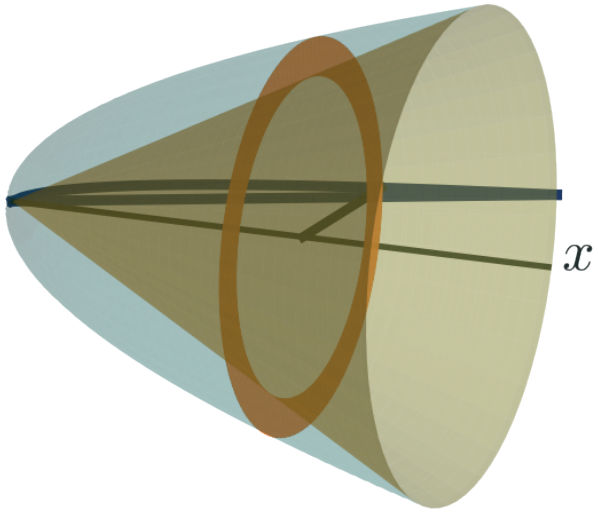
$$V = \pi \int_a^b \Big(\big[f(x)\big]^2 - \big[g(x)\big]^2 \Big) dx.$$

About the y-axis, with $x = f(y)$ outer and $x = g(y)$ inner: $V = \pi \displaystyle\int_c^d \Big(\big[f(y)\big]^2 - \big[g(y)\big]^2 \Big) dy.$

Square each radius, then subtract: $\pi \int (R^2 - r^2)$, not $\pi \int (R - r)^2$.

Example 1 Washers about the x-axis

The region between $y = \sqrt{x}$ and $y = \dfrac{x}{2}$ is revolved about the x-axis. Find the volume.



The solid is a bowl ($y = \sqrt{x}$) with a cone removed ($y = x/2$); each slice is a washer.

1. $\sqrt{x} = \frac{x}{2} \Rightarrow x = \frac{x^2}{4} \Rightarrow x = 0$ or $x = 4$. On $(0, 4)$, $\sqrt{x} > \frac{x}{2}$ (test $x = 1$: $1 > \frac{1}{2}$), so $R = \sqrt{x}$ and $r = \frac{x}{2}$.
2. $V = \pi \displaystyle\int_0^4 \left(x - \frac{x^2}{4} \right) dx = \pi \left[\frac{x^2}{2} - \frac{x^3}{12} \right]_0^4 = \pi \left(8 - \frac{16}{3} \right) = \frac{8\pi}{3} \approx 8.378.$

Example 2 Washers about the y-axis

The region in the first quadrant between $y = x$ and $y = x^2$ is revolved about the y-axis. Find the volume.

1. Slice horizontally. Solve for x: $y = x \Rightarrow x = y$ and $y = x^2 \Rightarrow x = \sqrt{y}$. For $0 \le y \le 1$, $\sqrt{y} \ge y$, so the outer radius is $R = \sqrt{y}$ and the inner radius is $r = y$.
2. $V = \pi \displaystyle\int_0^1 (y - y^2) dy = \pi \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = \frac{\pi}{6}.$

Example 3 The square-then-subtract rule

Find the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

The curves meet at $x = 0, 2$ and $2x \ge x^2$ between them. $V = \pi \displaystyle\int_0^2 \big[(2x)^2 - (x^2)^2\big] dx = \pi \int_0^2 (4x^2 - x^4)\, dx = \pi \left(\frac{32}{3} - \frac{32}{5} \right) = \frac{64\pi}{15} \approx 13.404$. A common error is $\pi \int (2x - x^2)^2 dx = \frac{16\pi}{15}$, which is the volume of a solid whose radius is the *gap*, not of the washer.

Common errors

(1) $\pi \int (R - r)^2$ instead of $\pi \int (R^2 - r^2)$. **(2)** Swapping outer and inner radius (a negative volume signals this). **(3)** About the y-axis, leaving R and r in terms of x.

AP Exam Tip

Label the outer and inner radius separately on your sketch and write "$R = \ldots$, $r = \ldots$" before the integral. The scoring guide usually gives one point for the integrand in the form $\pi(R^2 - r^2)$ and one for limits and answer.

Check yourself 8.11.

1. The region between $y = x$ and $y = x^3$ for $0 \le x \le 1$ is revolved about the x-axis. Find the volume.
2. The region between $y = \sqrt{x}$ and $y = x$ is revolved about the y-axis. Find the volume.
3. The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Find the volume.

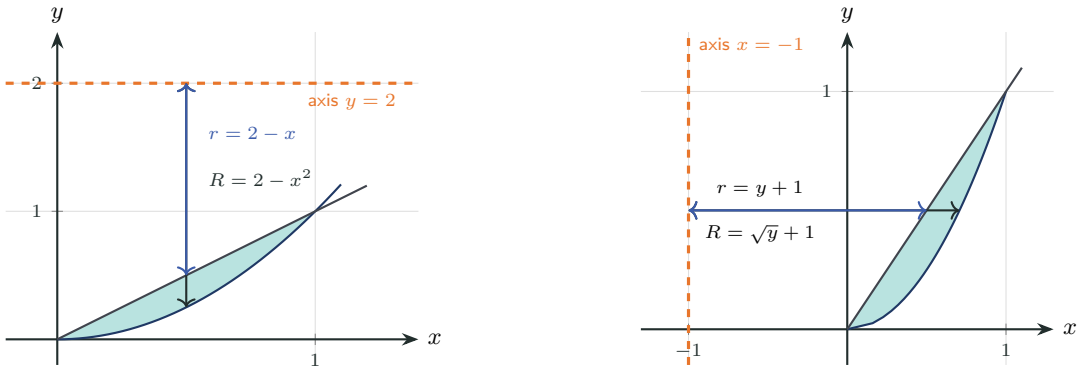
Topic 8.12 Volume with the Washer Method: Revolving Around Other Axes

Measure both radii from the new axis

For an axis $y = k$ or $x = k$ that does not touch the region, both radii are distances to the new axis. If the axis lies *below* (or to the left of) the region, the outer radius is $f(x) - k$ for the upper boundary f and the inner is $g(x) - k$ for the lower boundary g. If the axis lies *above* (or to the right of) the region, as in Example 1, the outer radius goes to the *lower* boundary: $R = k - g(x)$ and $r = k - f(x)$. In every case ask: *which boundary is farther from the axis?*

Washer method about $y = k$ or $x = k$

$V = \pi \int (R^2 - r^2)\, d(\text{variable along the axis})$, with $R =$ distance from the axis to the **farther** boundary and $r =$ distance to the **nearer** boundary.
Horizontal axis $y = k$: integrate dx. Vertical axis $x = k$: integrate dy and write the boundaries as $x = \cdots$.



Left: Example 1 (axis $y = 2$ above the region). Right: Example 2 (axis $x = -1$ to its left).

Example 1 Washers about $y = 2$

The region between $y = x$ and $y = x^2$ $(0 \le x \le 1)$ is revolved about the line $y = 2$. Find the volume.

1. The axis is above the region. The boundary farther from $y = 2$ is the lower curve $y = x^2$, so $R = 2 - x^2$; the nearer is $y = x$, so $r = 2 - x$.
2. $V = \pi \displaystyle\int_0^1 \left[(2 - x^2)^2 - (2 - x)^2\right] dx$. Expanding, $(2 - x^2)^2 - (2 - x)^2 = x^4 - 5x^2 + 4x$, so $V = \pi \left[\frac{x^5}{5} - \frac{5x^3}{3} + 2x^2\right]_0^1 = \pi\left(\frac{1}{5} - \frac{5}{3} + 2\right) = \frac{8\pi}{15} \approx 1.676$.

Example 2 Washers about $x = -1$

The region between $y = x$ and $y = x^2$ is revolved about the line $x = -1$. Find the volume.

1. Vertical axis: slice horizontally. Left boundary $x = y$, right boundary $x = \sqrt{y}$, $0 \le y \le 1$. The axis is to the left of the region, so the outer radius is to the right boundary: $R = \sqrt{y} + 1$ and $r = y + 1$.
2. $V = \pi \displaystyle\int_0^1 \left[(\sqrt{y} + 1)^2 - (y + 1)^2\right] dy = \pi \int_0^1 (2\sqrt{y} - y - y^2)\, dy = \pi\left(\frac{4}{3} - \frac{1}{2} - \frac{1}{3}\right) = \frac{\pi}{2} \approx 1.571$.

Example 3 Write an integral expression and evaluate

The region between $y = \sqrt{x}$ and $y = \dfrac{x}{2}$ is revolved about the line $y = -1$. Write an integral expression for the volume and evaluate it.

The axis is below the region. Distances from $y = -1$: outer (upper curve) $R = \sqrt{x} + 1$, inner (lower curve) $r = \frac{x}{2} + 1$. $V = \pi \displaystyle\int_0^4 \left[(\sqrt{x}+1)^2 - \left(\tfrac{x}{2}+1\right)^2\right] dx = \pi \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx = \pi\left(\frac{32}{3} - \frac{16}{3}\right) = \frac{16\pi}{3} \approx 16.755.$

Which volume method?

Cross sections of known shape	$V = \int A\, d(\cdot)$; A from the base length s
Region touches the axis	discs: $V = \pi \int R^2$, $R =$ distance from axis to curve
Gap between region and axis	washers: $V = \pi \int (R^2 - r^2)$, R farther, r nearer
Variable of integration	along the axis: x for a horizontal axis, y for a vertical axis

Common errors

(1) Measuring a radius from the x-axis instead of from $y = k$. **(2)** Using $r = k - f$ and $R = k - g$ in the wrong order when the axis is above the region (the farther boundary is the *lower* curve). **(3)** Forgetting that $(2 - x^2)^2 \neq 4 - x^4$.

AP Exam Tip

Mark the axis, then a typical slice, then *both* radii on your own sketch. A "write, but do not evaluate" part still needs the correct limits, π, and each radius squared.

Check yourself 8.12.

1. The region between $y = x$ and $y = x^2$ is revolved about the line $y = -1$. Find the volume.
2. The region between $x = y^2$ and $x = y$ is revolved about the line $x = -2$. Find the volume.

Unit 8 Practice – Part A: Multiple Choice

No calculator · 14 questions

1 What is the average value of $f(x) = 3x^2 - 2x$ on the interval $[1, 3]$?

(A) 10 **(B)** 9
(C) 11 **(D)** 18

2 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ for $0 \le t \le 3$. What is the total distance traveled by the particle?

(A) -4 **(B)** 0
(C) 4 **(D)** 8

3 What is the area of the region enclosed by the graphs of $y = x^2$ and $y = x + 2$?

(A) $\frac{9}{2}$ **(B)** $\frac{10}{3}$
(C) $-\frac{9}{2}$ **(D)** $\frac{21}{2}$

4 The region bounded by $y = x^2$, the x-axis and $x = 2$ is revolved about the x-axis. What is the volume of the solid generated?

(A) $\frac{8\pi}{3}$ **(B)** 32π
(C) $\frac{32\pi}{5}$ **(D)** $\frac{32}{5}$

5 The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the volume of the solid generated?

(A) $\frac{\pi}{30}$ **(B)** $\frac{2\pi}{15}$
(C) $\frac{\pi}{6}$ **(D)** $\frac{\pi}{3}$

6 The base of a solid is the region enclosed by $y = x^2$ and $y = 4$. Cross sections of the solid perpendicular to the y-axis are squares. What is the volume of the solid?

(A) 8 **(B)** $\frac{32}{3}$

(C) $\frac{512}{15}$ **(D)** 32

7 The rate $R(t)$, in people per minute, at which people enter a theatre is given by a differentiable function. Selected values are shown.

t (minutes)	0	3	5	9
$R(t)$	4	6	5	7

Using a trapezoidal sum with the three subintervals indicated by the table, what is the approximation of the number of people who enter during $0 \le t \le 9$?

(A) 44 **(B)** 56

(C) 50 **(D)** 100

8 Tickets for a concert are sold at a rate of $R(t)$ tickets per hour, t hours after sales open. What is the meaning of $\int_2^5 R(t)\,dt$?

(A) The total number of tickets sold between hour 2 and hour 5

(B) The average number of tickets sold per hour between hour 2 and hour 5

(C) The number of tickets sold at hour 5

(D) The rate at which tickets are being sold at hour 5, minus the rate at hour 2

9 If $\int_1^4 f(x)\,dx = 12$, what is the average value of $3f(x) + 2$ on $[1, 4]$?

(A) $\frac{38}{3}$ **(B)** 12

(C) 14 **(D)** 38

10 What is the area of the region in the first quadrant bounded by $y = \sqrt{x}$, $y = x - 2$ and the x-axis?

(A) $\frac{16}{3}$ **(B)** $\frac{10}{3}$

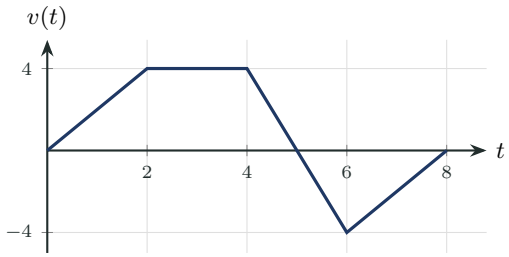
(C) $\frac{8}{3}$ **(D)** $\frac{26}{3}$

11 The region bounded by $y = \sqrt{x}$, the x-axis and $x = 4$ is revolved about the line $y = -1$. What is the volume of the solid generated?

(A) $\frac{68\pi}{3}$ **(B)** $\frac{4\pi}{3}$

(C) 8π **(D)** $\frac{56\pi}{3}$

12 The graph of the velocity $v(t)$ of a particle moving along a line, in meters per second, is shown for $0 \le t \le 8$; it consists of line segments. What is the total distance traveled by the particle, in meters, from $t = 0$ to $t = 8$?



(A) 20 **(B)** 14

(C) 6 **(D)** 8

13 A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. At which of the following times is the speed of the particle increasing?

(A) $t = 0.5$ **(B)** $t = 1.5$

(C) $t = 2$ **(D)** $t = 2.5$

14 Water leaks into a basement. The amount of water $W(t)$, in gallons, has rate of change $W'(t) = t^2 - 5t + 4$ for $0 \leq t \leq 6$ hours, and $W(0) = 0$. At what time t is the amount of water greatest?

(A) $t = 0$

(B) $t = 1$

(C) $t = 6$

(D) $t = 4$

Unit 8 Practice – Part B: Multiple Choice

Graphing calculator allowed · 7 questions

15 What is the average value of $f(x) = e^{-x^2}$ on the interval $[0, 2]$?

(A) 0.441 **(B)** 0.882

(C) -0.491 **(D)** 0.509

16 Let $f(x) = e^x$ and $g(x) = 4 - x^2$. What is the area of the region enclosed by the graphs of f and g?

(A) 1.957 **(B)** 9.442

(C) -6.428 **(D)** 6.428

17 The region bounded by $y = \ln x$, the x-axis and $x = e$ is revolved about the x-axis. What is the volume of the solid generated?

(A) 0.718 **(B)** 2.257

(C) 3.142 **(D)** 4.513

18 A particle moves along a line with velocity $v(t) = \sin\left(t^2\right)$ for $0 \leq t \leq 2$. What is the total distance traveled by the particle?

(A) 0.804 **(B)** -0.757

(C) 0.985 **(D)** 0.895

19 A tank holds 80 liters of water at $t = 0$. Water drains from the tank at the rate $r(t) = 10e^{-0.3t}$ liters per minute. How many liters of water are in the tank at $t = 5$?

(A) 54.104 **(B)** 25.896

(C) 105.896 **(D)** 77.769

20 The base of a solid is the region bounded by $y = \ln x$, the x-axis and $x = 3$. Cross sections perpendicular to the x-axis are squares. What is the volume of the solid?

(A) 1.296 **(B)** 1.679

(C) 3.233 **(D)** 1.029

21 Let R be the region enclosed by $y = x^2$ and $y = 3 - x$. What is the volume of the solid generated when R is revolved about the x-axis?

(A) 34.373 **(B)** 107.986

(C) 63.810 **(D)** 151.029

Unit 8 Practice – Part C: Free Response

Two questions, 9 points each

22 **[9 points] (Calculator active.)** Water flows into a tank at the rate $R(t) = 12 + 8\cos\left(\frac{t}{3}\right)$ gallons per minute and drains out at the rate $D(t) = t + 3$ gallons per minute, for $0 \le t \le 10$ minutes. At time $t = 0$ the tank contains 50 gallons of water. Show the work that leads to your answers; give decimal answers to three places.

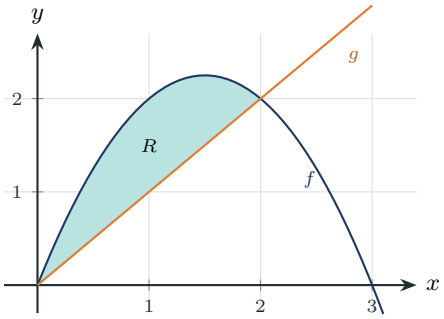
(a) Find the total number of gallons of water that flow into the tank during the 10 minutes. **[2 points]**

(b) Is the amount of water in the tank increasing or decreasing at $t = 7$? Give a reason for your answer. **[2 points]**

(c) At what time t, for $0 \le t \le 10$, is the amount of water in the tank greatest? Justify your answer. **[3 points]**

(d) Find the amount of water in the tank at the time found in part (c). **[2 points]**

23 **[9 points] (No calculator.)** Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 3x - x^2$ and $g(x) = x$, as shown in the figure.



(a) Find the area of R. **[2 points]**

(b) Region R is the base of a solid. For each x, the cross section of the solid perpendicular to the x-axis is a square. Find the volume of the solid. **[2 points]**

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is revolved about the x-axis. **[2 points]**

(d) Write an integral expression for the volume of the solid generated when R is revolved about the line $y = 4$, and find its value. **[3 points]**